

WetInk Next

Полный снимок исходного кода: P0-P31.5

P31.5: EngineRenderer разделён на оркестратор, UndoCompressionPipeline и StrokeCommitter. Единый commit для STAMP и RIBBON, упорядоченная tile-based Undo/Redo история и snapshot активного штриха сохранены.

Включено файлов: 127. Сгенерировано из рабочей копии проекта.

1. app/build.gradle.kts

```
1  plugins {
2      id("com.android.application")
3      id("org.jetbrains.kotlin.android")
4      id("org.jetbrains.kotlin.plugin.compose")
5      id("org.jetbrains.kotlin.plugin.serialization")
6  }
7
8  android {
9      namespace = "com.wetinknext"
10     compileSdk = 36
11
12     defaultConfig {
13         applicationId = "com.wetinknext"
14         minSdk = 26
15         targetSdk = 36
16         versionCode = 1
17         versionName = "0.1.0"
18         testInstrumentationRunner = "androidx.test.runner.AndroidJUnitRunner"
19     }
20
21     buildTypes {
22         release {
23             isMinifyEnabled = false
24         }
25     }
26
27     compileOptions {
28         sourceCompatibility = JavaVersion.VERSION_17
29         targetCompatibility = JavaVersion.VERSION_17
30     }
31
32     kotlinOptions {
33         jvmTarget = "17"
34     }
35
36     buildFeatures {
37         compose = true
38         buildConfig = true
39     }
40 }
41
42
43 dependencies {
44     val composeBom = platform("androidx.compose:compose-bom:2024.09.03")
45     implementation(composeBom)
46     androidTestImplementation(composeBom)
47
48     implementation("androidx.core:core-ktx:1.13.1")
49     implementation("androidx.activity:activity-compose:1.9.2")
50     implementation("androidx.compose.ui:ui")
51     implementation("androidx.compose.ui:ui-tooling-preview")
52     implementation("androidx.compose.material3:material3")
53     implementation("androidx.compose.material:material-icons-extended")
54     implementation("org.jetbrains.kotlinx:kotlinx-serialization-json:1.6.3")
55
56     testImplementation("junit:junit:4.13.2")
57
58     debugImplementation("androidx.compose.ui:ui-tooling")
59     debugImplementation("androidx.compose.ui:ui-test-manifest")
60 }
```

2. app/src/main/AndroidManifest.xml

```
1  <?xml version="1.0" encoding="utf-8"?>
2  <manifest xmlns:android="http://schemas.android.com/apk/res/android">
3      <application
4          android:allowBackup="true"
5          android:icon="@mipmap/ic_launcher"
6          android:roundIcon="@mipmap/ic_launcher_round"
7          android:label="@string/app_name"
8          android:supportRtl="true"
9          android:theme="@style/Theme.WetInkNext">
10         <activity
11             android:name=".app.MainActivity"
12             android:exported="true">
13             <intent-filter>
14                 <action android:name="android.intent.action.MAIN" />
15                 <category android:name="android.intent.category.LAUNCHER" />
16             </intent-filter>
17         </activity>
18     </application>
19 </manifest>
```

3. app/src/main/java/com/wetinknext/app/EditorScreen.kt

```
1 package com.wetinknext.app
2
3 import androidx.compose.foundation.background
4 import androidx.compose.foundation.border
5 import androidx.compose.foundation.clickable
6 import androidx.compose.foundation.layout.*
7 import androidx.compose.foundation.lazy.grid.GridCells
8 import androidx.compose.foundation.lazy.grid.LazyVerticalGrid
9 import androidx.compose.runtime.*
10 import androidx.compose.ui.Alignment
11 import androidx.compose.ui.Modifier
12 import androidx.compose.ui.graphics.Color
13 import androidx.compose.ui.graphics.toArgb
14 import androidx.compose.ui.draw.clip
15 import androidx.compose.foundation.shape.RoundedCornerShape
16 import androidx.compose.ui.unit.dp
17 import androidx.compose.ui.viewinterop.AndroidView
18 import androidx.compose.ui.platform.LocalContext
19 import com.wetinknext.engine.brush.BrushPreset
20 import com.wetinknext.engine.brush.BrushLibrary
21 import com.wetinknext.engine.core.EditorUiState
22 import com.wetinknext.engine.core.PaintSurfaceView
23 import com.wetinknext.ui.animation.AnimationTimelineToolbar
24 import com.wetinknext.ui.color.ColorPanel
25 import com.wetinknext.ui.color.GlesColorState
26 import com.wetinknext.ui.components.*
27 import com.wetinknext.ui.state.LayerState
28 import com.wetinknext.ui.theme.WetInkTheme
29 import com.wetinknext.ui.theme.AppTheme
30 import com.wetinknext.ui.theme.AppThemes
31 import com.wetinknext.ui.theme.ThemeController
32 import com.wetinknext.ui.theme.rememberThemeController
33
34 private enum class EditorPanel {
35     NONE,
36     BRUSH,
37     LAYERS,
38     COLOR,
39     SELECTION,
40     TRANSFORM,
41     ADJUSTMENTS,
42     ANIMATION,
43     SETTINGS,
44 }
45
46 @Composable
47 fun EditorScreen(
48     onBack: () -> Unit = {}
49 ) {
50     val context = LocalContext.current
51     val themeController = rememberThemeController()
52     val theme = themeController.current
53     var uiState by remember { mutableStateOf(EditorUiState.empty) }
54     var openPanel by remember { mutableStateOf(EditorPanel.NONE) }
55     var isEraser by remember { mutableStateOf(false) }
56
57     val colorState = remember { GlesColorState(context) }
58     var brushColor by remember { mutableStateOf(Color.Black) }
59     var selectedBrush by remember { mutableStateOf<BrushPreset?>(BrushLibrary.pencil6B) }
60
61     val layerState = remember { LayerState() }
62
63     val surface = remember {
64         PaintSurfaceView(context).also { view ->
65             view.onEditorStateChange = { uiState = it }
66         }
67     }
68
69     LaunchedEffect(uiState) {
70         layerState.syncFromEditor(uiState)
71     }
72
73     // Initialize state
74     LaunchedEffect(surface) {
75         surface.requestState()
76         surface.applyBrushPreset(BrushLibrary.pencil6B)
77         selectedBrush = BrushLibrary.pencil6B
78     }
79
80     LaunchedEffect(theme.id) {
81         surface.setCanvasBackdrop(theme.canvasBackdrop.toArgb(), theme.canvasGrid.toArgb())
82     }
83
84     WetInkTheme(theme = theme, fontMode = themeController.font) {
85         Box(
86             modifier = Modifier
87                 .fillMaxSize()
88                 .background(theme.appBg),
89         ) {
90             AndroidView(
```

```

91         modifier = Modifier.fillMaxSize(),
92         factory = { surface },
93     )
94
95     // Top Toolbar
96     Box(
97         modifier = Modifier
98             .align(Alignment.TopEnd)
99     ) {
100         TopToolbar(
101             theme = theme,
102             currentColor = brushColor,
103             canUndo = uiState.canUndo,
104             canRedo = uiState.canRedo,
105             isSelectionActive = openPanel == EditorPanel.SELECTION,
106             isTransformActive = openPanel == EditorPanel.TRANSFORM,
107             isAnimationActive = openPanel == EditorPanel.ANIMATION,
108             isAdjustmentsActive = openPanel == EditorPanel.ADJUSTMENTS,
109             onUndoClick = { surface.undo() },
110             onRedoClick = { surface.redo() },
111             onSelectionClick = {
112                 openPanel = if (openPanel == EditorPanel.SELECTION) EditorPanel.NONE else EditorPanel.SELECTION
113             },
114             onTransformClick = {
115                 openPanel = if (openPanel == EditorPanel.TRANSFORM) EditorPanel.NONE else EditorPanel.TRANSFORM
116             },
117             onAnimationClick = {
118                 openPanel = if (openPanel == EditorPanel.ANIMATION) EditorPanel.NONE else EditorPanel.ANIMATION
119             },
120             onAdjustmentsClick = {
121                 openPanel = if (openPanel == EditorPanel.ADJUSTMENTS) EditorPanel.NONE else EditorPanel.ADJUSTMENTS
122             },
123             onSettingsClick = {
124                 openPanel = if (openPanel == EditorPanel.SETTINGS) EditorPanel.NONE else EditorPanel.SETTINGS
125             },
126             onLayersClick = {
127                 openPanel = if (openPanel == EditorPanel.LAYERS) EditorPanel.NONE else EditorPanel.LAYERS
128             },
129             onColorClick = {
130                 openPanel = if (openPanel == EditorPanel.COLOR) EditorPanel.NONE else EditorPanel.COLOR
131             },
132         )
133     }
134
135     // Left controls keep the same compact stack as the reference editor.
136     Column(
137         modifier = Modifier
138             .align(Alignment.TopStart)
139             .padding(top = 100.dp),
140         horizontalAlignment = Alignment.Start,
141     ) {
142         SideToolbar(
143             theme = theme,
144             brushSize = uiState.brushSizePx,
145             brushOpacity = uiState.brushOpacity,
146             isEraser = isEraser,
147             onBrushSizeChange = { surface.setBrushSize(it) },
148             onBrushOpacityChange = { surface.setBrushOpacity(it) },
149             onBrushClick = {
150                 openPanel = if (openPanel == EditorPanel.BRUSH) EditorPanel.NONE else EditorPanel.BRUSH
151             },
152             onEraserClick = { isEraser = it }
153         )
154         LeftCanvasActions(
155             theme = theme,
156             onClearLayerClick = { surface.clearActiveLayer() },
157             onMirrorClick = { surface.toggleCanvasMirror() },
158         )
159     }
160
161     // Panel Host
162     EditorPanelHost(
163         panel = openPanel,
164         theme = theme,
165         surface = surface,
166         layerState = layerState,
167         colorState = colorState,
168         brushColor = brushColor,
169         selectedBrush = selectedBrush,
170         themeController = themeController,
171         onBrushSelected = { preset ->
172             selectedBrush = preset
173             surface.applyBrushPreset(preset)
174         },
175         onColorChange = {
176             brushColor = it
177             surface.setBrushColor(it)
178         },
179         onDismiss = { openPanel = EditorPanel.NONE }
180     )
181 }
182 }

```

```

183 }
184
185 @Composable
186 private fun EditorPanelHost(
187     panel: EditorPanel,
188     theme: com.wetinknext.ui.theme.AppTheme,
189     surface: PaintSurfaceView?,
190     layerState: LayerState,
191     colorState: GlesColorState,
192     brushColor: Color,
193     selectedBrush: BrushPreset?,
194     themeController: ThemeController,
195     onBrushSelected: (BrushPreset) -> Unit,
196     onColorChange: (Color) -> Unit,
197     onDismiss: () -> Unit
198 ) {
199     Box(modifier = Modifier.fillMaxSize()) {
200         when (panel) {
201             EditorPanel.NONE -> Unit
202
203             EditorPanel.SETTINGS -> {
204                 Box(modifier = Modifier.align(Alignment.TopEnd).padding(top = 70.dp, end = 16.dp)) {
205                     ThemeSettingsPanel(
206                         theme = theme,
207                         themeController = themeController,
208                         onDismiss = onDismiss,
209                     )
210                 }
211             }
212
213             EditorPanel.BRUSH -> {
214                 Box(modifier = Modifier.align(Alignment.CenterStart).padding(start = 72.dp)) {
215                     BrushPanel(
216                         currentBrush = selectedBrush ?: BrushLibrary.pencil6B,
217                         onBrushSelect = onBrushSelected,
218                         onBrushStudioOpen = { /* TODO */ },
219                         theme = theme,
220                         onDismiss = onDismiss,
221                         onDuplicate = { /* TODO */ },
222                         onDelete = { /* TODO */ }
223                     )
224                 }
225             }
226
227             EditorPanel.LAYERS -> {
228                 Box(modifier = Modifier.align(Alignment.TopEnd).padding(top = 80.dp, end = 16.dp)) {
229                     LayersPanel(
230                         state = layerState,
231                         theme = theme,
232                         onDismiss = onDismiss,
233                         onAddLayer = { surface?.addLayer() },
234                         onSelectLayer = { surface?.setActiveLayer(it) },
235                         onVisibleChange = { id, visible -> surface?.setLayerVisible(id, visible) },
236                         onOpacityChange = { id, opacity -> surface?.setLayerOpacity(id, opacity) },
237                         onRemoveLayer = { id -> surface?.removeLayer(id) }
238                     )
239                 }
240             }
241
242             EditorPanel.COLOR -> {
243                 Box(modifier = Modifier.align(Alignment.TopEnd).padding(top = 80.dp, end = 16.dp)) {
244                     ColorPanel(
245                         state = colorState,
246                         color = brushColor,
247                         theme = theme,
248                         onColorChange = onColorChange,
249                         onAddFromPhoto = { /* TODO */ },
250                         onDismiss = onDismiss
251                     )
252                 }
253             }
254
255             EditorPanel.SELECTION -> {
256                 Box(modifier = Modifier.align(Alignment.BottomCenter).padding(bottom = 32.dp)) {
257                     SelectionToolbar(
258                         theme = theme,
259                         currentShape = SelectionShapeUi.FREEHAND,
260                         onShapeChange = { /* TODO */ },
261                         onDone = onDismiss
262                     )
263                 }
264             }
265
266             EditorPanel.TRANSFORM -> {
267                 Box(modifier = Modifier.align(Alignment.BottomCenter).padding(bottom = 32.dp)) {
268                     TransformMenuView(
269                         theme = theme,
270                         currentMode = TransformModeUi.UNIFORM,
271                         actions = object : TransformActions {
272                             override fun reset() { /* TODO */ }
273                             override fun cancel() { onDismiss() }
274                             override fun apply() { onDismiss() }

```

```

275         override fun flipHorizontal() { /* TODO */ }
276         override fun flipVertical() { /* TODO */ }
277         override fun setMode(mode: TransformModeUi) { /* TODO */ }
278     }
279 )
280 }
281 }
282
283 EditorPanel.ADJUSTMENTS -> {
284     Box(modifier = Modifier.align(Alignment.TopEnd).padding(top = 80.dp, end = 16.dp)) {
285         AdjustmentsPanel(
286             theme = theme,
287             onClose = onDismiss,
288             onAction = { /* TODO */ }
289         )
290     }
291 }
292
293 EditorPanel.ANIMATION -> {
294     Box(modifier = Modifier.align(Alignment.BottomCenter)) {
295         AnimationTimelineToolbar(
296             theme = theme,
297             document = com.wetinknext.domain.animation.AnimationDocument(enabled = true),
298             currentFrameId = 0L,
299             onFrameSelect = {},
300             onFrameMove = { _, _ -> },
301             onAddFrame = {},
302             onDuplicateFrame = {},
303             onDeleteFrame = {},
304             onHoldChange = { _, _ -> },
305             onRoleToggle = { _, _ -> },
306             onAssembleLayers = {},
307             onLayersClick = {},
308             onClose = onDismiss
309         )
310     }
311 }
312 }
313 }
314 }
315
316 @Composable
317 private fun ThemeSettingsPanel(
318     theme: AppTheme,
319     themeController: ThemeController,
320     onDismiss: () -> Unit,
321 ) {
322     var preferencesOpen by remember { mutableStateOf(false) }
323     Column(
324         modifier = Modifier
325             .width(320.dp)
326             .clip(RoundedCornerShape(22.dp))
327             .background(theme.panelBg)
328             .padding(16.dp),
329         verticalArrangement = Arrangement.spacedBy(12.dp),
330     ) {
331         Row(verticalAlignment = Alignment.CenterVertically) {
332             androidx.compose.material3.Text(
333                 text = if (preferencesOpen) "Предпочтения" else "Меню",
334                 color = theme.textPrimary,
335                 style = androidx.compose.material3.MaterialTheme.typography.titleLarge,
336                 modifier = Modifier.weight(1f),
337             )
338             androidx.compose.material3.Text(
339                 text = "x",
340                 color = theme.textSecondary,
341                 style = androidx.compose.material3.MaterialTheme.typography.headlineSmall,
342                 modifier = Modifier.size(28.dp).clickable(onClick = onDismiss),
343             )
344         }
345
346         Row(
347             modifier = Modifier.fillMaxWidth().clip(RoundedCornerShape(10.dp))
348                 .background(theme.panelBgVariant.copy(alpha = .5f)).padding(3.dp),
349             horizontalArrangement = Arrangement.spacedBy(2.dp),
350         ) {
351             listOf("Меню", "Холст", "Экспорт", "Сведения").forEachIndexed { index, label ->
352                 Box(
353                     modifier = Modifier
354                         .weight(1f).clip(RoundedCornerShape(8.dp))
355                         .background(if (index == 0) theme.panelBgVariant else Color.Transparent)
356                         .padding(vertical = 8.dp),
357                     contentAlignment = Alignment.Center,
358                 ) {
359                     androidx.compose.material3.Text(label, color = if (index == 0) theme.textPrimary else theme.textSecondary,
360                     style = androidx.compose.material3.MaterialTheme.typography.labelSmall) }
361             }
362         }
363
364         if (!preferencesOpen) {
365             androidx.compose.material3.Text(
366                 text = "Предпочтения",
367                 color = theme.textPrimary,

```

```

366         modifier = Modifier.fillMaxWidth().clickable { preferencesOpen = true }.padding(vertical = 14.dp, horizontal
= 4.dp),
367     )
368     } else {
369         androidx.compose.material3.Text("Тема", color = theme.textSecondary, style = androidx.compose.material3.
MaterialTheme.typography.labelMedium)
370         val choices = AppThemes.all + themeController.customTheme.copy(id = "custom", displayName = "Своя")
371         choices.chunked(5).forEach { row ->
372             Row(modifier = Modifier.fillMaxWidth(), horizontalArrangement = Arrangement.SpaceBetween) {
373                 row.forEach { candidate ->
374                     val selected = candidate.id == themeController.current.id
375                     Box(
376                         modifier = Modifier.size(48.dp).clip(androidx.compose.foundation.shape.CircleShape)
377                         .background(if (selected) Color.White else Color.Transparent)
378                         .clickable { if (candidate.id == "custom") themeController.selectCustom() else
themeController.select(candidate) }
379                         .padding(2.dp),
380                         contentAlignment = Alignment.Center,
381                     ) {
382                         Box(
383                             modifier = Modifier.fillMaxSize().clip(androidx.compose.foundation.shape.CircleShape)
384                             .background(if (candidate.id == "custom") theme.panelBgVariant else candidate.appBg),
385                             contentAlignment = Alignment.Center,
386                         ) {
387                             if (candidate.id == "custom") androidx.compose.material3.Text("Своя", color = Color.White,
style = androidx.compose.material3.MaterialTheme.typography.labelSmall)
388                             else Box(Modifier.size(24.dp).clip(androidx.compose.foundation.shape.CircleShape).background(
candidate.accent))
389                         }
390                     }
391                 }
392                 repeat(5 - row.size) { Spacer(Modifier.size(48.dp)) }
393             }
394         }
395         androidx.compose.material3.Text(themeController.current.displayName, color = theme.textPrimary)
396     }
397 }
398 }
399
400 private fun PaintSurfaceView.applyBrushPreset(
401     preset: BrushPreset
402 ) {
403     val settings = preset.settings
404     applyBrush(settings)
405
406     val grainPath = settings.grainAssetPath
407         ?.trim()
408         ?.takeIf { it.isNotEmpty() }
409
410     if (grainPath == null) {
411         clearGrainTexture()
412     } else {
413         loadGrainTexture(
414             path = grainPath,
415             scale = settings.grainScale,
416             canvasLocked = settings.grainCanvasLocked,
417             depth = settings.textureDepth,
418             contrast = settings.textureContrast,
419         )
420     }
421 }
422
423 val shapePath = settings.shapeAssetPath?.trim()?.takeIf { it.isNotEmpty() }
424 if (shapePath == null) {
425     clearShapeTexture()
426 } else {
427     loadShapeTexture(path = shapePath, rgbToAlpha = settings.rgbToAlpha)
428 }
429 }

```


4. app/src/main/java/com/wetinknext/app/MainActivity.kt

```
1 package com.wetinknext.app
2
3 import android.os.Bundle
4 import androidx.activity.ComponentActivity
5 import androidx.activity.compose.setContent
6
7 class MainActivity : ComponentActivity() {
8     override fun onCreate(savedInstanceState: Bundle?) {
9         super.onCreate(savedInstanceState)
10        setContent { EditorScreen() }
11    }
12 }
```

5. app/src/main/java/com/wetinknext/domain/animation/AnimationModels.kt

```
1 package com.wetinknext.domain.animation
2
3 import kotlinx.serialization.Serializable
4
5 @Serializable
6 enum class AnimationPlaybackMode {
7     LOOP,
8     PING_PONG,
9     ONE_SHOT,
10 }
11
12 @Serializable
13 enum class AnimationFrameRole {
14     NORMAL,
15     BACKGROUND,
16     FOREGROUND,
17 }
18
19 @Serializable
20 data class OnionSkinSettings(
21     val enabled: Boolean = false, // Disabled by default until renderer supports it
22     val previousFrames: Int = 1,
23     val nextFrames: Int = 1,
24     val opacity: Float = 0.35f,
25     val previousColorArgb: Int = 0xFFFFF4FA3.toInt(),
26     val nextColorArgb: Int = 0xFF4A8DFF.toInt(),
27     val blendPrimaryFrame: Boolean = false,
28 )
29
30 @Serializable
31 data class AnimationFrame(
32     val id: Long,
33     val layerIds: List<Long>,
34     val holdFrames: Int = 1,
35     val role: AnimationFrameRole = AnimationFrameRole.NORMAL,
36 )
37
38 @Serializable
39 data class AnimationDocument(
40     val enabled: Boolean = false,
41     val framesPerSecond: Int = 12,
42     val playbackMode: AnimationPlaybackMode = AnimationPlaybackMode.LOOP,
43     val onionSkin: OnionSkinSettings = OnionSkinSettings(),
44     val frames: List<AnimationFrame> = emptyList(),
45 )
```

6. app/src/main/java/com/wetinknext/domain/animation/AnimationRenderPlan.kt

```
1 package com.wetinknext.domain.animation
2
3 /**
4  * Defines which frames and roles should be rendered in the current view.
5  */
6 data class AnimationRenderPlan(
7     val backgroundFrame: AnimationFrame?,
8     val currentFrame: AnimationFrame?,
9     val foregroundFrame: AnimationFrame?,
10    val previousOnionFrames: List<AnimationFrame>,
11    val nextOnionFrames: List<AnimationFrame>,
12 )
13
14 /**
15  * Logic to decide which layers are visible based on the animation document.
16  * This is a pure function for easy testing.
17  */
18 fun buildAnimationRenderPlan(
19     document: AnimationDocument,
20     currentFrameId: Long,
21 ): AnimationRenderPlan {
22     if (!document.enabled) {
23         return AnimationRenderPlan(null, null, null, emptyList(), emptyList())
24     }
25
26     val frames = document.frames
27     val currentIndex = frames.indexOfFirst { it.id == currentFrameId }
28
29     val bgFrame = frames.find { it.role == AnimationFrameRole.BACKGROUND }
30     val fgFrame = frames.find { it.role == AnimationFrameRole.FOREGROUND }
31     val current = if (currentIndex >= 0) frames[currentIndex] else null
32
33     val prevOnions = mutableListOf<AnimationFrame>()
34     val nextOnions = mutableListOf<AnimationFrame>()
35
36     if (document.onionSkin.enabled && currentIndex >= 0) {
37         // Previous frames
38         for (i in 1..document.onionSkin.previousFrames) {
39             val idx = currentIndex - i
40             if (idx >= 0) {
41                 val f = frames[idx]
42                 if (f.role == AnimationFrameRole.NORMAL) prevOnions.add(f)
43             }
44         }
45         // Next frames
46         for (i in 1..document.onionSkin.nextFrames) {
47             val idx = currentIndex + i
48             if (idx < frames.size) {
49                 val f = frames[idx]
50                 if (f.role == AnimationFrameRole.NORMAL) nextOnions.add(f)
51             }
52         }
53     }
54
55     return AnimationRenderPlan(
56         backgroundFrame = bgFrame,
57         currentFrame = current,
58         foregroundFrame = fgFrame,
59         previousOnionFrames = prevOnions,
60         nextOnionFrames = nextOnions
61     )
62 }
```

7. app/src/main/java/com/wetinknext/domain/animation/AnimationTimeline.kt

```
1 package com.wetinknext.domain.animation
2
3 /**
4  * Creates a default set of animation frames from a list of layer IDs.
5  * Each layer becomes its own frame. System "Background" (usually ID 1) should be filtered out by the caller.
6  */
7 fun createFramesFromLayers(layerIds: List<Long>): List<AnimationFrame> {
8     return layerIds.map { layerId ->
9         AnimationFrame(
10             id = layerId, // Use layerId as initial frame ID for simplicity in 1:1 mapping
11             layerIds = listOf(layerId)
12         )
13     }
14 }
15
16 /**
17  * Ensures the animation document is valid given the currently existing layers.
18  * Removes dead layer references, empty frames, and enforces range limits.
19  */
20 fun normalizedAnimationDocument(
21     document: AnimationDocument,
22     existingLayerIds: Set<Long>,
23 ): AnimationDocument {
24     return normalizedAnimationDocument(
25         document = document,
26         layersInDocumentOrder = existingLayerIds.map { AnimationLayerInfo(id = it) },
27     )
28 }
29
30 /** Minimal layer data needed by the timeline. Kept independent from OpenGL resources. */
31 data class AnimationLayerInfo(
32     val id: Long,
33     val isCanvasBackgroundLayer: Boolean = false,
34     val isClipping: Boolean = false,
35 )
36
37 /**
38  * Normalizes frames against the real document order. Canvas background never becomes a frame.
39  * Layer IDs inside every frame are also kept in document order, which is required for clipping.
40  */
41 fun normalizedAnimationDocument(
42     document: AnimationDocument,
43     layersInDocumentOrder: List<AnimationLayerInfo>,
44 ): AnimationDocument {
45     val order = layersInDocumentOrder.mapIndexed { index, layer -> layer.id to index }.toMap()
46     val animatableIds = layersInDocumentOrder
47         .filterNot { it.isCanvasBackgroundLayer }
48         .map { it.id }
49     val usedLayerIds = mutableSetOf<Long>()
50
51     val normalizedFrames = document.frames.mapNotNull { frame ->
52         val validLayerIds = frame.layerIds
53             .filter { id -> id in order && id !in usedLayerIds }
54             .sortedBy { order.getOrDefault(it, 0) }
55
56         if (validLayerIds.isEmpty()) {
57             null
58         } else {
59             usedLayerIds.addAll(validLayerIds)
60             frame.copy(
61                 layerIds = validLayerIds,
62                 holdFrames = frame.holdFrames.coerceIn(1, 120)
63             )
64         }
65     }
66     // A layer can belong to only one frame. Any newly created document layer gets its own frame.
67     val missingLayerIds = animatableIds.filter { it !in usedLayerIds }
68     val newFrames = missingLayerIds.map { layerId ->
69         AnimationFrame(id = layerId, layerIds = listOf(layerId))
70     }
71     return document.copy(
72         framesPerSecond = document.framesPerSecond.coerceIn(1, 60),
73         frames = normalizedFrames + newFrames
74     )
75 }
76
77 /**
78  * Combines multiple layers into a single animation frame.
79  */
80 fun groupLayersIntoFrame(
81     document: AnimationDocument,
82     layerIdsToGroup: List<Long>,
83     newFrameId: Long = System.nanoTime()
84 ): AnimationDocument {
85     val orderedIds = layerIdsToGroup.distinct()
86     val idsSet = orderedIds.toSet()
87     if (idsSet.isEmpty()) return document
88
89     val insertionIndex = document.frames.indexOfFirst { frame -> frame.layerIds.any { it in idsSet } }
90     val cleanedFrames = document.frames.map { frame ->
```

```

91     frame.copy(layerIds = frame.layerIds.filter { it !in idsSet })
92   }.filter { it.layerIds.isNotEmpty() }.toMutableList()
93   val newFrame = AnimationFrame(id = newFrameId, layerIds = orderedIds)
94   val targetIndex = if (insertionIndex < 0) cleanedFrames.size else insertionIndex.coerceAtMost(cleanedFrames.size)
95   cleanedFrames.add(targetIndex, newFrame)
96   return document.copy(frames = cleanedFrames)
97 }
98
99 sealed interface AnimationFrameGroupingValidation {
100   data class Valid(val layerIdsInDocumentOrder: List<Long>) : AnimationFrameGroupingValidation
101   data object EmptySelection : AnimationFrameGroupingValidation
102   data class MissingClippingBase(val clippingLayerId: Long) : AnimationFrameGroupingValidation
103   data class MissingClippingLayer(val baseLayerId: Long, val clippingLayerId: Long) : AnimationFrameGroupingValidation
104 }
105
106 /**
107  * A clipping layer and its base must live in one animation frame: otherwise a frame would render
108  * with a different compositing result than the document. The selection is returned in document order.
109  */
110 fun validateAnimationFrameGrouping(
111   selectedLayerIds: Set<Long>,
112   layersInDocumentOrder: List<AnimationLayerInfo>,
113 ): AnimationFrameGroupingValidation {
114   val layers = layersInDocumentOrder.filterNot { it.isCanvasBackgroundLayer }
115   val selected = selectedLayerIds.intersect(layers.map { it.id }.toSet())
116   if (selected.isEmpty()) return AnimationFrameGroupingValidation.EmptySelection
117
118   val baseByClippingId = mutableMapOf<Long, Long>()
119   layers.forEachIndexed { index, layer ->
120     if (layer.isClipping) {
121       val base = layers.subList(0, index).lastOrNull { !it.isClipping }
122       ?: return AnimationFrameGroupingValidation.MissingClippingBase(layer.id)
123       baseByClippingId[layer.id] = base.id
124       if (layer.id in selected && base.id !in selected) {
125         return AnimationFrameGroupingValidation.MissingClippingLayer(layer.id, base.id)
126       }
127     }
128   }
129   baseByClippingId.forEach { (clippingId, baseId) ->
130     if (baseId in selected && clippingId !in selected) {
131       return AnimationFrameGroupingValidation.MissingClippingLayer(baseId, clippingId)
132     }
133   }
134   return AnimationFrameGroupingValidation.Valid(layers.filter { it.id in selected }.map { it.id })
135 }
136
137 /**
138  * Expands frames into a flat list of frame IDs to be played back, accounting for holdFrames.
139  */
140 fun expandedPlaybackFrameIds(
141   frames: List<AnimationFrame>,
142 ): List<Long> {
143   return frames.flatMap { frame ->
144     List(frame.holdFrames) { frame.id }
145   }
146 }
147
148 data class PlaybackStep(
149   val nextIndex: Int,
150   val nextDirection: Int, // 1 for forward, -1 for backward
151   val shouldStop: Boolean = false
152 )
153
154 /**
155  * Calculates the next index in the playback sequence.
156  */
157 fun nextPlaybackIndex(
158   currentIndex: Int,
159   direction: Int,
160   frameCount: Int,
161   mode: AnimationPlaybackMode,
162 ): PlaybackStep {
163   if (frameCount <= 0) return PlaybackStep(0, 1, true)
164   if (frameCount == 1) return PlaybackStep(0, 1, mode == AnimationPlaybackMode.ONE_SHOT)
165
166   var nextDir = direction
167   var nextIdx = currentIndex + direction
168   var stop = false
169
170   when (mode) {
171     AnimationPlaybackMode.LOOP -> {
172       if (nextIdx >= frameCount) nextIdx = 0
173       if (nextIdx < 0) nextIdx = frameCount - 1
174     }
175     AnimationPlaybackMode.PING_PONG -> {
176       if (nextIdx >= frameCount) {
177         nextIdx = (frameCount - 2).coerceAtLeast(0)
178         nextDir = -1
179       } else if (nextIdx < 0) {
180         nextIdx = (1).coerceAtMost(frameCount - 1)
181         nextDir = 1
182       }
183     }
184   }

```

```
183     }
184     AnimationPlaybackMode.ONE_SHOT -> {
185         if (nextIdx >= frameCount) {
186             nextIdx = frameCount - 1
187             stop = true
188         }
189     }
190 }
191
192 return PlaybackStep(nextIdx, nextDir, stop)
193 }
```

8. app/src/main/java/com/wetinknext/engine/brush/BlendController.kt

```
1 package com.wetinknext.engine.brush
2
3 import android.opengl.GLES30
4
5 class BlendController {
6
7     fun begin(policy: BlendPolicy) {
8         GLES30.glEnable(GLES30.GL_BLEND)
9         // NON_BUILDUP needs a dedicated accumulation mask. Until that pass is
10        // introduced, both policies use the correct premultiplied source-over.
11        GLES30.glBlendEquation(GLES30.GL_FUNC_ADD)
12        GLES30.glBlendFunc(
13            GLES30.GL_ONE,
14            GLES30.GL_ONE_MINUS_SRC_ALPHA,
15        )
16    }
17
18    fun end() {
19        GLES30.glBlendEquation(GLES30.GL_FUNC_ADD)
20        GLES30.glBlendFunc(
21            GLES30.GL_ONE,
22            GLES30.GL_ONE_MINUS_SRC_ALPHA,
23        )
24        GLES30.glDisable(GLES30.GL_BLEND)
25    }
26 }
```

9. app/src/main/java/com/wetinknext/engine/brush/BrushDynamics.kt

```
1 package com.wetinknext.engine.brush
2
3 import kotlin.math.pow
4 import kotlin.math.sqrt
5
6 object BrushDynamics {
7
8     fun tiltMagnitude(
9         tiltX: Float,
10        tiltY: Float,
11    ): Float {
12        return sqrt(
13            tiltX * tiltX +
14            tiltY * tiltY,
15        ).coerceIn(0f, 1f)
16    }
17
18    /**
19     * Resolves the final radius and opacity for a single dab based on input.
20     */
21    fun resolve(
22        settings: BrushSettings,
23        pressure: Float,
24        tiltX: Float,
25        tiltY: Float,
26        out: ResolvedDab,
27    ) {
28        val p = pressure
29            .coerceIn(0f, 1f)
30            .let { it.toDouble().pow(settings.pressureGamma.toDouble()).toFloat() }
31
32        val tilt = tiltMagnitude(tiltX, tiltY)
33
34        val pressureSize = if (settings.pressureToSize) {
35            settings.minSizeRatio + (1f - settings.minSizeRatio) * p
36        } else {
37            1f
38        }
39        val tiltSize = 1f + settings.tiltToSize * tilt
40        out.radius = (
41            settings.baseRadiusPx * pressureSize * tiltSize
42        ).coerceAtLeast(0.25f)
43
44        val pressureOpacity = if (settings.pressureToOpacity) p else 1f
45
46        // Tilt must not make a dab more opaque.
47        val tiltOpacity = 1f - (0.45f * settings.tiltToOpacity * tilt)
48        out.coverage = (pressureOpacity * tiltOpacity).coerceIn(0f, 1f)
49
50        // Local dab alpha only. Global brush opacity is applied during stroke blit.
51        out.opacity = out.coverage
52    }
53 }
54
55 data class ResolvedDab(
56     var radius: Float = 0f,
57     var opacity: Float = 1f,
58     var coverage: Float = 1f,
59 )
```


10. app/src/main/java/com/wetinknext/engine/brush/BrushLibrary.kt

```
1 package com.wetinknext.engine.brush
2
3 import androidx.compose.material.icons.Icons
4 import androidx.compose.material.icons.filled.Brush
5 import androidx.compose.material.icons.filled.Create
6 import androidx.compose.material.icons.filled.Star
7 import androidx.compose.ui.graphics.vector.ImageVector
8
9 data class BrushUiCategory(
10     val name: String,
11     val icon: ImageVector,
12     val brushes: List<BrushPreset>,
13 )
14
15 object BrushLibrary {
16     val pencil6B = BrushPreset(
17         id = "pencil_6b",
18         settings = BrushSettings(
19             name = "Pencil 6B",
20             category = "Pencil",
21             renderMode = BrushRenderMode.STAMP,
22             baseRadiusPx = 8f,
23             opacity = 0.82f,
24             flow = 0.65f,
25             spacing = 0.08f,
26             spacingUsesDiameter = true,
27             hardness = 0.72f,
28             smoothing = 0.08f,
29             streamline = 0.04f,
30             pressureToSize = true,
31             pressureToOpacity = true,
32             pressureGamma = 1.25f,
33             minSizeRatio = 0.30f,
34             tiltToSize = 0.55f,
35             grainAssetPath = "asset:brush/pencil_6b_grain.png",
36             grainCanvasLocked = true,
37             grainScale = 5.5f,
38             textureDepth = 0.65f,
39             textureContrast = 1.35f,
40             blendPolicy = BlendPolicy.NORMAL_BUILDUP,
41         ),
42     )
43
44     val gPen = BrushPreset(
45         id = "g_pen",
46         settings = BrushSettings(
47             name = "G-Pen",
48             category = "Ink",
49             renderMode = BrushRenderMode.RIBBON,
50             baseRadiusPx = 4.5f,
51             opacity = 1f,
52             flow = 1f,
53             smoothing = 0.35f,
54             streamline = 0.22f,
55             pressureToSize = true,
56             pressureToOpacity = false,
57             pressureGamma = 1.7f,
58             minSizeRatio = 0.06f,
59             blendPolicy = BlendPolicy.NON_BUILDUP,
60         )
61     )
62
63     val allCategories = listOf(
64         BrushUiCategory(
65             name = "Pencil",
66             icon = Icons.Default.Create,
67             brushes = listOf(pencil6B)
68         ),
69         BrushUiCategory(
70             name = "Ink",
71             icon = Icons.Default.Brush,
72             brushes = listOf(gPen)
73         ),
74         BrushUiCategory(
75             name = "Favorites",
76             icon = Icons.Default.Star,
77             brushes = emptyList()
78         )
79     )
80 }
```

11. app/src/main/java/com/wetinknext/engine/brush/BrushPreset.kt

```
1 package com.wetinknext.engine.brush
2
3 import kotlinx.serialization.Serializable
4
5 @Serializable
6 data class BrushPreset(
7     val id: String,
8     val settings: BrushSettings,
9 )
```

12. app/src/main/java/com/wetinknext/engine/brush/BrushRepository.kt

```
1 package com.wetinknext.engine.brush
2
3 import kotlinx.serialization.encodeToString
4 import kotlinx.serialization.json.Json
5
6 class BrushRepository(
7     private val json: Json = defaultJson,
8 ) {
9     fun encode(preset: BrushPreset): String =
10         json.encodeToString(preset)
11
12     fun decode(value: String): BrushPreset =
13         json.decodeFromString<BrushPreset>(value).let { preset ->
14             preset.copy(settings = preset.settings.resolved())
15         }
16
17     companion object {
18         val defaultJson = Json {
19             encodeDefaults = true
20             ignoreUnknownKeys = true
21             explicitNulls = false
22         }
23     }
24 }
```

13. app/src/main/java/com/wetinknext/engine/brush/BrushSettings.kt

```
1 package com.wetinknext.engine.brush
2
3 import kotlinx.serialization.Serializable
4
5 @Serializable
6 enum class BrushRenderMode { STAMP, RIBBON, WET }
7
8 @Serializable
9 enum class BlendPolicy { NORMAL_BUILDUP, NON_BUILDUP }
10
11 @Serializable
12 enum class RibbonCap { ROUND, BUTT }
13
14 @Serializable
15 enum class RibbonJoin { MITER, ROUND, BEVEL }
16
17 @Serializable
18 data class RibbonSettings(
19     val minWidthRatio: Float = 0.06f,
20     val taperStartPx: Float = 8f,
21     val taperEndPx: Float = 14f,
22     val miterLimit: Float = 3f,
23     val aaWidthPx: Float = 1f,
24     val cap: RibbonCap = RibbonCap.ROUND,
25     val join: RibbonJoin = RibbonJoin.ROUND,
26     val minPointDistancePx: Float = 0.5f,
27 )
28
29 @Serializable
30 data class WetSettings(val wetness: Float = 0f, val spread: Float = 0f, val bleed: Float = 0f)
31
32 @Serializable
33 data class BrushSettings(
34     val name: String = "Debug Stamp",
35     val category: String = "Debug",
36     val renderMode: BrushRenderMode = BrushRenderMode.STAMP,
37     /**
38      * Internal brush radius in document pixels.
39      * UI displays diameter = baseRadiusPx * 2.
40      */
41     val baseRadiusPx: Float = 14f,
42     val opacity: Float = 1f,
43     val flow: Float = 1f,
44     val spacing: Float = 0.16f,
45     val spacingUsesDiameter: Boolean = true,
46     val hardness: Float = 1f,
47     val colorArgb: Long = 0xFF000000L,
48     val smoothing: Float = 0f,
49     val streamline: Float = 0f,
50     val antiAliasLevel: Int = 2,
51     val useTempStrokeBuffer: Boolean = false,
52     val pressureToSize: Boolean = true,
53     val pressureToOpacity: Boolean = true,
54     val pressureGamma: Float = 1f,
55     val minSizeRatio: Float = 0.05f,
56     val velocityToSize: Float = 0f,
57     val velocityToOpacity: Float = 0f,
58     val tiltToSize: Float = 0f,
59     val tiltToOpacity: Float = 0f,
60     val tiltToRotation: Float = 0f,
61     val rotationJitter: Float = 0f,
62     val sizeJitter: Float = 0f,
63     val opacityJitter: Float = 0f,
64     val scatter: Float = 0f,
65     val shapeAssetPath: String? = null,
66     val grainAssetPath: String? = null,
67     val grainScale: Float = 1f,
68     val grainCanvasLocked: Boolean = true,
69     val rgbToAlpha: Boolean = false,
70     val textureContrast: Float = 1f,
71     val textureDepth: Float = 1f,
72     val pixelPen: Boolean = false,
73     val squareStroke: Boolean = false,
74     val noAntialias: Boolean = false,
75     val blendPolicy: BlendPolicy = BlendPolicy.NORMAL_BUILDUP,
76     val ribbon: RibbonSettings = RibbonSettings(),
77     val wet: WetSettings = WetSettings(),
78 ) {
79     /** Returns a settings copy safe for every engine path. */
80     fun resolved(): BrushSettings {
81         val safeRadius = baseRadiusPx.coerceIn(0.25f, 4096f)
82         val safeOpacity = opacity.coerceIn(0f, 1f)
83         val safeFlow = flow.coerceIn(0f, 1f)
84         val safeSpacing = spacing.coerceIn(0.001f, 4f)
85         val safeGamma = pressureGamma.coerceIn(0.05f, 8f)
86         val safeMinSize = minSizeRatio.coerceIn(0.01f, 1f)
87
88         val safeVelocityToSize = velocityToSize.coerceIn(0f, 1f)
89         val safeVelocityToOpacity = velocityToOpacity.coerceIn(0f, 1f)
90         val safeTiltToSize = tiltToSize.coerceIn(0f, 1f)
```

```

91     val safeTiltToOpacity = tiltToOpacity.coerceIn(0f, 1f)
92     val safeTiltToRotation = tiltToRotation.coerceIn(0f, 1f)
93     val safeSizeJitter = sizeJitter.coerceIn(0f, 1f)
94     val safeOpacityJitter = opacityJitter.coerceIn(0f, 1f)
95     val safeRotationJitter = rotationJitter.coerceIn(0f, 1f)
96     val safeScatter = scatter.coerceAtLeast(0f)
97
98     val safeGrainScale = grainScale.coerceIn(0.0001f, 256f)
99     val safeTextureDepth = textureDepth.coerceIn(0f, 1f)
100    val safeTextureContrast = textureContrast.coerceIn(0f, 4f)
101    val safeAa = antiAliasLevel.coerceIn(0, 3)
102
103    val safeRibbon = ribbon.copy(
104        minWidthRatio = ribbon.minWidthRatio.coerceIn(0.01f, 1f),
105        taperStartPx = ribbon.taperStartPx.coerceAtLeast(0f),
106        taperEndPx = ribbon.taperEndPx.coerceAtLeast(0f),
107        miterLimit = ribbon.miterLimit.coerceIn(1f, 32f),
108        aaWidthPx = ribbon.aaWidthPx.coerceIn(0f, 16f),
109        minPointDistancePx = ribbon.minPointDistancePx.coerceIn(0.05f, 32f),
110    )
111    val safeWet = wet.copy(
112        wetness = wet.wetness.coerceIn(0f, 1f),
113        spread = wet.spread.coerceIn(0f, 1f),
114        bleed = wet.bleed.coerceIn(0f, 1f),
115    )
116
117    return copy(
118        baseRadiusPx = safeRadius,
119        opacity = safeOpacity,
120        flow = safeFlow,
121        spacing = safeSpacing,
122        pressureGamma = safeGamma,
123        minSizeRatio = safeMinSize,
124        velocityToSize = safeVelocityToSize,
125        velocityToOpacity = safeVelocityToOpacity,
126        tiltToSize = safeTiltToSize,
127        tiltToOpacity = safeTiltToOpacity,
128        tiltToRotation = safeTiltToRotation,
129        sizeJitter = safeSizeJitter,
130        opacityJitter = safeOpacityJitter,
131        rotationJitter = safeRotationJitter,
132        scatter = safeScatter,
133        grainScale = safeGrainScale,
134        textureDepth = safeTextureDepth,
135        textureContrast = safeTextureContrast,
136        antiAliasLevel = safeAa,
137        ribbon = safeRibbon,
138        wet = safeWet,
139    )
140 }
141 }

```

14. app/src/main/java/com/wetinknext/engine/brush/BrushTexture.kt

```
1 package com.wetinknext.engine.brush
2
3 import android.opengl.GLES30
4 import com.wetinknext.engine.gl.GlCheck
5 import java.nio.ByteBuffer
6
7 class BrushTexture {
8     var textureId: Int = 0
9     private set
10
11     var width: Int = 0
12     private set
13
14     var height: Int = 0
15     private set
16
17     fun createFromRgba(
18         width: Int,
19         height: Int,
20         rgba: ByteBuffer,
21     ) {
22         GlCheck.checkOnGThread()
23         release()
24
25         val ids = IntArray(1)
26         GLES30.glGenTextures(1, ids, 0)
27         textureId = ids[0]
28
29         check(textureId != 0) { "Failed to create brush texture" }
30
31         this.width = width
32         this.height = height
33
34         GLES30.glBindTexture(GLES30.GL_TEXTURE_2D, textureId)
35
36         GLES30.glTexParameteri(
37             GLES30.GL_TEXTURE_2D,
38             GLES30.GL_TEXTURE_MIN_FILTER,
39             GLES30.GL_LINEAR,
40         )
41         GLES30.glTexParameteri(
42             GLES30.GL_TEXTURE_2D,
43             GLES30.GL_TEXTURE_MAG_FILTER,
44             GLES30.GL_LINEAR,
45         )
46         GLES30.glTexParameteri(
47             GLES30.GL_TEXTURE_2D,
48             GLES30.GL_TEXTURE_WRAP_S,
49             GLES30.GL_REPEAT,
50         )
51         GLES30.glTexParameteri(
52             GLES30.GL_TEXTURE_2D,
53             GLES30.GL_TEXTURE_WRAP_T,
54             GLES30.GL_REPEAT,
55         )
56
57         GLES30.glPixelStorei(
58             GLES30.GL_UNPACK_ALIGNMENT,
59             1,
60         )
61
62         GLES30.glTexImage2D(
63             GLES30.GL_TEXTURE_2D,
64             0,
65             GLES30.GL_RGBA,
66             width,
67             height,
68             0,
69             GLES30.GL_RGBA,
70             GLES30.GL_UNSIGNED_BYTE,
71             rgba,
72         )
73
74         GLES30.glPixelStorei(
75             GLES30.GL_UNPACK_ALIGNMENT,
76             4,
77         )
78
79         GLES30.glBindTexture(
80             GLES30.GL_TEXTURE_2D,
81             0,
82         )
83     }
84
85     fun release() {
86         GlCheck.checkOnGThread()
87         if (textureId != 0) {
88             GLES30.glDeleteTextures(
89                 1,
90                 intArrayOf(textureId),
91             )
92         }
93     }
94 }
```

```
91         0,  
92     )  
93 }  
94  
95     textureId = 0  
96     width = 0  
97     height = 0  
98 }  
99  
100 }
```

15. app/src/main/java/com/wetinknext/engine/brush/CapsuleEmitter.kt

```

1 package com.wetinknext.engine.brush
2
3 import com.wetinknext.engine.input.InputBatch
4 import kotlin.math.hypot
5 import kotlin.math.max
6 import kotlin.math.pow
7
8 /**
9  * Преобразует поток InputSample в цепочку капсул.
10  *
11  * Один сегмент:
12  *   A = x0, y0, radius0
13  *   B = x1, y1, radius1
14  *
15  * Капсула с одинаковыми радиусами является обычной capsule.
16  * При разном давлении shader строит round-cone.
17  *
18  * Важно:
19  * - не хранит InputSample;
20  * - не создаёт List/FloatArray на MOVE;
21  * - reusable state принадлежит GL/render pipeline;
22  * - pointerId фиксируется на DOWN;
23  * - давление применяется к радиусу;
24  * - opacity намеренно не применяется здесь, она применяется одним blit-pass.
25  */
26 class CapsuleEmitter(
27     private var settings: BrushSettings,
28 ) {
29     private var strokeSettings: BrushSettings? = null
30     private val stabilizer = Stabilizer()
31     private val pressureFilter = PressureFilter()
32     private val resolvedDab = ResolvedDab()
33
34     private var active = false
35     private var pointerId = -1
36
37     private var hasLastPoint = false
38     private var lastX = 0f
39     private var lastY = 0f
40     private var lastRadius = 0f
41     private var lastCoverage = 1f
42
43     // Окно точек для интерполяции P0, P1, P2, P3
44     private var p0x = 0f; private var p0y = 0f; private var p0r = 0f
45     private var p1x = 0f; private var p1y = 0f; private var p1r = 0f
46     private var p2x = 0f; private var p2y = 0f; private var p2r = 0f
47     private var p3x = 0f; private var p3y = 0f; private var p3r = 0f
48     private var p0c = 1f; private var p1c = 1f
49     private var p2c = 1f; private var p3c = 1f
50     private var pointsInWindow = 0
51
52     private var emittedSegments = 0
53
54     var hasStroke: Boolean = false
55         private set
56
57     var totalLength: Float = 0f
58         private set
59
60     var minX: Float = 0f
61         private set
62
63     var minY: Float = 0f
64         private set
65
66     var maxX: Float = 0f
67         private set
68
69     var maxY: Float = 0f
70         private set
71
72     var hasBounds: Boolean = false
73         private set
74
75     fun updateSettings(value: BrushSettings) {
76         // Active stroke reads strokeSettings; this value is for the next begin().
77         settings = value.resolved()
78     }
79
80     private fun activeSettings(): BrushSettings {
81         check(active) { "Capsule emitter has no active stroke" }
82         return strokeSettings ?: error("Missing stroke settings")
83     }
84
85     /**
86     * Полный сброс текущего stroke.
87     * overflowCount рендера здесь не сбрасывается.
88     */
89     fun reset() {
90         active = false

```



```

91         strokeSettings = null
92         pointerId = -1
93
94         hasLastPoint = false
95         lastX = 0f
96         lastY = 0f
97         lastRadius = 0f
98         lastCoverage = 1f
99
100        pointsInWindow = 0
101
102        emittedSegments = 0
103        hasStroke = false
104        totalLength = 0f
105
106        minX = 0f
107        minY = 0f
108        maxX = 0f
109        maxY = 0f
110        hasBounds = false
111
112        stabilizer.reset()
113        pressureFilter.reset()
114    }
115
116    /**
117     * Начало stroke.
118     *
119     * Первый sample добавляется как вырожденная капсула.
120     * В fragment shader она превращается в чистый диск.
121     */
122    fun begin(
123        batch: InputBatch,
124        out: CapsuleStrokeRenderer,
125        strokeSettings: BrushSettings = settings,
126    ) {
127        reset()
128        out.beginStroke()
129        val resolvedSettings = strokeSettings.resolved()
130        this.strokeSettings = resolvedSettings
131
132        if (batch.sampleCount <= 0) return
133
134        val sample = batch.samples[0]
135
136        active = true
137        pointerId = sample.pointerId
138
139        stabilizer.strength = (
140            resolvedSettings.smoothing * SMOOTHING_WEIGHT +
141            resolvedSettings.streamline * STREAMLINE_WEIGHT
142        ).coerceIn(0f, 1f)
143
144        stabilizer.process(
145            timestampNanos = sample.timestampNanos,
146            rawX = sample.canvasX,
147            rawY = sample.canvasY,
148        )
149
150        val x = stabilizer.x
151        val y = stabilizer.y
152        val filteredPressure = pressureFilter.filter(sample.timestampNanos, sample.pressure)
153        BrushDynamics.resolve(resolvedSettings, filteredPressure, sample.tiltX, sample.tiltY, resolvedDab)
154        val radius = resolvedDab.radius
155        val coverage = resolvedDab.coverage
156
157        lastX = x
158        lastY = y
159        lastRadius = radius
160        lastCoverage = coverage
161        hasLastPoint = true
162        hasStroke = true
163
164        // Инициализация окна
165        p0x = x; p0y = y; p0r = radius
166        p1x = x; p1y = y; p1r = radius
167        p2x = x; p2y = y; p2r = radius
168        p0c = coverage; p1c = coverage; p2c = coverage
169        pointsInWindow = 1
170
171        // Вырожденный сегмент: круглый dab/tap.
172        out.addSegment(
173            x0 = x,
174            y0 = y,
175            radius0 = radius,
176            coverage0 = coverage,
177            x1 = x,
178            y1 = y,
179            radius1 = radius,
180            coverage1 = coverage,
181        )
182        emittedSegments++

```

```

183         includeBounds(x, y, radius)
184     }
185 }
186
187 /**
188  * Обрабатывает MOVE batch.
189  *
190  * Historical samples уже должны находиться внутри batch
191  * раньше текущего sample, это обеспечивает StrokeInputCapturer.
192  */
193 fun append(
194     batch: InputBatch,
195     out: CapsuleStrokeRenderer,
196 ) {
197     if (!active || batch.sampleCount <= 0) return
198     val settings = activeSettings()
199
200     for (index in 0 until batch.sampleCount) {
201         val sample = batch.samples[index]
202
203         if (sample.pointerId != pointerId) {
204             continue
205         }
206
207         stabilizer.process(
208             timestampNanos = sample.timestampNanos,
209             rawX = sample.canvasX,
210             rawY = sample.canvasY,
211         )
212
213         val x = stabilizer.x
214         val y = stabilizer.y
215         val filteredPressure = pressureFilter.filter(sample.timestampNanos, sample.pressure)
216         BrushDynamics.resolve(settings, filteredPressure, sample.tiltX, sample.tiltY, resolvedDab)
217         val radius = resolvedDab.radius
218         val coverage = resolvedDab.coverage
219
220         emitPoint(
221             x = x,
222             y = y,
223             radius = radius,
224             coverage = coverage,
225             out = out,
226         )
227     }
228 }
229
230 /**
231  * Завершает stroke.
232  *
233  * Последний текущий sample должен быть передан через append()
234  * до вызова finish().
235  */
236 fun finish(
237     out: CapsuleStrokeRenderer,
238     cancel: Boolean,
239 ) {
240     if (!active) return
241     val settings = activeSettings()
242
243     if (cancel) {
244         reset()
245         out.clearStrokeData()
246         return
247     }
248
249     // Принудительно завершаем кривую, используя последнюю точку как P3
250     if (pointsInWindow >= 2 && settings.smoothing > 1e-4f) {
251         interpolate(
252             p0x, p0y, p0r,
253             p0c,
254             p1x, p1y, p1r, p1c,
255             p2x, p2y, p2r, p2c,
256             p2x, p2y, p2r, p2c,
257             out,
258         )
259     }
260
261     active = false
262     pointerId = -1
263 }
264
265 private fun emitPoint(
266     x: Float,
267     y: Float,
268     radius: Float,
269     coverage: Float,
270     out: CapsuleStrokeRenderer,
271 ) {
272     val settings = activeSettings()
273     val dx = x - lastX
274     val dy = y - lastY

```

```

275     val distance = hypot(dx, dy)
276
277     if (distance < minimumPointDistance()) {
278         return
279     }
280
281     if (settings.smoothing <= 1e-4f) {
282         // Raw mode: прямые сегменты без задержки на окно
283         out.addSegment(lastX, lastY, lastRadius, lastCoverage, x, y, radius, coverage)
284         emittedSegments++
285         totalLength += distance
286         includeBounds(lastX, lastY, lastRadius)
287         includeBounds(x, y, radius)
288     } else {
289         // Interpolated mode: Catmull-Rom через окно
290         when (pointsInWindow) {
291             1 -> {
292                 p2x = x; p2y = y; p2r = radius; p2c = coverage
293                 pointsInWindow = 2
294             }
295             2 -> {
296                 p3x = x; p3y = y; p3r = radius; p3c = coverage
297                 interpolate(p0x, p0y, p0r, p0c, p1x, p1y, p1r, p1c, p2x, p2y, p2r, p2c, p3x, p3y, p3r, p3c, out)
298                 // Сдвиг окна
299                 p0x = p1x; p0y = p1y; p0r = p1r; p0c = p1c
300                 p1x = p2x; p1y = p2y; p1r = p2r; p1c = p2c
301                 p2x = p3x; p2y = p3y; p2r = p3r; p2c = p3c
302             }
303         }
304     }
305
306     hasStroke = true
307     lastX = x
308     lastY = y
309     lastRadius = radius
310     lastCoverage = coverage
311 }
312
313 /**
314  * Адаптивная интерполяция сегмента P1-P2 с использованием P0 и P3 как контрольных точек.
315  */
316 private fun interpolate(
317     p0x: Float, p0y: Float, p0r: Float, p0c: Float,
318     p1x: Float, p1y: Float, p1r: Float, p1c: Float,
319     p2x: Float, p2y: Float, p2r: Float, p2c: Float,
320     p3x: Float, p3y: Float, p3r: Float, p3c: Float,
321     out: CapsuleStrokeRenderer
322 ) {
323     val dist = hypot(p2x - p1x, p2y - p1y)
324     if (dist < 1e-4f) return
325
326     // Вычисляем контрольные точки для кубической кривой Безье (Catmull-Rom -> Bezier)
327     val c1x = p1x + (p2x - p0x) / 6f
328     val c1y = p1y + (p2y - p0y) / 6f
329     val c2x = p2x - (p3x - p1x) / 6f
330     val c2y = p2y - (p3y - p1y) / 6f
331
332     // Адаптивное количество шагов: примерно каждые 0.25..0.5 радиуса
333     val avgRadius = (p1r + p2r) * 0.5f
334     val stepPx = max(1f, avgRadius * INTERPOLATION_STEP_RATIO)
335     val steps = max(1, (dist / stepPx).toInt().coerceAtMost(MAX_INTERPOLATION_STEPS))
336
337     var prevX = p1x
338     var prevY = p1y
339     var prevR = p1r
340     var prevCoverage = p1c
341
342     for (i in 1..steps) {
343         val t = i.toFloat() / steps
344         val mt = 1f - t
345
346         // Кубическое Безье для позиции
347         val x = mt * mt * mt * p1x + 3f * mt * mt * t * c1x + 3f * mt * t * t * c2x + t * t * t * p2x
348         val y = mt * mt * mt * p1y + 3f * mt * mt * t * c1y + 3f * mt * t * t * c2y + t * t * t * p2y
349
350         // Линейная интерполяция радиуса
351         val r = p1r + (p2r - p1r) * t
352         val coverage = p1c + (p2c - p1c) * t
353
354         out.addSegment(prevX, prevY, prevR, prevCoverage, x, y, r, coverage)
355         emittedSegments++
356         totalLength += hypot(x - prevX, y - prevY)
357         includeBounds(x, y, r)
358
359         prevX = x
360         prevY = y
361         prevR = r
362         prevCoverage = coverage
363     }
364 }
365
366 /**

```

```

367     * Для капсул не нужен такой плотный шаг, как для stamp-кисти:
368     * segment shader сам покрывает весь отрезок между А и В.
369     */
370     private fun minimumPointDistance(): Float {
371         val settings = activeSettings()
372         val configured = settings.ribbon.minPointDistancePx
373             .coerceIn(MIN_POINT_DISTANCE_MIN, MAX_POINT_DISTANCE)
374
375         val radiusBased = settings.baseRadiusPx * POINT_STEP_RADIUS_RATIO
376
377         return max(
378             configured,
379             radiusBased.coerceAtMost(MAX_POINT_DISTANCE),
380         )
381     }
382
383     private fun includeBounds(
384         x: Float,
385         y: Float,
386         radius: Float,
387     ) {
388         val left = x - radius
389         val top = y - radius
390         val right = x + radius
391         val bottom = y + radius
392
393         if (!hasBounds) {
394             minX = left
395             minY = top
396             maxX = right
397             maxY = bottom
398             hasBounds = true
399             return
400         }
401
402         if (left < minX) minX = left
403         if (top < minY) minY = top
404         if (right > maxX) maxX = right
405         if (bottom > maxY) maxY = bottom
406     }
407
408     companion object {
409         private const val SMOOTHING_WEIGHT = 0.7f
410         private const val STREAMLINE_WEIGHT = 0.3f
411
412         private const val MIN_RADIUS_PX = 0.25f
413
414         private const val MIN_POINT_DISTANCE_MIN = 0.05f
415         private const val MAX_POINT_DISTANCE = 32f
416         private const val POINT_STEP_RADIUS_RATIO = 0.08f
417
418         private const val INTERPOLATION_STEP_RATIO = 0.4f
419         private const val MAX_INTERPOLATION_STEPS = 64
420     }
421 }

```

16. app/src/main/java/com/wetinknext/engine/brush/CapsuleStrokeRenderer.kt

```

1  package com.wetinknext.engine.brush
2
3  import android.opengl.GLES30
4  import com.wetinknext.BuildConfig
5  import com.wetinknext.engine.gl.GlCheck
6  import com.wetinknext.engine.gl.GlProgram
7  import com.wetinknext.engine.gl.RenderTarget
8  import com.wetinknext.engine.gl.ShaderLib
9  import java.nio.ByteBuffer
10 import java.nio.ByteOrder
11
12 /**
13  * GPU-renderer для круглого штриха.
14  *
15  * Один instance:
16  *   x0, y0, radius0,
17  *   x1, y1, radius1
18  *
19  * Геометрия каждого сегмента является AABB, а настоящая форма
20  * рассчитывается в capsuleFragment через signed distance.
21  *
22  * Важно:
23  * - target должен быть очищен перед началом нового штриха;
24  * - рисование использует premultiplied source-over blending;
25  * - цвет должен быть постоянным для всего strokeTarget;
26  * - выход shader: premultiplied linear RGBA.
27  */
28 class CapsuleStrokeRenderer(
29     private val maxSegments: Int = DEFAULT_MAX_SEGMENTS,
30 ) {
31     private var program: GlProgram? = null
32
33     private var vaoId = 0
34     private var cornerVboId = 0
35     private var instanceVboId = 0
36
37     private var uCanvasToClip = -1
38     private var uColorLinear = -1
39     private val blendController = BlendController()
40     private var uCanvasSize = -1
41     private var uGrainTex = -1
42     private var uGrainActive = -1
43     private var uGrainScale = -1
44     private var uGrainCanvasLocked = -1
45     private var uTextureDepth = -1
46     private var uTextureContrast = -1
47     private var uFlow = -1
48
49     private var grainTextureId = 0
50     private var grainScale = 1f
51     private var grainCanvasLocked = true
52     private var textureDepth = 1f
53     private var textureContrast = 1f
54
55     private val checkGlErrors: Boolean
56         get() = BuildConfig.DEBUG
57
58     /**
59     * Формат:
60     *   x0, y0, r0, x1, y1, r1
61     */
62     private val instanceData =
63         ByteBuffer
64             .allocateDirect(
65                 maxSegments * FLOATS_PER_SEGMENT * Float.SIZE_BYTES,
66             )
67             .order(ByteOrder.nativeOrder())
68             .asFloatBuffer()
69
70     var segmentCount: Int = 0
71         private set
72
73     private var uploadedSegmentCount = 0
74
75     /**
76     * Не сбрасывается на каждый новый stroke.
77     * Иначе диагностика потерь снова будет скрыта.
78     */
79     var overflowCount: Long = 0L
80         private set
81
82     val isEmpty: Boolean
83         get() = segmentCount == 0
84
85     val hasPendingSegments: Boolean
86         get() = segmentCount > uploadedSegmentCount
87
88     fun create() {
89         GlCheck.checkOnGlThread()
90         release()

```

```

91
92     program = GLProgram(
93         vertexSource = ShaderLib.capsuleVertex,
94         fragmentSource = ShaderLib.capsuleFragment,
95     )
96
97     val currentProgram = checkNotNull(program)
98     currentProgram.use()
99
100    uCanvasToClip = GLES30.glGetUniformLocation(
101        currentProgram.id,
102        "uCanvasToClip",
103    )
104    uColorLinear = GLES30.glGetUniformLocation(
105        currentProgram.id,
106        "uColorLinear",
107    )
108    uCanvasSize = GLES30.glGetUniformLocation(
109        currentProgram.id,
110        "uCanvasSize",
111    )
112    uGrainTex = GLES30.glGetUniformLocation(
113        currentProgram.id,
114        "uGrainTex",
115    )
116    uGrainScale = GLES30.glGetUniformLocation(
117        currentProgram.id,
118        "uGrainScale",
119    )
120    uGrainActive = GLES30.glGetUniformLocation(
121        currentProgram.id,
122        "uGrainActive",
123    )
124    uGrainCanvasLocked = GLES30.glGetUniformLocation(
125        currentProgram.id,
126        "uGrainCanvasLocked",
127    )
128    uTextureDepth = GLES30.glGetUniformLocation(
129        currentProgram.id,
130        "uTextureDepth",
131    )
132    uTextureContrast = GLES30.glGetUniformLocation(
133        currentProgram.id,
134        "uTextureContrast",
135    )
136    uFlow = GLES30.glGetUniformLocation(
137        currentProgram.id,
138        "uFlow",
139    )
140
141    check(uCanvasToClip >= 0) {
142        "Capsule shader uniform uCanvasToClip is missing"
143    }
144    check(uColorLinear >= 0) {
145        "Capsule shader uniform uColorLinear is missing"
146    }
147    check(uCanvasSize >= 0) {
148        "Capsule shader uniform uCanvasSize is missing"
149    }
150    check(uFlow >= 0) {
151        "Capsule shader uniform uFlow is missing"
152    }
153
154    val corners = floatArrayOf(
155        -1f, -1f,
156        1f, -1f,
157        -1f, 1f,
158        1f, 1f,
159    )
160
161    val cornerData =
162        ByteBuffer
163            .allocateDirect(corners.size * Float.SIZE_BYTES)
164            .order(ByteOrder.nativeOrder())
165            .asFloatBuffer()
166
167    cornerData.put(corners)
168    cornerData.position(0)
169
170    val ids = IntArray(1)
171
172    GLES30.glGenVertexArrays(1, ids, 0)
173    vaoId = ids[0]
174
175    GLES30.glGenBuffers(1, ids, 0)
176    cornerVboId = ids[0]
177
178    GLES30.glGenBuffers(1, ids, 0)
179    instanceVboId = ids[0]
180
181    check(vaoId != 0) { "Failed to create capsule VAO" }
182    check(cornerVboId != 0) { "Failed to create capsule corner VBO" }

```

```

183         check(instanceVboId != 0) { "Failed to create capsule instance VBO" }
184
185         GLES30.glBindVertexArray(vaoId)
186
187         // Углы AABB-квада.
188         GLES30.glBindBuffer(
189             GLES30.GL_ARRAY_BUFFER,
190             cornerVboId,
191         )
192         GLES30.glBufferData(
193             GLES30.GL_ARRAY_BUFFER,
194             corners.size * Float.SIZE_BYTES,
195             cornerData,
196             GLES30.GL_STATIC_DRAW,
197         )
198         GLES30.glEnableVertexAttribArray(ATTR_CORNER)
199         GLES30.glVertexAttribPointer(
200             ATTR_CORNER,
201             2,
202             GLES30.GL_FLOAT,
203             false,
204             2 * Float.SIZE_BYTES,
205             0,
206         )
207
208         // Данные сегментов.
209         GLES30.glBindBuffer(
210             GLES30.GL_ARRAY_BUFFER,
211             instanceVboId,
212         )
213         GLES30.glBufferData(
214             GLES30.GL_ARRAY_BUFFER,
215             maxSegments * STRIDE_BYTES,
216             null,
217             GLES30.GL_DYNAMIC_DRAW,
218         )
219
220         // A: x0, y0, radius0
221         GLES30.glEnableVertexAttribArray(ATTR_SEGMENT_A)
222         GLES30.glVertexAttribPointer(
223             ATTR_SEGMENT_A,
224             4,
225             GLES30.GL_FLOAT,
226             false,
227             STRIDE_BYTES,
228             0,
229         )
230         GLES30.glVertexAttribDivisor(
231             ATTR_SEGMENT_A,
232             1,
233         )
234
235         // B: x1, y1, radius1
236         GLES30.glEnableVertexAttribArray(ATTR_SEGMENT_B)
237         GLES30.glVertexAttribPointer(
238             ATTR_SEGMENT_B,
239             4,
240             GLES30.GL_FLOAT,
241             false,
242             STRIDE_BYTES,
243             4 * Float.SIZE_BYTES,
244         )
245         GLES30.glVertexAttribDivisor(
246             ATTR_SEGMENT_B,
247             1,
248         )
249
250         GLES30.glBindBuffer(
251             GLES30.GL_ARRAY_BUFFER,
252             0,
253         )
254         GLES30.glBindVertexArray(0)
255
256         GLES30.glUseProgram(0)
257
258         checkNoGLError("CapsuleStrokeRenderer.create")
259     }
260
261     /**
262     * Начинает новый stroke.
263     *
264     * Старый target должен быть очищен отдельно:
265     * strokeTarget.clear(0f, 0f, 0f, 0f)
266     */
267     fun beginStroke() {
268         segmentCount = 0
269         uploadedSegmentCount = 0
270
271         instanceData.clear()
272         instanceData.limit(
273             maxSegments * FLOATS_PER_SEGMENT,
274         )

```

```

275     }
276
277     /**
278     * Добавляет один сегмент в CPU-buffer.
279     *
280     * Если capacity исчерпана, сегмент не теряется молча:
281     * overflowCount увеличивается.
282     */
283     fun addSegment(
284         x0: Float,
285         y0: Float,
286         radius0: Float,
287         coverage0: Float,
288         x1: Float,
289         y1: Float,
290         radius1: Float,
291         coverage1: Float,
292     ): Boolean {
293         if (segmentCount >= maxSegments) {
294             overflowCount++
295             return false
296         }
297
298         val offset = segmentCount * FLOATS_PER_SEGMENT
299
300         instanceData.put(offset, x0)
301         instanceData.put(offset + 1, y0)
302         instanceData.put(offset + 2, radius0.coerceAtLeast(0.01f))
303         instanceData.put(offset + 3, coverage0.coerceIn(0f, 1f))
304
305         instanceData.put(offset + 4, x1)
306         instanceData.put(offset + 5, y1)
307         instanceData.put(offset + 6, radius1.coerceAtLeast(0.01f))
308         instanceData.put(offset + 7, coverage1.coerceIn(0f, 1f))
309
310         segmentCount++
311         return true
312     }
313
314     /**
315     * Загружает и рисует только новые сегменты.
316     *
317     * Используется для preview во время MOVE.
318     * Перед первым вызовом target должен быть очищен.
319     */
320     fun drawPending(
321         target: RenderTarget,
322         width: Int,
323         height: Int,
324         canvasToClip: FloatArray,
325         colorLinear: FloatArray,
326         blendPolicy: BlendPolicy,
327         flow: Float,
328     ) {
329         val first = uploadedSegmentCount
330         val count = segmentCount - uploadedSegmentCount
331
332         if (count <= 0) return
333
334         val drawn = drawRange(
335             target = target,
336             width = width,
337             height = height,
338             canvasToClip = canvasToClip,
339             colorLinear = colorLinear,
340             firstSegment = first,
341             count = count,
342             blendPolicy = blendPolicy,
343             flow = flow,
344         )
345
346         if (drawn) {
347             uploadedSegmentCount = segmentCount
348         }
349     }
350
351     /**
352     * Полностью рисует все накопленные сегменты.
353     *
354     * Используется при commit, если target был очищен.
355     */
356     fun drawAll(
357         target: RenderTarget,
358         width: Int,
359         height: Int,
360         canvasToClip: FloatArray,
361         colorLinear: FloatArray,
362         blendPolicy: BlendPolicy,
363         flow: Float,
364     ) {
365         if (segmentCount <= 0) return
366     }

```



```

367         drawRange(
368             target = target,
369             width = width,
370             height = height,
371             canvasToClip = canvasToClip,
372             colorLinear = colorLinear,
373             firstSegment = 0,
374             count = segmentCount,
375             blendPolicy = blendPolicy,
376             flow = flow,
377         )
378     }
379
380     /**
381     * Полностью перерисовывает сегменты в диапазоне [firstSegment, firstSegment + count).
382     *
383     * Важный момент: glVertexAttribPointer вызывается здесь повторно,
384     * потому что offset у instanced-атрибутов меняется для каждого диапазона.
385     */
386     private fun drawRange(
387         target: RenderTarget,
388         width: Int,
389         height: Int,
390         canvasToClip: FloatArray,
391         colorLinear: FloatArray,
392         firstSegment: Int,
393         count: Int,
394         blendPolicy: BlendPolicy,
395         flow: Float,
396     ): Boolean {
397         val currentProgram = program ?: return false
398
399         require(canvasToClip.size >= 16) {
400             "canvasToClip must contain at least 16 floats"
401         }
402         require(colorLinear.size >= 3) {
403             "colorLinear must contain at least 3 floats"
404         }
405         require(firstSegment >= 0)
406         require(count >= 0)
407         require(firstSegment + count <= segmentCount)
408
409         if (count == 0) return true
410
411         target.bind()
412
413         GLES30.glViewport(
414             0,
415             0,
416             width,
417             height,
418         )
419
420         GLES30.glDisable(GLES30.GL_SCISSOR_TEST)
421
422         blendController.begin(blendPolicy)
423
424         currentProgram.use()
425
426         GLES30.glUniformMatrix4fv(
427             uCanvasToClip,
428             1,
429             false,
430             canvasToClip,
431             0,
432         )
433
434         GLES30.glUniform3f(
435             uColorLinear,
436             colorLinear[0].coerceIn(0f, 1f),
437             colorLinear[1].coerceIn(0f, 1f),
438             colorLinear[2].coerceIn(0f, 1f),
439         )
440
441         GLES30.glUniform2f(
442             uCanvasSize,
443             width.toFloat(),
444             height.toFloat(),
445         )
446
447         GLES30.glUniform1f(
448             uGrainScale,
449             grainScale,
450         )
451
452         GLES30.glUniform1i(
453             uGrainActive,
454             if (grainTextureId != 0) 1 else 0,
455         )
456
457         GLES30.glUniform1i(
458             uGrainCanvasLocked,

```

```

459         if (grainCanvasLocked) 1 else 0,
460     )
461
462     GLES30.glUniform1f(
463         uTextureDepth,
464         textureDepth,
465     )
466
467     GLES30.glUniform1f(
468         uTextureContrast,
469         textureContrast,
470     )
471
472     GLES30.glUniform1f(
473         uFlow,
474         flow.coerceIn(0f, 1f),
475     )
476
477     if (grainTextureId != 0) {
478         GLES30.glActiveTexture(GLES30.GL_TEXTURE2)
479         GLES30.glBindTexture(
480             GLES30.GL_TEXTURE_2D,
481             grainTextureId,
482         )
483         GLES30.glUniform1i(uGrainTex, 2)
484     }
485
486     /*
487     * FloatBuffer должен содержать только передаваемый диапазон.
488     * glBufferSubData прочитает remaining() элементов.
489     */
490     val sourceStart = firstSegment * FLOATS_PER_SEGMENT
491     val sourceEnd =
492         sourceStart + count * FLOATS_PER_SEGMENT
493
494     instanceData.position(sourceStart)
495     instanceData.limit(sourceEnd)
496
497     GLES30.glBindVertexArray(vaoId)
498     GLES30.glBindBuffer(
499         GLES30.GL_ARRAY_BUFFER,
500         instanceVboId,
501     )
502
503     GLES30.glBufferSubData(
504         GLES30.GL_ARRAY_BUFFER,
505         firstSegment * STRIDE_BYTES,
506         count * STRIDE_BYTES,
507         instanceData,
508     )
509
510     /*
511     * Для ES 3.0 нет glDrawArraysInstancedBaseInstance.
512     * Поэтому offset атрибутов сдвигается вручную.
513     *
514     * VAO должен быть привязан во время glVertexAttribPointer.
515     */
516     GLES30.glVertexAttribPointer(
517         ATTR_SEGMENT_A,
518         4,
519         GLES30.GL_FLOAT,
520         false,
521         STRIDE_BYTES,
522         firstSegment * STRIDE_BYTES,
523     )
524
525     GLES30.glVertexAttribPointer(
526         ATTR_SEGMENT_B,
527         4,
528         GLES30.GL_FLOAT,
529         false,
530         STRIDE_BYTES,
531         firstSegment * STRIDE_BYTES + 4 * Float.SIZE_BYTES,
532     )
533
534     GLES30.glDrawArraysInstanced(
535         GLES30.GL_TRIANGLE_STRIP,
536         0,
537         CORNER_VERTEX_COUNT,
538         count,
539     )
540
541     if (grainTextureId != 0) {
542         GLES30.glActiveTexture(GLES30.GL_TEXTURE2)
543         GLES30.glBindTexture(
544             GLES30.GL_TEXTURE_2D,
545             0,
546         )
547     }
548
549     GLES30.glActiveTexture(GLES30.GL_TEXTURE0)
550

```

```

551     GLES30.glBindBuffer(
552         GLES30.GL_ARRAY_BUFFER,
553         0,
554     )
555     GLES30.glBindVertexArray(0)
556
557     blendController.end()
558
559     GLES30.glBindFramebuffer(
560         GLES30.GL_FRAMEBUFFER,
561         0,
562     )
563
564     instanceData.clear()
565     instanceData.limit(
566         maxSegments * FLOATS_PER_SEGMENT,
567     )
568
569     if (checkGLErrors) {
570         checkNoGLError("CapsuleStrokeRenderer.drawRange")
571     }
572     return true
573 }
574
575 /**
576  * Сбрасывает только GPU/CPU состояние текущего renderer.
577  * Счётчик overflow сохраняется.
578  */
579 fun clearStrokeData() {
580     segmentCount = 0
581     uploadedSegmentCount = 0
582     instanceData.clear()
583     instanceData.limit(
584         maxSegments * FLOATS_PER_SEGMENT,
585     )
586 }
587
588 fun setGrainTexture(
589     textureId: Int,
590     scale: Float,
591     canvasLocked: Boolean,
592     depth: Float,
593     contrast: Float,
594 ) {
595     grainTextureId = textureId
596     grainScale = scale.coerceAtLeast(0.0001f)
597     grainCanvasLocked = canvasLocked
598     textureDepth = depth.coerceIn(0f, 1f)
599     textureContrast = contrast.coerceIn(0f, 2f)
600 }
601
602 fun clearGrainTexture() {
603     grainTextureId = 0
604     grainScale = 1f
605     grainCanvasLocked = true
606     textureDepth = 1f
607     textureContrast = 1f
608 }
609
610 /**
611  * Все GL-объекты удаляются только на GL-потоке.
612  */
613 fun release() {
614     GLCheck.checkOnGLThread()
615     program?.release()
616     program = null
617
618     if (instanceVboId != 0) {
619         GLES30.glDeleteBuffers(
620             1,
621             intArrayOf(instanceVboId),
622             0,
623         )
624     }
625
626     if (cornerVboId != 0) {
627         GLES30.glDeleteBuffers(
628             1,
629             intArrayOf(cornerVboId),
630             0,
631         )
632     }
633
634     if (vaoId != 0) {
635         GLES30.glDeleteVertexArrays(
636             1,
637             intArrayOf(vaoId),
638             0,
639         )
640     }
641
642     vaoId = 0

```

```

643         cornerVboId = 0
644         instanceVboId = 0
645
646         uCanvasToClip = -1
647         uColorLinear = -1
648         uCanvasSize = -1
649         uGrainTex = -1
650         uGrainScale = -1
651         uGrainActive = -1
652         uGrainCanvasLocked = -1
653         uTextureDepth = -1
654         uTextureContrast = -1
655         uFlow = -1
656         grainTextureId = 0
657
658         segmentCount = 0
659         uploadedSegmentCount = 0
660
661         instanceData.clear()
662         instanceData.limit(
663             maxSegments * FLOATS_PER_SEGMENT,
664         )
665     }
666
667     private fun checkNoGLError(label: String) {
668         val error = GLES30.glGetError()
669         check(error == GLES30.GL_NO_ERROR) {
670             "$label: GL error 0x${error.toString(16)}"
671         }
672     }
673
674     companion object {
675
676         private const val ATTR_CORNER = 0
677         private const val ATTR_SEGMENT_A = 1
678         private const val ATTR_SEGMENT_B = 2
679
680         private const val CORNER_VERTEX_COUNT = 4
681
682         const val FLOATS_PER_SEGMENT = 8
683         const val STRIDE_BYTES =
684             FLOATS_PER_SEGMENT * Float.SIZE_BYTES
685
686         const val DEFAULT_MAX_SEGMENTS = 16_384
687     }
688 }

```

17. app/src/main/java/com/wetinknext/engine/brush/ClosureDebug.kt

```
1 package com.wetinknext.engine.brush
2
3 import android.util.Log
4 import com.wetinknext.BuildConfig
5
6 /** One debug log per finished stroke; never used from the render hot path. */
7 object ClosureDebug {
8     const val TAG = "RibbonClosure"
9     fun publish(distance: Float, threshold: Float, closed: Boolean, samples: Int) {
10         if (!BuildConfig.DEBUG) return
11         runCatching {
12             Log.d(TAG, "dist=%.2f thr=%.2f closed=%s n=%d".format(distance, threshold, closed, samples))
13         }
14     }
15 }
```

18. app/src/main/java/com/wetinknext/engine/brush/ColorSpaces.kt

```
1 package com.wetinknext.engine.brush
2
3 import kotlin.math.pow
4
5 object ColorSpaces {
6     fun srgbToLinear(value: Float): Float { val v=value.coerceIn(0f,1f); return if(v<=.04045f)v/12.92f else ((v+.055f)/1.
055f).pow(2.4f) }
7     fun linearToSrgb(value: Float): Float { val v=value.coerceIn(0f,1f); return if(v<=.0031308f)v*12.92f else 1.055f*v.pow(1f/
2.4f)-.055f }
8     fun srgb8ToLinear(Argb: Long,out:FloatArray){require(out.size>=3);out[0]=srgbToLinear(((Argb shr 16)and 0xFF)/255f);out[
1]=srgbToLinear(((Argb shr 8)and 0xFF)/255f);out[2]=srgbToLinear((Argb and 0xFF)/255f)}
9 }
```

19. app/src/main/java/com/wetinknext/engine/brush/DabBuffer.kt

```

1 package com.wetinknext.engine.brush
2
3 import com.wetinknext.engine.core.DirtyRect
4 import java.nio.ByteBuffer
5 import java.nio.ByteOrder
6
7 class DabBuffer(val capacity: Int = DEFAULT_CAPACITY) {
8     init { require(capacity > 0) }
9     val floats = ByteBuffer.allocateDirect(capacity * FLOATS_PER_DAB * Float.SIZE_BYTES).order(ByteOrder.nativeOrder()).
asFloatBuffer()
10    var count = 0; private set; var overflowCount = 0L; private set
11    fun clear() { count = 0; overflowCount = 0L; floats.clear() }
12    fun add(x: Float, y: Float, radius: Float, rotation: Float, coverage: Float, flow: Float): Boolean { if (count >=
capacity) { overflowCount++; return false }; floats.put(x); floats.put(y); floats.put(radius); floats.put(rotation); floats.put(
coverage.coerceIn(0f, 1f)); floats.put(flow.coerceIn(0f, 1f)); count++; return true }
13    fun prepareForUpload() { floats.position(0); floats.limit(count * FLOATS_PER_DAB) }
14    fun prepareForUpload(firstDab: Int, dabCount: Int) {
15        require(firstDab >= 0)
16        require(dabCount >= 0)
17        require(firstDab + dabCount <= count)
18        floats.position(firstDab * FLOATS_PER_DAB)
19        floats.limit((firstDab + dabCount) * FLOATS_PER_DAB)
20    }
21    /** Restores the direct buffer for appending after a GL upload. */
22    fun finishUpload() {
23        floats.limit(capacity * FLOATS_PER_DAB)
24        floats.position(count * FLOATS_PER_DAB)
25    }
26
27    /**
28     * Includes the exact circular footprint of each emitted dab plus a caller
29     * supplied safety fringe for analytic AA and textured-tip filtering.
30     */
31    fun includeDirtyRect(rect: DirtyRect, extraMargin: Float) {
32        val safeExtraMargin = extraMargin.coerceAtLeast(0f)
33        for (index in 0 until count) {
34            val offset = index * FLOATS_PER_DAB
35            val x = floats.get(offset)
36            val y = floats.get(offset + 1)
37            val radius = floats.get(offset + 2).coerceAtLeast(0f) + safeExtraMargin
38            rect.include(
39                left = x - radius,
40                top = y - radius,
41                right = x + radius,
42                bottom = y + radius,
43            )
44        }
45    }
46    companion object { const val FLOATS_PER_DAB = 6; const val DEFAULT_CAPACITY = 8192 }
47 }

```

20. app/src/main/java/com/wetinknext/engine/brush/DabRenderer.kt

```

1  package com.wetinknext.engine.brush
2
3  import android.opengl.GLES30
4  import com.wetinknext.engine.gl.GlCheck
5  import com.wetinknext.engine.gl.GlProgram
6  import com.wetinknext.engine.gl.RenderTarget
7  import com.wetinknext.engine.gl.ShaderLib
8  import java.nio.ByteBuffer
9  import java.nio.ByteOrder
10
11 /** Instanced circular dab renderer. All colour values are premultiplied linear output. */
12 class DabRenderer(private val maxDabs: Int) {
13     private var program: GlProgram? = null
14     private var vaoId = 0
15     private var quadBufferId = 0
16     private var instanceBufferId = 0
17     private var uCanvasToClip = -1
18     private var uColorLinear = -1
19     private var uCanvasSize = -1
20     private var uGrainTex = -1
21     private var uGrainActive = -1
22     private var uGrainScale = -1
23     private var uGrainCanvasLocked = -1
24     private var uTextureDepth = -1
25     private var uTextureContrast = -1
26     private var uStrokeOpacity = -1
27     private var uShapeTex = -1
28     private var uShapeActive = -1
29     private var uReverseShape = -1
30     private var uRgbToAlpha = -1
31
32     private var grainTextureId = 0
33     private var grainScale = 1f
34     private var grainCanvasLocked = true
35     private var textureDepth = 1f
36     private var textureContrast = 1f
37     private var shapeTextureId = 0
38     private var reverseShape = false
39     private var rgbToAlpha = false
40
41     private var uploadedDabCount = 0
42     private val blendController = BlendController()
43
44     /** Number of dabs already written into the current stroke target. */
45     val uploadedCount: Int get() = uploadedDabCount
46
47     fun create() {
48         GlCheck.checkOnGUIThread()
49         release()
50         uploadedDabCount = 0
51         program = GlProgram(ShaderLib.dabVertex, ShaderLib.dabFragment)
52         val currentProgram = checkNotNull(program)
53         currentProgram.use()
54         uCanvasToClip = GLES30.glGetUniformLocation(currentProgram.id, "uCanvasToClip")
55         uColorLinear = GLES30.glGetUniformLocation(currentProgram.id, "uColorLinear")
56         uCanvasSize = GLES30.glGetUniformLocation(currentProgram.id, "uCanvasSize")
57         uGrainTex = GLES30.glGetUniformLocation(currentProgram.id, "uGrainTex")
58         uGrainActive = GLES30.glGetUniformLocation(currentProgram.id, "uGrainActive")
59         uGrainScale = GLES30.glGetUniformLocation(currentProgram.id, "uGrainScale")
60         uGrainCanvasLocked = GLES30.glGetUniformLocation(currentProgram.id, "uGrainCanvasLocked")
61         uTextureDepth = GLES30.glGetUniformLocation(currentProgram.id, "uTextureDepth")
62         uTextureContrast = GLES30.glGetUniformLocation(currentProgram.id, "uTextureContrast")
63         uStrokeOpacity = GLES30.glGetUniformLocation(currentProgram.id, "uStrokeOpacity")
64         uShapeTex = GLES30.glGetUniformLocation(currentProgram.id, "uShapeTex")
65         uShapeActive = GLES30.glGetUniformLocation(currentProgram.id, "uShapeActive")
66         uReverseShape = GLES30.glGetUniformLocation(currentProgram.id, "uReverseShape")
67         uRgbToAlpha = GLES30.glGetUniformLocation(currentProgram.id, "uRgbToAlpha")
68
69         check(uCanvasToClip >= 0 && uColorLinear >= 0 && uStrokeOpacity >= 0) { "Dab shader uniforms missing" }
70
71         val quad = floatArrayOf(-1f, -1f, 1f, -1f, -1f, 1f, 1f, 1f)
72         val quadData = ByteBuffer.allocateDirect(quad.size * Float.SIZE_BYTES)
73             .order(ByteOrder.nativeOrder())
74             .asFloatBuffer()
75         quadData.put(quad).position(0)
76
77         val ids = IntArray(1)
78         GLES30.glGenVertexArrays(1, ids, 0)
79         vaoId = ids[0]
80         GLES30.glGenBuffers(1, ids, 0)
81         quadBufferId = ids[0]
82         GLES30.glGenBuffers(1, ids, 0)
83         instanceBufferId = ids[0]
84
85         GLES30.glBindVertexArray(vaoId)
86         GLES30.glBindBuffer(GLES30.GL_ARRAY_BUFFER, quadBufferId)
87         GLES30.glBufferData(
88             GLES30.GL_ARRAY_BUFFER,
89             quad.size * Float.SIZE_BYTES,
90             quadData,

```



```

91         GLES30.GL_STATIC_DRAW,
92     )
93     GLES30.glEnableVertexAttribArray(0)
94     GLES30.glVertexAttribPointer(0, 2, GLES30.GL_FLOAT, false, 0, 0)
95
96     GLES30.glBindBuffer(GLES30.GL_ARRAY_BUFFER, instanceBufferId)
97     GLES30.glBufferData(
98         GLES30.GL_ARRAY_BUFFER,
99         maxDabs * DabBuffer.FLOATS_PER_DAB * Float.SIZE_BYTES,
100         null,
101         GLES30.GL_DYNAMIC_DRAW,
102     )
103     GLES30.glEnableVertexAttribArray(1)
104     GLES30.glVertexAttribPointer(1, 4, GLES30.GL_FLOAT, false, DabBuffer.FLOATS_PER_DAB * Float.SIZE_BYTES, 0)
105     GLES30.glVertexAttribDivisor(1, 1)
106     GLES30.glEnableVertexAttribArray(2)
107     GLES30.glVertexAttribPointer(2, 2, GLES30.GL_FLOAT, false, DabBuffer.FLOATS_PER_DAB * Float.SIZE_BYTES, 16)
108     GLES30.glVertexAttribDivisor(2, 1)
109     GLES30.glBindBuffer(GLES30.GL_ARRAY_BUFFER, 0)
110     GLES30.glBindVertexArray(0)
111 }
112
113 /** Draws all buffered dabs into a document-space RenderTarget. */
114 fun drawInto(
115     target: RenderTarget,
116     width: Int,
117     height: Int,
118     canvasToFbo: FloatArray,
119     dabs: DabBuffer,
120     colorLinear: FloatArray,
121     blendPolicy: BlendPolicy,
122     strokeOpacity: Float = 1f,
123 ) {
124     if (dabs.count == 0) return
125     require(colorLinear.size >= 3)
126
127     dabs.prepareForUpload(firstDab = 0, dabCount = dabs.count)
128     try {
129         drawRange(
130             target = target,
131             width = width,
132             height = height,
133             canvasToFbo = canvasToFbo,
134             dabs = dabs,
135             firstDab = 0,
136             count = dabs.count,
137             colorLinear = colorLinear,
138             blendPolicy = blendPolicy,
139             strokeOpacity = strokeOpacity,
140         )
141     } finally {
142         dabs.finishUpload()
143     }
144 }
145
146 fun release() {
147     GLCheck.checkOnGLThread()
148     program?.release()
149     program = null
150     uploadedDabCount = 0
151     grainTextureId = 0
152     shapeTextureId = 0
153     reverseShape = false
154     rgbToAlpha = false
155     if (instanceBufferId != 0) GLES30.glDeleteBuffers(1, intArrayOf(instanceBufferId), 0)
156     if (quadBufferId != 0) GLES30.glDeleteBuffers(1, intArrayOf(quadBufferId), 0)
157     if (vaoId != 0) GLES30.glDeleteVertexArrays(1, intArrayOf(vaoId), 0)
158     vaoId = 0
159     quadBufferId = 0
160     instanceBufferId = 0
161     uCanvasToClip = -1
162     uColorLinear = -1
163     uStrokeOpacity = -1
164 }
165
166 fun beginStroke() {
167     uploadedDabCount = 0
168 }
169
170 fun clearStrokeData() {
171     uploadedDabCount = 0
172 }
173
174 fun setGrainTexture(
175     textureId: Int,
176     scale: Float,
177     canvasLocked: Boolean,
178     depth: Float,
179     contrast: Float,
180 ) {
181     grainTextureId = textureId
182     grainScale = scale.coerceAtLeast(0.0001f)

```

```

183         grainCanvasLocked = canvasLocked
184         textureDepth = depth.coerceIn(0f, 1f)
185         textureContrast = contrast.coerceIn(0f, 2f)
186     }
187
188     fun clearGrainTexture() {
189         grainTextureId = 0
190         grainScale = 1f
191         grainCanvasLocked = true
192         textureDepth = 1f
193         textureContrast = 1f
194     }
195
196     /** Sets a tip mask sampled once per dab in local dab coordinates. */
197     fun setShapeTexture(textureId: Int, reverse: Boolean, rgbToAlpha: Boolean) {
198         shapeTextureId = textureId
199         reverseShape = reverse
200         this.rgbToAlpha = rgbToAlpha
201     }
202
203     fun clearShapeTexture() {
204         shapeTextureId = 0
205         reverseShape = false
206         rgbToAlpha = false
207     }
208
209     fun drawPendingInto(
210         target: RenderTarget,
211         width: Int,
212         height: Int,
213         canvasToFbo: FloatArray,
214         dabs: DabBuffer,
215         colorLinear: FloatArray,
216         blendPolicy: BlendPolicy,
217         strokeOpacity: Float = 1f,
218     ) {
219         val first = uploadedDabCount
220         val count = dabs.count - first
221         if (count <= 0) return
222
223         dabs.prepareForUpload(first, count)
224         try {
225             drawRange(
226                 target = target,
227                 width = width,
228                 height = height,
229                 canvasToFbo = canvasToFbo,
230                 dabs = dabs,
231                 firstDab = first,
232                 count = count,
233                 colorLinear = colorLinear,
234                 blendPolicy = blendPolicy,
235                 strokeOpacity = strokeOpacity,
236             )
237             uploadedDabCount = dabs.count
238         } finally {
239             dabs.finishUpload()
240         }
241     }
242
243     private fun drawRange(
244         target: RenderTarget,
245         width: Int,
246         height: Int,
247         canvasToFbo: FloatArray,
248         dabs: DabBuffer,
249         firstDab: Int,
250         count: Int,
251         colorLinear: FloatArray,
252         blendPolicy: BlendPolicy,
253         strokeOpacity: Float,
254     ) {
255         val currentProgram = program ?: return
256         target.bind()
257         GLES30.glViewport(0, 0, width, height)
258         blendController.begin(blendPolicy)
259         currentProgram.use()
260         GLES30.glUniformMatrix4fv(uCanvasToClip, 1, false, canvasToFbo, 0)
261         GLES30.glUniform3f(uColorLinear, colorLinear[0], colorLinear[1], colorLinear[2])
262
263         GLES30.glUniform2f(uCanvasSize, width.toFloat(), height.toFloat())
264         GLES30.glUniform1i(uGrainActive, if (grainTextureId != 0) 1 else 0)
265         GLES30.glUniform1f(uGrainScale, grainScale)
266         GLES30.glUniform1i(uGrainCanvasLocked, if (grainCanvasLocked) 1 else 0)
267         GLES30.glUniform1f(uTextureDepth, textureDepth)
268         GLES30.glUniform1f(uTextureContrast, textureContrast)
269         GLES30.glUniform1f(uStrokeOpacity, strokeOpacity.coerceIn(0f, 1f))
270         GLES30.glUniform1i(uShapeActive, if (shapeTextureId != 0) 1 else 0)
271         GLES30.glUniform1i(uReverseShape, if (reverseShape) 1 else 0)
272         GLES30.glUniform1i(uRgbToAlpha, if (rgbToAlpha) 1 else 0)
273
274         if (grainTextureId != 0) {

```

```

275         GLES30.glActiveTexture(GLES30.GL_TEXTURE2)
276         GLES30.glBindTexture(GLES30.GL_TEXTURE_2D, grainTextureId)
277         GLES30.glUniform1i(uGrainTex, 2)
278     }
279     if (shapeTextureId != 0) {
280         GLES30.glActiveTexture(GLES30.GL_TEXTURE3)
281         GLES30.glBindTexture(GLES30.GL_TEXTURE_2D, shapeTextureId)
282         GLES30.glUniform1i(uShapeTex, 3)
283     }
284
285     GLES30.glBindVertexArray(vaoId)
286     GLES30.glBindBuffer(GLES30.GL_ARRAY_BUFFER, instanceBufferId)
287     GLES30.glBufferSubData(
288         GLES30.GL_ARRAY_BUFFER,
289         firstDab * DabBuffer.FLOATS_PER_DAB * Float.SIZE_BYTES,
290         count * DabBuffer.FLOATS_PER_DAB * Float.SIZE_BYTES,
291         dabs.floats,
292     )
293
294     // Reset offsets for instancing
295     GLES30.glVertexAttribPointer(
296         1, 4, GLES30.GL_FLOAT, false, DabBuffer.FLOATS_PER_DAB * Float.SIZE_BYTES, 0
297     )
298     GLES30.glVertexAttribPointer(
299         2, 2, GLES30.GL_FLOAT, false, DabBuffer.FLOATS_PER_DAB * Float.SIZE_BYTES, 16
300     )
301
302     // Manual attribute offset for ES 3.0 (since BaseInstance is 3.1+)
303     GLES30.glVertexAttribPointer(
304         1, 4, GLES30.GL_FLOAT, false, DabBuffer.FLOATS_PER_DAB * Float.SIZE_BYTES,
305         (firstDab * DabBuffer.FLOATS_PER_DAB * Float.SIZE_BYTES)
306     )
307     GLES30.glVertexAttribPointer(
308         2, 2, GLES30.GL_FLOAT, false, DabBuffer.FLOATS_PER_DAB * Float.SIZE_BYTES,
309         (firstDab * DabBuffer.FLOATS_PER_DAB * Float.SIZE_BYTES + 16)
310     )
311
312     GLES30.glDrawArraysInstanced(GLES30.GL_TRIANGLE_STRIP, 0, 4, count)
313
314     if (grainTextureId != 0) {
315         GLES30.glActiveTexture(GLES30.GL_TEXTURE2)
316         GLES30.glBindTexture(GLES30.GL_TEXTURE_2D, 0)
317     }
318     if (shapeTextureId != 0) {
319         GLES30.glActiveTexture(GLES30.GL_TEXTURE3)
320         GLES30.glBindTexture(GLES30.GL_TEXTURE_2D, 0)
321     }
322     GLES30.glActiveTexture(GLES30.GL_TEXTURE0)
323
324     GLES30.glBindBuffer(GLES30.GL_ARRAY_BUFFER, 0)
325     GLES30.glBindVertexArray(0)
326     blendController.end()
327     GLES30.glBindFramebuffer(GLES30.GL_FRAMEBUFFER, 0)
328 }
329 }

```

21. app/src/main/java/com/wetinknext/engine/brush/OneEuroFilter.kt

```
1 package com.wetinknext.engine.brush
2
3 import kotlin.math.PI
4 import kotlin.math.sqrt
5
6 class OneEuroFilter(private var minCutoff: Float = 1.6f, private var beta: Float = 0.25f, private var dCutoff: Float = 1f) {
7     private var initialized = false; private var lastTimestampNanos = 0L
8     private var x = 0f; private var y = 0f; private var dx = 0f; private var dy = 0f
9     fun reset() { initialized = false; lastTimestampNanos = 0L; x = 0f; y = 0f; dx = 0f; dy = 0f }
10    fun filter(timestampNanos: Long, rawX: Float, rawY: Float, out: FloatArray) {
11        require(out.size >= 2)
12        if (!initialized) { initialized = true; lastTimestampNanos = timestampNanos; x = rawX; y = rawY; out[0] = x; out[1] =
y; return }
13        var dt = (timestampNanos - lastTimestampNanos) / 1_000_000_000f
14        if (dt < MIN_DT) dt = MIN_DT
15        lastTimestampNanos = timestampNanos
16        val rate = 1f / dt; val alphaD = alpha(rate, dCutoff)
17        dx = alphaD * ((rawX - x) * rate) + (1f - alphaD) * dx; dy = alphaD * ((rawY - y) * rate) + (1f - alphaD) * dy
18        val alpha = alpha(rate, (minCutoff + beta * sqrt(dx * dx + dy * dy)).coerceAtLeast(0.01f))
19        x = alpha * rawX + (1f - alpha) * x; y = alpha * rawY + (1f - alpha) * y; out[0] = x; out[1] = y
20    }
21    private fun alpha(rate: Float, cutoff: Float): Float { val tau = 1f / (2f * PI.toFloat() * cutoff); return 1f / (1f + tau
* rate) }
22    private companion object { const val MIN_DT = 1f / 1000f }
23 }
```

22. app/src/main/java/com/wetinknext/engine/brush/PressureFilter.kt

```
1 package com.wetinknext.engine.brush
2
3 import kotlin.math.exp
4
5 /**
6  * Time-based stylus pressure smoothing.
7  * A zero pressure sample while the pointer is still in contact is treated as a
8  * device glitch; Android sends the real zero value on UP after the final point.
9  */
10 class PressureFilter {
11     private var initialized = false
12     private var lastTimestampNanos = 0L
13     private var value = 0f
14
15     fun reset() { initialized = false; lastTimestampNanos = 0L; value = 0f }
16
17     fun filter(timestampNanos: Long, rawPressure: Float): Float {
18         val raw = rawPressure.coerceIn(0f, 1f)
19         if (!initialized) { initialized = true; lastTimestampNanos = timestampNanos; value = raw; return value }
20         val dt = ((timestampNanos - lastTimestampNanos) / NANOS_PER_SECOND).coerceIn(MIN_DT, MAX_DT)
21         lastTimestampNanos = timestampNanos
22         val stableRaw = if (raw <= ZERO_GLITCH_THRESHOLD && value > ZERO_GLITCH_THRESHOLD) value else raw
23         val tau = if (stableRaw >= value) RISE_TAU_SECONDS else FALL_TAU_SECONDS
24         val alpha = 1f - exp((-dt / tau).toDouble()).toFloat()
25         value += (stableRaw - value) * alpha
26         return value.coerceIn(0f, 1f)
27     }
28
29     private companion object {
30         const val NANOS_PER_SECOND = 1_000_000_000f
31         const val MIN_DT = 1f / 1000f
32         const val MAX_DT = 1f / 20f
33         const val RISE_TAU_SECONDS = .018f
34         const val FALL_TAU_SECONDS = .055f
35         const val ZERO_GLITCH_THRESHOLD = .01f
36     }
37 }
```

23. app/src/main/java/com/wetinknext/engine/brush/Stabilizer.kt

```
1 package com.wetinknext.engine.brush
2
3 import kotlin.math.sqrt
4
5 class Stabilizer(private val filter: OneEuroFilter = OneEuroFilter()) {
6     var strength = 0f; var x = 0f; private set; var y = 0f; private set; var velocity = 0f; private set
7     private var hasLast = false; private var lastX = 0f; private var lastY = 0f; private var lastT = 0L; private val filtered
8     = FloatArray(2)
9     fun reset() { filter.reset(); x = 0f; y = 0f; velocity = 0f; hasLast = false; lastX = 0f; lastY = 0f; lastT = 0L }
10    fun process(timestampNanos: Long, rawX: Float, rawY: Float) {
11        if (strength <= 0f) { x = rawX; y = rawY } else { filter.filter(timestampNanos, rawX, rawY, filtered); x = filtered[
12            0]; y = filtered[1] }
13        if (hasLast) { var dt = (timestampNanos - lastT) / 1_000_000_000f; if (dt < 1f / 1000f) dt = 1f / 1000f; val d = sqrt(
14            (x-lastX)*(x-lastX)+(y-lastY)*(y-lastY)); velocity += 0.2f * (d / dt - velocity) }
15        hasLast = true; lastX = x; lastY = y; lastT = timestampNanos
16    }
17 }
```

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```

91     }
92   }
93 }
94 }
95
96 fun finish(out: DabBuffer, cancel: Boolean) {
97   if (!active) return
98   val settings = activeSettings()
99   if (cancel) {
100     reset()
101     return
102   }
103   if (settings.smoothing > .0001f && pointCount >= 3) {
104     flushSmoothedTail(out)
105   }
106   if (!movedDuringStroke || carriedDistance > .001f) {
107     addDab(
108       x = lastX,
109       y = lastY,
110       pressure = lastPressure,
111       tiltX = lastTiltX,
112       tiltY = lastTiltY,
113       out = out,
114     )
115   }
116   reset()
117 }
118
119 private fun appendSmoothedPoint(x: Float, y: Float, pressure: Float, out: DabBuffer) {
120   p0x = p1x; p0y = p1y; p0p = p1p
121   p1x = p2x; p1y = p2y; p1p = p2p
122   p2x = p3x; p2y = p3y; p2p = p3p
123   p3x = x; p3y = y; p3p = pressure
124   pointCount = (pointCount + 1).coerceAtMost(4)
125
126   when (pointCount) {
127     2 -> addPoint(x, y, pressure, lastTiltX, lastTiltY, out)
128     4 -> interpolateCatmullRom(out)
129   }
130 }
131
132 /** Processes the last delayed Catmull-Rom segment with a duplicated endpoint. */
133 private fun flushSmoothedTail(out: DabBuffer) {
134   p0x = p1x; p0y = p1y; p0p = p1p
135   p1x = p2x; p1y = p2y; p1p = p2p
136   p2x = p3x; p2y = p3y; p2p = p3p
137   p3x = p2x; p3y = p2y; p3p = p2p
138   pointCount = 4
139   interpolateCatmullRom(out)
140 }
141
142 private fun interpolateCatmullRom(out: DabBuffer) {
143   val settings = activeSettings()
144   val dx = p2x - p1x
145   val dy = p2y - p1y
146   val distance = sqrt(dx * dx + dy * dy)
147   if (distance < .0001f) return
148
149   val steps = (distance / maxOf(1f, settings.baseRadiusPx * .25f))
150     .toInt()
151     .coerceIn(2, 64)
152
153   for (index in 1..steps) {
154     val t = index.toFloat() / steps
155     val t2 = t * t
156     val t3 = t2 * t
157     val x = .5f * (
158       2f * p1x + (-p0x + p2x) * t +
159       (2f * p0x - 5f * p1x + 4f * p2x - p3x) * t2 +
160       (-p0x + 3f * p1x - 3f * p2x + p3x) * t3
161     )
162     val y = .5f * (
163       2f * p1y + (-p0y + p2y) * t +
164       (2f * p0y - 5f * p1y + 4f * p2y - p3y) * t2 +
165       (-p0y + 3f * p1y - 3f * p2y + p3y) * t3
166     )
167     val pressure = p1p + (p2p - p1p) * t
168     addPoint(x, y, pressure, lastTiltX, lastTiltY, out)
169   }
170 }
171
172 private fun addPoint(x: Float, y: Float, p: Float, tx: Float, ty: Float, out: DabBuffer) {
173   val settings = activeSettings()
174   val dx = x - lastX
175   val dy = y - lastY
176   val distance = sqrt(dx * dx + dy * dy)
177   if (distance <= .0001f) return
178   if (distance > 1.5f) movedDuringStroke = true
179
180   var travelled = 0f
181   while (travelled < distance) {
182     val tProbe = (travelled / distance).coerceIn(0f, 1f)

```



```

183         val probePressure = lastPressure + (p - lastPressure) * tProbe
184         val probeTiltX = lastTiltX + (tx - lastTiltX) * tProbe
185         val probeTiltY = lastTiltY + (ty - lastTiltY) * tProbe
186
187         BrushDynamics.resolve(
188             settings = settings,
189             pressure = probePressure,
190             tiltX = probeTiltX,
191             tiltY = probeTiltY,
192             out = resolvedDab,
193         )
194
195         val step = spacingForRadius(resolvedDab.radius)
196         // Radius can shrink with pressure, leaving more carried distance
197         // than the new step. Emit at the current point and reset it.
198         val needed = (step - carriedDistance).coerceAtLeast(0f)
199
200         if (travelled + needed > distance) {
201             carriedDistance += distance - travelled
202             break
203         }
204
205         travelled += needed
206         val t = (travelled / distance).coerceIn(0f, 1f)
207         addDab(
208             x = lastX + dx * t,
209             y = lastY + dy * t,
210             pressure = lastPressure + (p - lastPressure) * t,
211             tiltX = lastTiltX + (tx - lastTiltX) * t,
212             tiltY = lastTiltY + (ty - lastTiltY) * t,
213             out = out,
214         )
215         carriedDistance = 0f
216     }
217     lastX = x
218     lastY = y
219     lastPressure = p
220     lastTiltX = tx
221     lastTiltY = ty
222 }
223
224 private fun spacingForRadius(radius: Float): Float {
225     val settings = activeSettings()
226     val unit = if (settings.spacingUsesDiameter) radius * 2f else radius
227     return (unit * settings.spacing).coerceIn(.25f, 256f)
228 }
229
230 private fun addDab(
231     x: Float,
232     y: Float,
233     pressure: Float,
234     tiltX: Float,
235     tiltY: Float,
236     out: DabBuffer,
237 ) {
238     val settings = activeSettings()
239     BrushDynamics.resolve(
240         settings = settings,
241         pressure = pressure,
242         tiltX = tiltX,
243         tiltY = tiltY,
244         out = resolvedDab,
245     )
246     out.add(
247         x = x,
248         y = y,
249         radius = resolvedDab.radius,
250         rotation = 0f,
251         coverage = resolvedDab.coverage,
252         flow = settings.flow.coerceIn(0f, 1f),
253     )
254 }
255 }

```

25. app/src/main/java/com/wetinknext/engine/brush/TextureLoader.kt

```

1  package com.wetinknext.engine.brush
2
3  import android.content.Context
4  import android.graphics.Bitmap
5  import android.graphics.BitmapFactory
6  import java.nio.ByteBuffer
7  import java.nio.ByteOrder
8  import java.util.concurrent.ExecutorService
9  import java.util.concurrent.Executors
10
11 data class LoadedBrushTexture(
12     val path: String,
13     val width: Int,
14     val height: Int,
15     val rgba: ByteBuffer,
16 )
17
18 class TextureLoader(
19     private val context: Context,
20     private val executor: ExecutorService =
21         Executors.newSingleThreadExecutor(),
22 ) {
23     fun loadAsync(
24         path: String,
25         onLoad: (LoadedBrushTexture) -> Unit,
26         onError: (Throwable) -> Unit,
27     ) {
28         executor.execute {
29             try {
30                 val loaded = decode(path)
31                 onLoad(loaded)
32             } catch (error: Throwable) {
33                 onError(error)
34             }
35         }
36     }
37
38     private fun decode(path: String): LoadedBrushTexture {
39         val source = decodeBitmap(path)
40
41         val bitmap = if (source.config == Bitmap.Config.ARGB_8888) {
42             source
43         } else {
44             source.copy(Bitmap.Config.ARGB_8888, false)
45         }
46
47         val width = bitmap.width
48         val height = bitmap.height
49
50         val rgba = ByteBuffer
51             .allocateDirect(width * height * 4)
52             .order(ByteOrder.nativeOrder())
53
54         // Bitmap stores ARGB_8888 pixels as packed ARGB ints. Uploading that
55         // buffer directly makes the byte order platform-dependent; GLES expects
56         // the explicit RGBA byte sequence declared in BrushTexture.
57         val pixels = IntArray(width * height)
58         bitmap.getPixels(
59             pixels,
60             0,
61             width,
62             0,
63             0,
64             width,
65             height,
66         )
67         for (pixel in pixels) {
68             rgba.put(((pixel ushr 16) and 0xFF).toByte())
69             rgba.put(((pixel ushr 8) and 0xFF).toByte())
70             rgba.put((pixel and 0xFF).toByte())
71             rgba.put(((pixel ushr 24) and 0xFF).toByte())
72         }
73         rgba.flip()
74
75         if (bitmap != source && !bitmap.isRecycled) {
76             bitmap.recycle()
77         }
78
79         if (!source.isRecycled && source != bitmap) {
80             source.recycle()
81         }
82
83         return LoadedBrushTexture(
84             path = path,
85             width = width,
86             height = height,
87             rgba = rgba,
88         )
89     }
90 }

```

```
91     private fun decodeBitmap(path: String): Bitmap {
92         val bitmap = if (path.startsWith("asset:")) {
93             val assetPath = path.removePrefix("asset:")
94
95             context.assets.open(assetPath).use { input ->
96                 BitmapFactory.decodeStream(input)
97             }
98         } else {
99             BitmapFactory.decodeFile(path)
100         }
101
102         return requireNotNull(bitmap) {
103             "Cannot decode brush texture: $path"
104         }
105     }
106
107     fun shutdown() {
108         executor.shutdown()
109     }
110 }
```

26. app/src/main/java/com/wetinknext/engine/canvas/BlendMode.kt

```
1 package com.wetinknext.engine.canvas
2
3 /** Stored per layer now; shader implementations beyond NORMAL are scheduled after P6. */
4 enum class BlendMode(val id: Int) {
5     NORMAL(0),
6     MULTIPLY(1),
7     SCREEN(2),
8     OVERLAY(3),
9     DARKEN(4),
10    LIGHTEN(5),
11    ADD(6),
12 }
```

27. app/src/main/java/com/wetinknext/engine/canvas/Compositor.kt

```

1  package com.wetinknext.engine.canvas
2
3  import android.opengl.GLES30
4  import com.wetinknext.engine.gl.CanvasGeometry
5  import com.wetinknext.engine.gl.GlCheck
6  import com.wetinknext.engine.gl.GlProgram
7  import com.wetinknext.engine.gl.ShaderLib
8
9  /** Presents visible layers bottom-to-top, inserting the active preview at its layer position. */
10 class Compositor {
11     private var program: GlProgram? = null
12     private var uCanvasToClip = -1
13     private var uCanvasSize = -1
14     private var uLayerTex = -1
15     private var uStrokeTex = -1
16     private var uStrokeActive = -1
17     private var uOpacity = -1
18     private var uStrokeOpacity = -1
19
20     fun create() {
21         GlCheck.checkOnGltThread()
22         release()
23         program = GlProgram(ShaderLib.compositorVertex, ShaderLib.compositorFragment)
24         val currentProgram = checkNotNull(program)
25         currentProgram.use()
26         uCanvasToClip = GLES30.glGetUniformLocation(currentProgram.id, "uCanvasToClip")
27         uCanvasSize = GLES30.glGetUniformLocation(currentProgram.id, "uCanvasSize")
28         uLayerTex = GLES30.glGetUniformLocation(currentProgram.id, "uLayerTex")
29         uStrokeTex = GLES30.glGetUniformLocation(currentProgram.id, "uStrokeTex")
30         uStrokeActive = GLES30.glGetUniformLocation(currentProgram.id, "uStrokeActive")
31         uOpacity = GLES30.glGetUniformLocation(currentProgram.id, "uOpacity")
32         uStrokeOpacity = GLES30.glGetUniformLocation(currentProgram.id, "uStrokeOpacity")
33         check(uCanvasToClip >= 0 && uCanvasSize >= 0 && uLayerTex >= 0 &&
34             uStrokeTex >= 0 && uStrokeActive >= 0 && uOpacity >= 0 && uStrokeOpacity >= 0) {
35             "Compositor uniforms missing: canvasToClip=$uCanvasToClip, " +
36             "canvasSize=$uCanvasSize, layerTex=$uLayerTex, strokeTex=$uStrokeTex, " +
37             "strokeActive=$uStrokeActive, opacity=$uOpacity, strokeOpacity=$uStrokeOpacity"
38         }
39         GlCheck.noError("Compositor create")
40     }
41
42     fun render(
43         geometry: CanvasGeometry,
44         layers: LayerStack,
45         activeLayerId: Long,
46         strokeTextureId: Int,
47         canvasToClip: FloatArray,
48         strokeOpacity: Float,
49     ) {
50         val currentProgram = program ?: return
51         currentProgram.use()
52         GLES30.glUniformMatrix4fv(uCanvasToClip, 1, false, canvasToClip, 0)
53         GLES30.glUniform2f(uCanvasSize, layers.canvasWidth.toFloat(), layers.canvasHeight.toFloat())
54         GLES30.glEnable(GLES30.GL_BLEND)
55         GLES30.glBlendFunc(GLES30.GL_ONE, GLES30.GL_ONE_MINUS_SRC_ALPHA)
56
57         for (layer in layers.allLayers()) {
58             if (!layer.created || !layer.isVisible) continue
59             val hasStroke = layer.id == activeLayerId && strokeTextureId != 0
60
61             GLES30.glActiveTexture(GLES30.GL_TEXTURE0)
62             GLES30.glBindTexture(GLES30.GL_TEXTURE_2D, layer.target.textureId)
63             GLES30.glUniform1i(uLayerTex, 0)
64             if (hasStroke) {
65                 GLES30.glActiveTexture(GLES30.GL_TEXTURE1)
66                 GLES30.glBindTexture(GLES30.GL_TEXTURE_2D, strokeTextureId)
67                 GLES30.glUniform1i(uStrokeTex, 1)
68             }
69             GLES30.glUniform1i(uStrokeActive, if (hasStroke) 1 else 0)
70             GLES30.glUniform1f(uOpacity, layer.opacity.coerceIn(0f, 1f))
71             GLES30.glUniform1f(uStrokeOpacity, strokeOpacity.coerceIn(0f, 1f))
72             geometry.draw()
73         }
74
75         GLES30.glDisable(GLES30.GL_BLEND)
76         GLES30.glActiveTexture(GLES30.GL_TEXTURE1)
77         GLES30.glBindTexture(GLES30.GL_TEXTURE_2D, 0)
78         GLES30.glActiveTexture(GLES30.GL_TEXTURE0)
79         GLES30.glBindTexture(GLES30.GL_TEXTURE_2D, 0)
80     }
81
82     fun release() {
83         GlCheck.checkOnGltThread()
84         program?.release()
85         program = null
86         uCanvasToClip = -1
87         uCanvasSize = -1
88         uLayerTex = -1
89         uStrokeTex = -1
90         uStrokeActive = -1

```

```
91         uOpacity = -1
92         uStrokeOpacity = -1
93     }
94 }
```

28. app/src/main/java/com/wetinknext/engine/canvas/LayerStack.kt

```

1  package com.wetinknext.engine.canvas
2
3  import com.wetinknext.engine.core.Camera
4  import com.wetinknext.engine.gl.GlCaps
5
6  /**
7   * Render-thread-owned ordered collection of document layers.
8   * Index 0 is the bottom-most layer and the final item is the top-most layer.
9   */
10 class LayerStack {
11     val camera = Camera()
12
13     private val layers = mutableListOf<PaintLayer>()
14     private var nextId = 1L
15     private var documentUsesHalfFloat = false
16
17     var activeLayerId: Long = NO_LAYER
18     private set
19     var canvasWidth: Int = 0
20     private set
21     var canvasHeight: Int = 0
22     private set
23
24     val count: Int get() = layers.size
25
26     fun allLayers(): List<PaintLayer> = layers
27
28     fun activeLayer(): PaintLayer? = findLayerById(activeLayerId)
29
30     fun findLayerById(id: Long): PaintLayer? = layers.firstOrNull { it.id == id }
31
32     /** Creates the locked opaque white background and one transparent drawing layer. */
33     fun create(caps: GlCaps, width: Int, height: Int) {
34         require(width > 0 && height > 0)
35         release()
36         canvasWidth = width
37         canvasHeight = height
38         documentUsesHalfFloat = caps.supportsHalfFloatColorBuffer
39
40         val background = newLayer("Фон", documentUsesHalfFloat).also {
41             it.isLocked = true
42             it.target.clear(1f, 1f, 1f, 1f)
43         }
44         layers += background
45
46         val drawing = newLayer("Слой 1", documentUsesHalfFloat)
47         layers += drawing
48         activeLayerId = drawing.id
49     }
50
51     fun addLayer(name: String, insertAt: Int = layers.size): PaintLayer {
52         check(canvasWidth > 0 && canvasHeight > 0) { "LayerStack has not been created" }
53         val layer = newLayer(name, documentUsesHalfFloat)
54         layers.add(insertAt.coerceIn(0, layers.size), layer)
55         activeLayerId = layer.id
56         return layer
57     }
58
59     /** A document must retain at least one layer; locked layers cannot be removed. */
60     fun removeLayer(id: Long): PaintLayer? {
61         val index = layers.indexOfFirst { it.id == id }
62         if (index < 0 || layers.size <= 1) return null
63         val layer = layers[index]
64         if (layer.isLocked) return null
65
66         layers.removeAt(index)
67         layer.release()
68         if (activeLayerId == id) {
69             activeLayerId = layers[index.coerceAtMost(layers.lastIndex)].id
70         }
71         return layer
72     }
73
74     fun setActive(id: Long): Boolean {
75         if (findLayerById(id) == null) return false
76         activeLayerId = id
77         return true
78     }
79
80     fun moveLayer(id: Long, delta: Int): Boolean {
81         val index = layers.indexOfFirst { it.id == id }
82         if (index < 0) return false
83         val newIndex = (index + delta).coerceIn(0, layers.lastIndex)
84         if (newIndex == index) return false
85         val layer = layers.removeAt(index)
86         layers.add(newIndex, layer)
87         return true
88     }
89
90     fun release() {

```

```
91         layers.forEach(PaintLayer::release)
92         layers.clear()
93         activeLayerId = NO_LAYER
94         canvasWidth = 0
95         canvasHeight = 0
96         documentUsesHalfFloat = false
97     }
98
99     private fun newLayer(name: String, useHalfFloat: Boolean): PaintLayer =
100         PaintLayer(nextId++, name).also { it.create(canvasWidth, canvasHeight, useHalfFloat) }
101
102     private companion object {
103         const val NO_LAYER = -1L
104     }
105 }
```


29. app/src/main/java/com/wetinknext/engine/canvas/PaintLayer.kt

```
1 package com.wetinknext.engine.canvas
2
3 import com.wetinknext.engine.gl.RenderTarget
4
5 /** One paintable document layer and its private GPU target. */
6 class PaintLayer(
7     val id: Long,
8     var name: String,
9 ) {
10     val target = RenderTarget()
11
12     var isVisible = true
13     var isLocked = false
14     var opacity = 1f
15     var blendMode = BlendMode.NORMAL
16     var version = 0L
17
18     var created = false
19     private set
20
21     fun create(width: Int, height: Int, preferHalfFloat: Boolean) {
22         if (created && target.width == width && target.height == height) return
23         target.create(width, height, preferHalfFloat)
24         target.clear(0f, 0f, 0f, 0f)
25         created = true
26     }
27
28     fun clear() {
29         if (created) target.clear(0f, 0f, 0f, 0f)
30     }
31
32     fun release() {
33         target.release()
34         created = false
35     }
36 }
```

30. app/src/main/java/com/wetinknext/engine/canvas/StrokeBlitter.kt

```
1 package com.wetinknext.engine.canvas
2
3 import android.opengl.GLES30
4 import com.wetinknext.engine.gl.CanvasGeometry
5 import com.wetinknext.engine.gl.GlProgram
6 import com.wetinknext.engine.gl.RenderTarget
7 import com.wetinknext.engine.gl.ShaderLib
8
9 /** Переносит готовый штрих из strokeTarget в слой: premultiplied source-over * opacity. */
10 class StrokeBlitter {
11     private var program: GlProgram? = null
12     private var uCanvasToClip = -1
13     private var uCanvasSize = -1
14     private var uStrokeTex = -1
15     private var uOpacity = -1
16
17     fun create() {
18         release()
19         program = GlProgram(ShaderLib.compositorVertex, ShaderLib.strokeBlitFragment)
20         val p = checkNotNull(program)
21         p.use()
22         uCanvasToClip = GLES30.glGetUniformLocation(p.id, "uCanvasToClip")
23         uCanvasSize = GLES30.glGetUniformLocation(p.id, "uCanvasSize")
24         uStrokeTex = GLES30.glGetUniformLocation(p.id, "uStrokeTex")
25         uOpacity = GLES30.glGetUniformLocation(p.id, "uOpacity")
26         check(uCanvasToClip >= 0 && uCanvasSize >= 0 && uStrokeTex >= 0 && uOpacity >= 0) {
27             "StrokeBlitter uniforms missing"
28         }
29     }
30
31     fun blit(
32         layer: RenderTarget,
33         geometry: CanvasGeometry,
34         strokeTextureId: Int,
35         canvasToFbo: FloatArray,
36         width: Int,
37         height: Int,
38         opacity: Float,
39     ) {
40         val p = program ?: return
41         if (strokeTextureId == 0) return
42         layer.bind()
43         GLES30.glViewport(0, 0, width, height)
44         GLES30.glDisable(GLES30.GL_SCISSOR_TEST)
45         GLES30.glEnable(GLES30.GL_BLEND)
46         GLES30.glBlendEquation(GLES30.GL_FUNC_ADD)
47         GLES30.glBlendFunc(GLES30.GL_ONE, GLES30.GL_ONE_MINUS_SRC_ALPHA)
48         p.use()
49         GLES30.glUniformMatrix4fv(uCanvasToClip, 1, false, canvasToFbo, 0)
50         GLES30.glUniform2f(uCanvasSize, width.toFloat(), height.toFloat())
51         GLES30.glActiveTexture(GLES30.GL_TEXTURE0)
52         GLES30.glBindTexture(GLES30.GL_TEXTURE_2D, strokeTextureId)
53         GLES30.glUniform1i(uStrokeTex, 0)
54         GLES30.glUniform1f(uOpacity, opacity.coerceIn(0f, 1f))
55         geometry.draw()
56         GLES30.glDisable(GLES30.GL_BLEND)
57         GLES30.glBindTexture(GLES30.GL_TEXTURE_2D, 0)
58         GLES30.glBindFramebuffer(GLES30.GL_FRAMEBUFFER, 0)
59     }
60
61     fun release() {
62         program?.release()
63         program = null
64         uCanvasToClip = -1; uCanvasSize = -1; uStrokeTex = -1; uOpacity = -1
65     }
66 }
```

31. app/src/main/java/com/wetinknext/engine/core/Camera.kt

```
1 package com.wetinknext.engine.core
2
3 import java.util.concurrent.atomic.AtomicReference
4 import kotlin.math.min
5
6 /** Thread-safe camera state shared by the UI and GL threads. */
7 class Camera(initial: ViewTransform = ViewTransform()) {
8     private val state = AtomicReference(initial)
9
10    fun snapshot(): ViewTransform = state.get()
11
12    fun set(transform: ViewTransform) {
13        state.set(transform)
14    }
15
16    fun update(transform: (ViewTransform) -> ViewTransform) {
17        while (true) {
18            val current = state.get()
19            if (state.compareAndSet(current, transform(current))) return
20        }
21    }
22
23    fun reset() {
24        state.set(ViewTransform())
25    }
26
27    fun fitCanvas(
28        canvasWidth: Int,
29        canvasHeight: Int,
30        viewWidth: Int,
31        viewHeight: Int,
32        marginRatio: Float = 0.04f,
33    ) {
34        if (canvasWidth <= 0 || canvasHeight <= 0 || viewWidth <= 0 || viewHeight <= 0) return
35        val usableWidth = viewWidth * (1f - marginRatio * 2f)
36        val usableHeight = viewHeight * (1f - marginRatio * 2f)
37        val scale = min(usableWidth / canvasWidth, usableHeight / canvasHeight)
38        .coerceIn(ViewTransform.MIN_SCALE, ViewTransform.MAX_SCALE)
39        state.set(
40            ViewTransform(
41                scale = scale,
42                translateX = (viewWidth - canvasWidth * scale) * 0.5f,
43                translateY = (viewHeight - canvasHeight * scale) * 0.5f,
44            ),
45        )
46    }
47 }
```

32. app/src/main/java/com/wetinknext/engine/core/DirtyRect.kt

```
1 package com.wetinknext.engine.core
2
3 import kotlin.math.ceil
4 import kotlin.math.floor
5 import kotlin.math.max
6 import kotlin.math.min
7
8 /** Reusable mutable dirty rectangle in canvas coordinates. */
9 class DirtyRect {
10     var left = 0f; private set
11     var top = 0f; private set
12     var right = 0f; private set
13     var bottom = 0f; private set
14     var isEmpty = true; private set
15
16     fun clear() {
17         left = 0f; top = 0f; right = 0f; bottom = 0f; isEmpty = true
18     }
19
20     fun include(x: Float, y: Float, radius: Float) = include(x - radius, y - radius, x + radius, y + radius)
21
22     fun include(left: Float, top: Float, right: Float, bottom: Float) {
23         if (right < left || bottom < top) return
24         if (isEmpty) {
25             this.left = left; this.top = top; this.right = right; this.bottom = bottom; isEmpty = false
26             return
27         }
28         this.left = min(this.left, left); this.top = min(this.top, top)
29         this.right = max(this.right, right); this.bottom = max(this.bottom, bottom)
30     }
31
32     fun include(other: DirtyRect) {
33         if (!other.isEmpty) include(other.left, other.top, other.right, other.bottom)
34     }
35
36     fun expand(pixels: Float) {
37         if (!isEmpty && pixels > 0f) {
38             left -= pixels; top -= pixels; right += pixels; bottom += pixels
39         }
40     }
41
42     fun clamp(canvasWidth: Float, canvasHeight: Float) {
43         if (isEmpty) return
44         left = left.coerceIn(0f, canvasWidth); top = top.coerceIn(0f, canvasHeight)
45         right = right.coerceIn(0f, canvasWidth); bottom = bottom.coerceIn(0f, canvasHeight)
46         if (right <= left || bottom <= top) clear()
47     }
48
49     fun copyTo(out: DirtyRect) {
50         out.left = left; out.top = top; out.right = right; out.bottom = bottom; out.isEmpty = isEmpty
51     }
52
53     fun toPixelBounds(out: IntArray) {
54         require(out.size >= 4)
55         if (isEmpty) {
56             out[0] = 0; out[1] = 0; out[2] = 0; out[3] = 0
57             return
58         }
59         out[0] = floor(left).toInt(); out[1] = floor(top).toInt()
60         out[2] = ceil(right).toInt(); out[3] = ceil(bottom).toInt()
61     }
62 }
```

33. app/src/main/java/com/wetinknext/engine/core/EditorState.kt

```
1 package com.wetinknext.engine.core
2
3 /** Immutable state published by the GL thread for Compose to render. */
4 data class LayerUiModel(
5     val id: Long,
6     val name: String,
7     val isVisible: Boolean,
8     val isLocked: Boolean,
9     val opacity: Float,
10    val active: Boolean,
11    val canDelete: Boolean,
12 )
13
14 data class UndoDiagnostics(
15     val pendingJobs: Int = 0,
16     val staleResults: Int = 0,
17     val compressionFailures: Int = 0,
18     val restoreFailures: Int = 0,
19     val memoryBytes: Long = 0L,
20 )
21
22 data class EditorUiState(
23     val layers: List<LayerUiModel>,
24     val canUndo: Boolean,
25     val canRedo: Boolean,
26     val brushSizePx: Float,
27     val brushOpacity: Float,
28     val activeLayerId: Long,
29     val ready: Boolean,
30     val undoDiagnostics: UndoDiagnostics = UndoDiagnostics(),
31 ) {
32     companion object {
33         val empty = EditorUiState(
34             layers = emptyList(),
35             canUndo = false,
36             canRedo = false,
37             brushSizePx = 16f,
38             brushOpacity = 1f,
39             activeLayerId = -1L,
40             ready = false,
41         )
42     }
43 }
```

34. app/src/main/java/com/wetinknext/engine/core/EngineRenderer.kt

```

1  package com.wetinknext.engine.core
2
3  import android.opengl.GLES30
4  import android.opengl.GLSurfaceView
5  import android.util.Log
6  import android.view.MotionEvent
7  import androidx.compose.ui.graphics.toArgb
8  import com.wetinknext.BuildConfig
9  import com.wetinknext.engine.brush.BrushSettings
10 import com.wetinknext.engine.brush.BrushTexture
11 import com.wetinknext.engine.brush.LoadedBrushTexture
12 import com.wetinknext.engine.brush.BrushRenderMode
13 import com.wetinknext.engine.brush.CapsuleEmitter
14 import com.wetinknext.engine.brush.CapsuleStrokeRenderer
15 import com.wetinknext.engine.brush.ColorSpaces
16 import com.wetinknext.engine.brush.DabBuffer
17 import com.wetinknext.engine.brush.DabRenderer
18 import com.wetinknext.engine.brush.StampEmitter
19 import com.wetinknext.engine.canvas.Compositor
20 import com.wetinknext.engine.canvas.LayerStack
21 import com.wetinknext.engine.canvas.StrokeBlitter
22 import com.wetinknext.engine.gl.CanvasGeometry
23 import com.wetinknext.engine.gl.GlCaps
24 import com.wetinknext.engine.gl.GlCheck
25 import com.wetinknext.engine.gl.RenderTarget
26 import com.wetinknext.engine.input.InputAction
27 import com.wetinknext.engine.input.InputBatch
28 import com.wetinknext.engine.input.InputBatchPool
29 import com.wetinknext.engine.input.GestureRouter
30 import com.wetinknext.engine.input.StrokeInputCapturer
31 import com.wetinknext.engine.undo.TileSnapshotRestore
32 import com.wetinknext.engine.undo.UndoManager
33 import java.util.concurrent.ArrayBlockingQueue
34 import java.util.concurrent.atomic.AtomicBoolean
35 import javax.microedition.khronos.egl.EGLConfig
36 import javax.microedition.khronos.opengles.GL10
37
38 /**
39  * The P6 render-thread owner. Active dabs are accumulated only in strokeTarget;
40  * UP snapshots and merges them into the selected PaintLayer atomically.
41  */
42 class EngineRenderer(
43     private val documentWidth: Int = DEFAULT_CANVAS_WIDTH,
44     private val documentHeight: Int = DEFAULT_CANVAS_HEIGHT,
45 ) : GLSurfaceView.Renderer {
46     private var caps: GlCaps? = null
47     private val layerStack = LayerStack()
48     private val undoManager = UndoManager()
49     private val undoPipeline = UndoCompressionPipeline(
50         requestRender = { onInputRenderRequested?.invoke() },
51     )
52     private var undoRestoreFailures = 0
53     private var glThread: Thread? = null
54
55     private var geometry: CanvasGeometry? = null
56     private var compositor: Compositor? = null
57     private var dabRenderer: DabRenderer? = null
58     private var capsuleRenderer: CapsuleStrokeRenderer? = null
59     private var grainTexture: BrushTexture? = null
60     private var grainPath: String? = null
61     private var shapeTexture: BrushTexture? = null
62     private var shapePath: String? = null
63     private val strokeTarget = RenderTarget()
64     private val canvasToFboMatrix = FloatArray(16)
65     private val strokeCommitter = StrokeCommitter(
66         strokeTarget = strokeTarget,
67         canvasToFbo = canvasToFboMatrix,
68         undoPipeline = undoPipeline,
69         undoManager = undoManager,
70     )
71     private val canvasToClipMatrix = FloatArray(16)
72
73     private var screenWidth = 1
74     private var screenHeight = 1
75     private val canvasBackdropColor = floatArrayOf(0.08f, 0.09f, 0.12f)
76     private val canvasGridColor = floatArrayOf(0.16f, 0.18f, 0.22f)
77
78     private val inputPool = InputBatchPool(batchCount = 64, maxSamplesPerBatch = 256)
79     private val inputQueue = ArrayBlockingQueue<InputBatch>(64)
80     private val inputCapturer = StrokeInputCapturer(layerStack.camera, inputPool, inputQueue)
81     private val undoGestureRequested = AtomicBoolean(false)
82     private val redoGestureRequested = AtomicBoolean(false)
83     private val resetCameraGestureRequested = AtomicBoolean(false)
84     private val gestureRouter = GestureRouter(
85         camera = layerStack.camera,
86         drawingInput = inputCapturer,
87         callbacks = object : GestureRouter.Callbacks {
88             override fun cancelDrawing() {
89                 requestCancelFromInput()
90             }
91         }
92     )

```

```

91
92         override fun undoGesture() {
93             undoGestureRequested.set(true)
94         }
95
96         override fun redoGesture() {
97             redoGestureRequested.set(true)
98         }
99
100        override fun resetCamera() {
101            resetCameraGestureRequested.set(true)
102        }
103
104        override fun requestRender() {
105            onInputRenderRequested?.invoke()
106        }
107    },
108 )
109
110 private var brushSettings = BrushSettings(
111     name = "G-Pen",
112     renderMode = BrushRenderMode.RIBBON,
113     baseRadiusPx = 9f,
114     spacing = 0.12f,
115     colorArgb = 0xFF1B1F24L,
116     smoothing = 0.35f,
117     streamline = 0.22f,
118     pressureToOpacity = false,
119     pressureGamma = 1.7f,
120     minSizeRatio = 0.06f,
121     ribbon = com.wetinknext.engine.brush.RibbonSettings(
122         cap = com.wetinknext.engine.brush.RibbonCap.ROUND,
123         join = com.wetinknext.engine.brush.RibbonJoin.ROUND,
124         miterLimit = 2.5f,
125         minPointDistancePx = 1.25f,
126         aaWidthPx = 1f,
127         taperStartPx = 16f,
128         taperEndPx = 64f,
129     ),
130 )
131 private val stampEmitter = StampEmitter(brushSettings)
132 private val capsuleEmitter = CapsuleEmitter(brushSettings)
133 private val dabBuffer = DabBuffer()
134 private val strokeColorLinear = FloatArray(3)
135 private val activeStrokeColorLinear = FloatArray(3)
136 private val strokeDirtyRect = DirtyRect()
137 private val dirtyBounds = IntArray(4)
138
139 private var strokeActive = false
140 private var activeStrokeBrush: BrushSettings? = null
141 private var strokeBlitter: StrokeBlitter? = null
142 private var capsulePreviewInitialized = false
143 private var stampPreviewInitialized = false
144 private val cancelRequested = AtomicBoolean(false)
145
146 /** Assigned by PaintSurfaceView; invoked on the GL thread. */
147 var onStateChange: ((EditorUiState) -> Unit)? = null
148
149 /** Assigned by PaintSurfaceView; invoked from the UI touch thread. */
150 var onInputRenderRequested: (() -> Unit)? = null
151
152 fun setOnSecondaryPointerDown(listener: () -> Unit) {
153     inputCapturer.onSecondaryPointerDown = listener
154 }
155
156 fun onTouchEvent(event: MotionEvent): Boolean = gestureRouter.onTouchEvent(event)
157
158 fun requestCancelFromInput() {
159     cancelRequested.set(true)
160 }
161
162 /** Fits the document into the current GL surface. Must run on the GL thread. */
163 fun resetCamera() {
164     layerStack.camera.fitCanvas(
165         canvasWidth = layerStack.canvasWidth,
166         canvasHeight = layerStack.canvasHeight,
167         viewWidth = screenWidth,
168         viewHeight = screenHeight,
169     )
170 }
171
172 /** Applies theme colors to the area that surrounds the document. */
173 fun setCanvasBackdrop(backdropArgb: Int, gridArgb: Int) {
174     argbToRgb(backdropArgb, canvasBackdropColor)
175     argbToRgb(gridArgb, canvasGridColor)
176 }
177
178 /** These methods are called from GLSurfaceView.queueEvent by the UI layer. */
179 fun undo() {
180     if (undoPipeline.pendingCount > 0) return
181     resetStroke()
182     while (true) {

```

```

183         val entry = undoManager.peekUndo() ?: return
184         val layer = layerStack.findLayerById(entry.layerId)
185         if (layer == null) {
186             undoManager.dropUndo(entry)
187             continue
188         }
189         val restored = TileSnapshotRestore.restore(layer.target, entry.beforeTiles)
190         if (!restored) {
191             undoRestoreFailures++
192             publishState()
193             return
194         }
195         if (!undoManager.commitUndo(entry)) {
196             publishState()
197             return
198         }
199         layer.version++
200         publishState()
201         return
202     }
203 }
204
205 fun redo() {
206     if (undoPipeline.pendingCount > 0) return
207     resetStroke()
208     while (true) {
209         val entry = undoManager.peekRedo() ?: return
210         val layer = layerStack.findLayerById(entry.layerId)
211         if (layer == null) {
212             undoManager.dropRedo(entry)
213             continue
214         }
215         val restored = TileSnapshotRestore.restore(layer.target, entry.afterTiles)
216         if (!restored) {
217             undoRestoreFailures++
218             publishState()
219             return
220         }
221         if (!undoManager.commitRedo(entry)) {
222             publishState()
223             return
224         }
225         layer.version++
226         publishState()
227         return
228     }
229 }
230
231 fun setActiveLayer(id: Long): Boolean {
232     resetStroke()
233     return layerStack.setActive(id).also { changed ->
234         if (changed) publishState()
235     }
236 }
237
238 fun addLayer(): Long = addLayer(nextLayerName())
239
240 fun addLayer(name: String): Long {
241     resetStroke()
242     return layerStack.addLayer(name).id.also { publishState() }
243 }
244
245 fun removeLayer(id: Long): Boolean {
246     checkOnGUIThread()
247     resetStroke()
248     invalidatePendingUndoHistory()
249     val removed = layerStack.removeLayer(id) != null
250     if (removed) {
251         undoManager.removeEntriesForLayer(id)
252         publishState()
253     }
254     return removed
255 }
256
257 fun setLayerVisible(id: Long, visible: Boolean) {
258     resetStroke()
259     val layer = layerStack.findLayerById(id) ?: return
260     layer.isVisible = visible
261     publishState()
262 }
263
264 fun setLayerOpacity(id: Long, opacity: Float) {
265     val layer = layerStack.findLayerById(id) ?: return
266     layer.opacity = opacity.coerceIn(0f, 1f)
267     publishState()
268 }
269
270 /** Clears only the editable active layer. The locked background is protected. */
271 fun clearActiveLayer(): Boolean {
272     checkOnGUIThread()
273     resetStroke()
274     val layer = layerStack.activeLayer() ?: return false

```



```

275         val cleared = strokeCommitter.clearLayer(
276             layer = layer,
277             canvasWidth = layerStack.canvasWidth,
278             canvasHeight = layerStack.canvasHeight,
279         )
280         if (cleared) publishState()
281         return cleared
282     }
283
284     /** Mirrors only the camera view; document pixels and undo history stay unchanged. */
285     fun toggleCanvasMirror(): Boolean {
286         resetStroke()
287         val width = layerStack.canvasWidth
288         val height = layerStack.canvasHeight
289         if (width <= 0 || height <= 0) return false
290
291         // Flip around the document centre in screen space. Toggling the matrix
292         // sign alone flips around the camera origin and makes the canvas jump.
293         val camera = layerStack.camera
294         val current = camera.snapshot()
295         val canvasCenterX = width * .5f
296         val canvasCenterY = height * .5f
297         val before = FloatArray(2)
298         val after = FloatArray(2)
299         current.canvasToScreen(canvasCenterX, canvasCenterY, before)
300         val flipped = current.copy(flipX = !current.flipX)
301         flipped.canvasToScreen(canvasCenterX, canvasCenterY, after)
302         camera.set(
303             flipped.copy(
304                 translateX = flipped.translateX + before[0] - after[0],
305                 translateY = flipped.translateY + before[1] - after[1],
306             ),
307         )
308         return true
309     }
310
311     /** px = ДИАМЕТР кисти в canvas-пикселях, ровно то, что показывает UI. */
312     fun setBrushSize(px: Float) =
313         updateBrush(brushSettings.copy(baseRadiusPx = (px * .5f).coerceIn(.5f, 200f)))
314
315     fun setBrushOpacity(opacity: Float) =
316         updateBrush(brushSettings.copy(opacity = opacity.coerceIn(0f, 1f)))
317
318     fun setBrushColor(color: androidx.compose.ui.graphics.Color) =
319         updateBrush(brushSettings.copy(colorArgb = color.toArgb().toLong() and 0xFFFFFFFFL))
320
321     fun applyBrush(settings: BrushSettings) {
322         val nextSettings = settings
323             .copy(colorArgb = brushSettings.colorArgb)
324             .resolved()
325         updateBrush(nextSettings)
326     }
327
328     fun setBrushSettings(settings: BrushSettings) {
329         updateBrush(settings)
330     }
331
332     /** Правка кисти во время штриха больше не отменяет штрих: применяем на следующем DOWN. */
333     private fun updateBrush(next: BrushSettings) {
334         brushSettings = next.resolved()
335         if (!strokeActive) {
336             applyBrushToEmitters(brushSettings)
337             ColorSpaces.srgb8ToLinear(brushSettings.colorArgb, strokeColorLinear)
338         }
339         publishState()
340     }
341
342     private fun applyBrushToEmitters(settings: BrushSettings) {
343         stampEmitter.updateSettings(settings)
344         capsuleEmitter.updateSettings(settings)
345     }
346
347     fun requestState() {
348         publishState()
349     }
350
351     fun checkOnGThread() {
352         val current = Thread.currentThread()
353         check(current === gThread) {
354             "GL resource accessed outside GLSurfaceView render thread: current=$current, expected=$gThread"
355         }
356     }
357
358     override fun onSurfaceCreated(gl: GL10?, config: EGLConfig?) {
359         val currentThread = Thread.currentThread()
360         gThread = currentThread
361         GLCheck.setGThread(currentThread)
362         releaseGLObjects()
363         undoPipeline.ensureRunning()
364         val nextCaps = GLCaps.query()
365         caps = nextCaps
366         layerStack.create(nextCaps, documentWidth, documentHeight)

```

```

367         strokeTarget.create(documentWidth, documentHeight, nextCaps.supportsHalfFloatColorBuffer)
368         strokeTarget.clear(0f, 0f, 0f, 0f)
369         ViewTransform.buildCanvasToFbo(
370             documentWidth.toFloat(),
371             documentHeight.toFloat(),
372             canvasToFboMatrix,
373         )
374         geometry = CanvasGeometry().also { it.create(documentWidth, documentHeight) }
375         compositor = Compositor().also { it.create() }
376         dabRenderer = DabRenderer(dabBuffer.capacity).also { it.create() }
377         capsuleRenderer = CapsuleStrokeRenderer().also {
378             it.create()
379         }
380         strokeBlitter = StrokeBlitter().also { it.create() }
381         ColorSpaces.srgb8ToLinear(bushSettings.colorArgb, strokeColorLinear)
382         publishState()
383         GLCheck.noError("P6 surface creation")
384     }
385
386     override fun onSurfaceChanged(gl: GL10?, width: Int, height: Int) {
387         screenWidth = width.coerceAtLeast(1)
388         screenHeight = height.coerceAtLeast(1)
389         GLES30.glViewport(0, 0, screenWidth, screenHeight)
390         layerStack.camera.fitCanvas(layerStack.canvasWidth, layerStack.canvasHeight, screenWidth, screenHeight)
391         GLCheck.noError("P6 surface changed")
392     }
393
394     override fun onDrawFrame(gl: GL10?) {
395         if (cancelRequested.compareAndSet(true, false)) resetStroke()
396         if (undoGestureRequested.compareAndSet(true, false)) undo()
397         if (redoGestureRequested.compareAndSet(true, false)) redo()
398         if (resetCameraGestureRequested.compareAndSet(true, false)) resetCamera()
399         drainInput()
400         if (undoPipeline.process(undoManager)) publishState()
401
402         val frameStrokeBrush = activeStrokeBrush
403         if (strokeActive && frameStrokeBrush?.renderMode == BrushRenderMode.RIBBON) {
404             renderCapsulePreview(frameStrokeBrush)
405         } else if (strokeActive && frameStrokeBrush?.renderMode == BrushRenderMode.STAMP && dabBuffer.count > 0) {
406             // No longer clearing and drawing everything here, handled incrementally in drainInput
407         }
408
409         GLES30.glBindFramebuffer(GLES30.GL_FRAMEBUFFER, 0)
410         GLES30.glViewport(0, 0, screenWidth, screenHeight)
411         GLES30.glDisable(GLES30.GL_SCISSOR_TEST)
412         GLES30.glDisable(GLES30.GL_BLEND)
413         GLES30.glClearColor(canvasBackdropColor[0], canvasBackdropColor[1], canvasBackdropColor[2], 1f)
414         GLES30.glClear(GLES30.GL_COLOR_BUFFER_BIT)
415         drawBackdropGrid()
416
417         layerStack.camera.snapshot().buildCanvasToClip(
418             screenWidth.toFloat(),
419             screenHeight.toFloat(),
420             canvasToClipMatrix,
421         )
422         compositor?.render(
423             geometry = geometry ?: return,
424             layers = layerStack,
425             activeLayerId = layerStack.activeLayerId,
426             strokeTextureId = if (
427                 strokeActive &&
428                 (
429                     dabBuffer.count > 0 ||
430                     (
431                         frameStrokeBrush?.renderMode == BrushRenderMode.RIBBON &&
432                         capsuleEmitter.hasStroke
433                     )
434                 )
435             ) {
436                 strokeTarget.textureId
437             } else {
438                 0
439             },
440             // Both render modes write their local flow into strokeTarget. The
441             // compositor therefore applies only the snapshot's global opacity.
442             strokeOpacity = frameStrokeBrush?.opacity?.coerceIn(0f, 1f) ?: 1f,
443             canvasToClip = canvasToClipMatrix,
444         )
445     }
446
447     private fun drawBackdropGrid() {
448         val cellPx = 28
449         GLES30.glEnable(GLES30.GL_SCISSOR_TEST)
450         GLES30.glClearColor(canvasGridColor[0], canvasGridColor[1], canvasGridColor[2], 1f)
451         var x = 0
452         while (x < screenWidth) {
453             GLES30.glScissor(x, 0, 1, screenHeight)
454             GLES30.glClear(GLES30.GL_COLOR_BUFFER_BIT)
455             x += cellPx
456         }
457         var y = 0
458         while (y < screenHeight) {

```

```

459         GLES30.glScissor(0, y, screenWidth, 1)
460         GLES30.glClear(GLES30.GL_COLOR_BUFFER_BIT)
461         y += cellPx
462     }
463     GLES30.glDisable(GLES30.GL_SCISSOR_TEST)
464 }
465
466 private fun argbToRgb(argb: Int, out: FloatArray) {
467     out[0] = ((argb ushr 16) and 0xFF) / 255f
468     out[1] = ((argb ushr 8) and 0xFF) / 255f
469     out[2] = (argb and 0xFF) / 255f
470 }
471
472 /** Invalidates worker results created before a destructive document change. */
473 private fun invalidatePendingUndoHistory() {
474     checkOnGltThread()
475     undoPipeline.invalidate()
476 }
477
478 private fun shutdownUndoExecutor() {
479     undoPipeline.shutdown()
480 }
481
482 /** Safe to call from the UI thread; cancellation is executed on the next GL frame. */
483 fun cancelActiveStroke() {
484     cancelRequested.set(true)
485 }
486
487 /** Must run on the GL thread while the context is still current. */
488 fun releaseGlobjects() {
489     checkOnGltThread()
490     invalidatePendingUndoHistory()
491     discardPendingInput()
492     resetStroke()
493     undoManager.clear()
494     shutdownUndoExecutor()
495     dabRenderer?.release()
496     dabRenderer = null
497     grainTexture?.release()
498     grainTexture = null
499     grainPath = null
500     shapeTexture?.release()
501     shapeTexture = null
502     shapePath = null
503     capsuleRenderer?.release()
504     capsuleRenderer = null
505     strokeBlitter?.release()
506     strokeBlitter = null
507     compositor?.release()
508     compositor = null
509     geometry?.release()
510     geometry = null
511     strokeTarget.release()
512     layerStack.release()
513     caps = null
514 }
515
516 private fun drainInput() {
517     while (true) {
518         val batch = inputQueue.poll() ?: return
519         try {
520             when (batch.action) {
521                 InputAction.DOWN -> {
522                     if (!batch.isEmpty()) {
523                         resetStroke()
524                         val strokeBrush = brushSettings.resolved()
525                         activeStrokeBrush = strokeBrush
526                         ColorSpaces.srgb8ToLinear(strokeBrush.colorArgb, activeStrokeColorLinear)
527                         activeStrokeColorLinear.copyInto(strokeColorLinear)
528                         strokeActive = true
529
530                         if (strokeBrush.renderMode == BrushRenderMode.RIBBON) {
531                             capsuleEmitter.begin(
532                                 batch = batch,
533                                 out = checkNotNull(capsuleRenderer),
534                                 strokeSettings = strokeBrush,
535                             )
536                         } else {
537                             dabRenderer?.beginStroke()
538                             stampEmitter.begin(batch, dabBuffer, strokeBrush)
539                             strokeTarget.clear(0f, 0f, 0f, 0f)
540                             stampPreviewInitialized = true
541
542                             drawPendingStampPreview("DOWN")
543                         }
544                     }
545                 }
546                 InputAction.MOVE -> if (strokeActive) {
547                     val strokeBrush = activeStrokeBrush
548                     ?: error("Missing active stroke snapshot")
549                     if (strokeBrush.renderMode == BrushRenderMode.RIBBON) {

```

```

551         capsuleEmitter.append(
552             batch = batch,
553             out = checkNotNull(capsuleRenderer),
554         )
555     } else {
556         stampEmitter.append(batch, dabBuffer)
557         drawPendingStampPreview("MOVE")
558     }
559 }
560
561 InputAction.UP -> if (strokeActive) {
562     val strokeBrush = activeStrokeBrush
563     ?: error("Missing active stroke snapshot")
564     if (strokeBrush.renderMode == BrushRenderMode.RIBBON) {
565         val renderer = checkNotNull(capsuleRenderer)
566
567         // UP должен сначала обработать historical samples,
568         // затем текущую точку, которую уже положил capturer.
569         capsuleEmitter.append(
570             batch = batch,
571             out = renderer,
572         )
573         capsuleEmitter.finish(
574             out = renderer,
575             cancel = false,
576         )
577         commitCapsuleStroke(strokeBrush)
578     } else {
579         stampEmitter.append(batch, dabBuffer)
580         stampEmitter.finish(dabBuffer, cancel = false)
581         commitStroke(strokeBrush)
582     }
583 }
584 resetStroke()
585 }
586
587 InputAction.CANCEL -> resetStroke()
588 }
589 } finally {
590     inputPool.release(batch)
591 }
592 }
593 }
594
595 private fun commitStroke(strokeBrush: BrushSettings) {
596     val layer = layerStack.activeLayer() ?: return
597     if (layer.isLocked || dabBuffer.count == 0) return
598     if (!computeDirtyPixelBounds(dirtyBounds)) return
599     val blitter = strokeBlitter ?: return
600     val strokeGeometry = geometry ?: return
601
602     // Complete the isolated local-coverage mask before it is composited.
603     // Global brush opacity is deliberately applied only by the blit.
604     drawPendingStampPreview("UP")
605
606     if (strokeCommitter.commit(
607         layer = layer,
608         geometry = strokeGeometry,
609         blitter = blitter,
610         dirtyBounds = dirtyBounds,
611         canvasWidth = layerStack.canvasWidth,
612         canvasHeight = layerStack.canvasHeight,
613         opacity = strokeBrush.opacity,
614     )) {
615         publishState()
616     }
617 }
618
619 /** Returns a clamped pixel region covering the exact emitted dabs and their AA fringe. */
620 private fun computeDirtyPixelBounds(out: IntArray): Boolean {
621     if (dabBuffer.count == 0) return false
622     strokeDirtyRect.clear()
623     dabBuffer.includeDirtyRect(
624         rect = strokeDirtyRect,
625         extraMargin = STAMP_DIRTY_MARGIN_PX,
626     )
627     strokeDirtyRect.clamp(layerStack.canvasWidth.toFloat(), layerStack.canvasHeight.toFloat())
628     strokeDirtyRect.toPixelBounds(out)
629     return out[2] > out[0] && out[3] > out[1]
630 }
631
632 private fun drawPendingStampPreview(phase: String) {
633     val renderer = dabRenderer ?: return
634     val strokeBrush = activeStrokeBrush
635     ?: error("Missing active stroke snapshot")
636     check(renderer.uploadedCount <= dabBuffer.count) {
637         "Stamp preview rewind: uploaded=${renderer.uploadedCount}, dabs=${dabBuffer.count}"
638     }
639     if (BuildConfig.DEBUG) {
640         Log.d(
641             BRUSH_DIAG_TAG,
642             "phase=$phase dabs=${dabBuffer.count} pending=${dabBuffer.count - renderer.uploadedCount}",

```

```

643     )
644 }
645 renderer.drawPendingInto(
646     target = strokeTarget,
647     width = layerStack.canvasWidth,
648     height = layerStack.canvasHeight,
649     canvasToFbo = canvasToFboMatrix,
650     dabs = dabBuffer,
651     colorLinear = activeStrokeColorLinear,
652     blendPolicy = strokeBrush.blendPolicy,
653     strokeOpacity = 1f,
654 )
655 check(renderer.uploadedCount == dabBuffer.count) {
656     "Stamp preview did not consume all dabs: uploaded=${renderer.uploadedCount}, dabs=${dabBuffer.count}"
657 }
658 }
659
660 private fun resetStroke() {
661     strokeActive = false
662     activeStrokeBrush = null
663     stampEmitter.reset()
664     dabBuffer.clear()
665     capsuleEmitter.reset()
666     capsuleRenderer?.clearStrokeData()
667     capsulePreviewInitialized = false
668     stampPreviewInitialized = false
669     dabRenderer?.clearStrokeData()
670     if (strokeTarget.framebufferId != 0) {
671         strokeTarget.clear(0f, 0f, 0f, 0f)
672     }
673 }
674
675 /** Releases pooled batches without interpreting them during shutdown/context recreation. */
676 private fun discardPendingInput() {
677     while (true) {
678         val batch = inputQueue.poll() ?: return
679         inputPool.release(batch)
680     }
681 }
682
683 private fun publishState() {
684     val listener = onStateChange ?: return
685     val layers = layerStack.allLayers()
686     val activeLayerId = layerStack.activeLayerId
687     val canDeleteLayers = layers.size > 1
688     listener(
689         EditorUiState(
690             layers = layers.map { layer ->
691                 LayerUiModel(
692                     id = layer.id,
693                     name = layer.name,
694                     isVisible = layer.isVisible,
695                     isLocked = layer.isLocked,
696                     opacity = layer.opacity,
697                     active = layer.id == activeLayerId,
698                     canDelete = !layer.isLocked && canDeleteLayers,
699                 )
700             },
701             canUndo = undoManager.canUndo && undoPipeline.pendingCount == 0,
702             canRedo = undoManager.canRedo && undoPipeline.pendingCount == 0,
703             brushSizePx = brushSettings.baseRadiusPx * 2f,
704             brushOpacity = brushSettings.opacity,
705             activeLayerId = activeLayerId,
706             ready = layerStack.canvasWidth > 0 && layerStack.canvasHeight > 0,
707             undoDiagnostics = UndoDiagnostics(
708                 pendingJobs = undoPipeline.pendingCount,
709                 staleResults = undoPipeline.staleResultCount,
710                 compressionFailures = undoPipeline.compressionFailureCount,
711                 restoreFailures = undoRestoreFailures,
712                 memoryBytes = undoManager.memoryBytes,
713             ),
714         ),
715     )
716 }
717
718 /**
719  * Вызывается только через GLSurfaceView.queueEvent,
720  * то есть на GL-потоке с current EGL context.
721  */
722 fun applyLoadedGrain(
723     loaded: LoadedBrushTexture,
724     scale: Float,
725     canvasLocked: Boolean,
726     depth: Float,
727     contrast: Float,
728 ) {
729     val oldTexture = grainTexture
730     val newTexture = BrushTexture()
731
732     try {
733         newTexture.createFromRgba(
734             width = loaded.width,

```

```

735         height = loaded.height,
736         rgba = loaded.rgba,
737     )
738
739     capsuleRenderer?.setGrainTexture(
740         textureId = newTexture.textureId,
741         scale = scale,
742         canvasLocked = canvasLocked,
743         depth = depth,
744         contrast = contrast,
745     )
746
747     dabRenderer?.setGrainTexture(
748         textureId = newTexture.textureId,
749         scale = scale,
750         canvasLocked = canvasLocked,
751         depth = depth,
752         contrast = contrast,
753     )
754
755     grainTexture = newTexture
756     grainPath = loaded.path
757
758     oldTexture?.release()
759 } catch (error: Throwable) {
760     newTexture.release()
761     throw error
762 }
763 }
764
765 fun clearGrainTexture() {
766     capsuleRenderer?.clearGrainTexture()
767     dabRenderer?.clearGrainTexture()
768
769     grainTexture?.release()
770     grainTexture = null
771     grainPath = null
772 }
773
774 /** Must run on the GL thread. Shape masks affect STAMP dabs only. */
775 fun applyLoadedShape(
776     loaded: LoadedBrushTexture,
777     reverse: Boolean,
778     rgbToAlpha: Boolean,
779 ) {
780     val oldTexture = shapeTexture
781     val newTexture = BrushTexture()
782
783     try {
784         newTexture.createFromRgba(loaded.width, loaded.height, loaded.rgba)
785         dabRenderer?.setShapeTexture(newTexture.textureId, reverse, rgbToAlpha)
786         shapeTexture = newTexture
787         shapePath = loaded.path
788         oldTexture?.release()
789     } catch (error: Throwable) {
790         newTexture.release()
791         throw error
792     }
793 }
794
795 fun clearShapeTexture() {
796     dabRenderer?.clearShapeTexture()
797     shapeTexture?.release()
798     shapeTexture = null
799     shapePath = null
800 }
801
802 private fun nextLayerName(): String = "Слой ${layerStack.count + 1}"
803
804 private fun renderCapsulePreview(strokeBrush: BrushSettings) {
805     val renderer = capsuleRenderer ?: return
806
807     if (!capsulePreviewInitialized) {
808         strokeTarget.clear(
809             red = 0f,
810             green = 0f,
811             blue = 0f,
812             alpha = 0f,
813         )
814         capsulePreviewInitialized = true
815     }
816
817     renderer.drawPending(
818         target = strokeTarget,
819         width = layerStack.canvasWidth,
820         height = layerStack.canvasHeight,
821         canvasToClip = canvasToFboMatrix,
822         colorLinear = activeStrokeColorLinear,
823         blendPolicy = strokeBrush.blendPolicy,
824         flow = strokeBrush.flow,
825     )
826 }

```

```

827
828 private fun commitCapsuleStroke(strokeBrush: BrushSettings) {
829     val layer = layerStack.activeLayer() ?: return
830     val renderer = capsuleRenderer ?: return
831
832     if (layer.isLocked) return
833     if (!capsuleEmitter.hasStroke) return
834     if (!computeCapsuleDirtyBounds(dirtyBounds)) return
835     val blitter = strokeBlitter ?: return
836     val strokeGeometry = geometry ?: return
837
838     /*
839     * Сначала строим stroke в отдельном offscreen target.
840     * Старый слой и новый renderer не используют один FBO для рисования.
841     */
842     strokeTarget.clear(
843         red = 0f,
844         green = 0f,
845         blue = 0f,
846         alpha = 0f,
847     )
848
849     renderer.drawAll(
850         target = strokeTarget,
851         width = layerStack.canvasWidth,
852         height = layerStack.canvasHeight,
853         canvasToClip = canvasToFboMatrix,
854         colorLinear = activeStrokeColorLinear,
855         blendPolicy = strokeBrush.blendPolicy,
856         flow = strokeBrush.flow,
857     )
858
859     if (strokeCommitter.commit(
860         layer = layer,
861         geometry = strokeGeometry,
862         blitter = blitter,
863         dirtyBounds = dirtyBounds,
864         canvasWidth = layerStack.canvasWidth,
865         canvasHeight = layerStack.canvasHeight,
866         opacity = strokeBrush.opacity,
867     )) {
868         publishState()
869     }
870 }
871
872 private fun computeCapsuleDirtyBounds(
873     out: IntArray,
874 ): Boolean {
875     if (!capsuleEmitter.hasBounds) return false
876
877     strokeDirtyRect.clear()
878     strokeDirtyRect.include(
879         left = capsuleEmitter.minX,
880         top = capsuleEmitter.minY,
881         right = capsuleEmitter.maxX,
882         bottom = capsuleEmitter.maxY,
883     )
884     strokeDirtyRect.expand(AA_MARGIN_PX)
885     strokeDirtyRect.clamp(
886         layerStack.canvasWidth.toFloat(),
887         layerStack.canvasHeight.toFloat(),
888     )
889     strokeDirtyRect.toPixelBounds(out)
890
891     return out[2] > out[0] && out[3] > out[1]
892 }
893
894 companion object {
895     private const val BRUSH_DIAG_TAG = "BrushDiag"
896     const val DEFAULT_CANVAS_WIDTH = 1500
897     const val DEFAULT_CANVAS_HEIGHT = 2000
898     private const val AA_MARGIN_PX = 4f
899     private const val STAMP_DIRTY_MARGIN_PX = 4f
900 }
901 }

```

35. app/src/main/java/com/wetinknext/engine/core/PaintEngine.kt

```
1 package com.wetinknext.engine.core
2
3 import com.wetinknext.engine.canvas.LayerStack
4 import com.wetinknext.engine.gl.GlCaps
5 import com.wetinknext.engine.gl.RenderTarget
6
7 /** Lightweight document facade retained for engine callers outside the renderer. */
8 class PaintEngine {
9     val layerStack = LayerStack()
10    val camera: Camera get() = layerStack.camera
11    val canvasTarget: RenderTarget get() = checkNotNull(layerStack.activeLayer()).target
12
13    val canvasWidth: Int get() = layerStack.canvasWidth
14    val canvasHeight: Int get() = layerStack.canvasHeight
15    var initialized = false
16        private set
17
18    fun create(caps: GlCaps, width: Int, height: Int) {
19        layerStack.create(caps, width, height)
20        initialized = true
21    }
22
23    fun addLayer(name: String): Long = layerStack.addLayer(name).id
24
25    fun removeLayer(id: Long): Boolean = layerStack.removeLayer(id) != null
26
27    fun setActiveLayer(id: Long): Boolean = layerStack.setActive(id)
28
29    fun release() {
30        layerStack.release()
31        initialized = false
32    }
33 }
```


36. app/src/main/java/com/wetinknext/engine/core/PaintSurfaceView.kt

```

1  package com.wetinknext.engine.core
2
3  import android.content.Context
4  import android.opengl.GLSurfaceView
5  import android.util.AttributeSet
6  import android.view.MotionEvent
7  import com.wetinknext.engine.brush.BrushSettings
8  import com.wetinknext.engine.brush.TextureLoader
9  import com.wetinknext.engine.brush.LoadedBrushTexture
10
11 /** Thread boundary between Compose commands and the GL-owned paint engine. */
12 class PaintSurfaceView @JvmOverloads constructor(
13     context: Context,
14     attrs: AttributeSet? = null,
15 ) : GLSurfaceView(context, attrs) {
16     private val engineRenderer = EngineRenderer()
17
18     private val textureLoader = TextureLoader(context)
19     private var grainGeneration = 0L
20     private var shapeGeneration = 0L
21
22     var onTextureError: ((String, Throwable) -> Unit)? = null
23
24     var onEditorStateChange: ((EditorUiState) -> Unit)? = null
25
26     init {
27         setEGLContextClientVersion(3)
28         setRenderer(engineRenderer)
29         renderMode = RENDERMODE_WHEN_DIRTY
30         engineRenderer.onStateChange = { state ->
31             post { onEditorStateChange?.invoke(state) }
32             requestRender()
33         }
34         engineRenderer.onInputRenderRequested = { requestRender() }
35     }
36
37     override fun onTouchEvent(event: MotionEvent): Boolean {
38         val handled = engineRenderer.onTouchEvent(event)
39         if (handled) requestRender()
40         return handled || super.onTouchEvent(event)
41     }
42
43     fun requestState() = queueEvent {
44         engineRenderer.requestState()
45         requestRender()
46     }
47
48     fun undo() = queueEvent {
49         engineRenderer.undo()
50         requestRender()
51     }
52
53     fun redo() = queueEvent {
54         engineRenderer.redo()
55         requestRender()
56     }
57
58     fun setActiveLayer(id: Long) = queueEvent {
59         engineRenderer.setActiveLayer(id)
60         requestRender()
61     }
62
63     fun addLayer() = queueEvent {
64         engineRenderer.addLayer()
65         requestRender()
66     }
67
68     fun removeLayer(id: Long) = queueEvent {
69         engineRenderer.removeLayer(id)
70         requestRender()
71     }
72
73     fun setLayerVisible(id: Long, visible: Boolean) =
74         queueEvent {
75             engineRenderer.setLayerVisible(id, visible)
76             requestRender()
77         }
78
79     fun setLayerOpacity(id: Long, opacity: Float) =
80         queueEvent {
81             engineRenderer.setLayerOpacity(id, opacity)
82             requestRender()
83         }
84
85     fun clearActiveLayer() = queueEvent {
86         engineRenderer.clearActiveLayer()
87         requestRender()
88     }
89
90     fun toggleCanvasMirror() = queueEvent {

```

```

91         engineRenderer.toggleCanvasMirror()
92         requestRender()
93     }
94
95     fun setCanvasBackdrop(backdropArgb: Int, gridArgb: Int) = queueEvent {
96         engineRenderer.setCanvasBackdrop(backdropArgb, gridArgb)
97         requestRender()
98     }
99
100    fun setBrushSize(px: Float) = queueEvent {
101        engineRenderer.setBrushSize(px)
102        requestRender()
103    }
104
105    fun setBrushOpacity(opacity: Float) = queueEvent {
106        engineRenderer.setBrushOpacity(opacity)
107        requestRender()
108    }
109
110    fun applyBrush(settings: BrushSettings) = queueEvent {
111        engineRenderer.applyBrush(settings)
112        requestRender()
113    }
114
115    fun setBrushSettings(settings: BrushSettings) = queueEvent {
116        engineRenderer.setBrushSettings(settings)
117        requestRender()
118    }
119
120    fun setBrushColor(color: androidx.compose.ui.graphics.Color) = queueEvent {
121        engineRenderer.setBrushColor(color)
122        requestRender()
123    }
124
125    fun loadGrainTexture(
126        path: String,
127        scale: Float = 1f,
128        canvasLocked: Boolean = true,
129        depth: Float = 1f,
130        contrast: Float = 1f,
131    ) {
132        val generation = ++grainGeneration
133
134        textureLoader.loadAsync(
135            path = path,
136            onLoaded = { loaded ->
137                queueEvent {
138                    if (generation != grainGeneration) {
139                        return@queueEvent
140                    }
141
142                    engineRenderer.applyLoadedGrain(
143                        loaded = loaded,
144                        scale = scale,
145                        canvasLocked = canvasLocked,
146                        depth = depth,
147                        contrast = contrast,
148                    )
149                    requestRender()
150                }
151            },
152            onError = { error ->
153                post {
154                    onTextureError?.invoke(path, error)
155                }
156            },
157        )
158    }
159
160    fun clearGrainTexture() {
161        grainGeneration++
162
163        queueEvent {
164            engineRenderer.clearGrainTexture()
165            requestRender()
166        }
167    }
168
169    fun loadShapeTexture(
170        path: String,
171        reverse: Boolean = false,
172        rgbToAlpha: Boolean = false,
173    ) {
174        val generation = ++shapeGeneration
175        textureLoader.loadAsync(
176            path = path,
177            onLoaded = { loaded ->
178                queueEvent {
179                    if (generation != shapeGeneration) return@queueEvent
180                    engineRenderer.applyLoadedShape(loaded, reverse, rgbToAlpha)
181                    requestRender()
182                }
183            }

```

```

183         },
184         onError = { error -> post { onTextureError?.invoke(path, error) } },
185     )
186 }
187
188 fun clearShapeTexture() {
189     shapeGeneration++
190     queueEvent {
191         engineRenderer.clearShapeTexture()
192         requestRender()
193     }
194 }
195
196 override fun onPause() {
197     engineRenderer.cancelActiveStroke()
198     super.onPause()
199 }
200
201 override fun onDetachedFromWindow() {
202     grainGeneration++
203     shapeGeneration++
204     textureLoader.shutdown()
205
206     // Очищаем listeners, чтобы избежать утечек или вызовов в пустоту
207     onTextureError = null
208     onEditorStateChange = null
209     engineRenderer.onStateChange = null
210     engineRenderer.onInputRenderRequested = null
211     engineRenderer.setOnSecondaryPointerDown { }
212
213     // Пытаемся освободить GL-объекты, но не рассчитываем на 100% успех здесь
214     queueEvent {
215         // Отменяем активный штрих перед освобождением
216         engineRenderer.cancelActiveStroke()
217         engineRenderer.releaseGLObjects()
218     }
219     super.onDetachedFromWindow()
220 }
221 }

```

37. app/src/main/java/com/wetinknext/engine/core/StrokeCommitter.kt

```

1  package com.wetinknext.engine.core
2
3  import android.opengl.GLES30
4  import com.wetinknext.engine.canvas.PaintLayer
5  import com.wetinknext.engine.canvas.StrokeBlitter
6  import com.wetinknext.engine.gl.CanvasGeometry
7  import com.wetinknext.engine.gl.GlCheck
8  import com.wetinknext.engine.gl.RenderTarget
9  import com.wetinknext.engine.undo.TileSnapshot
10 import com.wetinknext.engine.undo.TileSnapshotCapture
11 import com.wetinknext.engine.undo.UndoManager
12 import com.wetinknext.engine.undo.UndoOperationType
13
14 /**
15  * GL-thread transaction boundary for a prepared stroke.
16  *
17  * The caller renders the active stroke into [strokeTarget]. This class then
18  * snapshots the affected layer tiles, performs the single premultiplied blit,
19  * captures the after-state, and sends the reversible transaction to history.
20  */
21 class StrokeCommitter(
22     private val strokeTarget: RenderTarget,
23     private val canvasToFbo: FloatArray,
24     private val undoPipeline: UndoCompressionPipeline,
25     private val undoManager: UndoManager,
26 ) {
27     /**
28      * Commits the already-rendered stroke target into [layer]. [dirtyBounds]
29      * is in canvas coordinates and is expanded to a deterministic tile set.
30      */
31     fun commit(
32         layer: PaintLayer,
33         geometry: CanvasGeometry,
34         blitter: StrokeBlitter,
35         dirtyBounds: IntArray,
36         canvasWidth: Int,
37         canvasHeight: Int,
38         opacity: Float,
39         operation: UndoOperationType = UndoOperationType.TILE_EDIT,
40         tag: String = "stroke",
41     ): Boolean {
42         GlCheck.checkOnGltThread()
43         if (!layer.created || layer.isLocked || strokeTarget.textureId == 0) return false
44         if (dirtyBounds.size < 4) return false
45
46         expandBoundsToTiles(dirtyBounds, canvasWidth, canvasHeight)
47         if (dirtyBounds[2] <= dirtyBounds[0] || dirtyBounds[3] <= dirtyBounds[1]) return false
48
49         val beforeRaw = TileSnapshotCapture.capture(layer.target, dirtyBounds)
50         blitter.blit(
51             layer = layer.target,
52             geometry = geometry,
53             strokeTextureId = strokeTarget.textureId,
54             canvasToFbo = canvasToFbo,
55             width = canvasWidth,
56             height = canvasHeight,
57             opacity = opacity.coerceIn(0f, 1f),
58         )
59         GLES30.glFlush()
60         layer.version++
61         val afterRaw = TileSnapshotCapture.capture(layer.target, dirtyBounds)
62         undoPipeline.enqueue(
63             undoManager = undoManager,
64             layerId = layer.id,
65             beforeRaw = beforeRaw,
66             afterRaw = afterRaw,
67             operation = operation,
68             tag = tag,
69         )
70         return true
71     }
72
73     /** Clears one editable layer and records the full canvas as one undo transaction. */
74     fun clearLayer(
75         layer: PaintLayer,
76         canvasWidth: Int,
77         canvasHeight: Int,
78     ): Boolean {
79         GlCheck.checkOnGltThread()
80         if (!layer.created || layer.isLocked) return false
81
82         val fullBounds = intArrayOf(0, 0, canvasWidth, canvasHeight)
83         val beforeRaw = TileSnapshotCapture.capture(layer.target, fullBounds)
84         layer.clear()
85         GLES30.glFlush()
86         layer.version++
87         val afterRaw = TileSnapshotCapture.capture(layer.target, fullBounds)
88         undoPipeline.enqueue(
89             undoManager = undoManager,
90             layerId = layer.id,

```

```

91         beforeRaw = beforeRaw,
92         afterRaw = afterRaw,
93         operation = UndoOperationType.CLEAR_LAYER,
94         tag = "clear_layer",
95     )
96     return true
97 }
98
99 private fun expandBoundsToTiles(
100     bounds: IntArray,
101     canvasWidth: Int,
102     canvasHeight: Int,
103 ) {
104     val tileSize = TileSnapshot.TILE_SIZE
105     bounds[0] = (bounds[0] / tileSize) * tileSize
106     bounds[1] = (bounds[1] / tileSize) * tileSize
107     bounds[2] = ((bounds[2] + tileSize - 1) / tileSize) * tileSize
108     bounds[3] = ((bounds[3] + tileSize - 1) / tileSize) * tileSize
109     bounds[0] = bounds[0].coerceIn(0, canvasWidth)
110     bounds[1] = bounds[1].coerceIn(0, canvasHeight)
111     bounds[2] = bounds[2].coerceIn(0, canvasWidth)
112     bounds[3] = bounds[3].coerceIn(0, canvasHeight)
113 }
114 }

```

38. app/src/main/java/com/wetinknext/engine/core/UndoCompressionPipeline.kt

```

1  package com.wetinknext.engine.core
2
3  import android.util.Log
4  import com.wetinknext.BuildConfig
5  import com.wetinknext.engine.undo.DeflateTileCompressor
6  import com.wetinknext.engine.undo.RawTileSnapshot
7  import com.wetinknext.engine.undo.TileCompressor
8  import com.wetinknext.engine.undo.UndoEntry
9  import com.wetinknext.engine.undo.UndoManager
10 import com.wetinknext.engine.undo.UndoOperationType
11 import java.util.HashMap
12 import java.util.concurrent.ConcurrentLinkedQueue
13 import java.util.concurrent.ExecutorService
14 import java.util.concurrent.Executors
15 import java.util.concurrent.RejectedExecutionException
16 import java.util.concurrent.TimeUnit
17
18 /**
19  * Orders tile-compression work without ever touching GL from a worker thread.
20  *
21  * Call [enqueue], [process], [invalidate], and [shutdown] on the GL thread.
22  * The worker only converts [RawTileSnapshot] objects into compressed [UndoEntry]
23  * instances and requests another frame once a result is ready.
24  */
25 class UndoCompressionPipeline(
26     private val requestRender: () -> Unit,
27     private val compressor: TileCompressor = DeflateTileCompressor(),
28     private val maxPendingJobs: Int = DEFAULT_MAX_PENDING_JOBS,
29     private val executorFactory: () -> ExecutorService = { Executors.newSingleThreadExecutor() },
30 ) {
31     private var executor = executorFactory()
32     private val pendingResults = ConcurrentLinkedQueue<UndoJobResult>()
33     private val completedResults = HashMap<Long, UndoJobResult>()
34
35     private var nextSequence = 0L
36     private var nextSequenceToApply = 0L
37     private var epoch = 0L
38
39     var pendingCount: Int = 0
40     private set
41     var compressionFailureCount: Int = 0
42     private set
43     var staleResultCount: Int = 0
44     private set
45
46     /**
47      * Schedules one ordered undo transaction. The synchronous fallback still
48      * travels through the same result queue, so it cannot overtake older work.
49      */
50     fun enqueue(
51         undoManager: UndoManager,
52         layerId: Long,
53         beforeRaw: List<RawTileSnapshot>,
54         afterRaw: List<RawTileSnapshot>,
55         operation: UndoOperationType = UndoOperationType.TILE_EDIT,
56         tag: String = "stroke",
57     ) {
58         val jobEpoch = epoch
59         val sequence = allocateSequence()
60         pendingCount++
61
62         if (pendingCount >= maxPendingJobs) {
63             compressSynchronously(
64                 epoch = jobEpoch,
65                 sequence = sequence,
66                 layerId = layerId,
67                 beforeRaw = beforeRaw,
68                 afterRaw = afterRaw,
69                 operation = operation,
70                 tag = tag,
71             )
72             process(undoManager)
73             return
74         }
75
76         submit(
77             epoch = jobEpoch,
78             sequence = sequence,
79             layerId = layerId,
80             beforeRaw = beforeRaw,
81             afterRaw = afterRaw,
82             operation = operation,
83             tag = tag,
84         )
85     }
86
87     /** Applies completed work in commit order. Must be called on the GL thread. */
88     fun process(undoManager: UndoManager): Boolean {
89         var changed = false
90

```

```

91         while (true) {
92             val result = pendingResults.poll() ?: break
93             if (result.sequence < nextSequenceToApply) {
94                 discard(result)
95                 decrementPending()
96                 changed = true
97             } else {
98                 completedResults.put(result.sequence, result)?.let(::disposeReady)
99             }
100         }
101
102         while (true) {
103             val result = completedResults.remove(nextSequenceToApply) ?: break
104             apply(result, undoManager)
105             decrementPending()
106             nextSequenceToApply++
107             changed = true
108         }
109
110         return changed
111     }
112
113     /** Rejects compression results that belong to a document state that no longer exists. */
114     fun invalidate() {
115         epoch++
116         nextSequenceToApply = nextSequence
117         val discardedCount = completedResults.size
118         completedResults.values.forEach(::disposeReady)
119         completedResults.clear()
120         pendingCount = (pendingCount - discardedCount).coerceAtLeast(0)
121         staleResultCount += discardedCount
122     }
123
124     /** Reopens the worker after an EGL/context recreation. */
125     fun ensureRunning() {
126         if (executor.isShutdown) executor = executorFactory()
127     }
128
129     /** Releases worker-owned results without changing the caller's [UndoManager]. */
130     fun shutdown() {
131         executor.shutdownNow()
132         try {
133             executor.awaitTermination(SHUTDOWN_TIMEOUT_MS, TimeUnit.MILLISECONDS)
134         } catch (error: InterruptedException) {
135             Thread.currentThread().interrupt()
136         }
137
138         while (true) {
139             val result = pendingResults.poll() ?: break
140             disposeReady(result)
141         }
142         completedResults.values.forEach(::disposeReady)
143         completedResults.clear()
144         pendingCount = 0
145         nextSequenceToApply = nextSequence
146     }
147
148     private fun allocateSequence(): Long = nextSequence++
149
150     private fun submit(
151         epoch: Long,
152         sequence: Long,
153         layerId: Long,
154         beforeRaw: List<RawTileSnapshot>,
155         afterRaw: List<RawTileSnapshot>,
156         operation: UndoOperationType,
157         tag: String,
158     ) {
159         try {
160             executor.execute {
161                 val result = runCatching {
162                     readyResult(
163                         epoch = epoch,
164                         sequence = sequence,
165                         layerId = layerId,
166                         beforeRaw = beforeRaw,
167                         afterRaw = afterRaw,
168                         operation = operation,
169                         tag = tag,
170                     )
171                 }.getOrElse { error ->
172                     UndoJobResult.Failed(epoch = epoch, sequence = sequence, error = error)
173                 }
174                 pendingResults.add(result)
175                 requestRender()
176             }
177         } catch (error: RejectedExecutionException) {
178             pendingResults.add(
179                 UndoJobResult.Failed(epoch = epoch, sequence = sequence, error = error),
180             )
181             requestRender()
182         }
183     }

```

```

183     }
184
185     private fun compressSynchronously(
186         epoch: Long,
187         sequence: Long,
188         layerId: Long,
189         beforeRaw: List<RawTileSnapshot>,
190         afterRaw: List<RawTileSnapshot>,
191         operation: UndoOperationType,
192         tag: String,
193     ) {
194         val result = runCatching {
195             readyResult(
196                 epoch = epoch,
197                 sequence = sequence,
198                 layerId = layerId,
199                 beforeRaw = beforeRaw,
200                 afterRaw = afterRaw,
201                 operation = operation,
202                 tag = tag,
203             )
204         }.getOrElse { error ->
205             UndoJobResult.Failed(epoch = epoch, sequence = sequence, error = error)
206         }
207         pendingResults.add(result)
208     }
209
210     private fun readyResult(
211         epoch: Long,
212         sequence: Long,
213         layerId: Long,
214         beforeRaw: List<RawTileSnapshot>,
215         afterRaw: List<RawTileSnapshot>,
216         operation: UndoOperationType,
217         tag: String,
218     ): UndoJobResult.Ready = UndoJobResult.Ready(
219         epoch = epoch,
220         sequence = sequence,
221         entry = UndoEntry(
222             layerId = layerId,
223             beforeTiles = beforeRaw.map { it.compress(compressor) },
224             afterTiles = afterRaw.map { it.compress(compressor) },
225             operation = operation,
226             tag = tag,
227         ),
228     )
229
230     private fun apply(result: UndoJobResult, undoManager: UndoManager) {
231         when (result) {
232             is UndoJobResult.Ready -> {
233                 if (isUndoResultCurrent(result.epoch, epoch)) {
234                     undoManager.push(result.entry)
235                 } else {
236                     result.entry.dispose()
237                     staleResultCount++
238                 }
239             }
240             is UndoJobResult.Failed -> {
241                 compressionFailureCount++
242                 if (BuildConfig.DEBUG) {
243                     Log.e(TAG, "Undo compression failed", result.error)
244                 }
245             }
246         }
247     }
248 }
249
250 private fun discard(result: UndoJobResult) {
251     disposeReady(result)
252     staleResultCount++
253 }
254
255 private fun disposeReady(result: UndoJobResult) {
256     if (result is UndoJobResult.Ready) result.entry.dispose()
257 }
258
259 private fun decrementPending() {
260     pendingCount = (pendingCount - 1).coerceAtLeast(0)
261 }
262
263 private companion object {
264     const val DEFAULT_MAX_PENDING_JOBS = 8
265     const val SHUTDOWN_TIMEOUT_MS = 100L
266     const val TAG = "TileUndo"
267 }
268 }

```


39. app/src/main/java/com/wetinknext/engine/core/UndoHistory.kt

```
1 package com.wetinknext.engine.core
2
3 /** Pure epoch contract used to reject undo results from a previous document state. */
4 fun isUndoResultCurrent(
5     resultEpoch: Long,
6     currentEpoch: Long,
7 ): Boolean = resultEpoch == currentEpoch
```

40. app/src/main/java/com/wetinknext/engine/core/UndoJobResult.kt

```
1 package com.wetinknext.engine.core
2
3 import com.wetinknext.engine.undo.UndoEntry
4
5 /** Terminal outcome of one tile-compression task. */
6 sealed interface UndoJobResult {
7     val epoch: Long
8     val sequence: Long
9
10     data class Ready(
11         override val epoch: Long,
12         override val sequence: Long,
13         val entry: UndoEntry,
14     ) : UndoJobResult
15
16     data class Failed(
17         override val epoch: Long,
18         override val sequence: Long,
19         val error: Throwable,
20     ) : UndoJobResult
21 }
```

41. app/src/main/java/com/wetinknext/engine/core/ViewTransform.kt

```

1 package com.wetinknext.engine.core
2
3 import kotlin.math.abs
4 import kotlin.math.cos
5 import kotlin.math.sin
6
7 /** Canonical canvas-to-view-local-screen transformation. */
8 data class ViewTransform(
9     val translateX: Float = 0f,
10    val translateY: Float = 0f,
11    val scale: Float = 1f,
12    val rotationRad: Float = 0f,
13    val flipX: Boolean = false,
14 ) {
15     private val flipSign: Float
16     get() = if (flipX) -1f else 1f
17
18     val m00: Float get() = cos(rotationRad) * scale * flipSign
19     val m01: Float get() = -sin(rotationRad) * scale
20     val m10: Float get() = sin(rotationRad) * scale * flipSign
21     val m11: Float get() = cos(rotationRad) * scale
22
23     fun canvasToScreen(canvasX: Float, canvasY: Float, out: FloatArray) {
24         require(out.size >= 2)
25         out[0] = m00 * canvasX + m01 * canvasY + translateX
26         out[1] = m10 * canvasX + m11 * canvasY + translateY
27     }
28
29     fun screenToCanvas(screenX: Float, screenY: Float, out: FloatArray) {
30         require(out.size >= 2)
31         val determinant = m00 * m11 - m01 * m10
32         if (abs(determinant) < 1e-8f) {
33             out[0] = 0f
34             out[1] = 0f
35             return
36         }
37         val inverseDeterminant = 1f / determinant
38         val dx = screenX - translateX
39         val dy = screenY - translateY
40         out[0] = (m11 * dx - m01 * dy) * inverseDeterminant
41         out[1] = (-m10 * dx + m00 * dy) * inverseDeterminant
42     }
43
44     fun zoomAround(anchorX: Float, anchorY: Float, newScale: Float): ViewTransform {
45         val safeScale = newScale.coerceIn(MIN_SCALE, MAX_SCALE)
46         val factor = safeScale / scale
47         return copy(
48             scale = safeScale,
49             translateX = anchorX - (anchorX - translateX) * factor,
50             translateY = anchorY - (anchorY - translateY) * factor,
51         )
52     }
53
54     fun rotateAround(anchorX: Float, anchorY: Float, deltaRad: Float): ViewTransform {
55         val c = cos(deltaRad)
56         val s = sin(deltaRad)
57         val dx = translateX - anchorX
58         val dy = translateY - anchorY
59         return copy(
60             rotationRad = rotationRad + deltaRad,
61             translateX = anchorX + dx * c - dy * s,
62             translateY = anchorY + dx * s + dy * c,
63         )
64     }
65
66     /** Applies one screen-space navigation update around a shared gesture anchor. */
67     fun applyGestureDelta(
68         panX: Float,
69         panY: Float,
70         anchorX: Float,
71         anchorY: Float,
72         zoomFactor: Float,
73         rotationDelta: Float,
74     ): ViewTransform {
75         val translated = copy(
76             translateX = translateX + panX,
77             translateY = translateY + panY,
78         )
79         val zoomed = translated.zoomAround(
80             anchorX = anchorX,
81             anchorY = anchorY,
82             newScale = scale * zoomFactor,
83         )
84         return zoomed.rotateAround(anchorX, anchorY, rotationDelta)
85     }
86
87     /** Builds a column-major matrix for canvas to OpenGL clip coordinates. */
88     fun buildCanvasToClip(viewWidth: Float, viewHeight: Float, out: FloatArray) {
89         require(out.size >= 16)
90         require(viewWidth > 0f)

```

```

91         require(viewHeight > 0f)
92         val sx = 2f / viewWidth
93         val sy = -2f / viewHeight
94         out[0] = m00 * sx; out[1] = m10 * sy; out[2] = 0f; out[3] = 0f
95         out[4] = m01 * sx; out[5] = m11 * sy; out[6] = 0f; out[7] = 0f
96         out[8] = 0f; out[9] = 0f; out[10] = 1f; out[11] = 0f
97         out[12] = translateX * sx - 1f; out[13] = translateY * sy + 1f
98         out[14] = 0f; out[15] = 1f
99     }
100
101     companion object {
102         const val MIN_SCALE = 0.02f
103         const val MAX_SCALE = 64f
104
105         fun buildCanvasToFbo(canvasWidth: Float, canvasHeight: Float, out: FloatArray) {
106             require(out.size >= 16); require(canvasWidth > 0f); require(canvasHeight > 0f)
107             out[0]=2f/canvasWidth;out[1]=0f;out[2]=0f;out[3]=0f;out[4]=0f;out[5]=2f/canvasHeight;out[6]=0f;out[7]=0f;out[
8]=0f;out[9]=0f;out[10]=1f;out[11]=0f;out[12]=-1f;out[13]=-1f;out[14]=0f;out[15]=1f
108         }
109     }
110 }

```

42. app/src/main/java/com/wetinknext/engine/gl/CanvasGeometry.kt

```
1 package com.wetinknext.engine.gl
2
3 import android.opengl.GLES30
4 import java.nio.ByteBuffer
5 import java.nio.ByteOrder
6
7 /** A document-sized quad; GL resources are owned by the render thread. */
8 class CanvasGeometry {
9     var vaoId = 0
10     private set
11     private var vboId = 0
12
13     fun create(width: Int, height: Int) {
14         release()
15         val vertices = floatArrayOf(0f, 0f, width.toFloat(), 0f, 0f, height.toFloat(), width.toFloat(), height.toFloat())
16         val data = ByteBuffer.allocateDirect(vertices.size * Float.SIZE_BYTES).order(ByteOrder.nativeOrder()).asFloatBuffer()
17         data.put(vertices).position(0)
18         val ids = IntArray(1)
19         GLES30.glGenVertexArrays(1, ids, 0); vaoId = ids[0]
20         GLES30.glGenBuffers(1, ids, 0); vboId = ids[0]
21         GLES30.glBindVertexArray(vaoId); GLES30.glBindBuffer(GLES30.GL_ARRAY_BUFFER, vboId)
22         GLES30.glBufferData(GLES30.GL_ARRAY_BUFFER, vertices.size * Float.SIZE_BYTES, data, GLES30.GL_STATIC_DRAW)
23         GLES30.glEnableVertexAttribArray(0); GLES30.glVertexAttribPointer(0, 2, GLES30.GL_FLOAT, false, 0, 0)
24         GLES30.glBindBuffer(GLES30.GL_ARRAY_BUFFER, 0); GLES30.glBindVertexArray(0)
25     }
26
27     fun draw() {
28         check(vaoId != 0)
29         GLES30.glBindVertexArray(vaoId); GLES30.glDrawArrays(GLES30.GL_TRIANGLE_STRIP, 0, 4); GLES30.glBindVertexArray(0)
30     }
31
32     fun release() {
33         if (vboId != 0) GLES30.glDeleteBuffers(1, intArrayOf(vboId), 0)
34         if (vaoId != 0) GLES30.glDeleteVertexArrays(1, intArrayOf(vaoId), 0)
35         vaoId = 0; vboId = 0
36     }
37 }
```

43. app/src/main/java/com/wetinknext/engine/gl/GeometryCache.kt

```
1 package com.wetinknext.engine.gl
2
3 import android.opengl.GLES30
4 import java.nio.ByteBuffer
5 import java.nio.ByteOrder
6
7 /** A single cached fullscreen triangle for future present passes. */
8 class GeometryCache {
9     private var vaoId = 0
10    private var vboId = 0
11    fun create() {
12        if (vaoId != 0) return
13        val vertices = floatArrayOf(-1f, -1f, 3f, -1f, -1f, 3f)
14        val data = ByteBuffer.allocateDirect(vertices.size * Float.SIZE_BYTES).order(ByteOrder.nativeOrder()).asFloatBuffer()
15        data.put(vertices).position(0)
16        val ids = IntArray(1); GLES30.glGenVertexArrays(1, ids, 0); vaoId = ids[0]
17        GLES30.glGenBuffers(1, ids, 0); vboId = ids[0]
18        GLES30.glBindVertexArray(vaoId); GLES30.glBindBuffer(GLES30.GL_ARRAY_BUFFER, vboId)
19        GLES30.glBufferData(GLES30.GL_ARRAY_BUFFER, vertices.size * Float.SIZE_BYTES, data, GLES30.GL_STATIC_DRAW)
20        GLES30.glEnableVertexAttribArray(0); GLES30.glVertexAttribPointer(0, 2, GLES30.GL_FLOAT, false, 0, 0)
21        GLES30.glBindBuffer(GLES30.GL_ARRAY_BUFFER, 0); GLES30.glBindVertexArray(0)
22    }
23    fun drawFullscreenTriangle() { check(vaoId != 0); GLES30.glBindVertexArray(vaoId); GLES30.glDrawArrays(GLES30.
GL_TRIANGLES, 0, 3); GLES30.glBindVertexArray(0) }
24    fun release() { if (vboId != 0) GLES30.glDeleteBuffers(1, intArrayOf(vboId), 0); if (vaoId != 0) GLES30.
glDeleteVertexArrays(1, intArrayOf(vaoId), 0); vboId = 0; vaoId = 0 }
25 }
```

44. app/src/main/java/com/wetinknext/engine/gl/GLCaps.kt

```
1 package com.wetinknext.engine.gl
2
3 import android.opengl.GLES30
4
5 /** Capabilities queried only after a GLES 3 context is current. */
6 data class GLCaps(
7     val renderer: String,
8     val version: String,
9     val supportsHalfFloatColorBuffer: Boolean,
10 ) {
11     companion object {
12         fun query(): GLCaps {
13             val extensions = GLES30.glGetString(GLES30.GL_EXTENSIONS).orEmpty()
14             val declared = extensions.contains("EXT_color_buffer_half_float") ||
15                 extensions.contains("EXT_color_buffer_float")
16             // Extension string сам по себе не доказательство: проверяем реальный FBO 4x4.
17             val supported = declared && RenderTarget().probeHalfFloatColorBuffer()
18             return GLCaps(
19                 renderer = GLES30.glGetString(GLES30.GL_RENDERER).orEmpty(),
20                 version = GLES30.glGetString(GLES30.GL_VERSION).orEmpty(),
21                 supportsHalfFloatColorBuffer = supported,
22             )
23         }
24     }
25 }
```

45. app/src/main/java/com/wetinknext/engine/gl/GLCheck.kt

```
1 package com.wetinknext.engine.gl
2
3 import android.opengl.GLES30
4
5 /** GL error checks intended for setup boundaries, never the render hot path. */
6 object GLCheck {
7     @Volatile
8     private var glThread: Thread? = null
9
10    fun setGLThread(thread: Thread) {
11        glThread = thread
12    }
13
14    fun checkOnGLThread() {
15        val current = Thread.currentThread()
16        val expected = glThread
17        check(current === expected) {
18            "GL resource accessed outside GLSurfaceView render thread: current=$current, expected=$expected"
19        }
20    }
21
22    fun noError(label: String) {
23        val error = GLES30.glGetError()
24        check(error == GLES30.GL_NO_ERROR) { "$label: GL error 0x${error.toString(16)}" }
25    }
26
27    fun framebufferComplete(label: String) {
28        val status = GLES30.glCheckFramebufferStatus(GLES30.GL_FRAMEBUFFER)
29        check(status == GLES30.GL_FRAMEBUFFER_COMPLETE) {
30            "$label: framebuffer incomplete (0x${status.toString(16)})"
31        }
32    }
33 }
```


46. app/src/main/java/com/wetinknext/engine/gl/GLProgram.kt

```
1 package com.wetinknext.engine.gl
2
3 import android.opengl.GLES30
4
5 /** Linked GLSL program. Construction and release must run on the GL thread. */
6 class GLProgram(vertexSource: String, fragmentSource: String) {
7     val id: Int = GLES30.glCreateProgram().also { program ->
8         check(program != 0) { "Unable to create GL program" }
9         val vertex = compile(GLES30.GL_VERTEX_SHADER, vertexSource)
10        val fragment = compile(GLES30.GL_FRAGMENT_SHADER, fragmentSource)
11        GLES30.glAttachShader(program, vertex); GLES30.glAttachShader(program, fragment); GLES30.glLinkProgram(program)
12        val linked = IntArray(1); GLES30.glGetProgramiv(program, GLES30.GL_LINK_STATUS, linked, 0)
13        GLES30.glDeleteShader(vertex); GLES30.glDeleteShader(fragment)
14        check(linked[0] == GLES30.GL_TRUE) { "Program link failed: ${GLES30.glGetProgramInfoLog(program)}" }
15    }
16    fun use() = GLES30.glUseProgram(id)
17    fun release() = GLES30.glDeleteProgram(id)
18
19    private fun compile(type: Int, source: String): Int = GLES30.glCreateShader(type).also { shader ->
20        check(shader != 0) { "Unable to create shader" }
21        GLES30.glShaderSource(shader, source); GLES30.glCompileShader(shader)
22        val compiled = IntArray(1); GLES30.glGetShaderiv(shader, GLES30.GL_COMPILE_STATUS, compiled, 0)
23        check(compiled[0] == GLES30.GL_TRUE) { "Shader compile failed: ${GLES30.glGetShaderInfoLog(shader)}" }
24    }
25 }
```

47. app/src/main/java/com/wetinknext/engine/gl/RenderTarget.kt

```

1  package com.wetinknext.engine.gl
2
3  import android.opengl.GLES30
4
5  /** Texture-backed FBO with RGBA16F probing and a guaranteed RGBA8 fallback. */
6  class RenderTarget {
7      var framebufferId: Int = 0
8      private set
9      var textureId: Int = 0
10     private set
11     var width: Int = 0
12     private set
13     var height: Int = 0
14     private set
15     var usesHalfFloat: Boolean = false
16     private set
17
18     fun create(width: Int, height: Int, preferHalfFloat: Boolean) {
19         GLCheck.checkOnGLThread()
20         require(width > 0 && height > 0)
21         release()
22         if (preferHalfFloat && createInternal(width, height, true)) return
23         check(createInternal(width, height, false)) { "RGBA8 render target creation failed" }
24     }
25
26     /** A 4x4 completeness probe used before allocating a document-sized target. */
27     fun probeHalfFloatColorBuffer(): Boolean {
28         release()
29         val result = createInternal(4, 4, true)
30         release()
31         return result
32     }
33
34     fun bind() {
35         check(framebufferId != 0) { "RenderTarget is not created" }
36         GLES30.glBindFramebuffer(GLES30.GL_FRAMEBUFFER, framebufferId)
37     }
38
39     fun clear(red: Float, green: Float, blue: Float, alpha: Float) {
40         bind()
41         // Clearing is constrained by the current viewport. Always restore the
42         // offscreen target's extent, not the last screen-sized present viewport.
43         GLES30.glViewport(0, 0, width, height)
44         GLES30.glDisable(GLES30.GL_SCISSOR_TEST)
45         GLES30.glDisable(GLES30.GL_BLEND)
46         GLES30.glColorMask(true, true, true, true)
47         GLES30.glClearColor(red, green, blue, alpha)
48         GLES30.glClear(GLES30.GL_COLOR_BUFFER_BIT)
49         GLES30.glBindFramebuffer(GLES30.GL_FRAMEBUFFER, 0)
50     }
51
52     fun release() {
53         GLCheck.checkOnGLThread()
54         if (framebufferId != 0) GLES30.glDeleteFramebuffers(1, intArrayOf(framebufferId), 0)
55         if (textureId != 0) GLES30.glDeleteTextures(1, intArrayOf(textureId), 0)
56         framebufferId = 0; textureId = 0; width = 0; height = 0; usesHalfFloat = false
57     }
58
59     private fun createInternal(targetWidth: Int, targetHeight: Int, halfFloat: Boolean): Boolean {
60         val textures = IntArray(1)
61         val framebuffers = IntArray(1)
62         GLES30.glGenTextures(1, textures, 0)
63         GLES30.glGenFramebuffers(1, framebuffers, 0)
64         val texture = textures[0]
65         val framebuffer = framebuffers[0]
66         if (texture == 0 || framebuffer == 0) return false
67         GLES30.glBindTexture(GLES30.GL_TEXTURE_2D, texture)
68         GLES30.glTexParameteri(GLES30.GL_TEXTURE_2D, GLES30.GL_TEXTURE_MIN_FILTER, GLES30.GL_LINEAR)
69         GLES30.glTexParameteri(GLES30.GL_TEXTURE_2D, GLES30.GL_TEXTURE_MAG_FILTER, GLES30.GL_LINEAR)
70         GLES30.glTexParameteri(GLES30.GL_TEXTURE_2D, GLES30.GL_TEXTURE_WRAP_S, GLES30.GL_CLAMP_TO_EDGE)
71         GLES30.glTexParameteri(GLES30.GL_TEXTURE_2D, GLES30.GL_TEXTURE_WRAP_T, GLES30.GL_CLAMP_TO_EDGE)
72         GLES30.glTexImage2D(GLES30.GL_TEXTURE_2D, 0, if (halfFloat) GLES30.GL_RGBA16F else GLES30.GL_RGBA8,
73             targetWidth, targetHeight, 0, GLES30.GL_RGBA, if (halfFloat) GLES30.GL_HALF_FLOAT else GLES30.GL_UNSIGNED_BYTE,
74             null)
75         GLES30.glBindFramebuffer(GLES30.GL_FRAMEBUFFER, framebuffer)
76         GLES30.glFramebufferTexture2D(GLES30.GL_FRAMEBUFFER, GLES30.GL_COLOR_ATTACHMENT0, GLES30.GL_TEXTURE_2D, texture, 0)
77         val complete = GLES30.glCheckFramebufferStatus(GLES30.GL_FRAMEBUFFER) == GLES30.GL_FRAMEBUFFER_COMPLETE
78         GLES30.glBindFramebuffer(GLES30.GL_FRAMEBUFFER, 0)
79         GLES30.glBindTexture(GLES30.GL_TEXTURE_2D, 0)
80         if (!complete) {
81             GLES30.glDeleteFramebuffers(1, framebuffers, 0); GLES30.glDeleteTextures(1, textures, 0)
82             return false
83         }
84         framebufferId = framebuffer; textureId = texture; width = targetWidth; height = targetHeight; usesHalfFloat =
85         halfFloat
86     }
87 }

```

48. app/src/main/java/com/wetinknext/engine/gl/ShaderLib.kt

```

1  package com.wetinknext.engine.gl
2
3  object ShaderLib {
4      const val compositorVertex = ""#version 300 es
5          layout(location=0) in vec2 aPosition;uniform mat4 uCanvasToClip;uniform vec2 uCanvasSize;out vec2 vUv;void main(){
6      const val compositorFragment = ""#version 300 es
7          precision highp float;
8
9          in vec2 vUv;
10         uniform sampler2D uLayerTex;
11         uniform sampler2D uStrokeTex;
12         uniform int uStrokeActive;
13         uniform float uOpacity;
14         uniform float uStrokeOpacity;
15
16         out vec4 fragColor;
17
18         vec3 linearToSrgb(vec3 color) {
19             vec3 c = clamp(color, 0.0, 1.0);
20             vec3 low = c * 12.92;
21             vec3 high = 1.055 * pow(c, vec3(1.0 / 2.4)) - 0.055;
22             return mix(low, high, step(vec3(0.0031308), c));
23         }
24
25         vec3 unpremultiply(vec4 color) {
26             return color.a <= 0.0001 ? vec3(0.0) : color.rgb / color.a;
27         }
28
29         void main() {
30             vec4 layer = texture(uLayerTex, vUv);
31             if (uStrokeActive == 1) {
32                 vec4 stroke = texture(uStrokeTex, vUv) * uStrokeOpacity;
33                 layer = stroke + layer * (1.0 - stroke.a);
34             }
35
36             float alpha = layer.a * uOpacity;
37             fragColor = vec4(linearToSrgb(unpremultiply(layer)) * alpha, alpha);
38         }
39     ""
40     const val dabVertex = ""#version 300 es
41         layout(location = 0) in vec2 aCorner;
42         layout(location = 1) in vec4 iDab0;
43         layout(location = 2) in vec2 iDab1;
44         uniform mat4 uCanvasToClip;
45         uniform vec2 uCanvasSize;
46         out vec2 vLocal;
47         out vec2 vCanvasUv;
48         flat out float vCoverage;
49         flat out float vFlow;
50         void main(){
51             float c=cos(iDab0.w), s=sin(iDab0.w);
52             vec2 r=vec2(c*aCorner.x-s*aCorner.y,s*aCorner.x+c*aCorner.y);
53             vLocal=aCorner;
54             vCoverage=iDab1.x;
55             vFlow=iDab1.y;
56             vec2 p = iDab0.xy+r*iDab0.z;
57             vCanvasUv = p / uCanvasSize;
58             gl_Position=uCanvasToClip*vec4(p,0.,1.);
59         }
60     ""
61     const val dabFragment = ""#version 300 es
62         precision highp float;
63         in vec2 vLocal;
64         in vec2 vCanvasUv;
65         flat in float vCoverage;
66         flat in float vFlow;
67         uniform vec3 uColorLinear;
68         uniform sampler2D uGrainTex;
69         uniform int uGrainActive;
70         uniform float uGrainScale;
71         uniform int uGrainCanvasLocked;
72         uniform float uTextureDepth;
73         uniform float uTextureContrast;
74         uniform float uStrokeOpacity;
75         uniform sampler2D uShapeTex;
76         uniform int uShapeActive;
77         uniform int uReverseShape;
78         uniform int uRgbToAlpha;
79         out vec4 fragColor;
80         float luminance(vec3 color) {
81             return dot(color, vec3(0.299, 0.587, 0.114));
82         }
83         float applyTextureLevels(float value, float contrast, float depth) {
84             float adjusted = clamp((value - 0.5) * contrast + 0.5, 0.0, 1.0);
85             return mix(1.0 - depth, 1.0, adjusted);
86         }
87         void main(){
88             float r=length(vLocal);
89             float aa=max(fwidth(r),.001);

```

```

90         float cov=1.-smoothstep(1.-aa,1.+aa,r);
91         if (cov <= 0.0) discard;
92
93         float shapeMask = 1.0;
94         if (uShapeActive == 1) {
95             vec2 shapeUv = vLocal * 0.5 + 0.5;
96             vec4 shapeColor = texture(uShapeTex, shapeUv);
97             shapeMask = uRgbToAlpha == 1
98                 ? dot(shapeColor.rgb, vec3(0.299, 0.587, 0.114))
99                 : shapeColor.a;
100             if (uReverseShape == 1) shapeMask = 1.0 - shapeMask;
101         }
102
103         float grainFactor = 1.0;
104         if (uGrainActive == 1) {
105             vec2 uv = uGrainCanvasLocked == 1
106                 ? vCanvasUv * max(uGrainScale, 0.0001)
107                 : (vLocal * 0.5 + 0.5) * max(uGrainScale, 0.0001);
108             vec4 grainColor = texture(uGrainTex, uv);
109             grainFactor = applyTextureLevels(
110                 luminance(grainColor.rgb), uTextureContrast, uTextureDepth
111             );
112         }
113
114         float a = vCoverage * vFlow * cov * shapeMask * grainFactor * uStrokeOpacity;
115         fragColor=vec4(uColorLinear*a,a);
116     }
117     ""
118     const val strokeCompositeFragment = ""#version 300 es
119     precision highp float; in vec2 vUv; uniform sampler2D uCanvasTex; uniform sampler2D uStrokeTex; uniform int
uStrokeActive; out vec4 fragColor;
120     vec3 toSrgb(vec3 c){c=clamp(c,0.,1.);return mix(c*12.92,1.055*pow(c,vec3(1./2.4))-0.55,step(vec3(.0031308),c));}
121     vec3 unpre(vec4 c){return c.a<=.0001?vec3(0.):c.rgb/c.a;}
122     void main(){vec4 c=texture(uCanvasTex,vUv);vec4 s=texture(uStrokeTex,vUv);float a=uStrokeActive==1?s.a+c.a*(1.-s.a):c
a;vec3 rgb=uStrokeActive==1&&a>.0001?(unpre(s)*s.a+unpre(c)*c.a*(1.-s.a))/a:unpre(c);fragColor=vec4(toSrgb(rgb),1.);}
123     ""
124     const val canvasPresentVertex = ""#version 300 es
125     layout(location = 0) in vec2 aPosition;
126     uniform mat4 uCanvasToClip;
127     uniform vec2 uCanvasSize;
128     out vec2 vUv;
129     void main() {
130         vUv = aPosition / uCanvasSize;
131         gl_Position = uCanvasToClip * vec4(aPosition, 0.0, 1.0);
132     }
133     ""
134     const val presentFragment = ""#version 300 es
135     precision highp float;
136     in vec2 vUv;
137     uniform sampler2D uTexture;
138     out vec4 fragColor;
139     void main() { fragColor = texture(uTexture, vUv); }
140     ""
141     const val fullscreenVertex = ""#version 300 es
142     layout(location = 0) in vec2 aPosition;
143     out vec2 vUv;
144     void main() { vUv = aPosition * 0.5 + 0.5; gl_Position = vec4(aPosition, 0.0, 1.0); }
145     ""
146     const val solidFragment = ""#version 300 es
147     precision highp float;
148     in vec2 vUv;
149     out vec4 fragColor;
150     void main() { fragColor = vec4(vUv, 0.0, 1.0); }
151     ""
152     const val ribbonVertex = ""#version 300 es
153     layout(location=0) in vec2 aPos; layout(location=1) in float aCoverage; layout(location=2) in float aAlpha;
154     uniform mat4 uCanvasToClip; out float vCoverage; out float vAlpha;
155     void main(){vCoverage=aCoverage;vAlpha=aAlpha;gl_Position=uCanvasToClip*vec4(aPos,0.,1.);}
156     ""
157     const val ribbonFragment = ""#version 300 es
158     precision highp float; in float vCoverage; in float vAlpha; uniform vec3 uColorLinear; uniform float uFlow;
159     uniform int uAntiAliasLevel; uniform int uNoAntialias; out vec4 fragColor;
160     void main(){float cov=vCoverage;if(uNoAntialias==1||uAntiAliasLevel==0)cov=step(.5,cov);else{float
e=uAntiAliasLevel==1?.35:uAntiAliasLevel==2?.5:.7;cov=smoothstep(.5-e,.5+e,cov);}float a=vAlpha*cov*uFlow;fragColor=vec4(
uColorLinear*a,a);}
161     ""
162
163     /**
164     * Один сегмент штриха = капсула (round cone): два круга с линейной интерполяцией радиуса.
165     * Вершинный шейдер строит только AABB сегмента, форма считается во фрагменте.
166     */
167     const val capsuleVertex = ""#version 300 es
168     layout(location = 0) in vec2 aCorner; // -1..1, TRIANGLE_STRIP из 4 вершин
169     layout(location = 1) in vec4 iA; // x, y, radius, coverage (начало сегмента)
170     layout(location = 2) in vec4 iB; // x, y, radius, coverage (конец сегмента)
171     uniform mat4 uCanvasToClip;
172     uniform vec2 uCanvasSize;
173     out vec2 vPos;
174     out vec2 vCanvasUv;
175     flat out vec4 vA;
176     flat out vec4 vB;
177     void main() {

```

```

178         // Запас 2 px на AA-переход, иначе край обрежется по границе квада.
179         vec2 lo = min(iA.xy - iA.z, iB.xy - iB.z) - 2.0;
180         vec2 hi = max(iA.xy + iA.z, iB.xy + iB.z) + 2.0;
181         vec2 p = mix(lo, hi, aCorner * 0.5 + 0.5);
182         vPos = p;
183         vCanvasUv = p / uCanvasSize;
184         vA = iA;
185         vB = iB;
186         gl_Position = uCanvasToClip * vec4(p, 0.0, 1.0);
187     }
188     ""
189
190     /**
191     * Аналитическое покрытие: нет каппов, джойнов, miter и AA-юбки как отдельных сущностей.
192     * Выход строго premultiplied linear: rgb = color * a, a = cov.
193     * Рисовать premultiplied source-over blending:
194     * GL_FUNC_ADD + GL_ONE, GL_ONE_MINUS_SRC_ALPHA.
195     */
196     const val capsuleFragment = ""#version 300 es
197         precision highp float;
198         in vec2 vPos;
199         in vec2 vCanvasUv;
200         flat in vec4 vA;
201         flat in vec4 vB;
202         uniform vec3 uColorLinear;
203         uniform sampler2D uGrainTex;
204         uniform int uGrainActive;
205         uniform float uGrainScale;
206         uniform int uGrainCanvasLocked;
207         uniform float uTextureDepth;
208         uniform float uTextureContrast;
209         uniform float uFlow;
210         out vec4 fragColor;
211
212         float sdRoundCone(vec2 p, vec2 a, vec2 b, float r1, float r2) {
213             vec2 ba = b - a;
214             float l2 = dot(ba, ba);
215             float rr = r1 - r2;
216
217             // Guard 1: вырожденный сегмент (тап, дубль сэмпла) -> обычный круг.
218             if (l2 < 1e-6) return length(p - a) - max(r1, r2);
219             // Guard 2: один круг поглощает другой (скачок давления на коротком шаге).
220             // Без этого a2 < 0 -> sqrt(отрицательного) -> NaN -> чёрные клинья.
221             if (rr * rr >= l2) return (r1 > r2) ? (length(p - a) - r1) : (length(p - b) - r2);
222
223             float a2 = l2 - rr * rr;
224             float il2 = 1.0 / l2;
225             vec2 pa = p - a;
226             float y = dot(pa, ba);
227             float z = y - l2;
228             vec2 xv = pa * l2 - ba * y;
229             float x2 = dot(xv, xv);
230             float y2 = y * y * l2;
231             float z2 = z * z * l2;
232             float k = sign(rr) * rr * rr * x2;
233
234             if (sign(z) * a2 * z2 > k) return sqrt(x2 + z2) * il2 - r2; // круглый конец у B
235             if (sign(y) * a2 * y2 < k) return sqrt(x2 + y2) * il2 - r1; // круглый конец у A
236             return (sqrt(x2 * a2 * il2) + y * rr) * il2 - r1; // касательная боковина
237         }
238
239         void main() {
240             float d = sdRoundCone(vPos, vA.xy, vB.xy, vA.z, vB.z);
241             float w = max(fwidth(d), 1e-4);
242             float cov = 1.0 - smoothstep(-w, w, d);
243             if (cov <= 0.0) discard;
244
245             float grainFactor = 1.0;
246             if (uGrainActive == 1) {
247                 vec2 uv = vCanvasUv * max(uGrainScale, 0.0001);
248                 float val = texture(uGrainTex, uv).r;
249                 val = clamp((val - 0.5) * uTextureContrast + 0.5, 0.0, 1.0);
250                 grainFactor = mix(1.0 - uTextureDepth, 1.0, val);
251             }
252
253             float segmentCoverage = mix(vA.w, vB.w, 0.5);
254             float finalAlpha = cov * segmentCoverage * grainFactor * uFlow;
255             fragColor = vec4(uColorLinear * finalAlpha, finalAlpha);
256         }
257     ""
258
259     const val strokeBlitFragment = ""#version 300 es
260         precision highp float;
261         in vec2 vUv;
262         uniform sampler2D uStrokeTex;
263         uniform float uOpacity;
264         out vec4 fragColor;
265         void main() { fragColor = texture(uStrokeTex, vUv) * uOpacity; }
266     ""
267 }

```

49. app/src/main/java/com/wetinknext/engine/input/GestureRecognizer.kt

```

1 package com.wetinknext.engine.input
2
3 import android.view.MotionEvent
4 import com.wetinknext.engine.core.Camera
5 import kotlin.math.atan2
6 import kotlin.math.hypot
7
8 /**
9  * Routes view touch input before it reaches the brush pipeline.
10  *
11  * One pointer draws. A second pointer cancels that stroke immediately and
12  * takes ownership of the interaction for canvas navigation. Two- and
13  * three-pointer taps trigger undo and redo respectively.
14  */
15 class GestureRouter(
16     private val camera: Camera,
17     private val drawingInput: StrokeInputCapturer,
18     private val callbacks: Callbacks,
19 ) {
20     interface Callbacks {
21         fun cancelDrawing()
22         fun undoGesture()
23         fun redoGesture()
24         fun resetCamera()
25         fun requestRender()
26     }
27
28     private var mode = InputMode.IDLE
29     private var firstX = 0f
30     private var firstY = 0f
31     private var secondX = 0f
32     private var secondY = 0f
33     private var previousDistance = 0f
34     private var previousAngle = 0f
35     private var gestureStartTime = 0L
36     private var gestureMoved = false
37     private var navigationPointerCount = 0
38     private var drawingStartTime = 0L
39     private var drawingStartX = 0f
40     private var drawingStartY = 0f
41     private var drawingMoved = false
42     private var lastTapTime = 0L
43     private var lastTapX = 0f
44     private var lastTapY = 0f
45
46     fun onTouchEvent(event: MotionEvent): Boolean = when (event.actionMasked) {
47         MotionEvent.ACTION_DOWN -> {
48             mode = InputMode.DRAWING
49             drawingStartTime = event.eventTime
50             drawingStartX = event.x
51             drawingStartY = event.y
52             drawingMoved = false
53             drawingInput.onTouchEvent(event)
54             true
55         }
56
57         MotionEvent.ACTION_POINTER_DOWN -> {
58             if (event.pointerCount >= 2) {
59                 beginNavigation(event)
60                 true
61             } else {
62                 false
63             }
64         }
65
66         MotionEvent.ACTION_MOVE -> when (mode) {
67             InputMode.DRAWING -> {
68                 if (hypot(event.x - drawingStartX, event.y - drawingStartY) > TAP_MOVE_THRESHOLD_PX) {
69                     drawingMoved = true
70                     lastTapTime = 0L
71                 }
72                 drawingInput.onTouchEvent(event)
73                 true
74             }
75
76             InputMode.NAVIGATING -> {
77                 updateNavigation(event)
78                 true
79             }
80
81             else -> false
82         }
83
84         MotionEvent.ACTION_POINTER_UP -> {
85             if (mode == InputMode.NAVIGATING) {
86                 handleNavigationPointerUp(event)
87                 true
88             } else {
89                 drawingInput.onTouchEvent(event)
90                 true

```

```

91     }
92 }
93
94 MotionEvent.ACTION_UP -> {
95     when (mode) {
96         InputMode.DRAWING -> {
97             drawingInput.onTouchEvent(event)
98             val duration = event.eventTime - drawingStartTime
99             if (!drawingMoved && duration <= TAP_MAX_DURATION_MS) {
100                 handleSingleFingerTap(event)
101             }
102         }
103         InputMode.NAVIGATING -> finishNavigation(event)
104         else -> Unit
105     }
106     mode = InputMode.IDLE
107     true
108 }
109
110 MotionEvent.ACTION_CANCEL -> {
111     drawingInput.cancelActiveStroke()
112     callbacks.cancelDrawing()
113     mode = InputMode.CANCELLED
114     callbacks.requestRender()
115     true
116 }
117
118 else -> false
119 }
120
121 private fun beginNavigation(event: MotionEvent) {
122     lastTapTime = 0L
123     if (mode == InputMode.DRAWING) {
124         // The queued CANCEL keeps the brush pipeline coherent; the callback
125         // clears an already-rendered preview on the GL thread.
126         drawingInput.cancelActiveStroke()
127         callbacks.cancelDrawing()
128     }
129
130     mode = InputMode.NAVIGATING
131     navigationPointerCount = event.pointerCount
132     gestureStartTime = event.eventTime
133     gestureMoved = false
134     captureNavigationPointers(event)
135     callbacks.requestRender()
136 }
137
138 private fun updateNavigation(event: MotionEvent) {
139     if (event.pointerCount < 2) return
140
141     val newFirstX = event.getX(0)
142     val newFirstY = event.getY(0)
143     val newSecondX = event.getX(1)
144     val newSecondY = event.getY(1)
145     val oldCenterX = (firstX + secondX) * .5f
146     val oldCenterY = (firstY + secondY) * .5f
147     val newCenterX = (newFirstX + newSecondX) * .5f
148     val newCenterY = (newFirstY + newSecondY) * .5f
149     val newDistance = distance(newFirstX, newFirstY, newSecondX, newSecondY)
150     val zoomFactor = if (previousDistance > MIN_DISTANCE) newDistance / previousDistance else 1f
151     val newAngle = angle(newFirstX, newFirstY, newSecondX, newSecondY)
152     val rotationDelta = normalizeAngle(newAngle - previousAngle)
153     val panX = newCenterX - oldCenterX
154     val panY = newCenterY - oldCenterY
155
156     camera.set(
157         camera.snapshot().applyGestureDelta(
158             panX = panX,
159             panY = panY,
160             anchorX = newCenterX,
161             anchorY = newCenterY,
162             zoomFactor = zoomFactor,
163             rotationDelta = rotationDelta,
164         ),
165     )
166
167     if (hypot(panX, panY) > TAP_MOVE_THRESHOLD_PX ||
168         kotlin.math.abs(zoomFactor - 1f) > TAP_ZOOM_THRESHOLD ||
169         kotlin.math.abs(rotationDelta) > TAP_ROTATION_THRESHOLD_RAD
170     ) {
171         gestureMoved = true
172     }
173
174     firstX = newFirstX
175     firstY = newFirstY
176     secondX = newSecondX
177     secondY = newSecondY
178     previousDistance = newDistance
179     previousAngle = newAngle
180     callbacks.requestRender()
181 }
182

```

```

183 private fun handleNavigationPointerUp(event: MotionEvent) {
184     // Once fewer than two pointers remain, navigation stays active until
185     // ACTION_UP so a two/three-finger tap can be recognised reliably.
186     if (event.pointerCount - 1 < 2) return
187     captureNavigationPointers(event, event.actionIndex)
188 }
189
190 private fun finishNavigation(event: MotionEvent) {
191     val duration = event.eventTime - gestureStartTime
192     if (!gestureMoved && duration <= TAP_MAX_DURATION_MS) {
193         when (navigationPointerCount) {
194             2 -> callbacks.undoGesture()
195             3 -> callbacks.redoGesture()
196         }
197     }
198     callbacks.requestRender()
199 }
200
201 private fun handleSingleFingerTap(event: MotionEvent) {
202     val now = event.eventTime
203     val dx = event.x - lastTapX
204     val dy = event.y - lastTapY
205     val isDoubleTap =
206         now - lastTapTime <= DOUBLE_TAP_TIMEOUT_MS &&
207         hypot(dx, dy) <= DOUBLE_TAP_DISTANCE_PX
208     if (isDoubleTap) {
209         callbacks.resetCamera()
210         lastTapTime = 0L
211     } else {
212         lastTapTime = now
213         lastTapX = event.x
214         lastTapY = event.y
215     }
216 }
217
218 private fun captureNavigationPointers(event: MotionEvent, excludedIndex: Int = -1) {
219     var firstIndex = -1
220     var secondIndex = -1
221     for (index in 0 until event.pointerCount) {
222         if (index == excludedIndex) continue
223         if (firstIndex < 0) firstIndex = index else {
224             secondIndex = index
225             break
226         }
227     }
228     if (firstIndex < 0 || secondIndex < 0) return
229     firstX = event.getX(firstIndex)
230     firstY = event.getY(firstIndex)
231     secondX = event.getX(secondIndex)
232     secondY = event.getY(secondIndex)
233     previousDistance = distance(firstX, firstY, secondX, secondY)
234     previousAngle = angle(firstX, firstY, secondX, secondY)
235 }
236
237 private fun distance(x0: Float, y0: Float, x1: Float, y1: Float): Float =
238     hypot(x1 - x0, y1 - y0)
239
240 private fun angle(x0: Float, y0: Float, x1: Float, y1: Float): Float =
241     atan2(y1 - y0, x1 - x0)
242
243 private fun normalizeAngle(value: Float): Float {
244     var result = value
245     while (result > Math.PI) result -= (Math.PI * 2.0).toFloat()
246     while (result < -Math.PI) result += (Math.PI * 2.0).toFloat()
247     return result
248 }
249
250 private companion object {
251     const val TAP_MAX_DURATION_MS = 220L
252     const val MIN_DISTANCE = .001f
253     const val TAP_MOVE_THRESHOLD_PX = 4f
254     const val TAP_ZOOM_THRESHOLD = .02f
255     const val TAP_ROTATION_THRESHOLD_RAD = .035f
256     const val DOUBLE_TAP_TIMEOUT_MS = 280L
257     const val DOUBLE_TAP_DISTANCE_PX = 48f
258 }
259 }

```


50. app/src/main/java/com/wetinknext/engine/input/InputAction.kt

```
1 package com.wetinknext.engine.input
2
3 enum class InputAction {
4     DOWN,
5     MOVE,
6     UP,
7     CANCEL,
8 }
```

51. app/src/main/java/com/wetinknext/engine/input/InputBatch.kt

```
1 package com.wetinknext.engine.input
2
3 /** Fixed-capacity container for passing input from the UI thread to the GL thread. */
4 class InputBatch(val maxSamples: Int) {
5     val samples = Array(maxSamples) { InputSample() }
6
7     var action: InputAction = InputAction.MOVE
8     private set
9     var sampleCount: Int = 0
10    private set
11    var prediction: Boolean = false
12    private set
13
14    fun begin(action: InputAction, prediction: Boolean = false) {
15        this.action = action
16        this.prediction = prediction
17        sampleCount = 0
18    }
19
20    fun addSample(
21        canvasX: Float, canvasY: Float, pressure: Float, tiltX: Float, tiltY: Float,
22        orientationRad: Float, timestampNanos: Long, pointerId: Int, tool: PointerTool,
23        historical: Boolean,
24    ): Boolean {
25        if (sampleCount >= maxSamples) return false
26        samples[sampleCount].set(
27            canvasX, canvasY, pressure.coerceIn(0f, 1f), tiltX.coerceIn(-1f, 1f),
28            tiltY.coerceIn(-1f, 1f), orientationRad, timestampNanos, pointerId, tool, historical,
29        )
30        sampleCount += 1
31        return true
32    }
33
34    fun isEmpty(): Boolean = sampleCount == 0
35
36    fun clear() {
37        for (index in 0 until sampleCount) samples[index].clear()
38        sampleCount = 0
39        prediction = false
40        action = InputAction.MOVE
41    }
42 }
```

52. app/src/main/java/com/wetinknext/engine/input/InputBatchPool.kt

```
1 package com.wetinknext.engine.input
2
3 import java.util.concurrent.atomic.AtomicInteger
4 import java.util.concurrent.atomic.AtomicReferenceArray
5
6 /** Fixed-size pool that returns null instead of allocating when exhausted. */
7 class InputBatchPool(
8     batchCount: Int = DEFAULT_BATCH_COUNT,
9     maxSamplesPerBatch: Int = DEFAULT_MAX_SAMPLES,
10 ) {
11     private val batches = AtomicReferenceArray<InputBatch>(batchCount)
12     private val available = AtomicInteger(batchCount)
13
14     init {
15         require(batchCount > 0)
16         require(maxSamplesPerBatch > 0)
17         for (index in 0 until batchCount) batches.set(index, InputBatch(maxSamplesPerBatch))
18     }
19
20     fun acquire(): InputBatch? {
21         while (true) {
22             val current = available.get()
23             if (current <= 0) return null
24             if (available.compareAndSet(current, current - 1)) {
25                 for (index in 0 until batches.length()) {
26                     val batch = batches.getAndSet(index, null)
27                     if (batch != null) return batch
28                 }
29                 available.incrementAndGet()
30                 return null
31             }
32         }
33     }
34
35     fun release(batch: InputBatch) {
36         batch.clear()
37         for (index in 0 until batches.length()) {
38             if (batches.compareAndSet(index, null, batch)) {
39                 available.incrementAndGet()
40                 return
41             }
42         }
43         error("InputBatchPool overflow: released batch does not belong to pool")
44     }
45
46     val freeCount: Int get() = available.get()
47
48     companion object {
49         const val DEFAULT_BATCH_COUNT = 96
50         const val DEFAULT_MAX_SAMPLES = 64
51     }
52 }
```

53. app/src/main/java/com/wetinknext/engine/input/InputMode.kt

```
1 package com.wetinknext.engine.input
2
3 /** Current ownership of touch input. Only DRAWING reaches the brush pipeline. */
4 enum class InputMode {
5     IDLE,
6     DRAWING,
7     NAVIGATING,
8     CANCELLED,
9 }
```

54. app/src/main/java/com/wetinknext/engine/input/InputSample.kt

```
1 package com.wetinknext.engine.input
2
3 /** Mutable input sample allocated only while a batch pool is initialized. */
4 class InputSample {
5     var canvasX = 0f
6     var canvasY = 0f
7     var pressure = 0f
8     var tiltX = 0f
9     var tiltY = 0f
10    var orientationRad = 0f
11    var timestampNanos = 0L
12    var pointerId = -1
13    var tool = PointerTool.UNKNOWN
14    var historical = false
15
16    fun set(
17        canvasX: Float, canvasY: Float, pressure: Float, tiltX: Float, tiltY: Float,
18        orientationRad: Float, timestampNanos: Long, pointerId: Int, tool: PointerTool,
19        historical: Boolean,
20    ) {
21        this.canvasX = canvasX; this.canvasY = canvasY; this.pressure = pressure
22        this.tiltX = tiltX; this.tiltY = tiltY; this.orientationRad = orientationRad
23        this.timestampNanos = timestampNanos; this.pointerId = pointerId; this.tool = tool
24        this.historical = historical
25    }
26
27    fun clear() {
28        canvasX = 0f; canvasY = 0f; pressure = 0f; tiltX = 0f; tiltY = 0f
29        orientationRad = 0f; timestampNanos = 0L; pointerId = -1
30        tool = PointerTool.UNKNOWN; historical = false
31    }
32 }
```

55. app/src/main/java/com/wetinknext/engine/input/PointerTool.kt

```
1 package com.wetinknext.engine.input
2
3 enum class PointerTool {
4     FINGER,
5     STYLUS,
6     ERASER,
7     UNKNOWN,
8 }
```

56. app/src/main/java/com/wetinknext/engine/input/StrokeInputCapturer.kt

```

1 package com.wetinknext.engine.input
2
3 import android.view.MotionEvent
4 import com.wetinknext.engine.core.Camera
5 import java.util.concurrent.ArrayBlockingQueue
6
7 /** Converts View-local MotionEvent samples to canvas coordinates on the UI thread. */
8 class StrokeInputCapturer(private val camera: Camera, private val pool: InputBatchPool, private val queue:
ArrayBlockingQueue<InputBatch>) {
9     private var activePointerId = -1
10    private val canvasPoint = FloatArray(2)
11    private val stylusAxes = StylusAxes()
12    private val tiltOut = FloatArray(2)
13
14    var onSecondaryPointerDown: (() -> Unit)? = null
15
16    /** Не скрываем потери: пул кончился => сэмплы утрачены. */
17    var droppedBatches = 0L
18    private set
19
20    /** Cancels the one pointer currently owned by the drawing pipeline. */
21    fun cancelActiveStroke(): Boolean {
22        if (activePointerId < 0) return false
23        activePointerId = -1
24        return enqueueEmpty(InputAction.CANCEL)
25    }
26
27    fun onTouchEvent(event: MotionEvent): Boolean = when (event.actionMasked) {
28        MotionEvent.ACTION_DOWN -> {
29            activePointerId = event.getPointerId(0)
30            enqueue(event, InputAction.DOWN, 0, false)
31        }
32        // Multi-touch is owned by GestureRouter. Drawing input only receives
33        // the one-pointer draw path and explicit cancellation from that router.
34        MotionEvent.ACTION_POINTER_DOWN -> false
35        MotionEvent.ACTION_MOVE -> {
36            val i = event.findPointerIndex(activePointerId)
37            i >= 0 && enqueue(event, InputAction.MOVE, i, true)
38        }
39        MotionEvent.ACTION_UP, MotionEvent.ACTION_POINTER_UP -> {
40            val i = if (event.actionMasked == MotionEvent.ACTION_UP) 0 else event.actionIndex
41            if (event.getPointerId(i) != activePointerId) true else {
42                activePointerId = -1
43                // historical = true: хвост быстрого штриха больше не теряется.
44                enqueue(event, InputAction.UP, i, true)
45            }
46        }
47        MotionEvent.ACTION_CANCEL -> { activePointerId = -1; enqueueEmpty(InputAction.CANCEL) }
48        else -> false
49    }
50
51    private fun enqueue(event: MotionEvent, action: InputAction, index: Int, historical: Boolean): Boolean {
52        val batch = pool.acquire() ?: run { droppedBatches++; return false }
53        batch.begin(action); val t=camera.snapshot(); val id=event.getPointerId(index); val tool=tool(event,index)
54        if(historical) {
55            for(h in 0 until event.historySize) {
56                t.screenToCanvas(event.getHistoricalX(index, h), event.getHistoricalY(index, h), canvasPoint)
57                stylusAxes.readTilt(event, index, h, tool, tiltOut)
58                val orientation = stylusAxes.readOrientation(event, index, h, tool)
59                if(!batch.addSample(
60                    canvasPoint[0], canvasPoint[1], pressure(event, tool, index, h),
61                    tiltOut[0], tiltOut[1], orientation,
62                    event.getHistoricalEventTime(h) * 1_000_000L, id, tool, true
63                )) break
64            }
65        }
66        t.screenToCanvas(event.getX(index), event.getY(index), canvasPoint)
67        stylusAxes.readTilt(event, index, -1, tool, tiltOut)
68        val orientation = stylusAxes.readOrientation(event, index, -1, tool)
69        batch.addSample(
70            canvasPoint[0], canvasPoint[1], pressure(event, tool, index, -1),
71            tiltOut[0], tiltOut[1], orientation,
72            event.eventTime * 1_000_000L, id, tool, false
73        )
74        if (!queue.offer(batch)) {
75            pool.release(batch)
76            droppedBatches++
77            if (action == InputAction.UP) {
78                enqueueEmpty(InputAction.CANCEL)
79            }
80            return false
81        }
82        return true
83    }
84
85    private fun enqueueEmpty(action: InputAction): Boolean {
86        val batch = pool.acquire() ?: run { droppedBatches++; return false }
87        batch.begin(action); if(!queue.offer(batch)){pool.release(batch);droppedBatches++;return false};return true }
88    private fun tool(e: MotionEvent, i: Int)=when(e.getToolType(i)){MotionEvent.TOOL_TYPE_STYLUS->PointerTool.STYLUS; MotionEvent.
TOOL_TYPE_ERASER->PointerTool.ERASER; MotionEvent.TOOL_TYPE_FINGER->PointerTool.FINGER; else->PointerTool.UNKNOWN}

```

```
89     private fun pressure(e:MotionEvent,t:PointerTool,i:Int,h:Int):Float=if(t==PointerTool.FINGER||t==PointerTool.UNKNOWN)
SYNTHETIC_FINGER_PRESSURE else (if(h>=0)e.getHistoricalPressure(i,h) else e.getPressure(i)).coerceIn(0f,1f)
90
91     companion object {
92         private const val SYNTHETIC_FINGER_PRESSURE = 0.62f
93     }
94 }
```


57. app/src/main/java/com/wetinknext/engine/input/StylusAxes.kt

```

1  package com.wetinknext.engine.input
2
3  import android.view.MotionEvent
4  import kotlin.math.cos
5  import kotlin.math.sin
6
7  class StylusAxes {
8
9      fun readTilt(
10         event: MotionEvent,
11         pointerIndex: Int,
12         historicalIndex: Int,
13         tool: PointerTool,
14         out: FloatArray,
15     ) {
16         if (tool == PointerTool.FINGER ||
17             tool == PointerTool.UNKNOWN) {
18             {
19                 out[0] = 0f
20                 out[1] = 0f
21                 return
22             }
23
24             val tilt = if (historicalIndex >= 0) {
25                 event.getHistoricalAxisValue(
26                     MotionEvent.AXIS_TILT,
27                     pointerIndex,
28                     historicalIndex,
29                 )
30             } else {
31                 event.getAxisValue(
32                     MotionEvent.AXIS_TILT,
33                     pointerIndex,
34                 )
35             }
36
37             val orientation = if (historicalIndex >= 0) {
38                 event.getHistoricalAxisValue(
39                     MotionEvent.AXIS_ORIENTATION,
40                     pointerIndex,
41                     historicalIndex,
42                 )
43             } else {
44                 event.getAxisValue(
45                     MotionEvent.AXIS_ORIENTATION,
46                     pointerIndex,
47                 )
48             }
49
50             val normalizedTilt =
51                 (tilt / (Math.PI.toFloat() * 0.5f))
52                 .coerceIn(0f, 1f)
53
54             out[0] =
55                 (cos(orientation) * normalizedTilt)
56                 .coerceIn(-1f, 1f)
57
58             out[1] =
59                 (sin(orientation) * normalizedTilt)
60                 .coerceIn(-1f, 1f)
61         }
62
63         fun readOrientation(
64             event: MotionEvent,
65             pointerIndex: Int,
66             historicalIndex: Int,
67             tool: PointerTool,
68         ): Float {
69             if (tool == PointerTool.FINGER || tool == PointerTool.UNKNOWN) return 0f
70             return if (historicalIndex >= 0) {
71                 event.getHistoricalAxisValue(MotionEvent.AXIS_ORIENTATION, pointerIndex, historicalIndex)
72             } else {
73                 event.getAxisValue(MotionEvent.AXIS_ORIENTATION, pointerIndex)
74             }
75         }
76     }

```

58. app/src/main/java/com/wetinknext/engine/undo/DeflateTileCompressor.kt

```

1  package com.wetinknext.engine.undo
2
3  import java.io.ByteArrayOutputStream
4  import java.util.zip.Deflater
5  import java.util.zip.Inflater
6
7  class DeflateTileCompressor(
8      private val level: Int = Deflater.BEST_SPEED,
9  ) : TileCompressor {
10
11      override fun compress(data: ByteArray): ByteArray {
12          val deflater = Deflater(level)
13          return try {
14              deflater.setInput(data)
15              deflater.finish()
16
17              val output = ByteArrayOutputStream(data.size)
18              val buffer = ByteArray(16 * 1024)
19
20              while (!deflater.finished()) {
21                  val count = deflater.deflate(buffer)
22                  output.write(buffer, 0, count)
23              }
24
25              output.toByteArray()
26          } finally {
27              deflater.end()
28          }
29      }
30
31      override fun decompress(
32          data: ByteArray,
33          expectedSize: Int,
34      ): ByteArray {
35          val inflater = Inflater()
36          return try {
37              inflater.setInput(data)
38
39              val output = ByteArrayOutputStream(expectedSize)
40              val buffer = ByteArray(16 * 1024)
41
42              while (!inflater.finished()) {
43                  val count = inflater.inflate(buffer)
44                  if (count == 0 && inflater.finished()) break
45                  check(count > 0) { "Decompression stalled or data corrupted" }
46                  output.write(buffer, 0, count)
47              }
48
49              output.toByteArray()
50          } finally {
51              inflater.end()
52          }
53      }
54  }

```

59. app/src/main/java/com/wetinknext/engine/undo/RawTileSnapshot.kt

```
1 package com.wetinknext.engine.undo
2
3 /** Uncompressed tile data captured on the GL thread. */
4 class RawTileSnapshot(
5     val coord: TileCoord,
6     val pixelLeft: Int,
7     val pixelTop: Int,
8     val pixelWidth: Int,
9     val pixelHeight: Int,
10    val bytesPerPixel: Int,
11    val rawBytes: ByteArray,
12 ) {
13     val rawSize: Int get() = rawBytes.size
14
15     fun compress(compressor: TileCompressor): TileSnapshot {
16         return TileSnapshot(
17             coord = coord,
18             pixelLeft = pixelLeft,
19             pixelTop = pixelTop,
20             pixelWidth = pixelWidth,
21             pixelHeight = pixelHeight,
22             bytesPerPixel = bytesPerPixel,
23             storedData = compressor.compress(rawBytes),
24             compressor = compressor,
25         )
26     }
27 }
```

60. app/src/main/java/com/wetinknext/engine/undo/TileCompressor.kt

```
1 package com.wetinknext.engine.undo
2
3 /**
4  * Extension point for P6.5 compression. P6 uses the identity implementation,
5  * so the 96 MB budget reflects the exact bytes that are currently retained.
6  */
7 interface TileCompressor {
8     fun compress(data: ByteArray): ByteArray
9
10    fun decompress(data: ByteArray, expectedSize: Int): ByteArray
11 }
12
13 /** P6 storage policy: no compression and no extra byte-array allocation. */
14 object IdentityTileCompressor : TileCompressor {
15     override fun compress(data: ByteArray): ByteArray = data
16
17     override fun decompress(data: ByteArray, expectedSize: Int): ByteArray {
18         check(data.size == expectedSize) {
19             "Raw tile has ${data.size} bytes, expected $expectedSize"
20         }
21         return data
22     }
23 }
```

61. app/src/main/java/com/wetinknext/engine/undo/TileSnapshot.kt

```
1 package com.wetinknext.engine.undo
2
3 data class TileCoord(val tx: Int, val ty: Int)
4
5 /** One full 256 px tile (or an edge tile) in the source layer's native format. */
6 class TileSnapshot(
7     val coord: TileCoord,
8     val pixelLeft: Int,
9     val pixelTop: Int,
10    val pixelWidth: Int,
11    val pixelHeight: Int,
12    val bytesPerPixel: Int,
13    private val storedData: ByteArray,
14    private val compressor: TileCompressor = IdentityTileCompressor,
15 ) {
16     val memorySize: Int get() = storedData.size
17     val rawSize: Int get() = pixelWidth * pixelHeight * bytesPerPixel
18
19     /** Returns raw bytes. */
20     fun decompress(): ByteArray = compressor.decompress(storedData, rawSize)
21
22     fun dispose() {
23         // P6 keeps heap ByteArrays; a later off-heap implementation can release here.
24     }
25
26     companion object {
27         const val TILE_SIZE = 256
28         const val BYTES_PER_PIXEL_RGBA16F = 8
29         const val BYTES_PER_PIXEL_RGBA8 = 4
30     }
31 }
```

62. app/src/main/java/com/wetinknext/engine/undo/TileSnapshotCapture.kt

```

1 package com.wetinknext.engine.undo
2 import android.opengl.GLES30
3 import android.util.Log
4 import com.wetinknext.BuildConfig
5 import com.wetinknext.engine.gl.GlCheck
6 import com.wetinknext.engine.gl.RenderTarget
7 import java.nio.ByteBuffer
8 import java.nio.ByteOrder
9 object TileSnapshotCapture {
10     private var reusable: ByteBuffer? = null
11
12     /** Captures every 256 px tile intersecting [bounds] = left, top, right, bottom. */
13     fun capture(target: RenderTarget, bounds: IntArray): List<RawTileSnapshot> {
14         GlCheck.checkOnGLThread()
15         require(target.framebufferId != 0)
16         require(target.textureId != 0)
17         require(bounds.size >= 4)
18         val left = bounds[0].coerceIn(0, target.width)
19         val top = bounds[1].coerceIn(0, target.height)
20         val right = bounds[2].coerceIn(0, target.width)
21         val bottom = bounds[3].coerceIn(0, target.height)
22         if (right <= left || bottom <= top) return emptyList()
23
24         if (BuildConfig.DEBUG) {
25             Log.d(
26                 TAG,
27                 "capture target=${target.textureId} fbo=${target.framebufferId} " +
28                 "canvasBounds=[$left,$top,$right,$bottom] " +
29                 "target=${target.width}x${target.height}",
30             )
31         }
32
33         val bytesPerPixel = if (target.usesHalfFloat) {
34             TileSnapshot.BYTES_PER_PIXEL_RGBA16F
35         } else {
36             TileSnapshot.BYTES_PER_PIXEL_RGBA8
37         }
38         val glType = if (target.usesHalfFloat) GLES30.GL_HALF_FLOAT else GLES30.GL_UNSIGNED_BYTE
39         val result = ArrayList<RawTileSnapshot>()
40         target.bind()
41         GLES30.glPixelStorei(GLES30.GL_PACK_ALIGNMENT, 1)
42         try {
43             for (tileY in top / TileSnapshot.TILE_SIZE..(bottom - 1) / TileSnapshot.TILE_SIZE) {
44                 for (tileX in left / TileSnapshot.TILE_SIZE..(right - 1) / TileSnapshot.TILE_SIZE) {
45                     val pixelLeft = tileX * TileSnapshot.TILE_SIZE
46                     // Tile coordinates are document/canvas coordinates (Y grows down).
47                     val canvasTop = tileY * TileSnapshot.TILE_SIZE
48                     val pixelWidth = minOf(TileSnapshot.TILE_SIZE, target.width - pixelLeft)
49                     val pixelHeight = minOf(TileSnapshot.TILE_SIZE, target.height - canvasTop)
50                     // buildCanvasToFbo maps canvas y=0 to the FBO's bottom row.
51                     // Canvas tile coordinates therefore already equal GL's bottom Y.
52                     val glBottom = canvasTop
53                     if (BuildConfig.DEBUG) {
54                         Log.d(
55                             TAG,
56                             "capture tile=($tileX,$tileY) " +
57                             "canvas=[$pixelLeft,$canvasTop,$pixelWidth,$pixelHeight] " +
58                             "gl=[$pixelLeft,$glBottom,$pixelWidth,$pixelHeight]",
59                         )
60                     }
61                     val byteCount = pixelWidth * pixelHeight * bytesPerPixel
62                     val buffer = obtainBuffer(byteCount)
63                     buffer.clear()
64                     GLES30.glReadPixels(
65                         pixelLeft, glBottom, pixelWidth, pixelHeight,
66                         GLES30.GL_RGBA, glType, buffer,
67                     )
68                     if (BuildConfig.DEBUG) {
69                         val error = GLES30.glGetError()
70                         check(error == GLES30.GL_NO_ERROR) {
71                             "TileSnapshotCapture.glReadPixels failed: 0x${error.toString(16)}"
72                         }
73                     }
74                     buffer.rewind()
75                     val rawBytes = ByteArray(byteCount)
76                     buffer.get(rawBytes)
77                     result += RawTileSnapshot(
78                         coord = TileCoord(tileX, tileY),
79                         pixelLeft = pixelLeft,
80                         pixelTop = canvasTop,
81                         pixelWidth = pixelWidth,
82                         pixelHeight = pixelHeight,
83                         bytesPerPixel = bytesPerPixel,
84                         rawBytes = rawBytes,
85                     )
86                 }
87             }
88             return result
89         } finally {
90             GLES30.glPixelStorei(GLES30.GL_PACK_ALIGNMENT, 4)
91         }
92     }
93 }

```

```
91         GLES30.glBindFramebuffer(GLES30.GL_FRAMEBUFFER, 0)
92     }
93 }
94
95 private fun obtainBuffer(size: Int): ByteBuffer {
96     val current = reusable
97     if (current != null && current.capacity() >= size) return current
98     return ByteBuffer.allocateDirect(size).order(ByteOrder.nativeOrder()).also { reusable = it }
99 }
100
101 private const val TAG = "TileUndo"
102 }
```

63. app/src/main/java/com/wetinknext/engine/undo/TileSnapshotRestore.kt

```

1  package com.wetinknext.engine.undo
2
3  import android.opengl.GLES30
4  import android.util.Log
5  import com.wetinknext.BuildConfig
6  import com.wetinknext.engine.gl.GlCheck
7  import com.wetinknext.engine.gl.RenderTarget
8  import java.nio.ByteBuffer
9  import java.nio.ByteOrder
10
11 /** Restores snapshot tiles into an already-created layer texture on the GL thread. */
12 object TileSnapshotRestore {
13     private var reusableBuffer: ByteBuffer? = null
14
15     fun restore(target: RenderTarget, tiles: List<TileSnapshot>): Boolean {
16         GlCheck.checkOnGLThread()
17         if (target.textureId == 0) return false
18         if (tiles.isEmpty()) return true
19
20         // Discard an unrelated previous error, then verify every upload below.
21         GLES30.glGetError()
22         GLES30.glActiveTexture(GLES30.GL_TEXTURE0)
23         GLES30.glBindTexture(GLES30.GL_TEXTURE_2D, target.textureId)
24         GLES30.glPixelStorei(GLES30.GL_UNPACK_ALIGNMENT, 1)
25         try {
26             for (tile in tiles) {
27                 if (!isValidTile(target, tile)) return false
28                 val data = tile.decompress()
29                 if (data.size != tile.rawSize) return false
30
31                 val glType = when (tile.bytesPerPixel) {
32                     TileSnapshot.BYTES_PER_PIXEL_RGBA16F -> GLES30.GL_HALF_FLOAT
33                     TileSnapshot.BYTES_PER_PIXEL_RGBA8 -> GLES30.GL_UNSIGNED_BYTE
34                     else -> return false
35                 }
36                 val buffer = obtainBuffer(data.size)
37                 buffer.clear()
38                 buffer.put(data)
39                 buffer.position(0)
40
41                 // buildCanvasToFbo maps canvas y=0 to the FBO's bottom row.
42                 val glBottom = tile.pixelTop
43                 if (BuildConfig.DEBUG) {
44                     Log.d(TAG, "restore tex=${target.textureId} tile=(${tile.coord.tx},${tile.coord.ty}) " +
45                         "canvas=[${tile.pixelLeft},${tile.pixelTop},${tile.pixelWidth},${tile.pixelHeight}]")
46                 }
47                 GLES30.glTexSubImage2D(
48                     GLES30.GL_TEXTURE_2D, 0, tile.pixelLeft, glBottom,
49                     tile.pixelWidth, tile.pixelHeight, GLES30.GL_RGBA, glType, buffer,
50                 )
51                 if (GLES30.glGetError() != GLES30.GL_NO_ERROR) return false
52             }
53             return true
54         } catch (error: Throwable) {
55             if (BuildConfig.DEBUG) Log.e(TAG, "Restore failed", error)
56             return false
57         } finally {
58             GLES30.glPixelStorei(GLES30.GL_UNPACK_ALIGNMENT, 4)
59             GLES30.glBindTexture(GLES30.GL_TEXTURE_2D, 0)
60             GLES30.glActiveTexture(GLES30.GL_TEXTURE0)
61             GLES30.glBindFramebuffer(GLES30.GL_FRAMEBUFFER, 0)
62         }
63     }
64
65     private fun isValidTile(target: RenderTarget, tile: TileSnapshot): Boolean {
66         if (tile.pixelWidth <= 0 || tile.pixelHeight <= 0) return false
67         if (tile.pixelLeft < 0 || tile.pixelTop < 0) return false
68         if (tile.pixelLeft + tile.pixelWidth > target.width ||
69             tile.pixelTop + tile.pixelHeight > target.height) return false
70         val expectedBytesPerPixel = if (target.usesHalfFloat) {
71             TileSnapshot.BYTES_PER_PIXEL_RGBA16F
72         } else {
73             TileSnapshot.BYTES_PER_PIXEL_RGBA8
74         }
75         return tile.bytesPerPixel == expectedBytesPerPixel
76     }
77
78     private fun obtainBuffer(size: Int): ByteBuffer {
79         val current = reusableBuffer
80         if (current != null && current.capacity() >= size) return current
81         return ByteBuffer.allocateDirect(size).order(ByteOrder.nativeOrder()).also { reusableBuffer = it }
82     }
83
84     private const val TAG = "TileUndo"
85 }

```


64. app/src/main/java/com/wetinknext/engine/undo/UndoEntry.kt

```
1 package com.wetinknext.engine.undo
2
3 /** One reversible history operation, retaining exact states from before and after its merge. */
4 data class UndoEntry(
5     val layerId: Long,
6     val beforeTiles: List<TileSnapshot>,
7     val afterTiles: List<TileSnapshot>,
8     val operation: UndoOperationType = UndoOperationType.TILE_EDIT,
9     val tag: String = "stroke",
10 ) {
11     val memorySize: Int
12     get() = beforeTiles.sumOf(TileSnapshot::memorySize) + afterTiles.sumOf(TileSnapshot::memorySize)
13
14     fun dispose() {
15         beforeTiles.forEach(TileSnapshot::dispose)
16         afterTiles.forEach(TileSnapshot::dispose)
17     }
18 }
```

65. app/src/main/java/com/wetinknext/engine/undo/UndoManager.kt

```

1 package com.wetinknext.engine.undo
2
3 /**
4  * Pure memory-bounded history. GPU reads and texture restores deliberately live
5  * in EngineRenderer, keeping this class deterministic and unit-testable.
6  */
7 class UndoManager(
8     private val maxMemoryBytes: Long = DEFAULT_MAX_MEMORY_BYTES,
9     private val maxSteps: Int = DEFAULT_MAX_STEPS,
10 ) {
11     private val undoStack = ArrayDeque<UndoEntry>()
12     private val redoStack = ArrayDeque<UndoEntry>()
13     private var undoMemoryBytes = 0L
14     private var redoMemoryBytes = 0L
15
16     val canUndo: Boolean get() = undoStack.isNotEmpty()
17     val canRedo: Boolean get() = redoStack.isNotEmpty()
18     val memoryBytes: Long get() = undoMemoryBytes + redoMemoryBytes
19     val undoCount: Int get() = undoStack.size
20     val redoCount: Int get() = redoStack.size
21
22     /** Records a new operation and invalidates the alternate redo history. */
23     fun push(entry: UndoEntry) {
24         undoStack.addLast(entry)
25         undoMemoryBytes += entry.memorySize
26         clearRedo()
27         trimToLimits()
28     }
29
30     /** Returns the candidate without changing history; restore it before committing. */
31     fun peekUndo(): UndoEntry? = undoStack.lastOrNull()
32
33     /** Moves the exact current undo candidate to redo after a successful restore. */
34     fun commitUndo(entry: UndoEntry): Boolean {
35         if (undoStack.lastOrNull() != entry) return false
36         undoStack.removeLast()
37         undoMemoryBytes -= entry.memorySize
38         redoStack.addLast(entry)
39         redoMemoryBytes += entry.memorySize
40         trimToLimits()
41         return true
42     }
43
44     /** Discards a no-longer-restorable undo record without moving it to redo. */
45     fun dropUndo(entry: UndoEntry): Boolean {
46         if (undoStack.lastOrNull() != entry) return false
47         undoStack.removeLast()
48         undoMemoryBytes -= entry.memorySize
49         entry.dispose()
50         return true
51     }
52
53     /** Returns the candidate without changing history; restore it before committing. */
54     fun peekRedo(): UndoEntry? = redoStack.lastOrNull()
55
56     /** Moves the exact current redo candidate to undo after a successful restore. */
57     fun commitRedo(entry: UndoEntry): Boolean {
58         if (redoStack.lastOrNull() != entry) return false
59         redoStack.removeLast()
60         redoMemoryBytes -= entry.memorySize
61         undoStack.addLast(entry)
62         undoMemoryBytes += entry.memorySize
63         trimToLimits()
64         return true
65     }
66
67     /** Discards a no-longer-restorable redo record without moving it to undo. */
68     fun dropRedo(entry: UndoEntry): Boolean {
69         if (redoStack.lastOrNull() != entry) return false
70         redoStack.removeLast()
71         redoMemoryBytes -= entry.memorySize
72         entry.dispose()
73         return true
74     }
75
76     fun clearRedo() {
77         while (redoStack.isNotEmpty()) {
78             val entry = redoStack.removeLast()
79             redoMemoryBytes -= entry.memorySize
80             entry.dispose()
81         }
82         redoMemoryBytes = 0L
83     }
84
85     fun clear() {
86         while (undoStack.isNotEmpty()) undoStack.removeLast().dispose()
87         undoMemoryBytes = 0L
88         clearRedo()
89     }
90

```

```

91  /** Removes history that can no longer be restored after deleting a layer. */
92  fun removeEntriesForLayer(layerId: Long) {
93      removeEntriesForLayer(undoStack, layerId) { entry -> undoMemoryBytes -= entry.memorySize }
94      removeEntriesForLayer(redoStack, layerId) { entry -> redoMemoryBytes -= entry.memorySize }
95      undoMemoryBytes = undoMemoryBytes.coerceAtLeast(0L)
96      redoMemoryBytes = redoMemoryBytes.coerceAtLeast(0L)
97  }
98
99  private fun removeEntriesForLayer(
100      stack: ArrayDeque<UndoEntry>,
101      layerId: Long,
102      subtractMemory: (UndoEntry) -> Unit,
103  ) {
104      if (stack.isEmpty()) return
105      val retained = ArrayDeque<UndoEntry>(stack.size)
106      while (stack.isNotEmpty()) {
107          val entry = stack.removeFirst()
108          if (entry.layerId == layerId) {
109              subtractMemory(entry)
110              entry.dispose()
111          } else {
112              retained.addLast(entry)
113          }
114      }
115      stack.addAll(retained)
116  }
117
118  private fun trimToLimits() {
119      while (undoStack.size > maxSteps) evictOldestUndo()
120      while (redoStack.size > maxSteps) evictOldestRedo()
121      while (memoryBytes > maxMemoryBytes && undoStack.size + redoStack.size > 1) {
122          when {
123              redoStack.size > 1 && redoMemoryBytes >= undoMemoryBytes -> evictOldestRedo()
124              undoStack.size > 1 -> evictOldestUndo()
125              redoStack.size > 1 -> evictOldestRedo()
126              else -> return
127          }
128      }
129  }
130
131  private fun evictOldestUndo() {
132      val entry = undoStack.removeFirstOrNull() ?: return
133      undoMemoryBytes -= entry.memorySize
134      entry.dispose()
135  }
136
137  private fun evictOldestRedo() {
138      val entry = redoStack.removeFirstOrNull() ?: return
139      redoMemoryBytes -= entry.memorySize
140      entry.dispose()
141  }
142
143  companion object {
144      const val DEFAULT_MAX_MEMORY_BYTES = 96L * 1024L * 1024L
145      const val DEFAULT_MAX_STEPS = 50
146  }
147 }

```

66. app/src/main/java/com/wetinknext/engine/undo/UndoOperationType.kt

```
1 package com.wetinknext.engine.undo
2
3 /**
4  * A lightweight operation discriminator for the current tile-snapshot history.
5  * Future history entries can replace this with full typed operations without
6  * changing existing stored tile edits.
7  */
8 enum class UndoOperationType {
9     TILE_EDIT,
10    CLEAR_LAYER,
11    LAYER_PROPERTIES,
12 }
```

67. app/src/main/java/com/wetinknext/ui/animation/AnimationSettingsMenu.kt

```

1  package com.wetinknext.ui.animation
2
3  import androidx.compose.foundation.background
4  import androidx.compose.foundation.border
5  import androidx.compose.foundation.clickable
6  import androidx.compose.foundation.layout.*
7  import androidx.compose.foundation.rememberScrollState
8  import androidx.compose.foundation.shape.CircleShape
9  import androidx.compose.foundation.verticalScroll
10 import androidx.compose.foundation.shape.RoundedCornerShape
11 import androidx.compose.material3.*
12 import androidx.compose.runtime.*
13 import androidx.compose.ui.Alignment
14 import androidx.compose.ui.Modifier
15 import androidx.compose.ui.draw.clip
16 import androidx.compose.ui.graphics.Color
17 import androidx.compose.ui.graphics.toArgb
18 import androidx.compose.ui.text.font.FontWeight
19 import androidx.compose.ui.unit.dp
20 import androidx.compose.ui.unit.sp
21 import com.wetinknext.domain.animation.AnimationDocument
22 import com.wetinknext.domain.animation.AnimationPlaybackMode
23 import com.wetinknext.domain.animation.OnionSkinSettings
24 import com.wetinknext.ui.theme.AppTheme
25
26 @Composable
27 fun AnimationSettingsMenu(
28     theme: AppTheme,
29     document: AnimationDocument,
30     onDocumentChange: (AnimationDocument) -> Unit,
31     onDismiss: () -> Unit,
32     modifier: Modifier = Modifier
33 ) {
34     Box(
35         modifier = modifier
36             .width(280.dp)
37             .heightIn(max = 360.dp)
38             .clip(RoundedCornerShape(24.dp))
39             .background(theme.panelBg.copy(alpha = 0.98f))
40             .border(1.dp, theme.panelStroke, RoundedCornerShape(24.dp))
41             .padding(16.dp)
42     ) {
43         Column(
44             modifier = Modifier.verticalScroll(rememberScrollState()),
45             verticalArrangement = Arrangement.spacedBy(10.dp),
46         ) {
47             Text("Настройки анимации", color = theme.textPrimary, fontSize = 16.sp, fontWeight = FontWeight.Bold)
48
49             // Playback Mode
50             Column(verticalArrangement = Arrangement.spacedBy(8.dp)) {
51                 Text("Режим", color = theme.textSecondary, fontSize = 12.sp)
52                 Row(
53                     modifier = Modifier.fillMaxWidth(),
54                     horizontalArrangement = Arrangement.spacedBy(8.dp)
55                 ) {
56                     PlaybackModeButton("Цикл", AnimationPlaybackMode.LOOP, document.playbackMode, theme) {
57                         onDocumentChange(document.copy(playbackMode = it))
58                     }
59                     PlaybackModeButton("Туда-сюда", AnimationPlaybackMode.PING_PONG, document.playbackMode, theme) {
60                         onDocumentChange(document.copy(playbackMode = it))
61                     }
62                     PlaybackModeButton("Один раз", AnimationPlaybackMode.ONE_SHOT, document.playbackMode, theme) {
63                         onDocumentChange(document.copy(playbackMode = it))
64                     }
65                 }
66             }
67
68             // FPS
69             Column(verticalArrangement = Arrangement.spacedBy(4.dp)) {
70                 Row(horizontalArrangement = Arrangement.SpaceBetween, modifier = Modifier.fillMaxWidth()) {
71                     Text("Скорость (FPS)", color = theme.textSecondary, fontSize = 12.sp)
72                     Text("${document.framesPerSecond}", color = theme.accent, fontSize = 12.sp, fontWeight = FontWeight.Bold)
73                 }
74                 Slider(
75                     value = document.framesPerSecond.toFloat(),
76                     onValueChange = { onDocumentChange(document.copy(framesPerSecond = it.toInt())) },
77                     valueRange = 1f..60f,
78                     colors = SliderDefaults.colors(thumbColor = theme.accent, activeTrackColor = theme.accent)
79                 )
80             }
81
82             HorizontalDivider(color = theme.panelStroke)
83
84             // Onion Skin Settings
85             Row(
86                 modifier = Modifier.fillMaxWidth(),
87                 horizontalArrangement = Arrangement.SpaceBetween,
88                 verticalAlignment = Alignment.CenterVertically
89             ) {
90                 Text("Калька (Onion Skin)", color = theme.textPrimary, fontSize = 14.sp)

```

```

91         Switch(
92             checked = document.onionSkin.enabled,
93             onCheckedChange = { onDocumentChange(document.copy(onionSkin = document.onionSkin.copy(enabled = it))) },
94             colors = SwitchDefaults.colors(checkedThumbColor = theme.accent, checkedTrackColor = theme.accent.copy(
alpha = 0.5f))
95         )
96     }
97
98     if (document.onionSkin.enabled) {
99         OnionCountSlider("Предыдущие", document.onionSkin.previousFrames, theme) {
100             onDocumentChange(document.copy(onionSkin = document.onionSkin.copy(previousFrames = it)))
101         }
102         OnionColorPicker(
103             label = "Цвет предыдущих",
104             selectedArgb = document.onionSkin.previousColorArgb,
105             theme = theme,
106             { color ->
107                 onDocumentChange(document.copy(onionSkin = document.onionSkin.copy(previousColorArgb = color.toArgb())))
108             }
109         )
110         OnionCountSlider("Следующие", document.onionSkin.nextFrames, theme) {
111             onDocumentChange(document.copy(onionSkin = document.onionSkin.copy(nextFrames = it)))
112         }
113         OnionColorPicker(
114             label = "Цвет следующих",
115             selectedArgb = document.onionSkin.nextColorArgb,
116             theme = theme,
117             { color ->
118                 onDocumentChange(document.copy(onionSkin = document.onionSkin.copy(nextColorArgb = color.toArgb())))
119             }
120         )
121         Column(verticalArrangement = Arrangement.spacedBy(4.dp)) {
122             Text("Прозрачность", color = theme.textSecondary, fontSize = 12.sp)
123             Slider(
124                 value = document.onionSkin.opacity,
125                 onValueChange = { onDocumentChange(document.copy(onionSkin = document.onionSkin.copy(opacity = it)))
126                 },
127                 valueRange = 0f..1f,
128                 colors = SliderDefaults.colors(thumbColor = theme.accent, activeTrackColor = theme.accent)
129             )
130         }
131     }
132 }
133
134 @Composable
135 private fun RowScope.PlaybackModeButton(
136     label: String,
137     mode: AnimationPlaybackMode,
138     current: AnimationPlaybackMode,
139     theme: AppTheme,
140     onClick: (AnimationPlaybackMode) -> Unit
141 ) {
142     val selected = mode == current
143     Box(
144         modifier = Modifier
145             .weight(1f)
146             .height(32.dp)
147             .clip(RoundedCornerShape(8.dp))
148             .background(if (selected) theme.accent else theme.panelInsetSoft)
149             .clickable { onClick(mode) },
150         contentAlignment = Alignment.Center
151     ) {
152         Text(label, color = if (selected) Color.White else theme.textSecondary, fontSize = 10.sp)
153     }
154 }
155
156 @Composable
157 private fun OnionCountSlider(
158     label: String,
159     value: Int,
160     theme: AppTheme,
161     onValueChange: (Int) -> Unit
162 ) {
163     Column(verticalArrangement = Arrangement.spacedBy(2.dp)) {
164         Row(horizontalArrangement = Arrangement.SpaceBetween, modifier = Modifier.fillMaxWidth()) {
165             Text(label, color = theme.textSecondary, fontSize = 11.sp)
166             Text("$value", color = theme.textPrimary, fontSize = 11.sp)
167         }
168         Slider(
169             value = value.toFloat(),
170             onValueChange = { onValueChange(it.toInt()) },
171             valueRange = 0f..12f,
172             colors = SliderDefaults.colors(thumbColor = theme.accent, activeTrackColor = theme.accent)
173         )
174     }
175 }
176
177 @Composable
178 private fun OnionColorPicker(
179     label: String,
180     selectedArgb: Int,

```

```

181     theme: AppTheme,
182     onColorChange: (Color) -> Unit,
183 ) {
184     val palette = listOf(
185         Color(0xFFFFF4FA3),
186         Color(0xFFFFF7A59),
187         Color(0xFFFFFC857),
188         Color(0xFF57D68D),
189         Color(0xFF4A8DFF),
190         Color(0xFFB079FF),
191     )
192     Column(verticalArrangement = Arrangement.spacedBy(6.dp)) {
193         Text(label, color = theme.textSecondary, fontSize = 12.sp)
194         Row(horizontalArrangement = Arrangement.spacedBy(8.dp)) {
195             palette.forEach { color ->
196                 val selected = color.toArgb() == selectedArgb
197                 Box(
198                     modifier = Modifier
199                         .size(24.dp)
200                         .clip(CircleShape)
201                         .background(color)
202                         .border(
203                             width = if (selected) 2.dp else 1.dp,
204                             color = if (selected) Color.White else theme.panelStroke,
205                             shape = CircleShape,
206                         )
207                     .clickable { onColorChange(color) }
208                 )
209             }
210         }
211     }
212 }

```

68. app/src/main/java/com/wetinknext/ui/animation/AnimationTimelineToolbar.kt

```

1 package com.wetinknext.ui.animation
2
3 import androidx.compose.foundation.Image
4 import androidx.compose.foundation.background
5 import androidx.compose.foundation.border
6 import androidx.compose.foundation.clickable
7 import androidx.compose.foundation.gestures.detectDragGesturesAfterLongPress
8 import androidx.compose.foundation.layout.*
9 import androidx.compose.foundation.lazy.LazyRow
10 import androidx.compose.foundation.lazy.itemsIndexed
11 import androidx.compose.foundation.lazy.rememberLazyListState
12 import androidx.compose.foundation.shape.CircleShape
13 import androidx.compose.foundation.shape.RoundedCornerShape
14 import androidx.compose.material.icons.Icons
15 import androidx.compose.material.icons.filled.*
16 import androidx.compose.material3.*
17 import androidx.compose.runtime.*
18 import androidx.compose.ui.Alignment
19 import androidx.compose.ui.Modifier
20 import androidx.compose.ui.draw.clip
21 import androidx.compose.ui.graphics.Color
22 import androidx.compose.ui.graphics.ImageBitmap
23 import androidx.compose.ui.graphics.graphicsLayer
24 import androidx.compose.ui.graphics.vector.ImageVector
25 import androidx.compose.ui.input.pointer.pointerInput
26 import androidx.compose.ui.layout.ContentScale
27 import androidx.compose.ui.platform.LocalDensity
28 import androidx.compose.ui.text.font.FontWeight
29 import androidx.compose.ui.unit.IntOffset
30 import androidx.compose.ui.unit.dp
31 import androidx.compose.ui.unit.sp
32 import androidx.compose.ui.window.Popup
33 import com.wetinknext.domain.animation.AnimationDocument
34 import com.wetinknext.domain.animation.AnimationFrame
35 import com.wetinknext.domain.animation.AnimationFrameRole
36 import com.wetinknext.ui.theme.AppTheme
37 import kotlin.math.roundToInt
38
39 data class FrameThumbnailLayer(
40     val image: ImageBitmap,
41     val opacity: Float,
42 )
43
44 @Composable
45 fun AnimationTimelineToolbar(
46     theme: AppTheme,
47     document: AnimationDocument,
48     currentFrameId: Long,
49     frameThumbnails: Map<Long, List<FrameThumbnailLayer>> = emptyMap(),
50     onFrameSelect: (Long) -> Unit,
51     onFrameMove: (frameId: Long, direction: Int) -> Unit,
52     onAddFrame: () -> Unit,
53     onDuplicateFrame: (Long) -> Unit,
54     onDeleteFrame: (Long) -> Unit,
55     onHoldChange: (Long, Int) -> Unit,
56     onRoleToggle: (Long, AnimationFrameRole) -> Unit,
57     onAssembleLayers: () -> Unit,
58     onLayersClick: () -> Unit,
59     onClose: () -> Unit,
60     onSettingsClick: () -> Unit = {},
61     isPlaying: Boolean = false,
62     onPlayToggle: () -> Unit = {},
63     modifier: Modifier = Modifier
64 ) {
65     val listState = rememberLazyListState()
66     val frames = document.frames
67     var showContextMenuForId by remember { mutableStateOf<Long?>(null) }
68
69     // Auto-scroll to current frame
70     LaunchedEffect(currentFrameId) {
71         val index = frames.indexOfFirst { it.id == currentFrameId }
72         if (index >= 0) {
73             listState.animateScrollToItem(index)
74         }
75     }
76
77     Column(
78         modifier = modifier
79             .fillMaxWidth()
80             .padding(bottom = 16.dp),
81         horizontalAlignment = Alignment.CenterHorizontally
82     ) {
83         Row(
84             modifier = Modifier
85                 .widthIn(min = 400.dp, max = 820.dp)
86                 .height(86.dp)
87                 .clip(RoundedCornerShape(28.dp))
88                 .background(theme.panelBg.copy(alpha = 0.94f))
89                 .border(1.2.dp, theme.panelStroke, RoundedCornerShape(28.dp))
90                 .padding(horizontal = 14.dp, vertical = 8.dp),

```



```

91         verticalAlignment = Alignment.CenterVertically
92     ) {
93         // Play toggle
94         Column(
95             horizontalAlignment = Alignment.CenterHorizontally,
96             modifier = Modifier.padding(end = 4.dp)
97         ) {
98             Box(
99                 modifier = Modifier
100                     .size(44.dp)
101                     .clip(CircleShape)
102                     .background(if (isPlaying) theme.accent.copy(alpha = 0.4f) else theme.accent)
103                     .clickable { onPlayToggle() },
104                 contentAlignment = Alignment.Center
105             ) {
106                 Icon(
107                     if (isPlaying) Icons.Default.Pause else Icons.Default.PlayArrow,
108                     null,
109                     tint = Color.White
110                 )
111             }
112             Text(if (isPlaying) "Пауза" else "Пуск", color = theme.textSecondary, fontSize = 9.sp)
113         }
114
115         Spacer(Modifier.width(10.dp))
116         Box(Modifier.width(1.dp).fillMaxHeight(0.6f).background(theme.panelStroke))
117         Spacer(Modifier.width(10.dp))
118
119         // Frames timeline
120         LazyRow(
121             state = listState,
122             modifier = Modifier.weight(1f),
123             horizontalArrangement = Arrangement.spacedBy(8.dp),
124             verticalAlignment = Alignment.CenterVertically
125         ) {
126             itemsIndexed(frames, key = { _, f -> f.id }) { _, frame ->
127                 TimelineFrameItem(
128                     frame = frame,
129                     isSelected = frame.id == currentFrameId,
130                     thumbnailLayers = frameThumbnails[frame.id].orEmpty(),
131                     theme = theme,
132                     canMoveLeft = frames.indexOf(frame) > 0,
133                     canMoveRight = frames.indexOf(frame) < frames.lastIndex,
134                     onMove = { direction -> onFrameMove(frame.id, direction) },
135                     onClick = {
136                         if (currentFrameId == frame.id) {
137                             showContextMenuForId = frame.id
138                         } else {
139                             onFrameSelect(frame.id)
140                         }
141                     }
142                 )
143             }
144         }
145
146         Spacer(Modifier.width(10.dp))
147         Box(Modifier.width(1.dp).fillMaxHeight(0.6f).background(theme.panelStroke))
148         Spacer(Modifier.width(10.dp))
149
150         // Actions
151         Row(
152             verticalAlignment = Alignment.CenterVertically,
153             horizontalArrangement = Arrangement.spacedBy(12.dp)
154         ) {
155             TimelineActionIcon(Icons.Default.Add, "Кадр", theme, onClick = onAddFrame)
156             TimelineActionIcon(Icons.Default.Layers, "Собрать", theme, onClick = onAssembleLayers)
157             TimelineActionIcon(Icons.Default.ContentCopy, "Копия", theme, onClick = { onDuplicateFrame(currentFrameId) })
158             TimelineActionIcon(Icons.Default.Delete, "Удалить", theme, enabled = frames.size > 1, onClick = {
onDeleteFrame(currentFrameId) })
159             TimelineActionIcon(Icons.Default.Settings, "Настр.", theme, onClick = onSettingsClick)
160             TimelineActionIcon(Icons.Default.Layers, "Слой", theme, onClick = onLayersClick)
161             TimelineActionIcon(Icons.Default.Close, "Закрыть", theme, onClick = onClose)
162         }
163     }
164
165     // Context Menu
166     showContextMenuForId?.let { frameId ->
167         val frame = frames.find { it.id == frameId }
168         if (frame != null) {
169             Popup(
170                 alignment = Alignment.TopCenter,
171                 offset = IntOffset(0, -220),
172                 onDismissRequest = { showContextMenuForId = null }
173             ) {
174                 FrameContextMenu(
175                     frame = frame,
176                     theme = theme,
177                     onDuplicate = { onDuplicateFrame(frameId); showContextMenuForId = null },
178                     onDelete = { onDeleteFrame(frameId); showContextMenuForId = null },
179                     onHoldChange = { onHoldChange(frameId, it) },
180                     onRoleToggle = { onRoleToggle(frameId, it); showContextMenuForId = null },
181                     canDelete = frames.size > 1

```

```

182     )
183     }
184     }
185     }
186     }
187 }
188
189 @Composable
190 private fun FrameContextMenu(
191     frame: AnimationFrame,
192     theme: AppTheme,
193     onDuplicate: () -> Unit,
194     onDelete: () -> Unit,
195     onHoldChange: (Int) -> Unit,
196     onRoleToggle: (AnimationFrameRole) -> Unit,
197     canDelete: Boolean
198 ) {
199     Surface(
200         modifier = Modifier
201             .width(220.dp)
202             .clip(RoundedCornerShape(16.dp))
203             .border(1.dp, theme.panelStroke, RoundedCornerShape(16.dp)),
204         color = theme.panelBg,
205         tonalElevation = 8.dp
206     ) {
207         Column(modifier = Modifier.padding(12.dp)) {
208             ContextMenuItem("Дублировать", Icons.Default.ContentCopy, theme, onClick = onDuplicate)
209             ContextMenuItem("Удалить", Icons.Default.Delete, theme, enabled = canDelete, onClick = onDelete)
210
211             HorizontalDivider(modifier = Modifier.padding(vertical = 8.dp), color = theme.panelStroke)
212
213             Text("Задержка: ${frame.holdFrames}", color = theme.textSecondary, fontSize = 11.sp, modifier = Modifier.padding(
horizontal = 8.dp))
214             Slider(
215                 value = frame.holdFrames.toFloat(),
216                 onValueChange = { onHoldChange(it.toInt()) },
217                 valueRange = 1f..120f,
218                 colors = SliderDefaults.colors(thumbColor = theme.accent, activeTrackColor = theme.accent)
219             )
220
221             HorizontalDivider(modifier = Modifier.padding(vertical = 8.dp), color = theme.panelStroke)
222
223             val isBg = frame.role == AnimationFrameRole.BACKGROUND
224             val isFg = frame.role == AnimationFrameRole.FOREGROUND
225
226             Row(verticalAlignment = Alignment.CenterVertically, modifier = Modifier.fillMaxWidth().clickable { onRoleToggle(
if (isBg) AnimationFrameRole.NORMAL else AnimationFrameRole.BACKGROUND) }.padding(8.dp)) {
227                 Checkbox(checked = isBg, onCheckedChange = null, colors = CheckboxDefaults.colors(checkedColor = theme.
accent))
228                 Spacer(Modifier.width(8.dp))
229                 Text("Фоновый слой", color = theme.textPrimary, fontSize = 13.sp)
230             }
231             Row(verticalAlignment = Alignment.CenterVertically, modifier = Modifier.fillMaxWidth().clickable { onRoleToggle(
if (isFg) AnimationFrameRole.NORMAL else AnimationFrameRole.FOREGROUND) }.padding(8.dp)) {
232                 Checkbox(checked = isFg, onCheckedChange = null, colors = CheckboxDefaults.colors(checkedColor = theme.
accent))
233                 Spacer(Modifier.width(8.dp))
234                 Text("Передний план", color = theme.textPrimary, fontSize = 13.sp)
235             }
236         }
237     }
238 }
239
240 @Composable
241 private fun ContextMenuItem(
242     label: String,
243     icon: ImageVector,
244     theme: AppTheme,
245     enabled: Boolean = true,
246     onClick: () -> Unit
247 ) {
248     Row(
249         modifier = Modifier
250             .fillMaxWidth()
251             .clickable(enabled = enabled) { onClick() }
252             .padding(8.dp)
253             .alphaIfDisabled(!enabled),
254         verticalAlignment = Alignment.CenterVertically
255     ) {
256         Icon(icon, null, tint = if (enabled) theme.textPrimary else theme.textSecondary.copy(alpha = 0.5f), modifier =
Modifier.size(18.dp))
257         Spacer(Modifier.width(12.dp))
258         Text(label, color = if (enabled) theme.textPrimary else theme.textSecondary.copy(alpha = 0.5f), fontSize = 14.sp)
259     }
260 }
261
262 @Composable
263 private fun TimelineFrameItem(
264     frame: AnimationFrame,
265     isSelected: Boolean,
266     thumbnailLayers: List<FrameThumbnailLayer>,
267     theme: AppTheme,

```

```

268     canMoveLeft: Boolean,
269     canMoveRight: Boolean,
270     onMove: (Int) -> Unit,
271     onClick: () -> Unit
272 ) {
273     val moveThresholdPx = with(LocalDensity.current) { 44.dp.toPx() }
274     var accumulatedDragX by remember(frame.id) { mutableFloatStateOf(0f) }
275     Column(
276         horizontalAlignment = Alignment.CenterHorizontally,
277         modifier = Modifier
278             .pointerInput(frame.id, canMoveLeft, canMoveRight) {
279                 detectDragGesturesAfterLongPress(
280                     onDragStart = { accumulatedDragX = 0f },
281                     onDrag = { change, dragAmount ->
282                         accumulatedDragX += dragAmount.x
283                         val direction = when {
284                             accumulatedDragX <= -moveThresholdPx && canMoveLeft -> -1
285                             accumulatedDragX >= moveThresholdPx && canMoveRight -> 1
286                             else -> 0
287                         }
288                         if (direction != 0) {
289                             change.consume()
290                             onMove(direction)
291                             accumulatedDragX = 0f
292                         }
293                     }
294             )
295     )
296     .clickable { onClick() }
297 ) {
298     Box(
299         modifier = Modifier
300             .size(54.dp, 40.dp)
301             .clip(RoundedCornerShape(4.dp))
302             .background(if (isSelected) theme.accent.copy(alpha = 0.2f) else theme.panelInsetSoft)
303             .border(
304                 width = if (isSelected) 2.dp else 1.dp,
305                 color = if (isSelected) theme.accent else theme.panelStroke,
306                 shape = RoundedCornerShape(4.dp)
307             ),
308         contentAlignment = Alignment.Center
309     ) {
310         if (thumbnailLayers.isNotEmpty()) {
311             thumbnailLayers.forEach { layer ->
312                 Image(
313                     bitmap = layer.image,
314                     contentDescription = null,
315                     contentScale = ContentScale.Fit,
316                     modifier = Modifier
317                         .fillMaxSize()
318                         .padding(2.dp)
319                         .graphicsLayer(alpha = layer.opacity.coerceIn(0f, 1f)),
320                 )
321             }
322         } else {
323             Icon(Icons.Default.Image, null, tint = theme.textSecondary.copy(alpha = 0.3f), modifier = Modifier.size(20.
324 dp))
325         }
326         if (frame.holdFrames > 1) {
327             Text("+${frame.holdFrames - 1}", color = theme.accent, fontSize = 9.sp, fontWeight = FontWeight.Bold)
328         } else {
329             Spacer(Modifier.height(11.dp))
330         }
331     }
332 }
333
334 @Composable
335 private fun TimelineActionIcon(
336     icon: ImageVector,
337     label: String,
338     theme: AppTheme,
339     enabled: Boolean = true,
340     onClick: () -> Unit
341 ) {
342     Column(
343         horizontalAlignment = Alignment.CenterHorizontally,
344         modifier = Modifier
345             .alphaIfDisabled(!enabled)
346             .clickable(enabled = enabled) { onClick() }
347     ) {
348         Icon(icon, null, tint = if (enabled) theme.textPrimary else theme.textSecondary.copy(alpha = 0.5f), modifier =
349 Modifier.size(22.dp))
350         Text(label, color = theme.textSecondary, fontSize = 9.sp)
351     }
352
353     private fun Modifier.alphaIfDisabled(disabled: Boolean): Modifier = if (disabled) this.then(Modifier.graphicsLayer { alpha =
354 0.5f }) else this

```

69. app/src/main/java/com/wetinknext/ui/color/ColorCollection.kt

```

1 package com.wetinknext.ui.color
2
3 import androidx.compose.ui.graphics.Color
4
5 data class ColorCollection(
6     val id: Long,
7     val name: String,
8     val colors: List<Int>,
9     val builtin: Boolean = false,
10 )
11
12 fun ColorCollection.composeColors(): List<Color> = colors.map { Color(it) }
13
14 private fun hex(s: String): Int {
15     val v = s.removePrefix("#").trim()
16     val r = v.substring(0, 2).toInt(16)
17     val g = v.substring(2, 4).toInt(16)
18     val b = v.substring(4, 6).toInt(16)
19     return (0xFF shl 24) or (r shl 16) or (g shl 8) or b
20 }
21
22 private fun palette(vararg hex: String): List<Int> = hex.map { hex(it) }
23
24 object BuiltinCollections {
25     val all: List<ColorCollection> = listOf(
26         ColorCollection(
27             id = -1L,
28             name = "Стандарт",
29             colors = palette(
30                 "#FFF3B0", "#FFD6A5", "#FFC2B0", "#FFCCD5", "#F1C6F2",
31                 "#D8C7F5", "#C5D7FF", "#B8E5FF", "#B6F0EE", "#C9F2D8",
32                 "#FFE066", "#FFB87A", "#FF9F8B", "#FF9CB0", "#E59FE6",
33                 "#B89BE8", "#9CB6F5", "#86CFF2", "#7FD9D5", "#8AD9A8",
34                 "#F7C548", "#F09870", "#E07A6E", "#D9748E", "#C383C6",
35                 "#9678D0", "#7798E0", "#5DB1E6", "#5CB8B5", "#65BD8A",
36             ),
37             builtin = true,
38         ),
39         ColorCollection(
40             id = -2L,
41             name = "Пастель",
42             colors = palette(
43                 "#F9E79F", "#F7D08A", "#F5B187", "#F0998A", "#EC9090",
44                 "#E68F9C", "#D88BAE", "#C58CC0", "#B391CD", "#9C95D6",
45                 "#A8D5B6", "#9FD2C9", "#9CCFD8", "#A0C9E0", "#A8C0E3",
46                 "#B5BBE3", "#C0B8DE", "#CCB5D3", "#D6B3C5", "#DDB3B6",
47                 "#C7E1A1", "#B7DAA3", "#A8D2AE", "#9CCABD", "#94C0CB",
48                 "#94B5D2", "#9DA9D1", "#B0A0CB", "#C19BBE", "#CC9AAB",
49             ),
50             builtin = true,
51         ),
52         ColorCollection(
53             id = -3L,
54             name = "Персонажи",
55             colors = palette(
56                 "#F4C9B5", "#F0D4BC", "#E8B89A", "#D89C7E", "#C0805F",
57                 "#9C5F3F", "#74442B", "#4F2C1C", "#311A11", "#1B0F0A",
58                 "#FBE3D4", "#F8DCC9", "#F2C5A5", "#E2A77F", "#C58557",
59                 "#A36739", "#7A4824", "#552E14", "#321A0A", "#FFFAF0",
60                 "#E8C7C1", "#D49C9C", "#B97070", "#8C4F4F", "#5C3030",
61                 "#C7BF81", "#9C9484", "#6E6757", "#A0B0C4", "#5A6878",
62             ),
63             builtin = true,
64         ),
65         ColorCollection(
66             id = -4L,
67             name = "Пейзаж",
68             colors = palette(
69                 "#AEDCF0", "#7FBFE1", "#56A0CB", "#3D80B0", "#2E608E",
70                 "#244670", "#1B335A", "#142545", "#0D1830", "#070D1E",
71                 "#CFE3B0", "#A8CF8C", "#7FB36A", "#5C9651", "#3F7A3F",
72                 "#2E5E33", "#214728", "#16321C", "#5C7B4A", "#3F5732",
73                 "#F4D58D", "#E3B26E", "#D08A4A", "#B5642E", "#923F1C",
74                 "#6E2A14", "#8B5A2B", "#5C3A1A", "#A38560", "#6F5B3D",
75             ),
76             builtin = true,
77         ),
78         ColorCollection(
79             id = -5L,
80             name = "Анимация",
81             colors = palette(
82                 "#111318", "#3B414A", "#7B8491", "#B8C0CC", "#FFFFFF",
83                 "#4A90E2", "#35B9D5", "#42B883", "#A5D646", "#F4D44D",
84                 "#F5A23D", "#F26B4F", "#E6536F", "#E66FAD", "#A368DB",
85                 "#6D8CFF", "#64D4E8", "#76D7A6", "#D8EA72", "#FFE58A",
86                 "#245EA8", "#167E98", "#20794F", "#6B8F20", "#B58B08",
87                 "#B65A12", "#B53832", "#A62E58", "#A23E86", "#633B9B",
88             ),
89             builtin = true,
90         ),
91     )
92 }

```

91)
92 }

70. app/src/main/java/com/wetinknext/ui/color/ColorMath.kt

```

1 package com.wetinknext.ui.color
2
3 import androidx.compose.ui.graphics.Color
4 import kotlin.math.abs
5
6 data class Hsl(val h: Float, val s: Float, val l: Float)
7
8 fun rgbToHsl(c: Color): Hsl {
9     val r = c.red
10    val g = c.green
11    val b = c.blue
12    val max = maxOf(r, g, b)
13    val min = minOf(r, g, b)
14    val l = (max + min) / 2f
15    val d = max - min
16    if (d == 0f) return Hsl(0f, 0f, l)
17    val s = if (l > 0.5f) d / (2f - max - min) else d / (max + min)
18    val h = when (max) {
19        r -> 60f * (((g - b) / d) % 6f)
20        g -> 60f * ((b - r) / d + 2f)
21        else -> 60f * ((r - g) / d + 4f)
22    }
23    return Hsl(((h % 360f) + 360f) % 360f, s, l)
24 }
25
26 fun hslToColor(h: Float, s: Float, l: Float): Color {
27     val hh = ((h % 360f) + 360f) % 360f
28     val c = (1f - abs(2f * l - 1f)) * s
29     val x = c * (1f - abs((hh / 60f) % 2f - 1f))
30     val m = l - c / 2f
31     val (r1, g1, b1) = when ((hh / 60f).toInt()) {
32         0 -> Triple(c, x, 0f)
33         1 -> Triple(x, c, 0f)
34         2 -> Triple(0f, c, x)
35         3 -> Triple(0f, x, c)
36         4 -> Triple(x, 0f, c)
37         else -> Triple(c, 0f, x)
38     }
39     return Color(
40         (r1 + m).coerceIn(0f, 1f),
41         (g1 + m).coerceIn(0f, 1f),
42         (b1 + m).coerceIn(0f, 1f),
43         1f,
44     )
45 }
46
47 fun hslToColor(hsl: Hsl): Color = hslToColor(hsl.h, hsl.s, hsl.l)
48
49 fun Color.toHex6(): String {
50     val r = (red * 255f).toInt().coerceIn(0, 255)
51     val g = (green * 255f).toInt().coerceIn(0, 255)
52     val b = (blue * 255f).toInt().coerceIn(0, 255)
53     return "%02X%02X%02X".format(r, g, b)
54 }
55
56 enum class HarmonyType(
57     val displayName: String,
58     val offsets: FloatArray,
59 ) {
60     Complementary("Комплементарная", floatArrayOf(0f, 180f)),
61     Analogous("Аналоговая", floatArrayOf(-30f, 0f, 30f)),
62     Triadic("Триадная", floatArrayOf(0f, 120f, 240f)),
63     SplitComplementary(
64         "Сплит-комплементарная",
65         floatArrayOf(0f, 150f, 210f),
66     ),
67     Tetradic("Тетрадная", floatArrayOf(0f, 60f, 180f, 240f)),
68     Square("Квадратная", floatArrayOf(0f, 90f, 180f, 270f)),
69     ;
70
71     fun derive(base: Hsl): List<Hsl> {
72         val h = base.h
73         return offsets.map { off ->
74             Hsl(((h + off) % 360f + 360f) % 360f, base.s, base.l)
75         }
76     }
77 }

```

71. app/src/main/java/com/wetinknext/ui/color/ColorPanel.kt

```

1  package com.wetinknext.ui.color
2
3  import androidx.compose.animation.core.animateFloatAsState
4  import androidx.compose.animation.core.tween
5  import androidx.compose.foundation.Canvas
6  import androidx.compose.foundation.ExperimentalFoundationApi
7  import androidx.compose.foundation.background
8  import androidx.compose.foundation.border
9  import androidx.compose.foundation.clickable
10 import androidx.compose.foundation.combinedClickable
11 import androidx.compose.foundation.gestures.detectDragGestures
12 import androidx.compose.foundation.gestures.detectHorizontalDragGestures
13 import androidx.compose.foundation.gestures.detectTapGestures
14 import androidx.compose.foundation.layout.*
15 import androidx.compose.foundation.rememberScrollState
16 import androidx.compose.foundation.shape.CircleShape
17 import androidx.compose.foundation.shape.RoundedCornerShape
18 import androidx.compose.foundation.text.BasicTextField
19 import androidx.compose.foundation.verticalScroll
20 import androidx.compose.material3.*
21 import androidx.compose.runtime.*
22 import androidx.compose.ui.Alignment
23 import androidx.compose.ui.Modifier
24 import androidx.compose.ui.draw.clip
25 import androidx.compose.ui.draw.shadow
26 import androidx.compose.ui.geometry.Offset
27 import androidx.compose.ui.geometry.Size
28 import androidx.compose.ui.graphics.Brush
29 import androidx.compose.ui.graphics.Color
30 import androidx.compose.ui.graphics.SolidColor
31 import androidx.compose.ui.graphics.drawscope.DrawScope
32 import androidx.compose.ui.graphics.drawscope.Stroke
33 import androidx.compose.ui.input.pointer.pointerInput
34 import androidx.compose.ui.platform.LocalDensity
35 import androidx.compose.ui.text.TextStyle
36 import androidx.compose.ui.text.font.FontWeight
37 import androidx.compose.ui.text.style.TextAlign
38 import androidx.compose.ui.unit.dp
39 import androidx.compose.ui.unit.IntOffset
40 import androidx.compose.ui.unit.sp
41 import androidx.compose.ui.window.Dialog
42 import com.wetinknext.ui.components.PanelSurface
43 import com.wetinknext.ui.theme.AppTheme
44 import kotlinx.coroutines.delay
45 import kotlin.math.atan2
46 import kotlin.math.cos
47 import kotlin.math.hypot
48 import kotlin.math.sin
49 import kotlin.math.roundToInt
50
51 private enum class ColorTab { Wheel, Map }
52
53 @Composable
54 fun ColorPanel(
55     state: GlesColorState,
56     color: Color,
57     theme: AppTheme,
58     onColorChange: (Color) -> Unit,
59     onAddFromPhoto: () -> Unit,
60     title: String = "Ғбер",
61     recordRecent: Boolean = true,
62     allowCollectionEditing: Boolean = true,
63     onPinCollection: (ColorCollection) -> Unit = {},
64     modifier: Modifier = Modifier,
65     onDismiss: () -> Unit,
66 ) {
67     var tab by remember { mutableStateOf(ColorTab.Wheel) }
68     var hsl by remember { mutableStateOf(rgbToHsl(color)) }
69     var harmonyType by remember { mutableStateOf(HarmonyType.Triadic) }
70     val latestHsl by rememberUpdatedState(hsl)
71
72     val wheelScrollState = rememberScrollState()
73     val mapScrollState = rememberScrollState()
74     val tabScrollState = when (tab) {
75         ColorTab.Wheel -> wheelScrollState
76         ColorTab.Map -> mapScrollState
77     }
78
79     LaunchedEffect(hsl, recordRecent) {
80         val selectedColor = hslToColor(hsl)
81         onColorChange(selectedColor)
82         if (recordRecent) {
83             delay(350)
84             state.pushRecentColor(selectedColor)
85         }
86     }
87
88     val dismissAndSave = {
89         if (recordRecent) state.pushRecentColor(hslToColor(hsl))
90         onDismiss()
91     }

```

```

91     }
92
93     DisposableEffect(Unit) {
94         onDispose {
95             if (recordRecent) state.pushRecentColor(hslToColor(latestHsl))
96         }
97     }
98
99     PanelSurface(
100         modifier = modifier.width(320.dp),
101     ) {
102         Column(
103             modifier = Modifier.padding(14.dp),
104         ) {
105             ColorHeader(
106                 preview = hslToColor(hsl),
107                 title = title,
108                 theme = theme,
109                 onConfirm = dismissAndSave,
110             )
111             Spacer(Modifier.height(10.dp))
112             TabBar(tab, theme) { tab = it }
113             Spacer(Modifier.height(12.dp))
114             Box(
115                 modifier = Modifier
116                     .height(TabContentHeight)
117                     .fillMaxWidth(),
118             ) {
119                 Column(
120                     modifier = Modifier
121                         .fillMaxSize()
122                         .verticalScroll(tabScrollState),
123                 ) {
124                     when (tab) {
125                         ColorTab.Wheel -> WheelTab(
126                             hsl = hsl,
127                             theme = theme,
128                             onHslChange = { hsl = it },
129                             harmonyType = harmonyType,
130                             onHarmonyTypeChange = { harmonyType = it },
131                             recent = state.recentColors.toList(),
132                             onPickRecent = { picked ->
133                                 val p = rgbToHsl(picked)
134                                 hsl = hsl.copy(h = p.h, s = p.s)
135                                 if (recordRecent) state.pushRecentColor(hslToColor(hsl))
136                             },
137                         )
138                         ColorTab.Map -> MapTab(
139                             userCollections = state.userCollections.toList(),
140                             currentColor = hslToColor(hsl),
141                             theme = theme,
142                             onPick = { picked ->
143                                 hsl = rgbToHsl(picked)
144                                 if (recordRecent) state.pushRecentColor(picked)
145                             },
146                             onDeleteUserCollection = { id ->
147                                 state.deleteUserCollection(id)
148                             },
149                             onAddFromPhoto = onAddFromPhoto,
150                             allowCollectionEditing = allowCollectionEditing,
151                             onCreatePalette = { name ->
152                                 state.addUserCollection(
153                                     ColorCollection(
154                                         id = System.currentTimeMillis(),
155                                         name = name.ifBlank { "Новая палитра" },
156                                         colors = emptyList(),
157                                         builtin = false,
158                                     ),
159                                 )
160                             },
161                             onAddColorToCollection = { id, color ->
162                                 state.addColorToCollection(id, color)
163                             },
164                             onReplaceColorInCollection = { id, slot, color ->
165                                 state.replaceColorInCollection(id, slot, color)
166                             },
167                             onRemoveColorFromCollection = { id, slot ->
168                                 state.removeColorFromCollection(id, slot)
169                             },
170                             onPinCollection = onPinCollection,
171                         )
172                     }
173                 }
174             }
175         }
176     }
177 }
178
179 private val TabContentHeight = 440.dp
180
181 @Composable
182 private fun ColorHeader(preview: Color, onConfirm: () -> Unit, title: String, theme: AppTheme) {

```



```

183     Row(verticalAlignment = Alignment.CenterVertically) {
184         Text(
185             title,
186             color = theme.textPrimary,
187             fontWeight = FontWeight.Bold,
188             fontSize = 18.sp,
189         )
190         Spacer(Modifier.weight(1f))
191         Box(
192             modifier = Modifier
193                 .size(28.dp)
194                 .clip(CircleShape)
195                 .background(preview)
196                 .border(1.dp, theme.panelStroke, CircleShape)
197                 .clickable(onClick = onConfirm),
198         )
199     }
200 }
201
202 @Composable
203 private fun TabBar(current: ColorTab, theme: AppTheme, onChange: (ColorTab) -> Unit) {
204     Row(
205         modifier = Modifier
206             .fillMaxWidth()
207             .clip(RoundedCornerShape(10.dp))
208             .background(theme.panelInset)
209             .padding(3.dp),
210         horizontalArrangement = Arrangement.spacedBy(3.dp),
211     ) {
212         TabChip(label = "Кольцо", selected = current == ColorTab.Wheel, theme = theme) { onChange(ColorTab.Wheel) }
213         TabChip(label = "Карта", selected = current == ColorTab.Map, theme = theme) { onChange(ColorTab.Map) }
214     }
215 }
216
217 @Composable
218 private fun RowScope.TabChip(
219     label: String,
220     selected: Boolean,
221     theme: AppTheme,
222     onClick: () -> Unit,
223 ) {
224     Box(
225         modifier = Modifier
226             .weight(1f)
227             .clip(RoundedCornerShape(8.dp))
228             .background(if (selected) theme.accent.copy(alpha = 0.25f) else Color.Transparent)
229             .clickable(onClick = onClick)
230             .padding(vertical = 7.dp),
231         contentAlignment = Alignment.Center,
232     ) {
233         Text(
234             label,
235             color = if (selected) theme.textPrimary else theme.textSecondary,
236             fontSize = 13.sp,
237             fontWeight = if (selected) FontWeight.SemiBold else FontWeight.Normal,
238         )
239     }
240 }
241
242 @Composable
243 private fun WheelTab(
244     hsl: Hsl,
245     theme: AppTheme,
246     onHslChange: (Hsl) -> Unit,
247     harmonyType: HarmonyType,
248     onHarmonyTypeChange: (HarmonyType) -> Unit,
249     recent: List<Color>,
250     onPickRecent: (Color) -> Unit,
251 ) {
252     HueSatWheel(
253         hsl = hsl,
254         harmonyType = harmonyType,
255         onHsChange = { newH, newS -> onHslChange(hsl.copy(h = newH, s = newS)) },
256         modifier = Modifier
257             .fillMaxWidth()
258             .height(160.dp),
259     )
260     Spacer(Modifier.height(8.dp))
261     LightnessSlider(
262         hsl = hsl,
263         theme = theme,
264         onChange = { newL -> onHslChange(hsl.copy(l = newL)) },
265     )
266     Spacer(Modifier.height(8.dp))
267     HexRow(color = hslToColor(hsl), theme = theme)
268     Spacer(Modifier.height(10.dp))
269     HarmonySection(
270         hsl = hsl,
271         theme = theme,
272         harmonyType = harmonyType,
273         onHarmonyTypeChange = onHarmonyTypeChange,
274         onPick = onPickRecent,

```

```

275     )
276     Spacer(Modifier.height(10.dp))
277     SectionTitle("Недавние", theme)
278     Spacer(Modifier.height(6.dp))
279     RecentRow(colors = recent.take(GRID_COLUMNS), theme = theme, onPick = onPickRecent)
280 }
281
282 @Composable
283 private fun HueSatWheel(
284     hsl: Hsl,
285     harmonyType: HarmonyType,
286     onHsChange: (Float, Float) -> Unit,
287     modifier: Modifier = Modifier,
288 ) {
289     val latestOnHsChange by rememberUpdatedState(onHsChange)
290     Canvas(
291         modifier = modifier
292             .clip(RoundedCornerShape(16.dp))
293             .pointerInput(Unit) {
294                 detectDragGestures { change, _ ->
295                     val (nh, ns) = pickHueSat(change.position, size.width.toFloat(), size.height.toFloat())
296                     latestOnHsChange(nh, ns)
297                 }
298             }
299             .pointerInput(Unit) {
300                 detectTapGestures(onTap = { offset ->
301                     val (nh, ns) = pickHueSat(offset, size.width.toFloat(), size.height.toFloat())
302                     latestOnHsChange(nh, ns)
303                 })
304             },
305     ) {
306         val w = size.width
307         val hgt = size.height
308         val cx = w / 2f
309         val cy = hgt / 2f
310         val outer = minOf(w, hgt) / 2f - 4f
311         if (outer <= 0f) return@Canvas
312
313         val steps = 120
314         for (i in 0 until steps) {
315             val a0 = i * 360f / steps
316             val a1 = (i + 1) * 360f / steps
317             val midA = (a0 + a1) / 2f
318             val hueColor = hslToColor(midA, 1f, 0.5f)
319             drawArc(
320                 color = hueColor,
321                 startAngle = a0 - 90f,
322                 sweepAngle = (a1 - a0) + 0.6f,
323                 useCenter = true,
324                 topLeft = Offset(cx - outer, cy - outer),
325                 size = Size(outer * 2, outer * 2),
326             )
327         }
328         drawCircle(
329             brush = Brush.radialGradient(
330                 colors = listOf(Color.White, Color.White.copy(alpha = 0f)),
331                 center = Offset(cx, cy),
332                 radius = outer,
333             ),
334             radius = outer,
335             center = Offset(cx, cy),
336         )
337
338         val derived = harmonyType.derive(hsl)
339         val verts = derived.map { d -> hslToWheelPoint(d, cx, cy, outer) }
340         drawHarmonyShape(harmonyType, verts)
341
342         val current = hslToWheelPoint(hsl, cx, cy, outer)
343         drawCircle(color = Color.White, radius = 7f, center = current, style = Stroke(width = 2f))
344         drawCircle(color = Color.Black, radius = 7f, center = current, style = Stroke(width = 1f))
345     }
346 }
347
348 private fun hslToWheelPoint(h: Hsl, cx: Float, cy: Float, outer: Float): Offset {
349     val a = Math.toRadians((h.h - 90.0))
350     val r = h.s.coerceIn(0f, 1f) * outer
351     return Offset(cx + (r * cos(a)).toFloat(), cy + (r * sin(a)).toFloat())
352 }
353
354 private fun DrawScope.drawHarmonyShape(type: HarmonyType, verts: List<Offset>) {
355     if (verts.size < 2) return
356     val edges: List<Pair<Int, Int>> = when (type) {
357         HarmonyType.Complementary -> listOf(0 to 1)
358         HarmonyType.Analogous -> listOf(0 to 1, 1 to 2)
359         HarmonyType.Triadic -> listOf(0 to 1, 1 to 2, 2 to 0)
360         HarmonyType.SplitComplementary -> listOf(0 to 1, 0 to 2, 1 to 2)
361         HarmonyType.Tetradic,
362         HarmonyType.Square -> listOf(0 to 1, 1 to 2, 2 to 3, 3 to 0)
363     }
364     edges.forEach { (a, b) ->
365         drawLine(color = Color.White.copy(alpha = 0.85f), start = verts[a], end = verts[b], strokeWidth = 2.5f)
366         drawLine(color = Color.Black.copy(alpha = 0.55f), start = verts[a], end = verts[b], strokeWidth = 1f)
367     }
368 }

```

```

367     }
368     verts.forEach { v ->
369         drawCircle(color = Color.White, radius = 4f, center = v)
370         drawCircle(color = Color.Black, radius = 4f, center = v, style = Stroke(width = 1f))
371     }
372 }
373
374 private fun pickHueSat(p: Offset, w: Float, h: Float): Pair<Float, Float> {
375     val cx = w / 2f
376     val cy = h / 2f
377     val outer = minOf(w, h) / 2f - 4f
378     val dx = p.x - cx
379     val dy = p.y - cy
380     val r = hypot(dx, dy)
381     val ang = Math.toDegrees(atan2(dy, dx).toDouble()) + 90.0
382     val nh = ((ang + 360.0) % 360.0).toFloat()
383     val ns = (r / outer).coerceIn(0f, 1f)
384     return nh to ns
385 }
386
387 @Composable
388 private fun LightnessSlider(hsl: Hsl, theme: AppTheme, onChange: (Float) -> Unit) {
389     val hue = hsl.h
390     val sat = hsl.s
391     val gradient = remember(hue, sat) {
392         Brush.horizontalGradient(
393             colors = listOf(
394                 hslToColor(hue, sat, 0f),
395                 hslToColor(hue, sat, 0.5f),
396                 hslToColor(hue, sat, 1f),
397             ),
398         )
399     }
400     val latestOnChange by rememberUpdatedState(onChange)
401     Row(verticalAlignment = Alignment.CenterVertically) {
402         Text("L", color = theme.textSecondary, fontSize = 12.sp, modifier = Modifier.width(16.dp))
403         Spacer(Modifier.width(8.dp))
404         Box(
405             modifier = Modifier
406                 .height(20.dp)
407                 .fillMaxWidth()
408                 .clip(RoundedCornerShape(10.dp))
409                 .background(gradient)
410                 .border(1.5.dp, theme.panelStroke, RoundedCornerShape(10.dp))
411                 .pointerInput(Unit) {
412                     detectDragGestures { change, _ ->
413                         val f = (change.position.x / size.width).coerceIn(0f, 1f)
414                         latestOnChange(f)
415                     }
416                 }
417                 .pointerInput(Unit) {
418                     detectTapGestures(onTap = { offset ->
419                         val f = (offset.x / size.width).coerceIn(0f, 1f)
420                         latestOnChange(f)
421                     })
422                 },
423         ) {
424             Canvas(modifier = Modifier.fillMaxSize()) {
425                 val knobR = (size.height / 2f - 2f).coerceAtMost(size.width / 2f).coerceAtLeast(0f)
426                 if (knobR <= 0f) return@Canvas
427                 val knobX = (hsl.l.coerceIn(0f, 1f) * size.width).coerceIn(knobR + 2f, size.width - knobR - 2f)
428
429                 drawCircle(color = Color.White, radius = knobR, center = Offset(knobX, size.height / 2f), style = Stroke(
width = 2.5f))
430                 drawCircle(color = theme.panelStroke, radius = knobR, center = Offset(knobX, size.height / 2f), style =
Stroke(width = 1f))
431             }
432         }
433     }
434 }
435
436 @Composable
437 private fun HexRow(color: Color, theme: AppTheme) {
438     Row(verticalAlignment = Alignment.CenterVertically) {
439         Text("Hex", color = theme.textSecondary, fontSize = 11.sp)
440         Spacer(Modifier.weight(1f))
441         Box(
442             modifier = Modifier
443                 .clip(RoundedCornerShape(8.dp))
444                 .background(theme.panelInset)
445                 .border(1.dp, theme.panelStroke, RoundedCornerShape(8.dp))
446                 .padding(horizontal = 10.dp, vertical = 4.dp),
447         ) {
448             Text(color.toHexString(), color = theme.textPrimary, fontSize = 12.sp, fontWeight = FontWeight.Medium)
449         }
450     }
451 }
452
453 @Composable
454 private fun HarmonySection(
455     hsl: Hsl,
456     theme: AppTheme,

```

```

457     harmonyType: HarmonyType,
458     onHarmonyTypeChange: (HarmonyType) -> Unit,
459     onPick: (Hsl) -> Unit,
460 ) {
461     SectionTitle("Тип гармонии", theme)
462     Spacer(Modifier.height(6.dp))
463     Row(
464         horizontalArrangement = Arrangement.spacedBy(6.dp),
465         modifier = Modifier.fillMaxWidth(),
466     ) {
467         HarmonyType.entries.forEach { type ->
468             HarmonyIcon(
469                 type = type,
470                 theme = theme,
471                 selected = type == harmonyType,
472                 modifier = Modifier.weight(1f),
473                 ) { onHarmonyTypeChange(type) }
474         }
475     }
476     Spacer(Modifier.height(6.dp))
477     Text(harmonyType.displayName, color = theme.textSecondary, fontSize = 11.sp, textAlign = TextAlign.Center, modifier =
478     Modifier.fillMaxWidth())
479     Spacer(Modifier.height(8.dp))
480     val derived = remember(harmonyType, hsl) { harmonyType.derive(hsl) }
481     Row(
482         horizontalArrangement = Arrangement.spacedBy(6.dp),
483         verticalAlignment = Alignment.CenterVertically,
484         modifier = Modifier.fillMaxWidth(),
485     ) {
486         derived.forEach { d ->
487             Box(
488                 modifier = Modifier
489                     .weight(1f)
490                     .height(32.dp)
491                     .clip(RoundedCornerShape(8.dp))
492                     .background(hslToColor(d))
493                     .border(1.dp, theme.panelInset, RoundedCornerShape(8.dp))
494                     .clickable { onPick(d) },
495             )
496         }
497         repeat(4 - derived.size) { Box(modifier = Modifier.weight(1f)) }
498     }
499 }
500 @Composable
501 private fun HarmonyIcon(
502     type: HarmonyType,
503     theme: AppTheme,
504     selected: Boolean,
505     modifier: Modifier = Modifier,
506     onClick: () -> Unit,
507 ) {
508     Box(
509         modifier = modifier
510             .aspectRatio(1f)
511             .clip(RoundedCornerShape(8.dp))
512             .background(if (selected) theme.accent.copy(alpha = 0.25f) else theme.panelInset)
513             .border(1.dp, if (selected) theme.accent else theme.panelInset, RoundedCornerShape(8.dp))
514             .clickable(onClick = onClick)
515             .padding(6.dp),
516         contentAlignment = Alignment.Center,
517     ) {
518         Canvas(modifier = Modifier.fillMaxSize()) {
519             val cx = size.width / 2f
520             val cy = size.height / 2f
521             val r = minOf(size.width, size.height) / 2f - 1f
522             val ringColor = (if (selected) Color.White else theme.textSecondary).copy(alpha = 0.6f)
523             val shapeColor = if (selected) Color.White else theme.textSecondary
524             drawCircle(color = ringColor, radius = r, center = Offset(cx, cy), style = Stroke(width = 1.2f))
525             val verts = type.offsets.map { off ->
526                 val a = Math.toRadians((off - 90.0).toDouble())
527                 Offset(cx + (r * 0.78f * cos(a)).toFloat(), cy + (r * 0.78f * sin(a)).toFloat())
528             }
529             val edges: List<Pair<Int, Int>> = when (type) {
530                 HarmonyType.Complementary -> listOf(0 to 1)
531                 HarmonyType.Analogous -> listOf(0 to 1, 1 to 2)
532                 HarmonyType.Triadic -> listOf(0 to 1, 1 to 2, 2 to 0)
533                 HarmonyType.SplitComplementary -> listOf(0 to 1, 0 to 2, 1 to 2)
534                 HarmonyType.Tetradic,
535                 HarmonyType.Square -> listOf(0 to 1, 1 to 2, 2 to 3, 3 to 0)
536             }
537             edges.forEach { (a, b) -> drawLine(color = shapeColor, start = verts[a], end = verts[b], strokeWidth = 1.8f) }
538             verts.forEach { v -> drawCircle(color = shapeColor, radius = 2.2f, center = v) }
539         }
540     }
541 }
542
543 private const val GRID_COLUMNS = 10
544
545 @Composable
546 private fun MapTab(
547     userCollections: List<ColorCollection>,

```

```

548     currentColor: Color,
549     theme: AppTheme,
550     onPick: (Color) -> Unit,
551     onDeleteUserCollection: (Long) -> Unit,
552     onAddFromPhoto: () -> Unit,
553     allowCollectionEditing: Boolean,
554     onCreatePalette: (String) -> Unit,
555     onAddColorToCollection: (Long, Color) -> Unit,
556     onReplaceColorInCollection: (Long, Int, Color) -> Unit,
557     onRemoveColorFromCollection: (Long, Int) -> Unit,
558     onPinCollection: (ColorCollection) -> Unit,
559 ) {
560     var showCreate by remember { mutableStateOf(false) }
561     Column(verticalArrangement = Arrangement.spacedBy(10.dp)) {
562         if (allowCollectionEditing) {
563             Row(horizontalArrangement = Arrangement.spacedBy(8.dp)) {
564                 MapActionButton(label = "+ Создать", theme = theme, modifier = Modifier.weight(1f), onClick = { showCreate =
true })
565                 MapActionButton(label = "+ Из фото", theme = theme, modifier = Modifier.weight(1f), onClick = onAddFromPhoto)
566             }
567         }
568         userCollections.forEach { collection ->
569             CollectionCard(collection = collection, theme = theme, onPick = onPick, onDelete = { onDeleteUserCollection(
collection.id) },
570                 onAddCurrentColor = { onAddColorToCollection(collection.id, currentColor) },
571                 onReplaceAt = { slot -> onReplaceColorInCollection(collection.id, slot, currentColor) },
572                 onRemoveAt = { slot -> onRemoveColorFromCollection(collection.id, slot) },
573                 onPin = { onPinCollection(collection) },
574             )
575         }
576         BuiltinCollections.all.forEach { collection ->
577             CollectionCard(collection = collection, theme = theme, onPick = onPick, onDelete = null, onAddCurrentColor = null,
onReplaceAt = null, onRemoveAt = null, onPin = { onPinCollection(collection) })
578         }
579     }
580     if (showCreate && allowCollectionEditing) {
581         CreatePaletteDialog(theme = theme, onConfirm = { name -> onCreatePalette(name); showCreate = false }, onDismiss = {
showCreate = false })
582     }
583 }
584
585 @Composable
586 private fun MapActionButton(label: String, theme: AppTheme, modifier: Modifier = Modifier, onClick: () -> Unit) {
587     Box(modifier = modifier.fillMaxWidth().clip(RoundedCornerShape(10.dp)).background(theme.accent.copy(alpha = 0.15f)).
border(1.2.dp, theme.accentSoft, RoundedCornerShape(10.dp)).clickable(onClick = onClick).padding(vertical = 10.dp, horizontal = 10.
dp), contentAlignment = Alignment.Center) {
588         Text(label, color = theme.textPrimary, fontSize = 13.sp, fontWeight = FontWeight.SemiBold)
589     }
590 }
591
592 @Composable
593 private fun CollectionCard(collection: ColorCollection, theme: AppTheme, onPick: (Color) -> Unit, onDelete: (() -> Unit)?,
onAddCurrentColor: (() -> Unit)?, onReplaceAt: ((Int) -> Unit)?, onRemoveAt: ((Int) -> Unit)?, onPin: () -> Unit) {
594     var editingIdx by remember { mutableStateOf<Int?>(null) }
595     val revealPx = with(LocalDensity.current) { 96.dp.toPx() }
596     var rawOffset by remember(collection.id) { mutableFloatStateOf(0f) }
597     val animatedOffset by animateFloatAsState(rawOffset, tween(140), label = "paletteSwipe")
598     val cardShape = RoundedCornerShape(10.dp)
599     Box(modifier = Modifier.fillMaxWidth().clip(cardShape)) {
600         Box(
601             modifier = Modifier.matchParentSize().background(theme.accent.copy(alpha = 0.9f)),
602             contentAlignment = Alignment.CenterEnd,
603         ) {
604             Text(
605                 "Закрепить",
606                 color = Color.White,
607                 fontSize = 12.sp,
608                 fontWeight = FontWeight.SemiBold,
609                 modifier = Modifier.width(96.dp).fillMaxHeight().clickable { onPin(); rawOffset = 0f }.wrapContentHeight(
Alignment.CenterVertically),
610                 textAlign = TextAlign.Center,
611             )
612         }
613         Column(
614             modifier = Modifier
615                 .fillMaxWidth()
616                 .offset { IntOffset(animatedOffset.roundToInt(), 0) }
617                 .background(theme.panelInset)
618                 .border(1.dp, theme.panelStroke, cardShape)
619                 .pointerInput(collection.id, revealPx) {
620                     detectHorizontalDragGestures(
621                         onHorizontalDrag = { change, dx ->
622                             if (dx < 0f || rawOffset < 0f) {
623                                 change.consume()
624                                 rawOffset = (rawOffset + dx).coerceIn(-revealPx, 0f)
625                             }
626                     },
627                     onDragEnd = { rawOffset = if (rawOffset < -revealPx * 0.4f) -revealPx else 0f },
628                     onDragCancel = { rawOffset = 0f },
629                 )
630             )
631         .padding(horizontal = 8.dp, vertical = 8.dp),

```

```

632         verticalArrangement = Arrangement.spacedBy(6.dp),
633     ) {
634         Row(verticalAlignment = Alignment.CenterVertically) {
635             Text(collection.name, color = theme.textPrimary, fontSize = 13.sp, fontWeight = FontWeight.SemiBold)
636             Spacer(Modifier.weight(1f))
637             if (onDelete != null) {
638                 Box(modifier = Modifier.size(22.dp).clip(CircleShape).background(theme.panelBg.copy(alpha = 0.7f)).border(1.
dp, theme.panelStroke, CircleShape).clickable(onClick = onDelete), contentAlignment = Alignment.Center) {
639                     Text("x", color = theme.textSecondary, fontSize = 14.sp, fontWeight = FontWeight.Bold)
640                 }
641             }
642         }
643         SwatchGrid(colors = collection.composeColors(), theme = theme, onPick = onPick, trailingPlusOnClick =
onAddCurrentColor, onLongPressIndex = if (onReplaceAt != null && onRemoveAt != null) { { idx -> editingIdx = idx } } else null)
644     }
645     }
646     val activeIdx = editingIdx
647     if (activeIdx != null && onReplaceAt != null && onRemoveAt != null) {
648         val swatchColor = collection.composeColors().getOrNull(activeIdx)
649         if (swatchColor != null) {
650             EditSwatchSheet(swatchColor = swatchColor, theme = theme, onReplace = { onReplaceAt(activeIdx); editingIdx = null
}, onRemove = { onRemoveAt(activeIdx); editingIdx = null }, onDismiss = { editingIdx = null })
651         } else editingIdx = null
652     }
653 }
654
655 @OptIn(ExperimentalFoundationApi::class)
656 @Composable
657 private fun SwatchGrid(colors: List<Color>, theme: AppTheme, onPick: (Color) -> Unit, trailingPlusOnClick: (() -> Unit)? =
null, onLongPressIndex: ((Int) -> Unit)? = null) {
658     val total = colors.size + if (trailingPlusOnClick != null) 1 else 0
659     if (total == 0) return
660     val itemRows = (0 until total).toList().chunked(GRID_COLUMNS)
661     Column(verticalArrangement = Arrangement.spacedBy(4.dp)) {
662         itemRows.forEach { row ->
663             Row(horizontalArrangement = Arrangement.spacedBy(4.dp), modifier = Modifier.fillMaxWidth()) {
664                 row.forEach { i ->
665                     if (i < colors.size) {
666                         val c = colors[i]
667                         val cellModifier = Modifier.weight(1f).aspectRatio(1f).clip(RoundedCornerShape(5.dp)).background(c).
border(1.dp, theme.panelInset, RoundedCornerShape(5.dp))
668                         if (onLongPressIndex != null) Box(modifier = cellModifier.combinedClickable(onClick = { onPick(c) },
onLongClick = { onLongPressIndex(i) }))
669                         else Box(modifier = cellModifier.clickable { onPick(c) })
670                     } else PlusTile(theme = theme, modifier = Modifier.weight(1f).aspectRatio(1f), onClick =
trailingPlusOnClick!!)
671                 }
672                 repeat(GRID_COLUMNS - row.size) { Box(modifier = Modifier.weight(1f)) }
673             }
674         }
675     }
676 }
677
678 @Composable
679 private fun EditSwatchSheet(swatchColor: Color, theme: AppTheme, onReplace: () -> Unit, onRemove: () -> Unit, onDismiss: () -
> Unit) {
680     Dialog(onDismissRequest = onDismiss) {
681         Column(modifier = Modifier.width(280.dp).clip(RoundedCornerShape(16.dp)).background(theme.panelBg).padding(16.dp),
verticalArrangement = Arrangement.spacedBy(12.dp)) {
682             Row(verticalAlignment = Alignment.CenterVertically) {
683                 Box(modifier = Modifier.size(28.dp).clip(RoundedCornerShape(6.dp)).background(swatchColor).border(1.dp, theme.
panelInset, RoundedCornerShape(6.dp)))
684                 Spacer(Modifier.width(10.dp)); Text("Цвет", color = theme.textPrimary, fontSize = 16.sp, fontWeight =
FontWeight.SemiBold)
685             }
686             DialogButton(label = "Заменить текущим", theme = theme, primary = true, modifier = Modifier.fillMaxWidth(),
onClick = onReplace)
687             DialogButton(label = "Удалить", theme = theme, primary = false, modifier = Modifier.fillMaxWidth(), onClick =
onRemove)
688             DialogButton(label = "Отмена", theme = theme, primary = false, modifier = Modifier.fillMaxWidth(), onClick =
onDismiss)
689         }
690     }
691 }
692
693 @Composable
694 private fun PlusTile(theme: AppTheme, modifier: Modifier = Modifier, onClick: () -> Unit) {
695     Box(modifier = modifier.clip(RoundedCornerShape(5.dp)).background(theme.panelBg.copy(alpha = 0.5f)).border(1.dp, theme.
accent.copy(alpha = 0.7f), RoundedCornerShape(5.dp)).clickable(onClick = onClick), contentAlignment = Alignment.Center) {
696         Text("+", color = theme.textPrimary, fontSize = 16.sp, fontWeight = FontWeight.Bold)
697     }
698 }
699
700 @Composable
701 private fun CreatePaletteDialog(theme: AppTheme, onConfirm: (String) -> Unit, onDismiss: () -> Unit) {
702     var name by remember { mutableStateOf("") }
703     Dialog(onDismissRequest = onDismiss) {
704         Column(modifier = Modifier.width(280.dp).clip(RoundedCornerShape(16.dp)).background(theme.panelBg).padding(16.dp),
verticalArrangement = Arrangement.spacedBy(12.dp)) {
705             Text("Новая палитра", color = theme.textPrimary, fontSize = 16.sp, fontWeight = FontWeight.SemiBold)
706             Box(modifier = Modifier.fillMaxWidth().clip(RoundedCornerShape(8.dp)).background(theme.panelInset).border(1.dp,
theme.panelInset, RoundedCornerShape(8.dp)).padding(horizontal = 10.dp, vertical = 8.dp)) {

```

```

707         BasicTextField(value = name, onValueChange = { name = it.take(40) }, singleLine = true, textStyle = TextStyle(
color = theme.textPrimary, fontSize = 14.sp), cursorBrush = SolidColor(theme.accent), decorationBox = { inner -> if (name.isEmpty(
)) Text("Название", color = theme.textSecondary, fontSize = 14.sp); inner() }, modifier = Modifier.fillMaxWidth())
708     }
709     Row(horizontalArrangement = Arrangement.spacedBy(8.dp), modifier = Modifier.fillMaxWidth()) {
710         DialogButton(label = "Отмена", theme = theme, primary = false, modifier = Modifier.weight(1f), onClick =
onDismiss)
711         DialogButton(label = "Создать", theme = theme, primary = true, modifier = Modifier.weight(1f), onClick = {
onConfirm(name) })
712     }
713 }
714 }
715 }
716
717 @Composable
718 private fun DialogButton(label: String, theme: AppTheme, primary: Boolean, modifier: Modifier = Modifier, onClick: () ->
Unit) {
719     Box(modifier = modifier.clip(RoundedCornerShape(8.dp)).background(if (primary) theme.accent.copy(alpha = 0.35f) else
theme.panelInset).border(1.dp, if (primary) theme.accent else theme.panelInset, RoundedCornerShape(8.dp)).clickable(onClick =
onClick).padding(vertical = 10.dp), contentAlignment = Alignment.Center) {
720         Text(label, color = theme.textPrimary, fontSize = 13.sp, fontWeight = FontWeight.SemiBold)
721     }
722 }
723
724 @Composable
725 private fun SectionTitle(text: String, theme: AppTheme) {
726     Text(text, color = theme.textSecondary, fontSize = 11.sp, fontWeight = FontWeight.Medium)
727 }
728
729 @Composable
730 private fun RecentRow(colors: List<Color>, theme: AppTheme, onPick: (Color) -> Unit) {
731     Row(horizontalArrangement = Arrangement.spacedBy(4.dp), verticalAlignment = Alignment.CenterVertically, modifier =
Modifier.fillMaxWidth()) {
732         repeat(GRID_COLUMNS) { i ->
733             val c = colors.getOrNull(i)
734             if (c != null) Box(modifier = Modifier.weight(1f).aspectRatio(1f).clip(RoundedCornerShape(5.dp)).background(c).
border(1.dp, theme.panelInset, RoundedCornerShape(5.dp)).clickable { onPick(c) })
735             else Box(modifier = Modifier.weight(1f).aspectRatio(1f).clip(RoundedCornerShape(5.dp)).background(theme.
panelInset).border(1.dp, theme.panelInset, RoundedCornerShape(5.dp)))
736         }
737     }
738 }

```

72. app/src/main/java/com/wetinknext/ui/color/GlesColorState.kt

```

1  package com.wetinknext.ui.color
2
3  import android.content.Context
4  import androidx.compose.runtime.mutableStateListOf
5  import androidx.compose.runtime.mutableLongStateOf
6  import androidx.compose.runtime.mutableStateOf
7  import androidx.compose.runtime.getValue
8  import androidx.compose.runtime.setValue
9  import androidx.compose.ui.graphics.Color
10 import androidx.compose.ui.graphics.toArgb
11 import androidx.compose.ui.geometry.Offset
12 import org.json.JSONArray
13 import org.json.JSONObject
14
15 class GlesColorState(context: Context? = null) {
16     private val prefs = context?.applicationContext?.getSharedPreferences(PrefsName, Context.MODE_PRIVATE)
17
18     val recentColors = mutableStateListOf<Color>()
19     val userCollections = mutableStateListOf<ColorCollection>()
20     var pinnedCollectionId by mutableLongStateOf(prefs?.getLong(PinnedCollectionKey, NoPinnedCollection) ?:
NoPinnedCollection)
21     private set
22     var pinnedPaletteOffset by mutableStateOf(
23         Offset(
24             prefs?.getFloat(PinnedPaletteOffsetXKey, DefaultPinnedOffsetX) ?: DefaultPinnedOffsetX,
25             prefs?.getFloat(PinnedPaletteOffsetYKey, DefaultPinnedOffsetY) ?: DefaultPinnedOffsetY,
26         ),
27     )
28     private set
29
30     init {
31         loadRecentColors()
32         loadUserCollections()
33     }
34
35     fun pushRecentColor(color: Color) {
36         val argb = color.toArgb()
37         val existingIndex = recentColors.indexOfFirst { it.toArgb() == argb }
38         if (existingIndex == 0) return
39         if (existingIndex > 0) recentColors.removeAt(existingIndex)
40         recentColors.add(0, color)
41         while (recentColors.size > MaxRecentColors) {
42             recentColors.removeAt(recentColors.lastIndex)
43         }
44         saveRecentColors()
45     }
46
47     private fun loadRecentColors() {
48         val encoded = prefs?.getString(RecentColorsKey, null) ?: return
49         encoded.split(',')
50             .mapNotNull { it.toIntOrNull() }
51             .take(MaxRecentColors)
52             .forEach { recentColors.add(Color(it)) }
53     }
54
55     private fun saveRecentColors() {
56         prefs?.edit()
57             ?.putString(RecentColorsKey, recentColors.joinToString(",") { it.toArgb().toString() })
58             ?.apply()
59     }
60
61     fun addUserCollection(collection: ColorCollection) {
62         userCollections.add(0, collection.copy(builtin = false))
63         saveUserCollections()
64     }
65
66     fun deleteUserCollection(id: Long) {
67         userCollections.removeAll { it.id == id }
68         if (pinnedCollectionId == id) unpinCollection()
69         saveUserCollections()
70     }
71
72     fun pinCollection(id: Long) {
73         pinnedCollectionId = id
74         prefs?.edit()?.putLong(PinnedCollectionKey, id)?.apply()
75     }
76
77     fun unpinCollection() {
78         pinnedCollectionId = NoPinnedCollection
79         prefs?.edit()?.remove(PinnedCollectionKey)?.apply()
80     }
81
82     fun persistPinnedPaletteOffset(offset: Offset) {
83         pinnedPaletteOffset = offset
84         prefs?.edit()
85             ?.putFloat(PinnedPaletteOffsetXKey, offset.x)
86             ?.putFloat(PinnedPaletteOffsetYKey, offset.y)
87             ?.apply()
88     }
89

```



```

90 fun addColorToCollection(id: Long, color: Color) {
91     updateUserCollection(id) { collection ->
92         collection.copy(colors = collection.colors + color.toArgb())
93     }
94 }
95
96 fun replaceColorInCollection(id: Long, slot: Int, color: Color) {
97     updateUserCollection(id) { collection ->
98         if (slot !in collection.colors.indices) return@updateUserCollection collection
99         val colors = collection.colors.toMutableList()
100         colors[slot] = color.toArgb()
101         collection.copy(colors = colors)
102     }
103 }
104
105 fun removeColorFromCollection(id: Long, slot: Int) {
106     updateUserCollection(id) { collection ->
107         if (slot !in collection.colors.indices) return@updateUserCollection collection
108         val colors = collection.colors.toMutableList()
109         colors.removeAt(slot)
110         collection.copy(colors = colors)
111     }
112 }
113
114 private fun updateUserCollection(id: Long, transform: (ColorCollection) -> ColorCollection) {
115     val index = userCollections.indexOfFirst { it.id == id }
116     if (index < 0) return
117     userCollections[index] = transform(userCollections[index]).copy(builtin = false)
118     saveUserCollections()
119 }
120
121 private fun loadUserCollections() {
122     val encoded = prefs?.getString(UserCollectionsKey, null) ?: return
123     runCatching {
124         val array = JSONArray(encoded)
125         for (i in 0 until array.length()) {
126             val item = array.optJSONObject(i) ?: continue
127             val colorsJson = item.optJSONArray("colors") ?: JSONArray()
128             val colors = buildList {
129                 for (j in 0 until colorsJson.length()) {
130                     add(colorsJson.getInt(j))
131                 }
132             }
133             userCollections.add(
134                 ColorCollection(
135                     id = item.optLong("id", System.currentTimeMillis() + i),
136                     name = item.optString("name", "Новая палитра"),
137                     colors = colors,
138                     builtin = false,
139                 ),
140             )
141         }
142     }
143 }
144
145 private fun saveUserCollections() {
146     val array = JSONArray()
147     userCollections.filterNot { it.builtin }.forEach { collection ->
148         val colors = JSONArray()
149         collection.colors.forEach { colors.put(it) }
150         array.put(
151             JSONObject()
152                 .put("id", collection.id)
153                 .put("name", collection.name)
154                 .put("colors", colors),
155         )
156     }
157     prefs?.edit()
158         ?.putString(UserCollectionsKey, array.toString())
159         ?.apply()
160 }
161
162 private companion object {
163     const val PrefsName = "wetinknext_color_state"
164     const val RecentColorsKey = "recent_colors"
165     const val UserCollectionsKey = "user_collections"
166     const val PinnedCollectionKey = "pinned_collection"
167     const val PinnedPaletteOffsetXKey = "pinned_palette_offset_x"
168     const val PinnedPaletteOffsetYKey = "pinned_palette_offset_y"
169     const val NoPinnedCollection = Long.MIN_VALUE
170     const val DefaultPinnedOffsetX = 84f
171     const val DefaultPinnedOffsetY = 128f
172     const val MaxRecentColors = 14
173 }
174 }

```

73. app/src/main/java/com/wetinknext/ui/components/AdjustmentsPanel.kt

```

1  package com.wetinknext.ui.components
2
3  import androidx.compose.foundation.background
4  import androidx.compose.foundation.border
5  import androidx.compose.foundation.clickable
6  import androidx.compose.foundation.layout.*
7  import androidx.compose.foundation.rememberScrollState
8  import androidx.compose.foundation.shape.RoundedCornerShape
9  import androidx.compose.foundation.verticalScroll
10 import androidx.compose.material3.HorizontalDivider
11 import androidx.compose.material3.Text
12 import androidx.compose.runtime.Composable
13 import androidx.compose.ui.Alignment
14 import androidx.compose.ui.Modifier
15 import androidx.compose.ui.draw.clip
16 import androidx.compose.ui.draw.shadow
17 import androidx.compose.ui.graphics.Color
18 import androidx.compose.ui.text.font.FontWeight
19 import androidx.compose.ui.unit.dp
20 import androidx.compose.ui.unit.sp
21 import com.wetinknext.ui.theme.AppTheme
22
23 enum class AdjustmentAction(val title: String) {
24     BRIGHTNESS_CONTRAST("Яркость и контраст"),
25     THRESHOLD("Бинаризация"),
26     BRIGHTNESS_TO_OPACITY("Преобразовать яркость в непрозрачность"),
27 }
28
29 @Composable
30 fun AdjustmentsPanel(
31     theme: AppTheme,
32     onClose: () -> Unit,
33     onAction: (AdjustmentAction) -> Unit,
34     modifier: Modifier = Modifier
35 ) {
36     Column(
37         modifier = modifier
38         .width(280.dp)
39         .shadow(8.dp, RoundedCornerShape(16.dp))
40         .clip(RoundedCornerShape(16.dp))
41         .background(theme.panelBg.copy(alpha = 0.98f))
42         .border(1.dp, theme.panelStroke, RoundedCornerShape(16.dp))
43         .padding(vertical = 8.dp)
44     ) {
45         Text(
46             text = "Коррекция",
47             color = theme.textPrimary,
48             fontSize = 16.sp,
49             fontWeight = FontWeight.Bold,
50             modifier = Modifier.padding(horizontal = 16.dp, vertical = 8.dp)
51         )
52
53         Column(
54             modifier = Modifier
55                 .fillMaxWidth()
56                 .verticalScroll(rememberScrollState())
57         ) {
58             AdjustmentAction.entries.forEachIndexed { index, action ->
59                 AdjustmentItem(
60                     title = action.title,
61                     theme = theme,
62                     onClick = {
63                         onClose()
64                         onAction(action)
65                     }
66                 )
67                 if (index < AdjustmentAction.entries.size - 1) {
68                     HorizontalDivider(
69                         modifier = Modifier.padding(horizontal = 16.dp),
70                         thickness = 0.5.dp,
71                         color = theme.panelStroke.copy(alpha = 0.5f)
72                     )
73                 }
74             }
75         }
76     }
77 }
78
79 @Composable
80 private fun AdjustmentItem(
81     title: String,
82     theme: AppTheme,
83     onClick: () -> Unit
84 ) {
85     Row(
86         modifier = Modifier
87             .fillMaxWidth()
88             .clickable { onClick() }
89             .padding(horizontal = 16.dp, vertical = 12.dp),
90         verticalAlignment = Alignment.CenterVertically,

```

```
91         horizontalArrangement = Arrangement.SpaceBetween
92     ) {
93         Text(
94             text = title,
95             color = theme.textPrimary,
96             fontSize = 14.sp,
97             modifier = Modifier.weight(1f)
98         )
99     }
100 }
```

74. app/src/main/java/com/wetinknext/ui/components/BrightnessContrastPanel.kt

```

1  package com.wetinknext.ui.components
2
3  import androidx.compose.foundation.background
4  import androidx.compose.foundation.border
5  import androidx.compose.foundation.clickable
6  import androidx.compose.foundation.layout.*
7  import androidx.compose.foundation.shape.CircleShape
8  import androidx.compose.foundation.shape.RoundedCornerShape
9  import androidx.compose.material.icons.Icons
10 import androidx.compose.material.icons.filled.Check
11 import androidx.compose.material.icons.filled.Close
12 import androidx.compose.material3.Icon
13 import androidx.compose.material3.Slider
14 import androidx.compose.material3.SliderDefaults
15 import androidx.compose.material3.Text
16 import androidx.compose.runtime.Composable
17 import androidx.compose.ui.Alignment
18 import androidx.compose.ui.Modifier
19 import androidx.compose.ui.draw.clip
20 import androidx.compose.ui.graphics.Color
21 import androidx.compose.ui.text.font.FontWeight
22 import androidx.compose.ui.unit.dp
23 import androidx.compose.ui.unit.sp
24 import com.wetinknext.ui.theme.AppTheme
25 import kotlin.math.roundToInt
26
27 @Composable
28 fun BrightnessContrastPanel(
29     theme: AppTheme,
30     brightness: Float, // -1..1
31     contrast: Float,   // -1..1
32     onBrightnessChange: (Float) -> Unit,
33     onContrastChange: (Float) -> Unit,
34     onApply: () -> Unit,
35     onCancel: () -> Unit,
36     isApplyEnabled: Boolean = true,
37     modifier: Modifier = Modifier
38 ) {
39     Column(
40         modifier = modifier
41             .width(360.dp)
42             .clip(RoundedCornerShape(28.dp))
43             .background(theme.panelBg.copy(alpha = 0.95f))
44             .border(1.2.dp, theme.panelStroke, RoundedCornerShape(28.dp))
45             .padding(horizontal = 20.dp, vertical = 16.dp),
46         verticalArrangement = Arrangement.spacedBy(16.dp)
47     ) {
48         Row(
49             modifier = Modifier.fillMaxWidth(),
50             horizontalArrangement = Arrangement.SpaceBetween,
51             verticalAlignment = Alignment.CenterVertically
52         ) {
53             Text(
54                 text = "Яркость и контраст",
55                 color = theme.textPrimary,
56                 fontSize = 15.sp,
57                 fontWeight = FontWeight.Bold
58             )
59
60             Row(
61                 verticalAlignment = Alignment.CenterVertically,
62                 horizontalArrangement = Arrangement.spacedBy(12.dp)
63             ) {
64                 Box(
65                     modifier = Modifier
66                         .size(36.dp)
67                         .clip(CircleShape)
68                         .background(theme.panelInset)
69                         .clickable { onCancel() },
70                     contentAlignment = Alignment.Center
71                 ) {
72                     Icon(Icons.Default.Close, null, tint = theme.textPrimary, modifier = Modifier.size(20.dp))
73                 }
74
75                 Box(
76                     modifier = Modifier
77                         .size(36.dp)
78                         .clip(CircleShape)
79                         .background(if (isApplyEnabled) theme.accent else theme.panelInset)
80                         .clickable(enabled = isApplyEnabled) { onApply() },
81                     contentAlignment = Alignment.Center
82                 ) {
83                     Icon(Icons.Default.Check, null, tint = Color.White, modifier = Modifier.size(20.dp))
84                 }
85             }
86         }
87
88         // Brightness
89         Column(verticalArrangement = Arrangement.spacedBy(4.dp)) {
90             Row(

```

```

91         modifier = Modifier.fillMaxWidth(),
92         horizontalArrangement = Arrangement.SpaceBetween
93     ) {
94         Text("Яркость", color = theme.textSecondary, fontSize = 13.sp)
95         Text("${(brightness * 100).roundToInt()}", color = theme.accent, fontSize = 13.sp, fontWeight = FontWeight.
Bold)
96     }
97     Slider(
98         value = brightness,
99         onValueChange = onBrightnessChange,
100        valueRange = -1f..1f,
101        colors = SliderDefaults.colors(
102            thumbColor = theme.accent,
103            activeTrackColor = theme.accent,
104            inactiveTrackColor = theme.panelInset
105        )
106    )
107 }
108
109 // Contrast
110 Column(verticalArrangement = Arrangement.spacedBy(4.dp)) {
111     Row(
112         modifier = Modifier.fillMaxWidth(),
113         horizontalArrangement = Arrangement.SpaceBetween
114     ) {
115         Text("Контраст", color = theme.textSecondary, fontSize = 13.sp)
116         Text("${(contrast * 100).roundToInt()}", color = theme.accent, fontSize = 13.sp, fontWeight = FontWeight.Bold)
117     }
118     Slider(
119         value = contrast,
120         onValueChange = onContrastChange,
121         valueRange = -1f..1f,
122         colors = SliderDefaults.colors(
123             thumbColor = theme.accent,
124             activeTrackColor = theme.accent,
125             inactiveTrackColor = theme.panelInset
126         )
127     )
128 }
129 }
130 }

```

75. app/src/main/java/com/wetinknext/ui/components/BrightnessToOpacityPanel.kt

```

1 package com.wetinknext.ui.components
2
3 import androidx.compose.foundation.background
4 import androidx.compose.foundation.border
5 import androidx.compose.foundation.clickable
6 import androidx.compose.foundation.layout.*
7 import androidx.compose.foundation.shape.CircleShape
8 import androidx.compose.foundation.shape.RoundedCornerShape
9 import androidx.compose.material.icons.Icons
10 import androidx.compose.material.icons.filled.Check
11 import androidx.compose.material.icons.filled.Close
12 import androidx.compose.material3.Icon
13 import androidx.compose.material3.Switch
14 import androidx.compose.material3.Text
15 import androidx.compose.runtime.Composable
16 import androidx.compose.ui.Alignment
17 import androidx.compose.ui.Modifier
18 import androidx.compose.ui.draw.clip
19 import androidx.compose.ui.graphics.Color
20 import androidx.compose.ui.text.font.FontWeight
21 import androidx.compose.ui.unit.dp
22 import androidx.compose.ui.unit.sp
23 import com.wetinknext.ui.theme.AppTheme
24
25 @Composable
26 fun BrightnessToOpacityPanel(
27     theme: AppTheme,
28     isInverted: Boolean,
29     onInvertChange: (Boolean) -> Unit,
30     onApply: () -> Unit,
31     onCancel: () -> Unit,
32     isApplyEnabled: Boolean = true,
33     modifier: Modifier = Modifier
34 ) {
35     Column(
36         modifier = modifier
37         .width(360.dp)
38         .clip(RoundedCornerShape(28.dp))
39         .background(theme.panelBg.copy(alpha = 0.95f))
40         .border(1.2.dp, theme.panelStroke, RoundedCornerShape(28.dp))
41         .padding(horizontal = 20.dp, vertical = 16.dp),
42         verticalArrangement = Arrangement.spacedBy(16.dp)
43     ) {
44         Row(
45             modifier = Modifier.fillMaxWidth(),
46             horizontalArrangement = Arrangement.SpaceBetween,
47             verticalAlignment = Alignment.CenterVertically
48         ) {
49             Text(
50                 text = "Яркость в непрозрачность",
51                 color = theme.textPrimary,
52                 fontSize = 15.sp,
53                 fontWeight = FontWeight.Bold
54             )
55
56             Row(
57                 verticalAlignment = Alignment.CenterVertically,
58                 horizontalArrangement = Arrangement.spacedBy(12.dp)
59             ) {
60                 Box(
61                     modifier = Modifier
62                     .size(36.dp)
63                     .clip(CircleShape)
64                     .background(theme.panelInset)
65                     .clickable { onCancel() },
66                     contentAlignment = Alignment.Center
67                 ) {
68                     Icon(Icons.Default.Close, null, tint = theme.textPrimary, modifier = Modifier.size(20.dp))
69                 }
70
71                 Box(
72                     modifier = Modifier
73                     .size(36.dp)
74                     .clip(CircleShape)
75                     .background(if (isApplyEnabled) theme.accent else theme.panelInset)
76                     .clickable(enabled = isApplyEnabled) { onApply() },
77                     contentAlignment = Alignment.Center
78                 ) {
79                     Icon(Icons.Default.Check, null, tint = Color.White, modifier = Modifier.size(20.dp))
80                 }
81             }
82         }
83
84         Text(
85             text = "Светлые области станут прозрачными, тёмные – непрозрачными",
86             color = theme.textSecondary,
87             fontSize = 12.sp,
88             lineHeight = 16.sp
89         )
90     }

```

```

91         Row(
92             modifier = Modifier.fillMaxWidth(),
93             verticalAlignment = Alignment.CenterVertically,
94             horizontalArrangement = Arrangement.SpaceBetween
95         ) {
96             Text(
97                 text = "Инвертировать",
98                 color = theme.textPrimary,
99                 fontSize = 14.sp
100            )
101            Switch(
102                checked = isInverted,
103                onCheckedChange = onInvertChange
104            )
105        }
106    }
107 }

```

76. app/src/main/java/com/wetinknext/ui/components/BrushPanel.kt

```

1  package com.wetinknext.ui.components
2
3  import androidx.compose.foundation.Canvas
4  import androidx.compose.foundation.Image
5  import androidx.compose.foundation.background
6  import androidx.compose.foundation.border
7  import androidx.compose.foundation.clickable
8  import androidx.compose.foundation.gestures.detectTapGestures
9  import androidx.compose.foundation.layout.*
10 import androidx.compose.foundation.lazy.LazyColumn
11 import androidx.compose.foundation.lazy.items
12 import androidx.compose.foundation.shape.CircleShape
13 import androidx.compose.foundation.shape.RoundedCornerShape
14 import androidx.compose.material.icons.Icons
15 import androidx.compose.material.icons.filled.*
16 import androidx.compose.material3.Icon
17 import androidx.compose.material3.Text
18 import androidx.compose.runtime.*
19 import androidx.compose.ui.Alignment
20 import androidx.compose.ui.Modifier
21 import androidx.compose.ui.draw.clip
22 import androidx.compose.ui.draw.drawBehind
23 import androidx.compose.ui.geometry.Offset
24 import androidx.compose.ui.geometry.Size
25 import androidx.compose.ui.graphics.Color
26 import androidx.compose.ui.graphics.ImageBitmap
27 import androidx.compose.ui.graphics.vector.ImageVector
28 import androidx.compose.ui.input.pointer.pointerInput
29 import androidx.compose.ui.layout.ContentScale
30 import androidx.compose.ui.text.font.FontWeight
31 import androidx.compose.ui.unit.dp
32 import androidx.compose.ui.unit.sp
33 import com.wetinknext.engine.brush.BrushLibrary
34 import com.wetinknext.engine.brush.BrushPreset
35 import com.wetinknext.engine.brush.BrushUiCategory
36 import com.wetinknext.ui.theme.AppTheme
37
38 private val BRUSH_SWIPE_REVEAL_WIDTH = 88.dp
39
40 @Composable
41 fun BrushPanel(
42     currentBrush: BrushPreset,
43     onBrushSelect: (BrushPreset) -> Unit,
44     onBrushStudioOpen: (BrushPreset) -> Unit,
45     theme: AppTheme,
46     onDismiss: () -> Unit,
47     onDuplicate: (BrushPreset) -> Unit,
48     onDelete: (BrushPreset) -> Unit,
49     modifier: Modifier = Modifier,
50     categories: List<BrushUiCategory> = BrushLibrary.allCategories,
51     previews: Map<String, ImageBitmap> = emptyMap(),
52     previewKey: (BrushPreset) -> String = { it.id },
53     onRequestPreview: (BrushPreset) -> Unit = {},
54     title: String = "Кисти",
55 ) {
56     fun categoryIndexForCurrentBrush(): Int {
57         return categories.indexOfFirst { category -> category.brushes.any { it.id == currentBrush.id } }
58     }
59     ?.takeIf { it >= 0 }
60     ?: 0
61
62     var selectedCategoryIndex by remember(categories, currentBrush.id) {
63         mutableIntStateOf(categoryIndexForCurrentBrush())
64     }
65     LaunchedEffect(categories, currentBrush.id) {
66         selectedCategoryIndex = categoryIndexForCurrentBrush()
67     }
68     val selectedCategory = categories.getOrNull(selectedCategoryIndex)
69
70     PanelSurface(
71         modifier = modifier
72             .width(420.dp)
73             .height(560.dp),
74     ) {
75         Row(Modifier.fillMaxSize()) {
76             Column(
77                 modifier = Modifier
78                     .width(70.dp)
79                     .fillMaxHeight()
80                     .clip(RoundedCornerShape(topStart = 22.dp, bottomStart = 22.dp))
81                     .background(theme.panelInset)
82                     .drawBehind {
83                         drawLine(
84                             color = theme.panelStroke,
85                             start = Offset(size.width, 0f),
86                             end = Offset(size.width, size.height),
87                             strokeWidth = 1.dp.toPx(),
88                         )
89                     }
90             )
91             .padding(vertical = 16.dp),
92             horizontalAlignment = Alignment.CenterHorizontally,

```



```

91         verticalArrangement = Arrangement.spacedBy(20.dp),
92     ) {
93         categories.forEachIndexed { index, category ->
94             val isSelected = index == selectedCategoryId
95             Box(
96                 modifier = Modifier
97                     .size(44.dp)
98                     .clip(CircleShape)
99                     .background(if (isSelected) theme.accent.copy(alpha = 0.3f) else Color.Transparent)
100                     .border(
101                         if (isSelected) 1.dp else 0.dp,
102                         if (isSelected) theme.accentSoft else Color.Transparent,
103                         CircleShape,
104                     )
105                 .clickable { selectedCategoryId = index },
106                 contentAlignment = Alignment.Center,
107             ) {
108                 Icon(
109                     category.icon,
110                     contentDescription = category.name,
111                     tint = if (isSelected) theme.accent else theme.iconInactive,
112                     modifier = Modifier.size(24.dp),
113                 )
114             }
115         }
116     }
117
118     Column(
119         modifier = Modifier
120             .weight(1f)
121             .fillMaxHeight()
122             .padding(14.dp),
123     ) {
124         Row(
125             verticalAlignment = Alignment.CenterVertically,
126             horizontalArrangement = Arrangement.spacedBy(12.dp)
127         ) {
128             Column(verticalArrangement = Arrangement.spacedBy(1.dp)) {
129                 Text(
130                     text = title,
131                     color = theme.textPrimary,
132                     fontSize = 18.sp,
133                     fontWeight = FontWeight.Bold,
134                 )
135                 selectedCategory?.let { category ->
136                     Text(
137                         text = category.name,
138                         color = theme.accent,
139                         fontSize = 11.sp,
140                         fontWeight = FontWeight.SemiBold,
141                     )
142                 }
143             }
144             Spacer(Modifier.weight(1f))
145
146             Icon(
147                 imageVector = Icons.Default.Close,
148                 contentDescription = "Закреть",
149                 tint = theme.textSecondary,
150                 modifier = Modifier.size(24.dp).clickable { onDismiss() }
151             )
152         }
153
154         Spacer(Modifier.height(14.dp))
155
156         LazyColumn(
157             verticalArrangement = Arrangement.spacedBy(8.dp),
158             modifier = Modifier.fillMaxSize(),
159         ) {
160             items(selectedCategory?.brushes.orEmpty(), key = { it.id }) { brush ->
161                 val isSelected = brush.id == currentBrush.id
162                 BrushItem(
163                     brush = brush,
164                     isSelected = isSelected,
165                     theme = theme,
166                     preview = previews[previewKey(brush)],
167                     onRequestPreview = { onRequestPreview(brush) },
168                     onStudioOpen = { onBrushStudioOpen(brush) },
169                     onDuplicate = { onDuplicate(brush) },
170                     onDelete = { onDelete(brush) },
171                     onClick = {
172                         if (isSelected) onBrushStudioOpen(brush)
173                         else onBrushSelect(brush)
174                     },
175                 )
176             }
177
178             if (selectedCategory?.brushes.isNullOrEmpty()) {
179                 item {
180                     Text(
181                         text = "Кисти отсутствуют",
182                         color = theme.textSecondary,

```

```

183                                     fontSize = 14.sp,
184                                     modifier = Modifier.padding(top = 20.dp),
185                                 )
186                             }
187                         }
188                     }
189                 }
190             }
191         }
192     }
193
194     @Composable
195     private fun BrushItem(
196         brush: BrushPreset,
197         isSelected: Boolean,
198         theme: AppTheme,
199         preview: ImageBitmap?,
200         onRequestPreview: () -> Unit,
201         onStudioOpen: () -> Unit,
202         onDuplicate: () -> Unit,
203         onDelete: () -> Unit,
204         onClick: () -> Unit,
205     ) {
206         val contentBg = if (isSelected) theme.panelBgSolid else theme.panelInset
207         LaunchedEffect(brush.id, brush.settings, preview) {
208             if (preview == null) {
209                 onRequestPreview()
210             }
211         }
212
213         SwipeRevealRow(
214             revealWidth = BRUSH_SWIPE_REVEAL_WIDTH,
215             onSwipeRight = onStudioOpen,
216             modifier = Modifier
217                 .fillMaxWidth()
218                 .clip(RoundedCornerShape(8.dp))
219                 .background(theme.panelInset),
220             actions = {
221                 BrushSwipeActions(
222                     theme = theme,
223                     onDuplicate = onDuplicate,
224                     onDelete = onDelete,
225                     canDelete = true // Simplified for now
226                 )
227             },
228         ) {
229             Box(
230                 modifier = Modifier
231                     .fillMaxWidth()
232                     .background(contentBg)
233                     .border(1.dp, if (isSelected) theme.accentSoft else theme.panelStroke, RoundedCornerShape(8.dp))
234                     .pointerInput(brush.id) {
235                         detectTapGestures(
236                             onTap = { onClick() },
237                         )
238                     }
239                     .padding(10.dp),
240             ) {
241                 Row(
242                     verticalAlignment = Alignment.CenterVertically,
243                     horizontalArrangement = Arrangement.spacedBy(12.dp),
244                 ) {
245                     BrushPreviewStrip(
246                         preview = preview,
247                         brush = brush,
248                         theme = theme,
249                         modifier = Modifier
250                             .width(142.dp)
251                             .height(46.dp),
252                     )
253                     Column(Modifier.weight(1f)) {
254                         Text(
255                             text = brush.settings.name,
256                             color = if (isSelected) theme.accent else theme.textPrimary,
257                             fontSize = 15.sp,
258                             fontWeight = if (isSelected) FontWeight.SemiBold else FontWeight.Medium,
259                             maxLines = 1,
260                         )
261                         Text(
262                             text = "Размер: ${((brush.settings.baseRadiusPx * 2f).toInt())} px",
263                             color = theme.textSecondary,
264                             fontSize = 12.sp,
265                             maxLines = 1,
266                         )
267                     }
268                     if (isSelected) {
269                         Box(
270                             Modifier
271                                 .size(8.dp)
272                                 .clip(CircleShape)
273                                 .background(theme.accent),
274                         )
275                     }
276                 }
277             }
278         }
279     }

```

```

275         }
276     }
277 }
278 }
279 }
280
281 @Composable
282 private fun BrushSwipeActions(
283     theme: AppTheme,
284     onDuplicate: () -> Unit,
285     onDelete: () -> Unit,
286     canDelete: Boolean
287 ) {
288     Row(
289         modifier = Modifier
290             .fillMaxWidth()
291             .fillMaxHeight(),
292         horizontalArrangement = Arrangement.End,
293         verticalAlignment = Alignment.CenterVertically,
294     ) {
295         BrushSwipeButton(
296             icon = Icons.Default.ContentCopy,
297             label = "Дубль",
298             color = theme.accent,
299             onClick = onDuplicate
300         )
301         BrushSwipeButton(
302             icon = Icons.Default.Delete,
303             label = "Удал.",
304             color = theme.danger,
305             enabled = canDelete,
306             onClick = onDelete
307         )
308     }
309 }
310
311 @Composable
312 private fun BrushSwipeButton(
313     icon: ImageVector,
314     label: String,
315     color: Color,
316     enabled: Boolean = true,
317     onClick: () -> Unit,
318 ) {
319     Column(
320         modifier = Modifier
321             .width(44.dp)
322             .fillMaxHeight()
323             .background(if (enabled) color else color.copy(alpha = 0.3f))
324             .clickable(enabled = enabled, onClick = onClick)
325             .padding(vertical = 4.dp),
326         horizontalAlignment = Alignment.CenterHorizontally,
327         verticalArrangement = Arrangement.Center,
328     ) {
329         Icon(icon, contentDescription = label, tint = Color.White, modifier = Modifier.size(18.dp))
330         Text(label, color = Color.White, fontSize = 8.sp, fontWeight = FontWeight.Medium)
331     }
332 }
333
334 @Composable
335 private fun BrushPreviewStrip(
336     preview: ImageBitmap?,
337     brush: BrushPreset,
338     theme: AppTheme,
339     modifier: Modifier = Modifier,
340 ) {
341     Box(
342         modifier = modifier
343             .clip(RoundedCornerShape(4.dp))
344             .background(Color.White)
345             .border(1.dp, theme.panelStroke, RoundedCornerShape(4.dp)),
346         contentAlignment = Alignment.Center,
347     ) {
348         PreviewCheckerboard(Modifier.fillMaxSize(), cellSize = 7f)
349         if (preview != null) {
350             Image(
351                 bitmap = preview,
352                 contentDescription = null,
353                 contentScale = ContentScale.Fit,
354                 modifier = Modifier.fillMaxSize(),
355             )
356         } else {
357             Canvas(Modifier.fillMaxSize()) {
358                 drawLine(
359                     color = theme.accent.copy(alpha = brush.settings.opacity.coerceIn(0.25f, 1f)),
360                     start = Offset(size.width * 0.12f, size.height * 0.5f),
361                     end = Offset(size.width * 0.88f, size.height * 0.5f),
362                     strokeWidth = (brush.settings.baseRadiusPx * 2f).coerceIn(2f, 12f),
363                 )
364             }
365         }
366     }

```

```

367 }
368
369 @Composable
370 private fun PreviewCheckerboard(
371     modifier: Modifier = Modifier,
372     cellSize: Float,
373 ) {
374     Canvas(modifier) {
375         val light = Color(0xFFE9E9E9)
376         val dark = Color(0xFFD0D0D0)
377         var y = 0f
378         var row = 0
379         while (y < size.height) {
380             var x = 0f
381             var col = 0
382             while (x < size.width) {
383                 drawRect(
384                     color = if ((row + col) % 2 == 0) light else dark,
385                     topLeft = Offset(x, y),
386                     size = Size(cellSize, cellSize),
387                 )
388                 x += cellSize
389                 col += 1
390             }
391             y += cellSize
392             row += 1
393         }
394     }
395 }

```

77. app/src/main/java/com/wetinknext/ui/components/LayersPanel.kt

```

1  package com.wetinknext.ui.components
2
3  import androidx.compose.foundation.background
4  import androidx.compose.foundation.border
5  import androidx.compose.foundation.clickable
6  import androidx.compose.foundation.layout.*
7  import androidx.compose.foundation.lazy.LazyColumn
8  import androidx.compose.foundation.lazy.items
9  import androidx.compose.foundation.shape.CircleShape
10 import androidx.compose.foundation.shape.RoundedCornerShape
11 import androidx.compose.material.icons.Icons
12 import androidx.compose.material.icons.filled.*
13 import androidx.compose.material3.*
14 import androidx.compose.runtime.*
15 import androidx.compose.ui.Alignment
16 import androidx.compose.ui.Modifier
17 import androidx.compose.ui.draw.clip
18 import androidx.compose.ui.graphics.Color
19 import androidx.compose.ui.graphics.vector.ImageVector
20 import androidx.compose.ui.text.font.FontWeight
21 import androidx.compose.ui.text.style.TextOverflow
22 import androidx.compose.ui.unit.dp
23 import androidx.compose.ui.unit.sp
24 import com.wetinknext.ui.state.LayerItem
25 import com.wetinknext.ui.state.LayerState
26 import com.wetinknext.ui.theme.AppTheme
27 import kotlin.math.roundToInt
28
29 private val PANEL_WIDTH = 320.dp
30
31 @Composable
32 fun LayersPanel(
33     state: LayerState,
34     theme: AppTheme,
35     modifier: Modifier = Modifier,
36     onDismiss: () -> Unit,
37     onAddLayer: () -> Unit,
38     onSelectLayer: (Long) -> Unit,
39     onVisibleChange: (Long, Boolean) -> Unit,
40     onOpacityChange: (Long, Float) -> Unit,
41     onRemoveLayer: (Long) -> Unit,
42 ) {
43     PanelSurface(
44         modifier = modifier.width(PANEL_WIDTH),
45     ) {
46         Column(
47             modifier = Modifier.padding(10.dp),
48             verticalArrangement = Arrangement.spacedBy(8.dp),
49         ) {
50             Row(
51                 verticalAlignment = Alignment.CenterVertically,
52                 horizontalArrangement = Arrangement.spacedBy(6.dp),
53             ) {
54                 Text(
55                     "Слой",
56                     color = theme.textPrimary,
57                     fontWeight = FontWeight.Bold,
58                     fontSize = 18.sp,
59                 )
60                 Spacer(Modifier.weight(1f))
61                 HeaderIconButton(
62                     theme = theme,
63                     icon = Icons.Default.Add,
64                     description = "Добавить слой",
65                     onClick = onAddLayer,
66                 )
67                 HeaderIconButton(
68                     theme = theme,
69                     icon = Icons.Default.Close,
70                     description = "Закрыть",
71                     onClick = onDismiss,
72                 )
73             }
74
75             LazyColumn(
76                 modifier = Modifier
77                     .fillMaxWidth()
78                     .heightIn(max = 400.dp),
79                 verticalArrangement = Arrangement.spacedBy(4.dp),
80             ) {
81                 items(
82                     items = state.layers.asReversed(),
83                     key = { it.id },
84                 ) { layerItem ->
85                     LayerRow(
86                         layer = layerItem,
87                         theme = theme,
88                         active = layerItem.id == state.activeLayerId,
89                         onSelect = { onSelectLayer(layerItem.id) },
90                         onVisibleChange = { onVisibleChange(layerItem.id, it) },

```

```

91         onOpacityChange = { onOpacityChange(layerItem.id, it) },
92         onDelete = { onRemoveLayer(layerItem.id) },
93         canDelete = state.layers.size > 1 && !layerItem.locked
94     )
95     }
96 }
97 }
98 }
99 }
100
101 @Composable
102 private fun LayerRow(
103     layer: LayerItem,
104     theme: AppTheme,
105     active: Boolean,
106     onSelect: () -> Unit,
107     onVisibleChange: (Boolean) -> Unit,
108     onOpacityChange: (Float) -> Unit,
109     onDelete: () -> Unit,
110     canDelete: Boolean,
111 ) {
112     val cardBg = if (active) theme.panelBgSolid else theme.panelBg
113
114     Column(
115         modifier = Modifier
116             .fillMaxWidth()
117             .clip(RoundedCornerShape(12.dp))
118             .background(cardBg)
119             .border(
120                 width = if (active) 1.2.dp else 0.dp,
121                 color = if (active) theme.accent else Color.Transparent,
122                 shape = RoundedCornerShape(12.dp),
123             )
124         .clickable { onSelect() }
125         .padding(8.dp),
126         verticalArrangement = Arrangement.spacedBy(4.dp)
127     ) {
128         Row(
129             modifier = Modifier.fillMaxWidth(),
130             verticalAlignment = Alignment.CenterVertically,
131         ) {
132             HeaderIconButton(
133                 theme = theme,
134                 icon = if (layer.visible) Icons.Default.Visibility else Icons.Default.VisibilityOff,
135                 description = "Видимость",
136                 onClick = { onVisibleChange(!layer.visible) },
137                 tint = if (layer.visible) (if (active) theme.accent else theme.textPrimary) else theme.iconInactive,
138             )
139
140             Spacer(Modifier.width(8.dp))
141
142             Text(
143                 layer.name,
144                 modifier = Modifier.weight(1f),
145                 color = theme.textPrimary,
146                 fontSize = 14.sp,
147                 fontWeight = if (active) FontWeight.SemiBold else FontWeight.Normal,
148                 maxLines = 1,
149                 overflow = TextOverflow.Ellipsis,
150             )
151
152             if (canDelete) {
153                 HeaderIconButton(
154                     theme = theme,
155                     icon = Icons.Default.Delete,
156                     description = "Удалить",
157                     onClick = onDelete,
158                     tint = theme.danger
159                 )
160             }
161         }
162
163         if (active) {
164             Column(modifier = Modifier.fillMaxWidth().padding(horizontal = 4.dp)) {
165                 Row(verticalAlignment = Alignment.CenterVertically) {
166                     Text("Непрозрачность", color = theme.textSecondary, fontSize = 11.sp)
167                     Spacer(Modifier.weight(1f))
168                     Text(
169                         "${(layer.opacity * 100f).roundToInt()}%",
170                         color = theme.textSecondary,
171                         fontSize = 11.sp,
172                     )
173                 }
174                 Slider(
175                     value = layer.opacity.coerceIn(0f, 1f),
176                     onValueChange = onOpacityChange,
177                     valueRange = 0f..1f,
178                     colors = SliderDefaults.colors(
179                         thumbColor = theme.accent,
180                         activeTrackColor = theme.accent,
181                         inactiveTrackColor = theme.accentMuted,
182                     ),

```

```

183         )
184     }
185 }
186 }
187 }
188
189 @Composable
190 private fun HeaderIconButton(
191     theme: AppTheme,
192     icon: ImageVector,
193     description: String,
194     onClick: () -> Unit,
195     enabled: Boolean = true,
196     tint: Color = theme.textPrimary,
197 ) {
198     Box(
199         modifier = Modifier
200             .size(32.dp)
201             .clip(CircleShape)
202             .background(if (enabled) theme.panelInsetSoft else theme.panelInset)
203             .clickable(enabled = enabled, onClick = onClick),
204         contentAlignment = Alignment.Center,
205     ) {
206         Icon(
207             icon,
208             contentDescription = description,
209             tint = if (enabled) tint else theme.iconInactive,
210             modifier = Modifier.size(18.dp),
211         )
212     }
213 }

```

78. app/src/main/java/com/wetinknext/ui/components/PanelSurface.kt

```

1  package com.wetinknext.ui.components
2
3  import androidx.compose.foundation.background
4  import androidx.compose.foundation.border
5  import androidx.compose.foundation.layout.Box
6  import androidx.compose.foundation.layout.BoxScope
7  import androidx.compose.foundation.shape.RoundedCornerShape
8  import androidx.compose.runtime.Composable
9  import androidx.compose.ui.Modifier
10 import androidx.compose.ui.draw.clip
11 import androidx.compose.ui.draw.shadow
12 import androidx.compose.ui.input.pointer.PointerEventPass
13 import androidx.compose.ui.input.pointer.pointerInput
14 import androidx.compose.ui.unit.dp
15 import com.wetinknext.ui.theme.LocalAppTheme
16
17 @Composable
18 fun PanelSurface(
19     modifier: Modifier = Modifier,
20     content: @Composable BoxScope.() -> Unit,
21 ) {
22     val theme = LocalAppTheme.current
23     val shape = RoundedCornerShape(22.dp)
24     Box(
25         modifier = modifier
26             .shadow(12.dp, shape, ambientColor = theme.panelInset.copy(alpha = 0.5f), spotColor = theme.panelInset.copy(alpha
= 0.5f))
27             .clip(shape)
28             .background(theme.panelBg)
29             .border(1.2.dp, theme.panelStroke, shape)
30     ) {
31         Box(modifier = Modifier.matchParentSize().swallowChrome())
32         content()
33     }
34 }
35
36 private fun Modifier.swallowChrome(): Modifier = pointerInput(Unit) {
37     awaitPointerEventScope {
38         while (true) {
39             val event = awaitPointerEvent(PointerEventPass.Initial)
40             for (change in event.changes) {
41                 if (change.pressed && !change.previousPressed) {
42                     change.consume()
43                 }
44             }
45         }
46     }
47 }

```


79. app/src/main/java/com/wetinknext/ui/components/SelectionToolbar.kt

```

1 package com.wetinknext.ui.components
2
3 import androidx.compose.foundation.BorderStroke
4 import androidx.compose.foundation.background
5 import androidx.compose.foundation.clickable
6 import androidx.compose.foundation.layout.*
7 import androidx.compose.foundation.shape.CircleShape
8 import androidx.compose.foundation.shape.RoundedCornerShape
9 import androidx.compose.material.icons.Icons
10 import androidx.compose.material.icons.automirrored.filled.CallMerge
11 import androidx.compose.material.icons.filled.*
12 import androidx.compose.material3.Icon
13 import androidx.compose.material3.Surface
14 import androidx.compose.material3.Text
15 import androidx.compose.runtime.Composable
16 import androidx.compose.ui.Alignment
17 import androidx.compose.ui.Modifier
18 import androidx.compose.ui.draw.clip
19 import androidx.compose.ui.graphics.Color
20 import androidx.compose.ui.graphics.vector.ImageVector
21 import androidx.compose.ui.text.font.FontWeight
22 import androidx.compose.ui.unit.dp
23 import androidx.compose.ui.unit.sp
24 import com.wetinknext.ui.theme.AppTheme
25
26 enum class SelectionShapeUi {
27     FREEHAND,
28     RECTANGLE,
29     ELLIPSE,
30 }
31
32 @Composable
33 fun SelectionToolbar(
34     theme: AppTheme,
35     currentShape: SelectionShapeUi,
36     onShapeChange: (SelectionShapeUi) -> Unit,
37     onDone: () -> Unit,
38     modifier: Modifier = Modifier,
39 ) {
40     val panelBorder = BorderStroke(1.dp, theme.panelStroke)
41
42     Column(
43         modifier = modifier
44             .fillMaxWidth()
45             .navigationBarsPadding()
46             .padding(horizontal = 16.dp, vertical = 20.dp),
47         horizontalAlignment = Alignment.CenterHorizontally,
48         verticalArrangement = Arrangement.spacedBy(12.dp),
49     ) {
50         // Row 1 – shape modes
51         Surface(
52             color = theme.panelBg.copy(alpha = 0.94f),
53             shape = RoundedCornerShape(24.dp),
54             border = panelBorder,
55         ) {
56             Row(
57                 modifier = Modifier
58                     .width(340.dp)
59                     .padding(4.dp)
60                     .height(32.dp)
61                     .clip(RoundedCornerShape(20.dp))
62                     .background(theme.panelInsetSoft),
63                 verticalAlignment = Alignment.CenterVertically,
64             ) {
65                 val shapes = listOf(
66                     SelectionShapeUi.FREEHAND to "От руки",
67                     SelectionShapeUi.RECTANGLE to "Прямоугольник",
68                     SelectionShapeUi.ELLIPSE to "Эллипс",
69                 )
70                 shapes.forEach { (shape, label) ->
71                     val isSelected = currentShape == shape
72                     Box(
73                         modifier = Modifier
74                             .weight(1f)
75                             .fillMaxHeight()
76                             .clip(RoundedCornerShape(20.dp))
77                             .background(if (isSelected) theme.accent else Color.Transparent)
78                             .clickable { onShapeChange(shape) },
79                         contentAlignment = Alignment.Center,
80                     ) {
81                         Text(
82                             text = label,
83                             color = if (isSelected) Color.White else theme.textPrimary,
84                             fontSize = 13.sp,
85                             fontWeight = if (isSelected) FontWeight.Bold else FontWeight.Normal,
86                             maxLines = 1,
87                         )
88                     }
89                 }
90             }
91         }
92     }
93 }

```

```

91     }
92
93     // Row 2 – Action icons (mostly disabled for now)
94     Surface(
95         color = theme.panelBg.copy(alpha = 0.94f),
96         shape = RoundedCornerShape(24.dp),
97         border = panelBorder,
98     ) {
99         Row(
100             modifier = Modifier.padding(horizontal = 4.dp, vertical = 4.dp),
101             horizontalArrangement = Arrangement.spacedBy(4.dp),
102             verticalAlignment = Alignment.CenterVertically,
103         ) {
104             SelectionIconButton(Icons.Default.Add, "Добавить", theme, isSelected = true, enabled = false)
105             SelectionIconButton(Icons.Default.Delete, "Удалить", theme, enabled = false)
106             SelectionIconButton(Icons.AutoMirrored.Filled.CallMerge, "Пересечь", theme, enabled = false)
107             SelectionIconButton(Icons.Default.Flip, "Инверсия", theme, enabled = false)
108             SelectionIconButton(Icons.Default.FormatColorFill, "Залить", theme, enabled = false)
109             SelectionIconButton(Icons.Default.ContentCopy, "Копировать", theme, enabled = false)
110             SelectionIconButton(Icons.Default.ContentCut, "Вырезать", theme, enabled = false)
111             SelectionIconButton(Icons.Default.AddToPhotos, "Вставить", theme, enabled = false)
112         }
113     }
114
115     // Row 3 – Reset / Delete / Done
116     Row(
117         horizontalArrangement = Arrangement.spacedBy(16.dp),
118         verticalAlignment = Alignment.CenterVertically,
119     ) {
120         SelectionRoundButton(Icons.Default.Refresh, "Сбросить", theme, enabled = false)
121         SelectionRoundButton(Icons.Default.Delete, "Удалить", theme, enabled = false)
122         Surface(
123             color = theme.accent,
124             shape = CircleShape,
125             modifier = Modifier.size(48.dp),
126         ) {
127             Box(
128                 modifier = Modifier
129                     .fillMaxSize()
130                     .clickable { onDone() },
131                 contentAlignment = Alignment.Center,
132             ) {
133                 Icon(Icons.Default.Check, "Готово", tint = Color.White)
134             }
135         }
136     }
137 }
138
139 @Composable
140 private fun SelectionIconButton(
141     icon: ImageVector,
142     description: String,
143     theme: AppTheme,
144     isSelected: Boolean = false,
145     enabled: Boolean = true,
146     onClick: () -> Unit = {},
147 ) {
148     Box(
149         modifier = Modifier
150             .size(48.dp)
151             .clip(RoundedCornerShape(8.dp))
152             .background(if (isSelected && enabled) theme.accent.copy(alpha = 0.30f) else Color.Transparent)
153             .clickable(enabled = enabled) { onClick() },
154         contentAlignment = Alignment.Center,
155     ) {
156         Icon(
157             imageVector = icon,
158             contentDescription = description,
159             tint = if (!enabled) theme.iconInactive.copy(alpha = 0.5f) else if (isSelected) theme.accent else theme.
160 textPrimary,
161             modifier = Modifier.size(24.dp),
162         )
163     }
164 }
165
166 @Composable
167 private fun SelectionRoundButton(
168     icon: ImageVector,
169     description: String,
170     theme: AppTheme,
171     enabled: Boolean = true,
172     onClick: () -> Unit = {},
173 ) {
174     Surface(
175         color = if (enabled) theme.panelBg.copy(alpha = 0.9f) else theme.panelBg.copy(alpha = 0.4f),
176         shape = CircleShape,
177         border = BorderStroke(1.dp, if (enabled) theme.panelStroke else theme.panelStroke.copy(alpha = 0.4f)),
178         modifier = Modifier.size(48.dp),
179     ) {
180         Box(
181             modifier = Modifier

```

```
182         .fillMaxSize()
183         .clickable(enabled = enabled) { onClick() },
184         contentAlignment = Alignment.Center,
185     ) {
186         Icon(
187             icon,
188             description,
189             tint = if (enabled) theme.textPrimary else theme.textPrimary.copy(alpha = 0.4f)
190         )
191     }
192 }
193 }
```

80. app/src/main/java/com/wetinknext/ui/components/SwipeRevealRow.kt

```

1  package com.wetinknext.ui.components
2
3  import androidx.compose.animation.core.animateFloatAsState
4  import androidx.compose.foundation.gestures.detectHorizontalDragGestures
5  import androidx.compose.foundation.layout.Arrangement
6  import androidx.compose.foundation.layout.Box
7  import androidx.compose.foundation.layout.Row
8  import androidx.compose.foundation.layout.RowScope
9  import androidx.compose.foundation.layout.fillMaxWidth
10 import androidx.compose.foundation.layout.offset
11 import androidx.compose.runtime.Composable
12 import androidx.compose.runtime.getValue
13 import androidx.compose.runtime.mutableStateOf
14 import androidx.compose.runtime.remember
15 import androidx.compose.runtime.setValue
16 import androidx.compose.ui.Modifier
17 import androidx.compose.ui.input.pointer.pointerInput
18 import androidx.compose.ui.platform.LocalDensity
19 import androidx.compose.ui.unit.Dp
20 import androidx.compose.ui.unit.IntOffset
21 import androidx.compose.ui.unit.dp
22 import kotlin.math.roundToInt
23
24 @Composable
25 fun SwipeRevealRow(
26     revealWidth: Dp,
27     actions: @Composable RowScope() -> Unit,
28     modifier: Modifier = Modifier,
29     rightSwipeThreshold: Dp = 56.dp,
30     onSwipeRight: (() -> Unit)? = null,
31     content: @Composable () -> Unit,
32 ) {
33     var rawOffset by remember { mutableStateOf(0f) }
34     var rightDragOffset by remember { mutableStateOf(0f) }
35     val revealPx = with(LocalDensity.current) { revealWidth.toPx() }
36     val rightSwipeThresholdPx = with(LocalDensity.current) { rightSwipeThreshold.toPx() }
37     val animatedOffset by animateFloatAsState(
38         targetValue = rawOffset.coerceIn(-revealPx, 0f),
39         label = "swipeRevealRow",
40     )
41
42     Box(modifier = modifier.fillMaxWidth()) {
43         Row(
44             modifier = Modifier.fillMaxWidth(),
45             horizontalArrangement = Arrangement.End,
46             content = actions,
47         )
48
49         Box(
50             modifier = Modifier
51                 .fillMaxWidth()
52                 .offset { IntOffset(animatedOffset.roundToInt(), 0) }
53                 .pointerInput(revealWidth) {
54                     detectHorizontalDragGestures(
55                         onHorizontalDrag = { change, dx ->
56                             if (dx < 0f || rawOffset < 0f) {
57                                 change.consume()
58                                 rawOffset += dx
59                                 rightDragOffset = 0f
60                             } else if (dx > 0f && rawOffset == 0f && onSwipeRight != null) {
61                                 change.consume()
62                                 rightDragOffset += dx
63                             }
64                         },
65                         onDragEnd = {
66                             if (rightDragOffset > rightSwipeThresholdPx) {
67                                 onSwipeRight?.invoke()
68                             }
69                             rawOffset = if (rawOffset < -revealPx * 0.4f) -revealPx else 0f
70                             rightDragOffset = 0f
71                         },
72                         onDragCancel = {
73                             rawOffset = 0f
74                             rightDragOffset = 0f
75                         },
76                     )
77                 },
78     ) {
79         content()
80     }
81 }
82 }

```

81. app/src/main/java/com/wetinknext/ui/components/ThresholdPanel.kt

```

1  package com.wetinknext.ui.components
2
3  import androidx.compose.foundation.background
4  import androidx.compose.foundation.border
5  import androidx.compose.foundation.clickable
6  import androidx.compose.foundation.layout.*
7  import androidx.compose.foundation.shape.CircleShape
8  import androidx.compose.foundation.shape.RoundedCornerShape
9  import androidx.compose.material.icons.Icons
10 import androidx.compose.material.icons.filled.Check
11 import androidx.compose.material.icons.filled.Close
12 import androidx.compose.material3.Icon
13 import androidx.compose.material3.Slider
14 import androidx.compose.material3.SliderDefaults
15 import androidx.compose.material3.Switch
16 import androidx.compose.material3.Text
17 import androidx.compose.runtime.Composable
18 import androidx.compose.ui.Alignment
19 import androidx.compose.ui.Modifier
20 import androidx.compose.ui.draw.clip
21 import androidx.compose.ui.graphics.Color
22 import androidx.compose.ui.text.font.FontWeight
23 import androidx.compose.ui.unit.dp
24 import androidx.compose.ui.unit.sp
25 import com.wetinknext.ui.theme.AppTheme
26 import kotlin.math.roundToInt
27
28 @Composable
29 fun ThresholdPanel(
30     theme: AppTheme,
31     threshold: Float, // 0..1
32     preserveTransparency: Boolean,
33     onThresholdChange: (Float) -> Unit,
34     onPreserveTransparencyChange: (Boolean) -> Unit,
35     onApply: () -> Unit,
36     onCancel: () -> Unit,
37     isApplyEnabled: Boolean = true,
38     modifier: Modifier = Modifier
39 ) {
40     Column(
41         modifier = modifier
42             .width(360.dp)
43             .clip(RoundedCornerShape(28.dp))
44             .background(theme.panelBg.copy(alpha = 0.95f))
45             .border(1.2.dp, theme.panelStroke, RoundedCornerShape(28.dp))
46             .padding(horizontal = 20.dp, vertical = 16.dp),
47         verticalArrangement = Arrangement.spacedBy(16.dp)
48     ) {
49         Row(
50             modifier = Modifier.fillMaxWidth(),
51             horizontalArrangement = Arrangement.SpaceBetween,
52             verticalAlignment = Alignment.CenterVertically
53         ) {
54             Text(
55                 text = "Бинаризация",
56                 color = theme.textPrimary,
57                 fontSize = 15.sp,
58                 fontWeight = FontWeight.Bold
59             )
60
61             Row(
62                 verticalAlignment = Alignment.CenterVertically,
63                 horizontalArrangement = Arrangement.spacedBy(12.dp)
64             ) {
65                 Box(
66                     modifier = Modifier
67                         .size(36.dp)
68                         .clip(CircleShape)
69                         .background(theme.panelInset)
70                         .clickable { onCancel() },
71                     contentAlignment = Alignment.Center
72                 ) {
73                     Icon(Icons.Default.Close, null, tint = theme.textPrimary, modifier = Modifier.size(20.dp))
74                 }
75
76                 Box(
77                     modifier = Modifier
78                         .size(36.dp)
79                         .clip(CircleShape)
80                         .background(if (isApplyEnabled) theme.accent else theme.panelInset)
81                         .clickable(enabled = isApplyEnabled) { onApply() },
82                     contentAlignment = Alignment.Center
83                 ) {
84                     Icon(Icons.Default.Check, null, tint = Color.White, modifier = Modifier.size(20.dp))
85                 }
86             }
87         }
88
89         Text(
90             text = "Пиксели светлее порога станут белыми, остальные – чёрными",

```

```

91         color = theme.textSecondary,
92         fontSize = 12.sp,
93         lineHeight = 16.sp
94     )
95
96     // Threshold
97     Column(verticalArrangement = Arrangement.spacedBy(4.dp)) {
98         Row(
99             modifier = Modifier.fillMaxWidth(),
100             horizontalArrangement = Arrangement.SpaceBetween
101         ) {
102             Text("Попор", color = theme.textSecondary, fontSize = 13.sp)
103             Text("${(threshold * 255).roundToInt()}", color = theme.accent, fontSize = 13.sp, fontWeight = FontWeight.
Bold)
104         }
105         Slider(
106             value = threshold,
107             onValueChange = onThresholdChange,
108             valueRange = 0f..1f,
109             colors = SliderDefaults.colors(
110                 thumbColor = theme.accent,
111                 activeTrackColor = theme.accent,
112                 inactiveTrackColor = theme.panelInset
113             )
114         )
115     }
116
117     Row(
118         modifier = Modifier.fillMaxWidth(),
119         verticalAlignment = Alignment.CenterVertically,
120         horizontalArrangement = Arrangement.SpaceBetween
121     ) {
122         Text(
123             text = "Сохранять прозрачность",
124             color = theme.textPrimary,
125             fontSize = 14.sp
126         )
127         Switch(
128             checked = preserveTransparency,
129             onCheckedChange = onPreserveTransparencyChange
130         )
131     }
132 }
133 }

```

82. app/src/main/java/com/wetinknext/ui/components/Toolbars.kt

```

1  package com.wetinknext.ui.components
2
3  import androidx.compose.animation.core.Spring
4  import androidx.compose.animation.core.animateFloatAsState
5  import androidx.compose.animation.core.spring
6  import androidx.compose.foundation.background
7  import androidx.compose.foundation.border
8  import androidx.compose.foundation.clickable
9  import androidx.compose.foundation.gestures.detectDragGestures
10 import androidx.compose.foundation.gestures.detectTapGestures
11 import androidx.compose.foundation.layout.*
12 import androidx.compose.foundation.shape.CircleShape
13 import androidx.compose.foundation.shape.RoundedCornerShape
14 import androidx.compose.material.icons.Icons
15 import androidx.compose.material.icons.automirrored.filled.Redo
16 import androidx.compose.material.icons.automirrored.filled.Undo
17 import androidx.compose.material.icons.filled.*
18 import androidx.compose.material3.Icon
19 import androidx.compose.runtime.*
20 import androidx.compose.ui.Alignment
21 import androidx.compose.ui.Modifier
22 import androidx.compose.ui.draw.clip
23 import androidx.compose.ui.graphics.Brush
24 import androidx.compose.ui.graphics.Color
25 import androidx.compose.ui.graphics.graphicsLayer
26 import androidx.compose.ui.graphics.vector.ImageVector
27 import androidx.compose.ui.input.pointer.pointerInput
28 import androidx.compose.ui.layout.onSizeChanged
29 import androidx.compose.ui.platform.LocalDensity
30 import androidx.compose.ui.res.vectorResource
31 import androidx.compose.ui.unit.IntOffset
32 import androidx.compose.ui.unit.dp
33 import com.wetinknext.R
34 import com.wetinknext.ui.theme.AppTheme
35
36 @Composable
37 fun TopToolbar(
38     theme: AppTheme,
39     currentColor: Color,
40     canUndo: Boolean,
41     canRedo: Boolean,
42     isSelectionActive: Boolean,
43     isTransformActive: Boolean,
44     isAnimationActive: Boolean,
45     isAdjustmentsActive: Boolean = false,
46     onUndoClick: () -> Unit,
47     onRedoClick: () -> Unit,
48     onSelectionClick: () -> Unit,
49     onTransformClick: () -> Unit,
50     onAnimationClick: () -> Unit,
51     onAdjustmentsClick: () -> Unit = {},
52     onSettingsClick: () -> Unit,
53     onLayersClick: () -> Unit,
54     onColorClick: () -> Unit,
55 ) {
56     Row(
57         modifier = Modifier
58             .padding(top = 12.dp, end = 12.dp)
59             .clip(RoundedCornerShape(25.dp))
60             .background(theme.panelBg.copy(alpha = 0.9f))
61             .border(1.2.dp, theme.panelStroke, RoundedCornerShape(25.dp))
62             .padding(horizontal = 8.dp, vertical = 6.dp),
63         verticalAlignment = Alignment.CenterVertically,
64         horizontalArrangement = Arrangement.spacedBy(10.dp)
65     ) {
66         ToolbarIcon(Icons.AutoMirrored.Filled.Undo, theme, isSelected = false, enabled = canUndo, onClick = onUndoClick)
67         ToolbarIcon(Icons.AutoMirrored.Filled.Redo, theme, isSelected = false, enabled = canRedo, onClick = onRedoClick)
68
69         // SelectAll has a visually denser glyph than Fullscreen; its 18 dp drawing size
70         // matches Transform's perceived 20 dp footprint while both buttons stay 34 dp.
71         ToolbarIcon(Icons.Default.SelectAll, theme, isSelected = isSelectionActive, iconSize = 18.dp, onClick =
onSelectionClick)
72         ToolbarIcon(Icons.Default.Movie, theme, isSelected = isAnimationActive, onClick = onAnimationClick)
73         ToolbarIcon(Icons.Default.Fullscreen, theme, isSelected = isTransformActive, onClick = onTransformClick)
74         ToolbarIcon(Icons.Default.Tune, theme, isSelected = isAdjustmentsActive, onClick = onAdjustmentsClick)
75         ToolbarIcon(Icons.Default.Layers, theme, onClick = onLayersClick)
76         ToolbarIcon(Icons.Default.Settings, theme, contentDescription = "Настройки", onClick = onSettingsClick)
77
78         ColorPickerDot(
79             color = currentColor,
80             theme = theme,
81             onClick = onColorClick
82         )
83     }
84 }
85
86 /** Compact actions placed directly under the left brush toolbar. */
87 @Composable
88 fun LeftCanvasActions(
89     theme: AppTheme,

```

```

90     onClearLayerClick: () -> Unit,
91     onMirrorClick: () -> Unit,
92 ) {
93     Column(
94         modifier = Modifier.padding(start = 12.dp, top = 10.dp),
95         verticalArrangement = Arrangement.spacedBy(10.dp),
96         horizontalAlignment = Alignment.CenterHorizontally,
97     ) {
98         CanvasActionButton(
99             icon = Icons.Default.DeleteSweep,
100            description = "Очистить активный слой",
101            theme = theme,
102            onClick = onClearLayerClick,
103        )
104        CanvasActionButton(
105            icon = Icons.Default.Flip,
106            description = "Зеркальный просмотр холста",
107            theme = theme,
108            onClick = onMirrorClick,
109        )
110    }
111 }
112
113 @Composable
114 private fun CanvasActionButton(
115     icon: ImageVector,
116     description: String,
117     theme: AppTheme,
118     onClick: () -> Unit,
119 ) {
120     Box(
121         modifier = Modifier
122             .size(44.dp)
123             .clip(CircleShape)
124             .background(theme.panelBg.copy(alpha = 0.9f))
125             .border(1.2.dp, theme.panelStroke, CircleShape)
126             .clickable(onClick = onClick),
127         contentAlignment = Alignment.Center,
128     ) {
129         Icon(icon, contentDescription = description, tint = theme.textPrimary, modifier = Modifier.size(20.dp))
130     }
131 }
132
133 @Composable
134 fun ColorPickerDot(
135     color: Color,
136     theme: AppTheme,
137     onClick: () -> Unit
138 ) {
139     var isPressed by remember { mutableStateOf(false) }
140
141     val scale by animateFloatAsState(
142         targetValue = if (isPressed) 1.15f else 1f,
143         animationSpec = spring(dampingRatio = Spring.DampingRatioMediumBouncy, stiffness = Spring.StiffnessLow),
144         label = "scale"
145     )
146
147     Box(
148         modifier = Modifier
149             .graphicsLayer {
150                 scaleX = scale
151                 scaleY = scale
152             }
153             .size(30.dp)
154             .clip(CircleShape)
155             .background(color)
156             .border(1.5.dp, theme.panelStroke, CircleShape)
157             .clickable { onClick() }
158     )
159 }
160
161 @Composable
162 fun SideToolbar(
163     theme: AppTheme,
164     brushSize: Float,
165     brushOpacity: Float,
166     isEraser: Boolean,
167     onBrushSizeChange: (Float) -> Unit,
168     onBrushOpacityChange: (Float) -> Unit,
169     onBrushClick: () -> Unit,
170     onEraserClick: (Boolean) -> Unit,
171 ) {
172     Column(
173         modifier = Modifier
174             .padding(start = 12.dp)
175             .width(44.dp)
176             .clip(RoundedCornerShape(22.dp))
177             .background(theme.panelBg.copy(alpha = 0.9f))
178             .border(1.2.dp, theme.panelStroke, RoundedCornerShape(22.dp))
179             .padding(vertical = 10.dp),
180         horizontalAlignment = Alignment.CenterHorizontally,
181         verticalArrangement = Arrangement.spacedBy(12.dp)

```



```

182     ) {
183         ToolbarIcon(Icons.Default.Brush, theme, isSelected = !isEraser, onClick = {
184             onEraserClick(false)
185             onBrushClick()
186         })
187         ToolbarIcon(ImageVector.vectorResource(R.drawable.ic_eraser), theme, isSelected = isEraser, onClick = { onEraserClick(
true) })
188
189         Spacer(Modifier.height(4.dp))
190
191         VerticalSlider(
192             value = brushSize,
193             onValueChange = onBrushSizeChange,
194             valueRange = 2f..400f,
195             theme = theme
196         )
197
198         VerticalSlider(
199             value = brushOpacity,
200             onValueChange = onBrushOpacityChange,
201             valueRange = 0f..1f,
202             theme = theme
203         )
204     }
205 }
206
207 @Composable
208 private fun VerticalSlider(
209     value: Float,
210     onValueChange: (Float) -> Unit,
211     valueRange: ClosedFloatingPointRange<Float> = 0f..1f,
212     theme: AppTheme,
213     modifier: Modifier = Modifier
214 ) {
215     val range = valueRange.endInclusive - valueRange.start
216     var trackHeightPx by remember { mutableFloatStateOf(1f) }
217     val density = LocalDensity.current
218
219     val norm = ((value - valueRange.start) / range).coerceIn(0f, 1f)
220
221     fun updateFromY(y: Float) {
222         val n = (1f - (y / trackHeightPx)).coerceIn(0f, 1f)
223         onValueChange(valueRange.start + n * range)
224     }
225
226     Box(
227         modifier = modifier
228             .width(28.dp)
229             .height(120.dp)
230             .clip(RoundedCornerShape(14.dp))
231             .background(theme.accentMuted.copy(alpha = 0.15f))
232             .border(1.dp, theme.panelStroke, RoundedCornerShape(14.dp))
233             .onSizeChanged { trackHeightPx = it.height.toFloat() }
234             .pointerInput(valueRange) {
235                 detectTapGestures { offset -> updateFromY(offset.y) }
236             }
237             .pointerInput(valueRange) {
238                 detectDragGestures { change, _ ->
239                     change.consume()
240                     updateFromY(change.position.y)
241                 }
242             }
243     ) {
244         Box(
245             Modifier
246                 .align(Alignment.BottomCenter)
247                 .fillMaxWidth()
248                 .fillMaxHeight(norm)
249                 .background(Brush.verticalGradient(listOf(theme.accentSoft, theme.accent)))
250         )
251         Box(
252             Modifier
253                 .align(Alignment.TopCenter)
254                 .offset {
255                     val thumbSizePx = with(density) { 22.dp.toPx() }
256                     IntOffset(0, ((1f - norm) * (trackHeightPx - thumbSizePx)).toInt())
257                 }
258                 .padding(2.dp)
259                 .size(22.dp)
260                 .clip(CircleShape)
261                 .background(theme.textPrimary)
262                 .border(1.dp, theme.panelStroke, CircleShape)
263         )
264     }
265 }
266
267 @Composable
268 private fun ToolbarIcon(
269     icon: ImageVector,
270     theme: AppTheme,
271     isSelected: Boolean = false,
272     enabled: Boolean = true,

```

```

273     contentDescription: String? = null,
274     iconSize: androidx.compose.ui.unit.Dp = 20.dp,
275     onClick: () -> Unit = {}
276 ) {
277     Box(
278         modifier = Modifier
279             .size(34.dp)
280             .clip(CircleShape)
281             .background(if (isSelected) theme.accentMuted.copy(alpha = 0.3f) else Color.Transparent)
282             .border(if (isSelected) 1.dp else 0.dp, if (isSelected) theme.accentSoft else Color.Transparent, CircleShape)
283             .clickable(enabled = enabled) { onClick() },
284         contentAlignment = Alignment.Center
285     ) {
286         Icon(
287             icon,
288             contentDescription = contentDescription,
289             tint = if (!enabled) theme.iconInactive.copy(alpha = 0.5f) else if (isSelected) theme.accent else theme.
textPrimary,
290             modifier = Modifier.size(iconSize)
291         )
292     }
293 }

```

83. app/src/main/java/com/wetinknext/ui/components/TransformMenuView.kt

```

1  package com.wetinknext.ui.components
2
3  import androidx.compose.foundation.BorderStroke
4  import androidx.compose.foundation.background
5  import androidx.compose.foundation.clickable
6  import androidx.compose.foundation.layout.*
7  import androidx.compose.foundation.shape.CircleShape
8  import androidx.compose.foundation.shape.RoundedCornerShape
9  import androidx.compose.material.icons.Icons
10 import androidx.compose.material.icons.filled.*
11 import androidx.compose.material3.*
12 import androidx.compose.runtime.Composable
13 import androidx.compose.ui.Alignment
14 import androidx.compose.ui.Modifier
15 import androidx.compose.ui.draw.clip
16 import androidx.compose.ui.draw.rotate
17 import androidx.compose.ui.graphics.Color
18 import androidx.compose.ui.graphics.vector.ImageVector
19 import androidx.compose.ui.text.font.FontWeight
20 import androidx.compose.ui.unit.dp
21 import androidx.compose.ui.unit.sp
22 import com.wetinknext.ui.theme.AppTheme
23
24 enum class TransformModeUi {
25     FREEFORM,
26     UNIFORM,
27     DISTORT,
28     WARP
29 }
30
31 interface TransformActions {
32     fun reset()
33     fun cancel()
34     fun apply()
35     fun flipHorizontal()
36     fun flipVertical()
37     fun setMode(mode: TransformModeUi)
38 }
39
40 @Composable
41 fun TransformMenuView(
42     theme: AppTheme,
43     actions: TransformActions,
44     currentMode: TransformModeUi,
45     modifier: Modifier = Modifier
46 ) {
47     val panelBorder = BorderStroke(1.dp, theme.panelStroke)
48
49     Column(
50         modifier = modifier
51             .fillMaxWidth()
52             .navigationBarPadding()
53             .padding(horizontal = 16.dp, vertical = 20.dp),
54         horizontalAlignment = Alignment.CenterHorizontally,
55         verticalArrangement = Arrangement.spacedBy(12.dp)
56     ) {
57         // Row 1: Modes
58         Surface(
59             color = theme.panelBg.copy(alpha = 0.94f),
60             shape = RoundedCornerShape(24.dp),
61             border = panelBorder
62         ) {
63             Row(
64                 modifier = Modifier
65                     .width(340.dp)
66                     .height(32.dp)
67                     .padding(4.dp),
68                 horizontalArrangement = Arrangement.spacedBy(4.dp)
69             ) {
70                 TransformModeButton("Свободно", TransformModeUi.FREEFORM, currentMode, theme, actions::setMode)
71                 TransformModeButton("Пропорц.", TransformModeUi.UNIFORM, currentMode, theme, actions::setMode)
72                 TransformModeButton("Исказить", TransformModeUi.DISTORT, currentMode, theme, actions::setMode)
73                 TransformModeButton("Деформ.", TransformModeUi.WARP, currentMode, theme, actions::setMode)
74             }
75         }
76
77         // Row 2: Actions
78         Surface(
79             color = theme.panelBg.copy(alpha = 0.94f),
80             shape = RoundedCornerShape(24.dp),
81             border = panelBorder
82         ) {
83             Row(
84                 modifier = Modifier.padding(horizontal = 4.dp, vertical = 4.dp),
85                 horizontalArrangement = Arrangement.spacedBy(4.dp),
86                 verticalAlignment = Alignment.CenterVertically
87             ) {
88                 ActionIconButton(Icons.Default.Flip, "Отразить Г", theme, enabled = false) { actions.flipHorizontal() }
89                 ActionIconButton(Icons.Default.Flip, "Отразить В", theme, iconModifier = Modifier.rotate(90f), enabled =
false) { actions.flipVertical() }

```

```

90         IconButton(Icons.Default.RotateRight, "Поворот 45", theme, enabled = false) { /* ... */ }
91         IconButton(Icons.Default.FitScreen, "Вписать", theme, enabled = false) { actions.reset() }
92
93         VerticalDivider(modifier = Modifier.height(24.dp).padding(horizontal = 4.dp), color = theme.panelStroke)
94
95         // Placeholder for interpolation/snapping
96         Box(modifier = Modifier.size(48.dp), contentAlignment = Alignment.Center) {
97             Icon(Icons.Default.LinkOff, null, tint = theme.textSecondary.copy(alpha = 0.4f), modifier = Modifier.size(
24.dp))
98         }
99     }
100 }
101
102 // Row 3: Main Controls
103 Row(
104     modifier = Modifier.fillMaxWidth(),
105     horizontalArrangement = Arrangement.spacedBy(16.dp, Alignment.CenterHorizontally),
106     verticalAlignment = Alignment.CenterVertically
107 ) {
108     // Reset
109     TransformRoundButton(Icons.Default.Refresh, "Сбросить", theme, enabled = false) { actions.reset() }
110
111     // Cancel
112     TransformRoundButton(Icons.Default.Close, "Отмена", theme) { actions.cancel() }
113
114     // Apply
115     Surface(
116         color = theme.accent.copy(alpha = 0.4f),
117         shape = CircleShape,
118         modifier = Modifier.size(48.dp)
119     ) {
120         Box(
121             modifier = Modifier.fillMaxSize(),
122             contentAlignment = Alignment.Center
123         ) {
124             Icon(Icons.Default.Check, "Применить", tint = Color.White.copy(alpha = 0.4f))
125         }
126     }
127 }
128 }
129 }
130
131 @Composable
132 private fun RowScope.TransformModeButton(
133     label: String,
134     mode: TransformModeUi,
135     currentMode: TransformModeUi,
136     theme: AppTheme,
137     onClick: (TransformModeUi) -> Unit
138 ) {
139     val isSelected = mode == currentMode
140     val enabled = false // Transform engine not ready
141     Box(
142         modifier = Modifier
143             .weight(1f)
144             .fillMaxHeight()
145             .clip(RoundedCornerShape(20.dp))
146             .background(if (isSelected) theme.accent.copy(alpha = 0.4f) else Color.Transparent)
147             .clickable(enabled = enabled) { onClick(mode) },
148         contentAlignment = Alignment.Center
149     ) {
150         Text(
151             text = label,
152             color = if (isSelected) Color.White.copy(alpha = 0.4f) else theme.textPrimary.copy(alpha = 0.4f),
153             fontSize = 13.sp,
154             fontWeight = if (isSelected) FontWeight.Bold else FontWeight.Normal
155         )
156     }
157 }
158
159 @Composable
160 private fun ActionIconButton(
161     icon: ImageVector,
162     description: String,
163     theme: AppTheme,
164     iconModifier: Modifier = Modifier,
165     enabled: Boolean = true,
166     onClick: () -> Unit
167 ) {
168     Box(
169         modifier = Modifier
170             .size(48.dp)
171             .clickable(enabled = enabled) { onClick() },
172         contentAlignment = Alignment.Center
173     ) {
174         Icon(
175             imageVector = icon,
176             contentDescription = description,
177             tint = if (enabled) theme.textPrimary else theme.textPrimary.copy(alpha = 0.4f),
178             modifier = iconModifier.size(24.dp)
179         )
180     }

```

```

181 }
182
183 @Composable
184 private fun TransformRoundButton(
185     icon: ImageVector,
186     description: String,
187     theme: AppTheme,
188     enabled: Boolean = true,
189     onClick: () -> Unit
190 ) {
191     Surface(
192         color = if (enabled) theme.panelBg.copy(alpha = 0.9f) else theme.panelBg.copy(alpha = 0.4f),
193         shape = CircleShape,
194         border = BorderStroke(1.dp, if (enabled) theme.panelStroke else theme.panelStroke.copy(alpha = 0.4f)),
195         modifier = Modifier.size(48.dp)
196     ) {
197         Box(
198             modifier = Modifier.fillMaxSize().clickable(enabled = enabled) { onClick() },
199             contentAlignment = Alignment.Center
200         ) {
201             Icon(icon, description, tint = if (enabled) theme.textPrimary else theme.textPrimary.copy(alpha = 0.4f))
202         }
203     }
204 }

```

84. app/src/main/java/com/wetinknext/ui/home/HomeScreen.kt

```

1  package com.wetinknext.ui.home
2
3  import androidx.compose.foundation.background
4  import androidx.compose.foundation.border
5  import androidx.compose.foundation.clickable
6  import androidx.compose.foundation.Image
7  import androidx.compose.foundation.layout.*
8  import androidx.compose.foundation.lazy.grid.GridCells
9  import androidx.compose.foundation.lazy.grid.LazyVerticalGrid
10 import androidx.compose.foundation.lazy.grid.items
11 import androidx.compose.foundation.rememberScrollState
12 import androidx.compose.foundation.shape.RoundedCornerShape
13 import androidx.compose.foundation.verticalScroll
14 import androidx.compose.material.icons.Icons
15 import androidx.compose.material.icons.outlined.Add
16 import androidx.compose.material.icons.outlined.Delete
17 import androidx.compose.material.icons.outlined.Edit
18 import androidx.compose.material.icons.outlined.MoreVert
19 import androidx.compose.material3.*
20 import androidx.compose.runtime.*
21 import androidx.compose.ui.Alignment
22 import androidx.compose.ui.Modifier
23 import androidx.compose.ui.draw.clip
24 import androidx.compose.ui.graphics.Color
25 import androidx.compose.ui.graphics.asImageBitmap
26 import androidx.compose.ui.graphics.graphicsLayer
27 import androidx.compose.ui.text.font.FontWeight
28 import androidx.compose.ui.text.style.TextAlign
29 import androidx.compose.ui.text.style.TextOverflow
30 import androidx.compose.ui.unit.dp
31 import androidx.compose.ui.unit.sp
32 import com.wetinknext.ui.theme.AppTheme
33 import java.text.SimpleDateFormat
34 import java.util.Date
35 import java.util.Locale
36
37 // Minimal models for Home UI shell
38 data class HomeProject(
39     val id: String,
40     val name: String,
41     val canvasWidth: Int,
42     val canvasHeight: Int,
43     val updatedAtMs: Long,
44 )
45
46 data class DeletedHomeProject(
47     val project: HomeProject,
48     val deletedAtMs: Long,
49 )
50
51 @Composable
52 fun HomeScreen(
53     theme: AppTheme,
54     projects: List<HomeProject>,
55     deletedProjects: List<DeletedHomeProject> = emptyList(),
56     onOpenProject: (HomeProject) -> Unit,
57     onCreateProject: (name: String, width: Int, height: Int, dpi: Int, colorProfile: String) -> Unit,
58     onRenameProject: (HomeProject, String) -> Unit,
59     onDeleteProject: (HomeProject) -> Unit,
60     onRestoreProject: (DeletedHomeProject) -> Unit,
61     onDeleteForever: (DeletedHomeProject) -> Unit,
62 ) {
63     var showNewDialog by remember { mutableStateOf(false) }
64     var renaming by remember { mutableStateOf<HomeProject?>(null) }
65     var deleting by remember { mutableStateOf<HomeProject?>(null) }
66     var showTrash by remember { mutableStateOf(false) }
67
68     Box(
69         modifier = Modifier
70             .fillMaxSize()
71             .background(theme.panelBgSolid),
72     ) {
73         Column(modifier = Modifier.fillMaxSize()) {
74             HomeTopBar(
75                 theme = theme,
76                 total = projects.size,
77                 onCreate = { showNewDialog = true },
78                 onTrash = { showTrash = true },
79             )
80             LazyVerticalGrid(
81                 columns = GridCells.Adaptive(minSize = 180.dp),
82                 contentPadding = PaddingValues(16.dp),
83                 horizontalArrangement = Arrangement.spacedBy(12.dp),
84                 verticalArrangement = Arrangement.spacedBy(12.dp),
85                 modifier = Modifier.fillMaxSize(),
86             ) {
87                 items(projects, key = { it.id }) { project ->
88                     ProjectTile(
89                         project = project,
90                         theme = theme,

```

```

91             onOpen = { onOpenProject(project) },
92             onRename = { renaming = project },
93             onDelete = { deleting = project },
94         )
95     }
96 }
97 }
98 }
99
100 if (showNewDialog) {
101     NewCanvasDialog(
102         theme = theme,
103         defaultWidth = 2800,
104         defaultHeight = 1840,
105         onDismiss = { showNewDialog = false },
106         onConfirm = { name, width, height, dpi, colorProfile ->
107             showNewDialog = false
108             onCreateProject(name, width, height, dpi, colorProfile)
109         },
110     )
111 }
112
113 renaming?.let { project ->
114     RenameDialog(
115         theme = theme,
116         initial = project.name,
117         onDismiss = { renaming = null },
118         onConfirm = { newName ->
119             onRenameProject(project, newName)
120             renaming = null
121         },
122     )
123 }
124
125 deleting?.let { project ->
126     AlertDialog(
127         onDismissRequest = { deleting = null },
128         confirmButton = {
129             TextButton(
130                 onClick = {
131                     onDeleteProject(project)
132                     deleting = null
133                 },
134             ) {
135                 Text("Удалить", color = theme.danger, fontWeight = FontWeight.SemiBold)
136             }
137         },
138         dismissButton = {
139             TextButton(onClick = { deleting = null }) {
140                 Text("Отмена", color = theme.textSecondary)
141             }
142         },
143         title = { Text("Удалить холст?", color = theme.textPrimary) },
144         text = { Text(project.name, color = theme.textSecondary) },
145         containerColor = theme.panelBgSolid,
146     )
147 }
148
149 if (showTrash) {
150     TrashDialog(
151         theme = theme,
152         deletedProjects = deletedProjects,
153         onRestore = { deleted ->
154             onRestoreProject(deleted)
155         },
156         onDeleteForever = { deleted ->
157             onDeleteForever(deleted)
158         },
159         onDismiss = { showTrash = false },
160     )
161 }
162 }
163
164 @Composable
165 private fun HomeTopBar(
166     theme: AppTheme,
167     total: Int,
168     onCreate: () -> Unit,
169     onTrash: () -> Unit,
170 ) {
171     Row(
172         modifier = Modifier
173             .fillMaxWidth()
174             .padding(horizontal = 16.dp, vertical = 14.dp),
175         verticalAlignment = Alignment.CenterVertically,
176     ) {
177         Column(modifier = Modifier.weight(1f)) {
178             Text(
179                 "Мои работы",
180                 color = theme.textPrimary,
181                 fontSize = 22.sp,
182                 fontWeight = FontWeight.SemiBold,

```

```

183         )
184         Spacer(Modifier.height(3.dp))
185         Text(
186             "Bcero: $total",
187             color = theme.textSecondary,
188             fontSize = 13.sp,
189             fontWeight = FontWeight.Medium,
190         )
191     }
192     TopIconButton(theme = theme, onClick = onCreate) {
193         Icon(Icons.Outlined.Add, null, tint = theme.textPrimary, modifier = Modifier.size(22.dp))
194     }
195     Spacer(Modifier.size(10.dp))
196     TopIconButton(theme = theme, onClick = onDelete) {
197         Icon(Icons.Outlined.Delete, null, tint = theme.textPrimary, modifier = Modifier.size(22.dp))
198     }
199 }
200 }
201
202 @Composable
203 private fun TopIconButton(
204     theme: AppTheme,
205     onClick: () -> Unit,
206     content: @Composable () -> Unit,
207 ) {
208     Box(
209         modifier = Modifier
210             .size(44.dp)
211             .clip(RoundedCornerShape(14.dp))
212             .background(theme.panelBg)
213             .border(1.dp, theme.panelStroke, RoundedCornerShape(14.dp))
214             .clickable(onClick = onClick),
215         contentAlignment = Alignment.Center,
216     ) {
217         content()
218     }
219 }
220
221 @Composable
222 private fun ProjectTile(
223     project: HomeProject,
224     theme: AppTheme,
225     onOpen: () -> Unit,
226     onRename: () -> Unit,
227     onDelete: () -> Unit,
228 ) {
229     var menuExpanded by remember { mutableStateOf(false) }
230
231     Column(
232         modifier = Modifier
233             .fillMaxWidth()
234             .clip(RoundedCornerShape(8.dp))
235             .background(theme.panelBg)
236             .border(1.2.dp, theme.panelStroke, RoundedCornerShape(8.dp))
237             .clickable(onClick = onOpen)
238             .padding(10.dp),
239     ) {
240         ProjectPreview(project = project, theme = theme)
241         Spacer(Modifier.height(10.dp))
242         Row(verticalAlignment = Alignment.CenterVertically) {
243             Column(modifier = Modifier.weight(1f)) {
244                 Text(
245                     project.name,
246                     color = theme.textPrimary,
247                     fontSize = 14.sp,
248                     fontWeight = FontWeight.SemiBold,
249                     maxLines = 1,
250                     overflow = TextOverflow.Ellipsis,
251                 )
252                 Text(
253                     "${project.canvasWidth} x ${project.canvasHeight} px · ${formatDate(project.updatedAtMs)}",
254                     color = theme.textSecondary,
255                     fontSize = 11.sp,
256                     maxLines = 1,
257                     overflow = TextOverflow.Ellipsis,
258                 )
259             }
260             Box {
261                 Box(
262                     modifier = Modifier
263                         .size(34.dp)
264                         .clip(RoundedCornerShape(12.dp))
265                         .clickable { menuExpanded = true },
266                     contentAlignment = Alignment.Center,
267                 ) {
268                     Icon(Icons.Outlined.MoreVert, null, tint = theme.textSecondary, modifier = Modifier.size(20.dp))
269                 }
270                 DropdownMenu(
271                     expanded = menuExpanded,
272                     onDismissRequest = { menuExpanded = false },
273                     modifier = Modifier.background(theme.panelBg)
274                 ) {

```



```

275         DropdownMenuItem(
276             text = { Text("Переименовать", color = theme.textPrimary) },
277             leadingIcon = { Icon(Icons.Outlined.Edit, null, tint = theme.textSecondary) },
278             onClick = {
279                 menuExpanded = false
280                 onRename()
281             },
282         )
283         DropdownMenuItem(
284             text = { Text("Удалить", color = theme.danger) },
285             leadingIcon = { Icon(Icons.Outlined.Delete, null, tint = theme.danger) },
286             onClick = {
287                 menuExpanded = false
288                 onDelete()
289             },
290         )
291     }
292 }
293
294 }
295 }
296
297 @Composable
298 private fun ProjectPreview(
299     project: HomeProject,
300     theme: AppTheme,
301 ) {
302     // Placeholder for actual preview loading
303     Box(
304         modifier = Modifier
305             .fillMaxWidth()
306             .aspectRatio(1.15f)
307             .clip(RoundedCornerShape(6.dp))
308             .background(theme.panelInset)
309             .border(1.dp, theme.panelStroke, RoundedCornerShape(6.dp)),
310         contentAlignment = Alignment.Center,
311     ) {
312         val ratio = project.canvasWidth.toFloat() / project.canvasHeight.coerceAtLeast(1).toFloat()
313         val swatchModifier = if (ratio >= 1f) {
314             Modifier
315                 .fillMaxWidth(0.74f)
316                 .aspectRatio(ratio)
317         } else {
318             Modifier
319                 .fillMaxWidth((0.54f * ratio).coerceIn(0.24f, 0.54f))
320                 .aspectRatio(ratio)
321         }
322
323         Box(
324             modifier = swatchModifier
325                 .clip(RoundedCornerShape(4.dp))
326                 .background(Color(0xFFFAFAFA))
327                 .border(1.dp, theme.panelStroke, RoundedCornerShape(4.dp)),
328         )
329     }
330 }
331
332 @Composable
333 private fun RenameDialog(
334     theme: AppTheme,
335     initial: String,
336     onDismiss: () -> Unit,
337     onConfirm: (String) -> Unit,
338 ) {
339     var name by remember(initial) { mutableStateOf(initial) }
340     AlertDialog(
341         onDismissRequest = onDismiss,
342         confirmButton = {
343             TextButton(
344                 onClick = {
345                     val cleaned = name.trim()
346                     if (cleaned.isNotEmpty()) onConfirm(cleaned)
347                 },
348             ) {
349                 Text("Готово", color = theme.accent, fontWeight = FontWeight.SemiBold)
350             }
351         },
352         dismissButton = {
353             TextButton(onClick = onDismiss) {
354                 Text("Отмена", color = theme.textSecondary)
355             }
356         },
357         title = { Text("Название", color = theme.textPrimary) },
358         text = {
359             OutlinedTextField(
360                 value = name,
361                 onValueChange = { name = it.take(48) },
362                 singleLine = true,
363                 colors = OutlinedTextFieldDefaults.colors(
364                     focusedBorderColor = theme.accent,
365                     unfocusedBorderColor = theme.panelStroke,
366                     focusedTextColor = theme.textPrimary,

```

```

367         unfocusedTextColor = theme.textPrimary
368     )
369 )
370 },
371     containerColor = theme.panelBgSolid,
372 )
373 }
374
375 @Composable
376 private fun TrashDialog(
377     theme: AppTheme,
378     deletedProjects: List<DeletedHomeProject>,
379     onRestore: (DeletedHomeProject) -> Unit,
380     onDeleteForever: (DeletedHomeProject) -> Unit,
381     onDismiss: () -> Unit,
382 ) {
383     AlertDialog(
384         onDismissRequest = onDismiss,
385         confirmButton = {
386             TextButton(onClick = onDismiss) {
387                 Text("Готово", color = theme.accent, fontWeight = FontWeight.SemiBold)
388             }
389         },
390         title = { Text("Корзина", color = theme.textPrimary) },
391         text = {
392             if (deletedProjects.isEmpty()) {
393                 Text("Удалённые работы хранятся здесь 3 дня.", color = theme.textSecondary, fontSize = 13.sp)
394             } else {
395                 Column(
396                     modifier = Modifier
397                         .fillMaxWidth()
398                         .height(360.dp)
399                         .verticalScroll(rememberScrollState()),
400                     verticalArrangement = Arrangement.spacedBy(10.dp),
401                 ) {
402                     Text(
403                         "Работы хранятся 3 дня.",
404                         color = theme.textSecondary,
405                         fontSize = 12.sp,
406                     )
407                     deletedProjects.sortedByDescending { it.deletedAtMs }.forEach { deleted ->
408                         DeletedProjectRow(
409                             deleted = deleted,
410                             theme = theme,
411                             onRestore = { onRestore(deleted) },
412                             onDeleteForever = { onDeleteForever(deleted) },
413                         )
414                     }
415                 }
416             }
417         },
418         containerColor = theme.panelBgSolid,
419     )
420 }
421
422 @Composable
423 private fun DeletedProjectRow(
424     deleted: DeletedHomeProject,
425     theme: AppTheme,
426     onRestore: () -> Unit,
427     onDeleteForever: () -> Unit,
428 ) {
429     val project = deleted.project
430     Row(
431         modifier = Modifier
432             .fillMaxWidth()
433             .clip(RoundedCornerShape(12.dp))
434             .background(theme.panelBg)
435             .border(1.dp, theme.panelStroke, RoundedCornerShape(12.dp))
436             .padding(10.dp),
437         verticalAlignment = Alignment.CenterVertically,
438     ) {
439         Box(
440             modifier = Modifier
441                 .size(54.dp)
442                 .clip(RoundedCornerShape(6.dp))
443                 .background(theme.panelInset)
444                 .border(1.dp, theme.panelStroke, RoundedCornerShape(6.dp)),
445             contentAlignment = Alignment.Center,
446         ) {
447             Box(
448                 modifier = Modifier
449                     .fillMaxWidth(0.5f)
450                     .aspectRatio(project.canvasWidth.toFloat() / project.canvasHeight.coerceAtLeast(1).toFloat())
451                     .background(Color(0xFFFAFAFA)),
452             )
453         }
454         Column(
455             modifier = Modifier
456                 .weight(1f)
457                 .padding(start = 10.dp),
458         ) {

```

```

459         Text(
460             project.name,
461             color = theme.textPrimary,
462             fontSize = 13.sp,
463             fontWeight = FontWeight.SemiBold,
464             maxLines = 1,
465             overflow = TextOverflow.Ellipsis,
466         )
467         Text(
468             "Осталось: ${trashDaysLeft(deleted.deletedAtMs)} дн.",
469             color = theme.textSecondary,
470             fontSize = 11.sp,
471         )
472         Row(horizontalArrangement = Arrangement.spacedBy(8.dp)) {
473             TextButton(onClick = onRestore) {
474                 Text("Восстановить", color = theme.accent, fontSize = 12.sp)
475             }
476             TextButton(onClick = onDeleteForever) {
477                 Text("Удалить", color = theme.textSecondary, fontSize = 12.sp)
478             }
479         }
480     }
481 }
482 }
483
484 private fun formatDate(timeMs: Long): String =
485     SimpleDateFormat("d MMM, HH:mm", Locale.forLanguageTag("ru")).format(Date(timeMs))
486
487 private fun trashDaysLeft(deletedAtMs: Long): Int {
488     val retentionMs = 3L * 24L * 60L * 60L * 1000L
489     val remainingMs = (retentionMs - (System.currentTimeMillis() - deletedAtMs)).coerceAtLeast(0L)
490     return ((remainingMs + 24L * 60L * 60L * 1000L - 1L) / (24L * 60L * 60L * 1000L)).toInt().coerceAtLeast(1)
491 }

```

85. app/src/main/java/com/wetinknext/ui/home/NewCanvasDialog.kt

```

1  package com.wetinknext.ui.home
2
3  import androidx.compose.foundation.background
4  import androidx.compose.foundation.border
5  import androidx.compose.foundation.clickable
6  import androidx.compose.foundation.horizontalScroll
7  import androidx.compose.foundation.layout.*
8  import androidx.compose.foundation.rememberScrollState
9  import androidx.compose.foundation.shape.RoundedCornerShape
10 import androidx.compose.foundation.text.KeyboardOptions
11 import androidx.compose.material3.*
12 import androidx.compose.runtime.*
13 import androidx.compose.ui.Alignment
14 import androidx.compose.ui.Modifier
15 import androidx.compose.ui.draw.clip
16 import androidx.compose.ui.graphics.Color
17 import androidx.compose.ui.text.font.FontWeight
18 import androidx.compose.ui.text.input.KeyboardType
19 import androidx.compose.ui.text.style.TextAlign
20 import androidx.compose.ui.unit.dp
21 import androidx.compose.ui.unit.sp
22 import com.wetinknext.ui.theme.AppTheme
23 import kotlin.math.max
24
25 private data class CanvasPreset(
26     val label: String,
27     val width: Int,
28     val height: Int,
29     val description: String,
30 )
31
32 private val QuickPresets = listOf(
33     CanvasPreset("Экран", 2800, 1840, "2800 x 1840 px"),
34     CanvasPreset("Квадрат", 2048, 2048, "2048 x 2048 px"),
35     CanvasPreset("4K", 4096, 2160, "4096 x 2160 px"),
36     CanvasPreset("A4", 2480, 3508, "210 x 297 мм"),
37     CanvasPreset("1080P", 1920, 1080, "1920 x 1080 px"),
38     CanvasPreset("9:16", 1080, 1920, "1080 x 1920 px"),
39 )
40
41 private val DpiOptions = listOf(72, 144, 300, 350, 600, 1200)
42 private val ColorProfiles = listOf("sRGB", "Display P3")
43
44 @Composable
45 fun NewCanvasDialog(
46     theme: AppTheme,
47     defaultWidth: Int,
48     defaultHeight: Int,
49     onDismiss: () -> Unit,
50     onConfirm: (name: String, width: Int, height: Int, dpi: Int, colorProfile: String) -> Unit,
51 ) {
52     val initialIndex = QuickPresets.indexOfFirst {
53         it.width == defaultWidth && it.height == defaultHeight
54     }
55     var selected by remember { mutableIntStateOf(initialIndex) }
56     var customW by remember { mutableStateOf(defaultWidth.toString()) }
57     var customH by remember { mutableStateOf(defaultHeight.toString()) }
58     var name by remember { mutableStateOf("Новый холст") }
59     var dpi by remember { mutableIntStateOf(300) }
60     var colorProfile by remember { mutableStateOf("sRGB") }
61     var dpiExpanded by remember { mutableStateOf(false) }
62     var profileExpanded by remember { mutableStateOf(false) }
63     val isCustom = selected !in QuickPresets.indices
64
65     AlertDialog(
66         onDismissRequest = onDismiss,
67         title = {
68             Text("Новый холст", color = theme.textPrimary, fontWeight = FontWeight.SemiBold)
69         },
70         text = {
71             Column {
72                 OutlinedTextField(
73                     value = name,
74                     onChange = { name = it.take(48) },
75                     label = { Text("Имя") },
76                     singleLine = true,
77                     modifier = Modifier.fillMaxWidth(),
78                     colors = OutlinedTextFieldDefaults.colors(
79                         focusedBorderColor = theme.accent,
80                         unfocusedBorderColor = theme.panelStroke,
81                         focusedLabelColor = theme.accent,
82                         unfocusedLabelColor = theme.textSecondary
83                     )
84                 )
85                 Spacer(Modifier.height(12.dp))
86                 Text("Размеры", color = theme.textSecondary, fontSize = 13.sp)
87                 Spacer(Modifier.height(8.dp))
88                 Row(
89                     horizontalArrangement = Arrangement.spacedBy(8.dp),
90                     modifier = Modifier.horizontalScroll(rememberScrollState()),

```

```

91         ) {
92             QuickPresets.forEachIndexed { index, preset ->
93                 QuickPresetCard(
94                     preset = preset,
95                     selected = selected == index,
96                     theme = theme,
97                     onClick = {
98                         selected = index
99                         customW = preset.width.toString()
100                        customH = preset.height.toString()
101                    },
102                )
103            }
104        }
105
106        Spacer(Modifier.height(12.dp))
107        TextButton(onClick = { selected = -1 }) {
108            Text(
109                "Свой размер",
110                color = if (isCustom) theme.accent else theme.textSecondary,
111                fontWeight = if (isCustom) FontWeight.SemiBold else FontWeight.Normal,
112                fontSize = 13.sp,
113            )
114        }
115
116        if (isCustom) {
117            Spacer(Modifier.height(4.dp))
118            Row(verticalAlignment = Alignment.CenterVertically) {
119                OutlinedTextField(
120                    value = customW,
121                    onValueChange = { v -> customW = v.filter { it.isDigit() }.take(5) },
122                    label = { Text("Ширина") },
123                    singleLine = true,
124                    keyboardOptions = KeyboardOptions(keyboardType = KeyboardType.Number),
125                    modifier = Modifier.weight(1f),
126                    colors = OutlinedTextFieldDefaults.colors(
127                        focusedBorderColor = theme.accent,
128                        unfocusedBorderColor = theme.panelStroke,
129                        focusedLabelColor = theme.accent,
130                        unfocusedLabelColor = theme.textSecondary
131                    )
132                )
133                Spacer(Modifier.width(8.dp))
134                Text("x", color = theme.textSecondary)
135                Spacer(Modifier.width(8.dp))
136                OutlinedTextField(
137                    value = customH,
138                    onValueChange = { v -> customH = v.filter { it.isDigit() }.take(5) },
139                    label = { Text("Высота") },
140                    singleLine = true,
141                    keyboardOptions = KeyboardOptions(keyboardType = KeyboardType.Number),
142                    modifier = Modifier.weight(1f),
143                    colors = OutlinedTextFieldDefaults.colors(
144                        focusedBorderColor = theme.accent,
145                        unfocusedBorderColor = theme.panelStroke,
146                        focusedLabelColor = theme.accent,
147                        unfocusedLabelColor = theme.textSecondary
148                    )
149                )
150            }
151        }
152
153        Spacer(Modifier.height(14.dp))
154        OptionRow(label = "DPI", value = "$dpi dpi", theme = theme, expanded = dpiExpanded, onExpand = { dpiExpanded
155            = true }) {
156            DpiOptions.forEach { option ->
157                DropdownMenuItem(
158                    text = { Text("$option dpi") },
159                    onClick = {
160                        dpi = option
161                        dpiExpanded = false
162                    },
163                )
164            }
165        }
166        OptionRow(label = "Цветовой профиль", value = colorProfile, theme = theme, expanded = profileExpanded,
167            onExpand = { profileExpanded = true }) {
168            ColorProfiles.forEach { option ->
169                DropdownMenuItem(
170                    text = { Text(option) },
171                    onClick = {
172                        colorProfile = option
173                        profileExpanded = false
174                    },
175                )
176            }
177        }
178        confirmButton = {
179            TextButton(
180                onClick = {

```

```

181         val (w, h) = if (isCustom) {
182             val cw = customW.toIntOrNull()?.coerceIn(MinDimension, MaxDimension) ?: return@TextButton
183             val ch = customH.toIntOrNull()?.coerceIn(MinDimension, MaxDimension) ?: return@TextButton
184             cw to ch
185         } else {
186             val preset = QuickPresets[selected]
187             preset.width to preset.height
188         }
189         onConfirm(name.trim().ifBlank { "Новый холст" }, w, h, dpi, colorProfile)
190     },
191     ) {
192         Text("Создать", color = theme.accent, fontWeight = FontWeight.SemiBold)
193     }
194 },
195 dismissButton = {
196     TextButton(onClick = onDismiss) {
197         Text("Отмена", color = theme.textSecondary)
198     }
199 },
200 containerColor = theme.panelBgSolid,
201 )
202 }
203
204 @Composable
205 private fun OptionRow(
206     label: String,
207     value: String,
208     theme: AppTheme,
209     expanded: Boolean,
210     onExpand: () -> Unit,
211     menu: @Composable () -> Unit,
212 ) {
213     Row(verticalAlignment = Alignment.CenterVertically, modifier = Modifier.fillMaxWidth()) {
214         Text(label, color = theme.textPrimary, fontSize = 13.sp)
215         Spacer(Modifier.weight(1f))
216         Box {
217             TextButton(onClick = onExpand) {
218                 Text(value, color = theme.accent, fontWeight = FontWeight.SemiBold)
219             }
220             DropdownMenu(
221                 expanded = expanded,
222                 onDismissRequest = { /* onExpand is handled by parent */ },
223                 modifier = Modifier.background(theme.panelBg)
224             ) {
225                 menu()
226             }
227         }
228     }
229 }
230
231 @Composable
232 private fun QuickPresetCard(
233     preset: CanvasPreset,
234     selected: Boolean,
235     theme: AppTheme,
236     onClick: () -> Unit,
237 ) {
238     Column(
239         horizontalAlignment = Alignment.CenterHorizontally,
240         modifier = Modifier
241             .width(100.dp)
242             .clip(RoundedCornerShape(12.dp))
243             .background(if (selected) theme.accent.copy(alpha = 0.15f) else theme.panelBg)
244             .border(1.5.dp, if (selected) theme.accent else theme.panelStroke, RoundedCornerShape(12.dp))
245             .clickable(onClick = onClick)
246             .padding(vertical = 12.dp, horizontal = 8.dp),
247     ) {
248         PresetSwitch(preset.width, preset.height, selected, theme)
249         Spacer(Modifier.height(8.dp))
250         Text(preset.label, color = theme.textPrimary, fontWeight = FontWeight.Medium, fontSize = 13.sp, textAlign = TextAlign.Center)
251         Text(preset.description, color = theme.textSecondary, fontSize = 10.sp, textAlign = TextAlign.Center, maxLines = 1)
252     }
253 }
254
255 @Composable
256 private fun PresetSwitch(width: Int, height: Int, selected: Boolean, theme: AppTheme) {
257     val scale = 40f / max(width.coerceAtLeast(1), height.coerceAtLeast(1))
258     val previewW = (width * scale).coerceAtLeast(8f)
259     val previewH = (height * scale).coerceAtLeast(8f)
260     Box(modifier = Modifier.size(48.dp), contentAlignment = Alignment.Center) {
261         Box(
262             modifier = Modifier
263                 .width(previewW.dp)
264                 .height(previewH.dp)
265                 .clip(RoundedCornerShape(4.dp))
266                 .background(if (selected) theme.accent else Color(0xFF5A5A60))
267                 .border(1.dp, if (selected) Color.White else theme.panelStroke, RoundedCornerShape(4.dp)),
268         )
269     }
270 }
271

```

```
272 private const val MinDimension = 256
273 private const val MaxDimension = 8192
```

86. app/src/main/java/com/wetinknext/ui/state/EditorActions.kt

```
1 package com.wetinknext.ui.state
2
3 /**
4  * Interface defining all possible user actions from the UI that affect the paint engine state.
5  */
6 interface EditorActions {
7     fun undo()
8     fun redo()
9     fun setBrushSize(px: Float)
10    fun setBrushOpacity(opacity: Float)
11
12    fun addLayer()
13    fun removeLayer(id: Long)
14    fun setActiveLayer(id: Long)
15    fun setLayerVisible(id: Long, visible: Boolean)
16    fun setLayerOpacity(id: Long, opacity: Float)
17 }
```


87. app/src/main/java/com/wetinknext/ui/state/LayerState.kt

```

1 package com.wetinknext.ui.state
2
3 import androidx.compose.runtime.*
4 import com.wetinknext.engine.core.EditorUiState
5
6 class LayerItem(
7     val id: Long,
8     name: String,
9     visible: Boolean,
10    opacity: Float,
11    locked: Boolean,
12 ) {
13     var name by mutableStateOf(name)
14     var visible by mutableStateOf(visible)
15     var opacity by mutableFloatStateOf(opacity)
16     var locked by mutableStateOf(locked)
17 }
18
19 class LayerState {
20     val layers = mutableStateListOf<LayerItem>()
21     var activeLayerId by mutableStateOf(-1L)
22
23     fun syncFromEditor(state: EditorUiState) {
24         val ids = state.layers.map { it.id }.toSet()
25         layers.removeAll { it.id !in ids }
26
27         state.layers.forEachIndexed { index, model ->
28             val item = layers.firstOrNull { it.id == model.id }
29
30             if (item == null) {
31                 layers.add(
32                     index.coerceAtMost(layers.size),
33                     LayerItem(
34                         id = model.id,
35                         name = model.name,
36                         visible = model.isVisible,
37                         opacity = model.opacity,
38                         locked = model.isLocked,
39                     ),
40                 )
41             } else {
42                 item.name = model.name
43                 item.visible = model.isVisible
44                 item.opacity = model.opacity
45                 item.locked = model.isLocked
46
47                 // If the item needs to be moved to match the index in state
48                 val currentIndex = layers.indexOf(item)
49                 if (currentIndex != index && index < layers.size) {
50                     layers.removeAt(currentIndex)
51                     layers.add(index, item)
52                 }
53             }
54         }
55
56         activeLayerId = state.activeLayerId
57     }
58 }

```

88. app/src/main/java/com/wetinknext/ui/theme/AppThemes.kt

```

1  package com.wetinknext.ui.theme
2
3  import androidx.compose.runtime.Immutable
4  import androidx.compose.ui.graphics.Color
5
6  @Immutable
7  data class AppTheme(
8      val id: String,
9      val displayName: String,
10     val isLight: Boolean,
11
12     // Core Layout
13     val appBg: Color,
14     val gridLine: Color,
15     val panelBg: Color,
16     val panelBgSolid: Color,
17     val panelStroke: Color,
18     val panelInset: Color,
19     val panelInsetSoft: Color,
20
21     // Canvas
22     val canvasBackdrop: Color,
23     val canvasGrid: Color,
24
25     // Typography & Icons
26     val textPrimary: Color,
27     val textSecondary: Color,
28     val iconInactive: Color,
29
30     // Accents
31     val accent: Color,
32     val accentSoft: Color,
33     val accentMuted: Color,
34     val danger: Color
35 ) {
36     val panelBgVariant: Color get() = panelInset
37 }
38
39 object AppThemes {
40     val Dark = AppTheme(
41         id = "dark", displayName = "Dark", isLight = false,
42         appBg = Color(0xFF1A0C12), gridLine = Color(0xFF2A1418),
43         panelBg = Color(0xFF2A141C), panelBgSolid = Color(0xFF2D1620),
44         panelStroke = Color(0xFF3D1F2A), panelInset = Color(0xFF1F1015), panelInsetSoft = Color(0xFF2A1820),
45         canvasBackdrop = Color(0xFF1A0C12), canvasGrid = Color(0xFF3A1C24),
46         textPrimary = Color(0xFFFF5E8E), textSecondary = Color(0xFF9A7F8A), iconInactive = Color(0xFFC8B0B8),
47         accent = Color(0xFFFF4D7A), accentSoft = Color(0xFFFF7AA0), accentMuted = Color(0x55FF4D7A), danger = Color(
0xFFFF3B5C)
48     )
49
50     val Light = AppTheme(
51         id = "light", displayName = "Light", isLight = true,
52         appBg = Color(0xFFFFF4E9), gridLine = Color(0xFFF5D0D7),
53         panelBg = Color(0xFFFFF1F4), panelBgSolid = Color(0xFFFFFFFF),
54         panelStroke = Color(0xFFF0C8D0), panelInset = Color(0xFFFFE0E6), panelInsetSoft = Color(0xFFFFD0DA),
55         canvasBackdrop = Color(0xFFFFF4E9), canvasGrid = Color(0xFFF5D0D7),
56         textPrimary = Color(0xFF3D1B26), textSecondary = Color(0xFF8A5A6A), iconInactive = Color(0xFFB07A88),
57         accent = Color(0xFFFF4D7A), accentSoft = Color(0xFFFFA0B7), accentMuted = Color(0x33FF4D7A), danger = Color(
0xFFFF3B5C)
58     )
59
60     val Red = AppTheme(
61         id = "red", displayName = "Red", isLight = false,
62         appBg = Color(0xFF1A0606), gridLine = Color(0xFF2A0A0A),
63         panelBg = Color(0xFF2A0C0C), panelBgSolid = Color(0xFF2D0E0E),
64         panelStroke = Color(0xFF3D1414), panelInset = Color(0xFF1F0808), panelInsetSoft = Color(0xFF2A1010),
65         canvasBackdrop = Color(0xFF1A0606), canvasGrid = Color(0xFF3A1414),
66         textPrimary = Color(0xFFFF5E8E), textSecondary = Color(0xFF9A7F7F), iconInactive = Color(0xFFC8B0B0),
67         accent = Color(0xFFFF3D3D), accentSoft = Color(0xFFFF7A7A), accentMuted = Color(0x55FF3D3D), danger = Color(
0xFFFF8888)
68     )
69
70     val Vampire = AppTheme(
71         id = "vampire", displayName = "Vampire", isLight = false,
72         appBg = Color(0xFF1B1C1B), gridLine = Color(0xFF252525),
73         panelBg = Color(0xFF262626), panelBgSolid = Color(0xFF2D2D2D),
74         panelStroke = Color(0xFF4B2529), panelInset = Color(0xFF171717), panelInsetSoft = Color(0xFF2A1B1E),
75         canvasBackdrop = Color(0xFF1B1C1B), canvasGrid = Color(0xFF2B2B2B),
76         textPrimary = Color(0xFFF2E0EC), textSecondary = Color(0xFFCDA2A6), iconInactive = Color(0xFFE15761),
77         accent = Color(0xFFFF3547), accentSoft = Color(0xFFFFA3A8), accentMuted = Color(0xA0FF3547), danger = Color(
0xFFFF4A5F)
78     )
79
80     val Olive = AppTheme(
81         id = "olive", displayName = "Olive", isLight = false,
82         appBg = Color(0xFF171B0D), gridLine = Color(0xFF252A16),
83         panelBg = Color(0xFF202711), panelBgSolid = Color(0xFF263014),
84         panelStroke = Color(0xFF566032), panelInset = Color(0xFF121806), panelInsetSoft = Color(0xFF2B3418),
85         canvasBackdrop = Color(0xFF171B0D), canvasGrid = Color(0xFF2F31C),
86         textPrimary = Color(0xFF1E7D0), textSecondary = Color(0xFFC7A77E), iconInactive = Color(0xFFE3A15E),

```

```

87         accent = Color(0xFFFFF982F), accentSoft = Color(0xFFFFC46A), accentMuted = Color(0x88FF982F), danger = Color(
0xFFFF624D)
88     )
89
90     val Blue = AppTheme(
91         id = "blue", displayName = "Blue", isLight = false,
92         appBg = Color(0xFF06101A), gridLine = Color(0xFF0A1828),
93         panelBg = Color(0xFF0C1A2A), panelBgSolid = Color(0xFF0E1D2E),
94         panelStroke = Color(0xFF14283D), panelInset = Color(0xFF08141F), panelInsetSoft = Color(0xFF101F2A),
95         canvasBackdrop = Color(0xFF06101A), canvasGrid = Color(0xFF14283A),
96         textPrimary = Color(0xFFE8EEF5), textSecondary = Color(0xFF7F8C9A), iconInactive = Color(0xFFB0BBC8),
97         accent = Color(0xFF4D7AFF), accentSoft = Color(0xFF7AA0FF), accentMuted = Color(0x554D7AFF), danger = Color(
0xFFFF3B5C)
98     )
99
100    val Studio = AppTheme(
101        id = "studio", displayName = "Studio", isLight = true,
102        appBg = Color(0xFFE1E1E1), gridLine = Color(0xFFD1D6DA),
103        panelBg = Color(0xFFFFF0F2F4), panelBgSolid = Color(0xFFFFBFCFD),
104        panelStroke = Color(0xFFD4DAE0), panelInset = Color(0xFFE7EBEF), panelInsetSoft = Color(0xFFDCE8F3),
105        canvasBackdrop = Color(0xFFDADADA), canvasGrid = Color(0xFFCFCFCF),
106        textPrimary = Color(0xFF20252A), textSecondary = Color(0xFF65737F), iconInactive = Color(0xFF5C8DB3),
107        accent = Color(0xFF4A9AD8), accentSoft = Color(0xFF7FBCB8), accentMuted = Color(0x334A9AD8), danger = Color(
0xFFE65C73)
108    )
109
110    val Neon = AppTheme(
111        id = "neon", displayName = "Neon", isLight = false,
112        appBg = Color(0xFF2B123E), gridLine = Color(0xFF32184A),
113        panelBg = Color(0xFF4A2366), panelBgSolid = Color(0xFF522A70),
114        panelStroke = Color(0xFF6C3A8D), panelInset = Color(0xFF321344), panelInsetSoft = Color(0xFF3C1853),
115        canvasBackdrop = Color(0xFF2B123E), canvasGrid = Color(0xFF3C1D55),
116        textPrimary = Color(0xFFFF3EAFE), textSecondary = Color(0xFFC6A8D8), iconInactive = Color(0xFF2FE0EA),
117        accent = Color(0xFF25DDE8), accentSoft = Color(0xFF73F3FA), accentMuted = Color(0x5525DDE8), danger = Color(
0xFFFF6E9C)
118    )
119
120    val Cyan = AppTheme(
121        id = "cyan", displayName = "Cyan", isLight = false,
122        appBg = Color(0xFF06181A), gridLine = Color(0xFF0A2628),
123        panelBg = Color(0xFF0C282A), panelBgSolid = Color(0xFF0E2C2E),
124        panelStroke = Color(0xFF143C3D), panelInset = Color(0xFF081F20), panelInsetSoft = Color(0xFF102A2A),
125        canvasBackdrop = Color(0xFF06181A), canvasGrid = Color(0xFF143A3A),
126        textPrimary = Color(0xFFE8F5F5), textSecondary = Color(0xFF7F9A9A), iconInactive = Color(0xFFB0C8C8),
127        accent = Color(0xFF26D9D9), accentSoft = Color(0xFF7AE0E0), accentMuted = Color(0x5526D9D9), danger = Color(
0xFFFF3B5C)
128    )
129
130    val Green = AppTheme(
131        id = "green", displayName = "Green", isLight = false,
132        appBg = Color(0xFF071A0C), gridLine = Color(0xFF0A2814),
133        panelBg = Color(0xFF0C2A14), panelBgSolid = Color(0xFF0E2D16),
134        panelStroke = Color(0xFF143D1F), panelInset = Color(0xFF081F0C), panelInsetSoft = Color(0xFF102A14),
135        canvasBackdrop = Color(0xFF071A0C), canvasGrid = Color(0xFF143A1F),
136        textPrimary = Color(0xFFE8F5EC), textSecondary = Color(0xFF7F9A88), iconInactive = Color(0xFFB0C8B8),
137        accent = Color(0xFF3DDB6E), accentSoft = Color(0xFF7AE0A0), accentMuted = Color(0x553DDB6E), danger = Color(
0xFFFF3B5C)
138    )
139
140    val Yellow = AppTheme(
141        id = "yellow", displayName = "Yellow", isLight = true,
142        appBg = Color(0xFFFFF8E0), gridLine = Color(0xFFFF5E5A0),
143        panelBg = Color(0xFFFFF4D0), panelBgSolid = Color(0xFFFFF0CF0),
144        panelStroke = Color(0xFFE6CC60), panelInset = Color(0xFFFFF0B8), panelInsetSoft = Color(0xFFFFF890),
145        canvasBackdrop = Color(0xFFFFF8E0), canvasGrid = Color(0xFFFF0DA80),
146        textPrimary = Color(0xFF3D2E08), textSecondary = Color(0xFF8A7A40), iconInactive = Color(0xFFB09860),
147        accent = Color(0xFFFF5B800), accentSoft = Color(0xFFFFD050), accentMuted = Color(0x555B800), danger = Color(
0xFFFF3B5C)
148    )
149
150    val Orange = AppTheme(
151        id = "orange", displayName = "Orange", isLight = true,
152        appBg = Color(0xFFFFF8D5), gridLine = Color(0xFFFF5C898),
153        panelBg = Color(0xFFFFF0C5), panelBgSolid = Color(0xFFFFF1E5),
154        panelStroke = Color(0xFFE69C58), panelInset = Color(0xFFFFD8B5), panelInsetSoft = Color(0xFFFFF890),
155        canvasBackdrop = Color(0xFFFFF8D5), canvasGrid = Color(0xFFFF0BC80),
156        textPrimary = Color(0xFF3D1E08), textSecondary = Color(0xFF8A5A30), iconInactive = Color(0xFFB07A50),
157        accent = Color(0xFFFFF8800), accentSoft = Color(0xFFFFD050), accentMuted = Color(0x55FF8800), danger = Color(
0xFFFF3B5C)
158    )
159
160    val Gray = AppTheme(
161        id = "gray", displayName = "Gray", isLight = false,
162        appBg = Color(0xFF1A1A1A), gridLine = Color(0xFF2A2A2A),
163        panelBg = Color(0xFF2A2A2A), panelBgSolid = Color(0xFF303030),
164        panelStroke = Color(0xFF404040), panelInset = Color(0xFF1F1F1F), panelInsetSoft = Color(0xFF2B2B2B),
165        canvasBackdrop = Color(0xFF1A1A1A), canvasGrid = Color(0xFF3A3A3A),
166        textPrimary = Color(0xFFFFF0F0), textSecondary = Color(0xFF909090), iconInactive = Color(0xFFB8B8B8),
167        accent = Color(0xFFB0B0B0), accentSoft = Color(0xFFD0D0D0), accentMuted = Color(0x55B0B0B0), danger = Color(
0xFFFF3B5C)
168    )
169

```

```

170     val White = AppTheme(
171         id = "white", displayName = "White", isLight = true,
172         appBg = Color(0xFFFFFFFF), gridLine = Color(0xFFE8E8E8),
173         panelBg = Color(0xFFF5F5F5), panelBgSolid = Color(0xFFFFFFFF),
174         panelStroke = Color(0xFFD8D8D8), panelInset = Color(0xFFEEEEEE), panelInsetSoft = Color(0xFFE0E0E0),
175         canvasBackdrop = Color(0xFFFFFFFF), canvasGrid = Color(0xFFE0E0E0),
176         textPrimary = Color(0xFF202020), textSecondary = Color(0xFF707070), iconInactive = Color(0xFF989898),
177         accent = Color(0xFF606060), accentSoft = Color(0xFF888888), accentMuted = Color(0x55606060), danger = Color(
0xFFFF3B5C)
178     )
179
180     val all = listOf(
181         Dark, Light, Red, Vampire, Olive, Blue, Studio, Neon, Cyan, Green, Yellow, Orange, Gray, White
182     )
183
184     val default = Gray
185 }
186
187 enum class UIFont { Classic, Italic }

```

89. app/src/main/java/com/wetinknext/ui/theme/Theme.kt

```

1 package com.wetinknext.ui.theme
2
3 import androidx.compose.material3.MaterialTheme
4 import androidx.compose.material3.Typography
5 import androidx.compose.material3.darkColorScheme
6 import androidx.compose.material3.lightColorScheme
7 import androidx.compose.runtime.Composable
8 import androidx.compose.runtime.CompositionLocalProvider
9 import androidx.compose.ui.graphics.Color
10 import androidx.compose.ui.text.TextStyle
11 import androidx.compose.ui.text.font.FontFamily
12 import androidx.compose.ui.text.font.FontStyle
13
14 @Composable
15 fun WetInkTheme(
16     theme: AppTheme,
17     fontMode: UIFont,
18     content: @Composable () -> Unit
19 ) {
20     val fontStyle = if (fontMode == UIFont.Italic) FontStyle.Italic else FontStyle.Normal
21     val fontFamily = if (fontMode == UIFont.Italic) FontFamily.Serif else FontFamily.SansSerif
22
23     val typography = Typography(
24         bodyLarge = TextStyle(fontStyle = fontStyle, fontFamily = fontFamily),
25         bodyMedium = TextStyle(fontStyle = fontStyle, fontFamily = fontFamily),
26         bodySmall = TextStyle(fontStyle = fontStyle, fontFamily = fontFamily),
27         titleLarge = TextStyle(fontStyle = fontStyle, fontFamily = fontFamily),
28         titleMedium = TextStyle(fontStyle = fontStyle, fontFamily = fontFamily),
29         titleSmall = TextStyle(fontStyle = fontStyle, fontFamily = fontFamily),
30         labelLarge = TextStyle(fontStyle = fontStyle, fontFamily = fontFamily),
31         labelMedium = TextStyle(fontStyle = fontStyle, fontFamily = fontFamily),
32         labelSmall = TextStyle(fontStyle = fontStyle, fontFamily = fontFamily),
33     )
34
35     val scheme = if (!theme.isLight) {
36         darkColorScheme(
37             primary = theme.accent,
38             onPrimary = theme.textPrimary,
39             background = theme.appBg, // Цвет стола
40             onBackground = theme.textPrimary,
41             surface = theme.panelBg, // Цвет панелей
42             onSurface = theme.textPrimary,
43             surfaceVariant = theme.panelBgVariant,
44             onSurfaceVariant = theme.textSecondary,
45         )
46     } else {
47         lightColorScheme(
48             primary = theme.accent,
49             onPrimary = Color.White,
50             background = theme.appBg,
51             onBackground = theme.textPrimary,
52             surface = theme.panelBg,
53             onSurface = theme.textPrimary,
54             surfaceVariant = theme.panelBgVariant,
55             onSurfaceVariant = theme.textSecondary,
56         )
57     }
58
59     CompositionLocalProvider(LocalAppTheme provides theme) {
60         MaterialTheme(
61             colorScheme = scheme,
62             typography = typography,
63             content = content,
64         )
65     }
66 }

```

90. app/src/main/java/com/wetinknext/ui/theme/ThemeController.kt

```

1  package com.wetinknext.ui.theme
2
3  import android.content.Context
4  import androidx.compose.runtime.*
5  import androidx.compose.runtime.saveable.Saver
6  import androidx.compose.runtime.saveable.rememberSaveable
7  import androidx.compose.ui.graphics.Color
8  import androidx.compose.ui.platform.LocalContext
9  import kotlin.math.pow
10
11  val LocalAppTheme = compositionLocalOf { AppThemes.Gray }
12
13  data class CustomThemeColors(
14      val appBackground: Int = 0xFF101826.toInt(),
15      val grid: Int = 0xFF38424C.toInt(),
16      val panel: Int = 0xFF172033.toInt(),
17      val accent: Int = 0xFF58A6FF.toInt(),
18      val text: Int = 0xFFFF4F7FB.toInt(),
19  )
20
21  class ThemeController(
22      initialThemeId: String = "gray",
23      initialFontMode: String = "Classic",
24      initialCustomColors: CustomThemeColors = CustomThemeColors()
25  ) {
26      var current by mutableStateOf(resolveTheme(initialThemeId, initialCustomColors))
27      private set
28      var font by mutableStateOf(if (initialFontMode == "Italic") UiFont.Italic else UiFont.Classic)
29      private set
30      var customColors by mutableStateOf(initialCustomColors)
31      private set
32      val customTheme: AppTheme
33      get() = createCustomTheme(customColors)
34
35      fun select(theme: AppTheme) {
36          current = if (theme.id == "custom") createCustomTheme(customColors) else theme
37      }
38
39      fun selectCustom() {
40          current = createCustomTheme(customColors)
41      }
42
43      fun selectFont(f: UiFont) { font = f }
44
45      fun updateCustomColors(newColors: CustomThemeColors) {
46          customColors = newColors
47          if (current.id == "custom") current = createCustomTheme(newColors)
48      }
49
50      private fun resolveTheme(id: String, colors: CustomThemeColors): AppTheme {
51          if (id == "custom") return createCustomTheme(colors)
52          return AppThemes.all.find { it.id == id } ?: AppThemes.Gray
53      }
54
55      private fun createCustomTheme(colors: CustomThemeColors): AppTheme {
56          val appBg = Color(colors.appBackground)
57          val panel = Color(colors.panel)
58          val accent = Color(colors.accent)
59          val text = Color(colors.text)
60          val grid = Color(colors.grid)
61          val isDark = luminanceOf(appBg) < 0.5f
62
63          return AppTheme(
64              id = "custom", displayName = "Своя", isLight = !isDark,
65              appBg = appBg, gridLine = grid,
66              panelBg = panel, panelBgSolid = panel.mix(if (isDark) Color.White else Color.Black, 0.08f),
67              panelStroke = panel.mix(accent, 0.3f),
68              panelInset = panel.mix(if (isDark) Color.Black else Color.White, 0.2f),
69              panelInsetSoft = panel.mix(accent, 0.1f),
70              canvasBackdrop = appBg, canvasGrid = grid,
71              textPrimary = text, textSecondary = text.copy(alpha = 0.6f),
72              iconInactive = text.copy(alpha = 0.4f),
73              accent = accent, accentSoft = accent.copy(alpha = 0.7f),
74              accentMuted = accent.copy(alpha = 0.3f),
75              danger = Color(0xFFFF4A5F)
76          )
77      }
78
79      private fun Color.mix(target: Color, amount: Float): Color {
80          return Color(
81              red = red * (1f - amount) + target.red * amount,
82              green = green * (1f - amount) + target.green * amount,
83              blue = blue * (1f - amount) + target.blue * amount,
84              alpha = alpha
85          )
86      }
87
88      private fun luminanceOf(color: Color): Float {
89          fun channel(value: Float): Float =
90              if (value <= 0.03928f) value / 12.92f else ((value + 0.055f) / 1.055f).pow(2.4f)

```

```

91         return 0.2126f * channel(color.red) + 0.7152f * channel(color.green) + 0.0722f * channel(color.blue)
92     }
93
94     companion object {
95         val Saver: Saver<ThemeController, *> = Saver(
96             save = { listOf(it.current.id, it.font.name, it.customColors.appBackground, it.customColors.grid, it.customColors.
panel, it.customColors.accent, it.customColors.text) },
97             restore = { list ->
98                 val l = list as List<*>
99                 ThemeController(
100                     initialThemeId = l[0] as String,
101                     initialFontMode = l[1] as String,
102                     initialCustomColors = CustomThemeColors(
103                         l[2] as Int, l[3] as Int, l[4] as Int, l[5] as Int, l[6] as Int
104                     )
105                 )
106             }
107         )
108     }
109 }
110
111 @Composable
112 fun rememberThemeController(): ThemeController {
113     val context = LocalContext.current
114     val prefs = remember { context.getSharedPreferences("wetinknext_theme_prefs", Context.MODE_PRIVATE) }
115
116     val controller = rememberSaveable(saver = ThemeController.Saver) {
117         val themeId = prefs.getString("theme_id", "gray") ?: "gray"
118         val fontMode = prefs.getString("font_mode", "Classic") ?: "Classic"
119         val customColors = CustomThemeColors(
120             appBackground = prefs.getInt("custom_bg", 0xFF101826.toInt()),
121             grid = prefs.getInt("custom_grid", 0xFF38424C.toInt()),
122             panel = prefs.getInt("custom_panel", 0xFF172033.toInt()),
123             accent = prefs.getInt("custom_accent", 0xFF58A6FF.toInt()),
124             text = prefs.getInt("custom_text", 0xFFF4F7FB.toInt())
125         )
126         ThemeController(themeId, fontMode, customColors)
127     }
128
129     LaunchedEffect(controller.current.id, controller.font, controller.customColors) {
130         prefs.edit()
131             .putString("theme_id", controller.current.id)
132             .putString("font_mode", controller.font.name)
133             .putInt("custom_bg", controller.customColors.appBackground)
134             .putInt("custom_grid", controller.customColors.grid)
135             .putInt("custom_panel", controller.customColors.panel)
136             .putInt("custom_accent", controller.customColors.accent)
137             .putInt("custom_text", controller.customColors.text)
138             .apply()
139     }
140
141     return controller
142 }

```

91. app/src/main/java/com/wetinknext/ui/theme/WetInkNextPalette.kt

```
1 package com.wetinknext.ui.theme
2
3 import androidx.compose.ui.graphics.Color
4
5 data class WetInkNextPalette(
6     val appBg: Color,
7     val panelBg: Color,
8     val panelStroke: Color,
9     val textPrimary: Color,
10    val textSecondary: Color,
11    val accent: Color,
12 ) {
13     companion object {
14         val default = WetInkNextPalette(
15             appBg = Color(0xFF17181C),
16             panelBg = Color(0xFF25272D),
17             panelStroke = Color(0xFF3A3D46),
18             textPrimary = Color(0xFFF3F4F6),
19             textSecondary = Color(0xFFA4A8B3),
20             accent = Color(0xFF4D7AFF),
21         )
22     }
23 }
```


92. app/src/main/res/drawable/ic_eraser.xml

```
1  <?xml version="1.0" encoding="utf-8"?>
2  <vector xmlns:android="http://schemas.android.com/apk/res/android"
3      android:width="24dp"
4      android:height="24dp"
5      android:viewportWidth="24"
6      android:viewportHeight="24">
7      <path
8          android:fillColor="#FFFFFF"
9          android:pathData="M14.59,3.41a2,2 0,0 1,2.83 0l3.17,3.17a2,2 0,0 1,0 2.83l11.41,18.59a2,2 0,0 1,-2.83 0l4.41,14.41a2,
2 0,0 1,0 -2.83l10.18,-8.17zM3,20h18v2H3z" />
10     <path
11         android:fillColor="#FFFFFF"
12         android:fillAlpha="0.55"
13         android:pathData="M7.05,11l4.95,4.95l3.18,-3.18l-4.95,-4.95z" />
14 </vector>
```

93. app/src/main/res/drawable/ic_flip_horizontal.xml

```
1  <?xml version="1.0" encoding="utf-8"?>
2  <vector xmlns:android="http://schemas.android.com/apk/res/android"
3      android:width="24dp"
4      android:height="24dp"
5      android:viewportWidth="24"
6      android:viewportHeight="24">
7      <!-- Central Axis (Centered at 12) -->
8      <path
9          android:fillColor="#FFFFFF"
10         android:pathData="M11.25,4h1.5v16h-1.5V4z"/>
11
12     <!-- Left solid bracket (Symmetrical) -->
13     <path
14         android:fillColor="#FFFFFF"
15         android:pathData="M9,6.5H5v11h4V16H6.5V8H9V6.5z"/>
16
17     <!-- Right dashed bracket (Mirror of left, perfectly aligned) -->
18     <!-- Top dash -->
19     <path android:fillColor="#FFFFFF" android:pathData="M15,6.5h2.5V8H15V6.5z"/>
20     <path android:fillColor="#FFFFFF" android:pathData="M18.5,6.5h0.5V8h-0.5V6.5z"/>
21     <!-- Side dashes -->
22     <path android:fillColor="#FFFFFF" android:pathData="M17.5,9.5h1.5v2h-1.5V9.5z"/>
23     <path android:fillColor="#FFFFFF" android:pathData="M17.5,12.5h1.5v2h-1.5V12.5z"/>
24     <!-- Bottom dash -->
25     <path android:fillColor="#FFFFFF" android:pathData="M15,16h2.5v1.5H15V16z"/>
26     <path android:fillColor="#FFFFFF" android:pathData="M18.5,16h0.5v1.5h-0.5V16z"/>
27 </vector>
```

94. app/src/main/res/drawable/ic_launcher_background.xml

```
1 <?xml version="1.0" encoding="utf-8"?>
2 <vector xmlns:android="http://schemas.android.com/apk/res/android"
3     android:width="108dp"
4     android:height="108dp"
5     android:viewportWidth="108"
6     android:viewportHeight="108">
7     <path
8         android:fillColor="#FF211019"
9         android:pathData="M0,0h108v108h-108z" />
10 </vector>
```

95. app/src/main/res/drawable/ic_launcher_foreground.xml

```

1  <?xml version="1.0" encoding="utf-8"?>
2  <vector xmlns:android="http://schemas.android.com/apk/res/android"
3      xmlns:aapt="http://schemas.android.com/aapt"
4      android:width="108dp"
5      android:height="108dp"
6      android:viewportWidth="108"
7      android:viewportHeight="108">
8      <group
9          android:pivotX="54"
10         android:pivotY="54"
11         android:scaleX="0.6"
12         android:scaleY="0.6">
13         <path
14             android:pathData="M14,88h80"
15             android:strokeWidth="5"
16             android:strokeLineCap="round">
17             <aapt:attr name="android:strokeColor">
18                 <gradient
19                     android:startX="78"
20                     android:startY="20"
21                     android:endX="30"
22                     android:endY="80"
23                     android:type="linear">
24                     <item android:offset="0" android:color="#FFFFB1C5" />
25                     <item android:offset="1" android:color="#FFF3D70" />
26                 </gradient>
27             </aapt:attr>
28         </path>
29         <path
30             android:pathData="M66,16c3.3,-3.3 8.7,-3.3 12,0l4,4c3.3,3.3 3.3,8.7 0,12l34,80l-16,4l4,-16l66,16z"
31             android:fillType="nonZero">
32             <aapt:attr name="android:fillColor">
33                 <gradient
34                     android:startX="78"
35                     android:startY="20"
36                     android:endX="30"
37                     android:endY="80"
38                     android:type="linear">
39                     <item android:offset="0" android:color="#FFFFB1C5" />
40                     <item android:offset="1" android:color="#FFF3D70" />
41                 </gradient>
42             </aapt:attr>
43         </path>
44     </group>
45 </vector>

```

96. app/src/main/res/mipmap-anydpi-v26/ic_launcher.xml

```
1 <?xml version="1.0" encoding="utf-8"?>
2 <adaptive-icon xmlns:android="http://schemas.android.com/apk/res/android">
3     <background android:drawable="@drawable/ic_launcher_background" />
4     <foreground android:drawable="@drawable/ic_launcher_foreground" />
5 </adaptive-icon>
```

97. app/src/main/res/mipmap-anydpi-v26/ic_launcher_round.xml

```
1 <?xml version="1.0" encoding="utf-8"?>
2 <adaptive-icon xmlns:android="http://schemas.android.com/apk/res/android">
3     <background android:drawable="@drawable/ic_launcher_background" />
4     <foreground android:drawable="@drawable/ic_launcher_foreground" />
5 </adaptive-icon>
```

98. app/src/main/res/mipmap-anydpi/ic_launcher.xml

```
1 <?xml version="1.0" encoding="utf-8"?>
2 <layer-list xmlns:android="http://schemas.android.com/apk/res/android">
3     <item android:drawable="@drawable/ic_launcher_background" />
4     <item android:drawable="@drawable/ic_launcher_foreground" />
5 </layer-list>
```

99. app/src/main/res/mipmap-anydpi/ic_launcher_round.xml

```
1 <?xml version="1.0" encoding="utf-8"?>
2 <layer-list xmlns:android="http://schemas.android.com/apk/res/android">
3     <item android:drawable="@drawable/ic_launcher_background" />
4     <item android:drawable="@drawable/ic_launcher_foreground" />
5 </layer-list>
```


100. app/src/main/res/values/strings.xml

```
1 <resources>
2     <string name="app_name">WetInk Next</string>
3 </resources>
```

101. app/src/main/res/values/themes.xml

```
1 <resources>
2     <style name="Theme.WetInkNext" parent="android:style/Theme.Material.NoActionBar" />
3 </resources>
```

102. app/src/test/java/com/wetinknext/engine/brush/BrushDynamicsTest.kt

```
1 package com.wetinknext.engine.brush
2
3 import org.junit.Assert.assertEquals
4 import org.junit.Test
5
6 class BrushDynamicsTest {
7     @Test
8     fun resolve_separatesLocalCoverageFromGlobalOpacity() {
9         val settings = BrushSettings(
10             baseRadiusPx = 10f,
11             opacity = 0.2f,
12             pressureGamma = 2f,
13             minSizeRatio = 0.1f,
14             tiltToSize = 0.5f,
15             tiltToOpacity = 1f,
16         )
17         val result = ResolvedDab()
18
19         BrushDynamics.resolve(
20             settings = settings,
21             pressure = 0.5f,
22             tiltX = 1f,
23             tiltY = 0f,
24             out = result,
25         )
26
27         //  $p = 0.5^2 = 0.25$ ; pressure size = .325; tilt size = 1.5.
28         assertEquals(4.875f, result.radius, 0.0001f)
29         // Tilt attenuates, rather than amplifies, local coverage.
30         assertEquals(0.1375f, result.coverage, 0.0001f)
31         assertEquals(result.coverage, result.opacity, 0.0001f)
32     }
33 }
```

103. app/src/test/java/com/wetinknext/engine/brush/BrushSerializationTest.kt

```
1 package com.wetinknext.engine.brush
2
3 import kotlinx.serialization.encodeToString
4 import org.junit.Assert.assertEquals
5 import org.junit.Test
6
7 class BrushSerializationTest {
8
9     @Test
10     fun settingsRoundTripKeepsRenderFields() {
11         val json = BrushRepository.defaultJson
12         val original = BrushSettings(
13             name = "Pencil 6B",
14             grainAssetPath = "asset:brush/pencil_6b.png",
15             grainScale = 5.5f,
16             textureDepth = .65f,
17             tiltToSize = .55f,
18             blendPolicy = BlendPolicy.NORMAL_BUILDUP,
19         )
20
21         val encoded = json.encodeToString(original)
22         val restored = json.decodeFromString<BrushSettings>(encoded)
23
24         assertEquals(original, restored)
25     }
26
27     @Test
28     fun presetRoundTrip() {
29         val repo = BrushRepository()
30         val original = BrushPreset(
31             id = "test-preset",
32             settings = BrushSettings(name = "Test Brush")
33         )
34
35         val encoded = repo.encode(original)
36         val restored = repo.decode(encoded)
37
38         assertEquals(original, restored)
39     }
40 }
```

104. app/src/test/java/com/wetinknext/engine/brush/BrushSettingsTest.kt

```
1 package com.wetinknext.engine.brush
2
3 import org.junit.Assert.assertEquals
4 import org.junit.Test
5
6 class BrushSettingsTest {
7     @Test
8     fun resolved_clampsBrushValues() {
9         val result = BrushSettings(
10             baseRadiusPx = -10f,
11             opacity = 3f,
12             flow = -1f,
13             spacing = 100f,
14             pressureGamma = 0f,
15             minSizeRatio = 4f,
16         ).resolved()
17
18         assertEquals(0.25f, result.baseRadiusPx, 0.001f)
19         assertEquals(1f, result.opacity, 0.001f)
20         assertEquals(0f, result.flow, 0.001f)
21         assertEquals(4f, result.spacing, 0.001f)
22         assertEquals(0.05f, result.pressureGamma, 0.001f)
23         assertEquals(1f, result.minSizeRatio, 0.001f)
24     }
25 }
```

105. app/src/test/java/com/wetinknext/engine/brush/CapsuleEmitterTest.kt

```

1  package com.wetinknext.engine.brush
2
3  import com.wetinknext.engine.input.InputAction
4  import com.wetinknext.engine.input.InputBatch
5  import com.wetinknext.engine.input.PointerTool
6  import org.junit.Assert.assertTrue
7  import org.junit.Test
8
9  class CapsuleEmitterTest {
10
11     @Test
12     fun finishFlushesTheLastInterpolatedSection() {
13         val settings = BrushSettings(
14             renderMode = BrushRenderMode.RIBBON,
15             baseRadiusPx = 10f,
16             smoothing = 0.5f,
17             streamline = 0f,
18         )
19         val emitter = CapsuleEmitter(settings)
20         // We need a dummy renderer to receive segments
21         // Since we can't easily mock GL context here, we just use a small maxSegments
22         val renderer = CapsuleStrokeRenderer(maxSegments = 128)
23
24         emitter.begin(batch(InputAction.DOWN, 0f, 0f, 0L), renderer)
25         emitter.append(batch(InputAction.MOVE, 100f, 0f, 16_000_000L), renderer)
26         emitter.append(batch(InputAction.MOVE, 200f, 30f, 32_000_000L), renderer)
27
28         val segmentsBeforeFinish = renderer.segmentCount
29         emitter.finish(renderer, cancel = false)
30         val segmentsAfterFinish = renderer.segmentCount
31
32         assertTrue("Should have some segments after append", segmentsBeforeFinish > 0)
33         assertTrue("Finish should have flushed more segments", segmentsAfterFinish > segmentsBeforeFinish)
34         assertTrue(emitter.hasStroke)
35     }
36
37     private fun batch(action: InputAction, x: Float, y: Float, t: Long) = InputBatch(8).apply {
38         begin(action)
39         addSample(x, y, .5f, 0f, 0f, 0f, t, 0, PointerTool.STYLUS, false)
40     }
41 }

```

106. app/src/test/java/com/wetinknext/engine/brush/CapsuleStrokeRendererTest.kt

```

1 package com.wetinknext.engine.brush
2
3 import org.junit.Assert.assertEquals
4 import org.junit.Assert.assertFalse
5 import org.junit.Assert.assertTrue
6 import org.junit.Test
7 import com.wetinknext.engine.gl.ShaderLib
8
9 class CapsuleStrokeRendererTest {
10
11     @Test
12     fun capsuleShaderAppliesFlowInsideTheStrokeTarget() {
13         assertTrue(ShaderLib.capsuleFragment.contains("uniform float uFlow"))
14         assertTrue(ShaderLib.capsuleFragment.contains("grainFactor * uFlow"))
15     }
16
17     @Test
18     fun initiallyEmpty() {
19         val renderer = CapsuleStrokeRenderer(maxSegments = 100)
20         assertTrue(renderer.isEmpty)
21         assertEquals(0, renderer.segmentCount)
22     }
23
24     @Test
25     fun addsSegmentsWithinCapacity() {
26         val renderer = CapsuleStrokeRenderer(maxSegments = 10)
27         renderer.beginStroke()
28
29         for (i in 0 until 10) {
30             val added = renderer.addSegment(0f, 0f, 5f, 1f, 10f, 10f, 5f, 1f)
31             assertTrue("Segment $i should be added", added)
32         }
33
34         assertEquals(10, renderer.segmentCount)
35         assertFalse(renderer.isEmpty)
36
37         val addedOverflow = renderer.addSegment(0f, 0f, 5f, 1f, 10f, 10f, 5f, 1f)
38         assertFalse("Overflow segment should not be added", addedOverflow)
39         assertEquals(1, renderer.overflowCount)
40     }
41
42     @Test
43     fun clearResetsStateButNotOverflow() {
44         val renderer = CapsuleStrokeRenderer(maxSegments = 10)
45         renderer.beginStroke()
46         renderer.addSegment(0f, 0f, 5f, 1f, 10f, 10f, 5f, 1f)
47
48         // Overflow it
49         for (i in 0 until 10) renderer.addSegment(0f, 0f, 1f, 1f, 1f, 1f, 1f, 1f)
50         assertEquals(1, renderer.overflowCount)
51
52         renderer.clearStrokeData()
53         assertTrue(renderer.isEmpty)
54         assertEquals(0, renderer.segmentCount)
55         assertEquals(1, renderer.overflowCount)
56     }
57 }

```

107. app/src/test/java/com/wetinknext/engine/brush/ColorSpacesTest.kt

```
1 package com.wetinknext.engine.brush
2 import org.junit.Assert.*
3 import org.junit.Test
4 class ColorSpacesTest { @Test fun blackAndWhiteAreStable(){val b=FloatArray(3);val w=FloatArray(3);ColorSpaces.srgb8ToLinear(
0xFF000000L,b);ColorSpaces.srgb8ToLinear(0xFFFFFFFFL,w);assertEquals(0f,b[0],.0001f);assertEquals(1f,w[0],.0001f)} @Test fun
roundTrip(){assertEquals(.25f,ColorSpaces.srgbToLinear(ColorSpaces.linearToSrgb(.25f)),.0001f)} }
```


108. app/src/test/java/com/wetinknext/engine/brush/DabBufferTest.kt

```
1 package com.wetinknext.engine.brush
2
3 import com.wetinknext.engine.core.DirtyRect
4 import org.junit.Assert.*
5 import org.junit.Test
6
7 class DabBufferTest {
8     @Test fun overflowIsCounted(){val b=DabBuffer(2);assertTrue(b.add(0f,0f,1f,0f,1f,1f));assertTrue(b.add(1f,1f,1f,0f,1f,
9 1f));assertFalse(b.add(2f,2f,1f,0f,1f,1f));assertEquals(1L,b.overflowCount)}
10    @Test fun clearResets(){val b=DabBuffer(1);b.add(0f,0f,1f,0f,1f,1f);b.clear();assertEquals(0,b.count)}
11    @Test fun dirtyRectIncludesTextureMargin(){
12        val buffer = DabBuffer(1)
13        buffer.add(10f, 20f, 5f, 0f, 1f, 1f)
14        val rect = DirtyRect()
15        buffer.includeDirtyRect(rect, extraMargin = 4f)
16        assertEquals(1f, rect.left, 0.001f)
17        assertEquals(19f, rect.right, 0.001f)
18        assertEquals(11f, rect.top, 0.001f)
19        assertEquals(29f, rect.bottom, 0.001f)
20    }
```

109. app/src/test/java/com/wetinknext/engine/brush/OneEuroFilterTest.kt

```
1 package com.wetinknext.engine.brush
2
3 import org.junit.Assert.assertEquals
4 import org.junit.Assert.assertFalse
5 import org.junit.Test
6
7 class OneEuroFilterTest {
8     @Test fun firstSamplePassesThrough() { val f=OneEuroFilter();val out=FloatArray(2);f.filter(0,10f,20f,out);assertEquals(
10f,out[0],.0001f);assertEquals(20f,out[1],.0001f) }
9     @Test fun samplesRemainFinite() { val f=OneEuroFilter();val out=FloatArray(2);f.filter(0,0f,0f,out);f.filter(8_000_000,
10f,5f,out);assertFalse(out[0].isNaN());assertFalse(out[1].isInfinite()) }
10 }
```

110. app/src/test/java/com/wetinknext/engine/brush/PressureFilterTest.kt

```
1 package com.wetinknext.engine.brush
2
3 import org.junit.Assert.assertTrue
4 import org.junit.Test
5
6 class PressureFilterTest {
7     @Test fun singleZeroPressureGlitchDoesNotCollapseWidth() {
8         val filter = PressureFilter()
9         filter.filter(0L, .6f)
10        val filtered = filter.filter(16_000_000L, 0f)
11        assertTrue("zero glitch must be ignored while stroke is active", filtered > .55f)
12    }
13
14    @Test fun realPressureChangeIsSmoothed() {
15        val filter = PressureFilter()
16        filter.filter(0L, .8f)
17        val filtered = filter.filter(16_000_000L, .2f)
18        assertTrue(filtered in .2f.. .8f)
19    }
20 }
```

111. app/src/test/java/com/wetinknext/engine/brush/StabilizerTest.kt

```
1 package com.wetinknext.engine.brush
2
3 import org.junit.Assert.assertEquals
4 import org.junit.Assert.assertTrue
5 import org.junit.Test
6
7 class StabilizerTest {
8     @Test fun strengthZeroPassesRawThrough(){val s=Stabilizer();s.process(0,100f,200f);assertEquals(100f,s.x,.0001f);
assertEquals(200f,s.y,.0001f)}
9     @Test fun velocityIsPositiveAfterMovement(){val s=Stabilizer();s.process(0,0f,0f);s.process(1_000_000_000,100f,0f);
assertTrue(s.velocity>0f)}
10 }
```

112. app/src/test/java/com/wetinknext/engine/brush/StampEmitterTest.kt

```

1 package com.wetinknext.engine.brush
2
3 import com.wetinknext.engine.input.*
4 import org.junit.Assert.assertEquals
5 import org.junit.Test
6
7 class StampEmitterTest {
8     @Test fun movingStrokeHasNoExtraDownDab(){val e=StampEmitter(BrushSettings(baseRadiusPx=10f,spacing=1f,
9 spacingUsesDiameter=false,pressureToSize=false,pressureToOpacity=false));val d=DabBuffer(16);val down=InputBatch(1).apply{begin(
10 InputAction.DOWN);addSample(0f,0f,1f,0f,0f,0f,0,0,PointerTool.STYLUS,false));val move=InputBatch(1).apply{begin(InputAction.MOVE);
11 addSample(20f,0f,1f,0f,0f,0f,1,0,PointerTool.STYLUS,false));e.begin(down,d);e.append(move,d);e.finish(d,false);assertEquals(2,d.
12 count)}}
13
14     @Test
15     fun activeStrokeKeepsItsSettingsAfterUiBrushChanges() {
16         val strokeSettings = BrushSettings(
17             baseRadiusPx = 5f,
18             spacing = 1f,
19             spacingUsesDiameter = false,
20             smoothing = 0f,
21             pressureToSize = false,
22             pressureToOpacity = false,
23         )
24         val emitter = StampEmitter(strokeSettings)
25         val dabs = DabBuffer(16)
26         val down = InputBatch(1).apply {
27             begin(InputAction.DOWN)
28             addSample(0f, 0f, 1f, 0f, 0f, 0f, 0L, 0, PointerTool.STYLUS, false)
29         }
30         val move = InputBatch(1).apply {
31             begin(InputAction.MOVE)
32             addSample(20f, 0f, 1f, 0f, 0f, 0f, 1L, 0, PointerTool.STYLUS, false)
33         }
34
35         emitter.begin(down, dabs, strokeSettings)
36         emitter.updateSettings(strokeSettings.copy(baseRadiusPx = 20f))
37         emitter.append(move, dabs)
38
39         assertEquals(5f, dabs.floats.get(2), 0.001f)
40     }
41 }

```

113. app/src/test/java/com/wetinknext/engine/core/DirtyRectTest.kt

```
1 package com.wetinknext.engine.core
2
3 import org.junit.Assert.assertArrayEquals
4 import org.junit.Assert.assertTrue
5 import org.junit.Test
6
7 class DirtyRectTest {
8     @Test fun includesDabsAndConvertsToPixelBounds() {
9         val rect = DirtyRect()
10        rect.include(100f, 120f, 10f)
11        rect.include(150f, 140f, 5f)
12        val pixels = IntArray(4)
13        rect.toPixelBounds(pixels)
14        assertArrayEquals(intArrayOf(90, 110, 155, 145), pixels)
15    }
16
17    @Test fun clampCanMakeRectEmpty() {
18        val rect = DirtyRect()
19        rect.include(-100f, -100f, -10f)
20        rect.clamp(100f, 100f)
21        assertTrue(rect.isEmpty)
22    }
23
24    @Test fun clearResetsRect() {
25        val rect = DirtyRect()
26        rect.include(0f, 0f, 10f)
27        rect.clear()
28        assertTrue(rect.isEmpty)
29    }
30 }
```

114. app/src/test/java/com/wetinknext/engine/core/UndoCompressionPipelineTest.kt

```

1 package com.wetinknext.engine.core
2
3 import com.wetinknext.engine.undo.IdentityTileCompressor
4 import com.wetinknext.engine.undo.RawTileSnapshot
5 import com.wetinknext.engine.undo.TileCompressor
6 import com.wetinknext.engine.undo.TileCoord
7 import com.wetinknext.engine.undo.UndoManager
8 import org.junit.Assert.assertEquals
9 import org.junit.Assert.assertFalse
10 import org.junit.Assert.assertTrue
11 import org.junit.Test
12 import java.util.concurrent.CountDownLatch
13 import java.util.concurrent.Executors
14 import java.util.concurrent.TimeUnit
15 import java.util.concurrent.atomic.AtomicInteger
16
17 class UndoCompressionPipelineTest {
18     @Test
19     fun synchronousFallbackUsesTheSameOrderedHistoryPath() {
20         val manager = UndoManager()
21         val pipeline = UndoCompressionPipeline(
22             requestRender = {},
23             compressor = IdentityTileCompressor,
24             maxPendingJobs = 1,
25         )
26
27         pipeline.enqueue(
28             undoManager = manager,
29             layerId = 7L,
30             beforeRaw = listOf(raw(byteArrayOf(1, 2, 3, 4))),
31             afterRaw = listOf(raw(byteArrayOf(5, 6, 7, 8))),
32         )
33
34         assertEquals(0, pipeline.pendingCount)
35         assertEquals(1, manager.undoCount)
36         assertEquals(0, manager.redoCount)
37         assertTrue(manager.canUndo)
38         pipeline.shutdown()
39     }
40
41     @Test
42     fun laterWorkerResultWaitsForTheEarlierSequence() {
43         val firstCompressionStarted = CountDownLatch(1)
44         val allowFirstCompression = CountDownLatch(1)
45         val secondResultReady = CountDownLatch(1)
46         val allResultsReady = CountDownLatch(2)
47         val resultCount = AtomicInteger(0)
48         val executor = Executors.newFixedThreadPool(2)
49         val compressor = object : TileCompressor {
50             override fun compress(data: ByteArray): ByteArray {
51                 if (data.first() == 0.toByte()) {
52                     firstCompressionStarted.countDown()
53                     check(allowFirstCompression.await(2, TimeUnit.SECONDS))
54                 }
55                 return data
56             }
57
58             override fun decompress(data: ByteArray, expectedSize: Int): ByteArray = data
59         }
60         val manager = UndoManager()
61         val pipeline = UndoCompressionPipeline(
62             requestRender = {
63                 if (resultCount.incrementAndGet() == 1) secondResultReady.countDown()
64                 allResultsReady.countDown()
65             },
66             compressor = compressor,
67             executorFactory = { executor },
68         )
69
70         pipeline.enqueue(
71             undoManager = manager,
72             layerId = 1L,
73             beforeRaw = listOf(raw(byteArrayOf(0, 0, 0, 0))),
74             afterRaw = listOf(raw(byteArrayOf(2, 0, 0, 0))),
75         )
76         assertTrue(firstCompressionStarted.await(1, TimeUnit.SECONDS))
77
78         pipeline.enqueue(
79             undoManager = manager,
80             layerId = 2L,
81             beforeRaw = listOf(raw(byteArrayOf(1, 0, 0, 0))),
82             afterRaw = listOf(raw(byteArrayOf(3, 0, 0, 0))),
83         )
84         assertTrue(secondResultReady.await(1, TimeUnit.SECONDS))
85
86         assertFalse(pipeline.process(manager))
87         assertEquals(0, manager.undoCount)
88         assertEquals(2, pipeline.pendingCount)
89
90         allowFirstCompression.countDown()

```

```

91         assertTrue(allResultsReady.await(1, TimeUnit.SECONDS))
92
93         assertTrue(pipeline.process(manager))
94         assertEquals(2, manager.undoCount)
95         assertEquals(0, pipeline.pendingCount)
96         pipeline.shutdown()
97     }
98
99     private fun raw(bytes: ByteArray) = RawTileSnapshot(
100         coord = TileCoord(0, 0),
101         pixelLeft = 0,
102         pixelTop = 0,
103         pixelWidth = 1,
104         pixelHeight = 1,
105         bytesPerPixel = 4,
106         rawBytes = bytes,
107     )
108 }

```


115. app/src/test/java/com/wetinknext/engine/core/UndoHistoryTest.kt

```
1 package com.wetinknext.engine.core
2
3 import org.junit.Assert.assertEquals
4 import org.junit.Assert.assertFalse
5 import org.junit.Assert.assertTrue
6 import org.junit.Test
7
8 class UndoHistoryTest {
9     @Test
10     fun staleResultIsRejected() {
11         assertFalse(isUndoResultCurrent(resultEpoch = 3L, currentEpoch = 4L))
12     }
13
14     @Test
15     fun failedCompressionDoesNotLeavePendingCountLocked() {
16         var pending = 1
17         val result: UndoJobResult =
18             UndoJobResult.Failed(epoch = 1L, sequence = 0L, error = IllegalStateException())
19
20         pending = (pending - 1).coerceAtLeast(0)
21
22         assertEquals(0, pending)
23         assertTrue(result is UndoJobResult.Failed)
24     }
25 }
```

116. app/src/test/java/com/wetinknext/engine/core/ViewTransformFboTest.kt

```
1 package com.wetinknext.engine.core
2 import org.junit.Assert.assertEquals
3 import org.junit.Test
4 class ViewTransformFboTest { @Test fun orthoValues(){val m=FloatArray(16);ViewTransform.buildCanvasToFbo(100f,200f,m);
assertEquals(.02f,m[0],.0001f);assertEquals(.01f,m[5],.0001f);assertEquals(-1f,m[12],.0001f);assertEquals(-1f,m[13],.0001f)} }
```

117. app/src/test/java/com/wetinknext/engine/core/ViewTransformTest.kt

```

1 package com.wetinknext.engine.core
2
3 import org.junit.Assert.assertEquals
4 import org.junit.Test
5
6 class ViewTransformTest {
7     @Test fun canvasAndScreenCoordinatesAreInverse() {
8         val transform = ViewTransform(120f, 80f, 2.5f, 0.35f, true)
9         val screen = FloatArray(2)
10        val canvas = FloatArray(2)
11        transform.canvasToScreen(240f, 310f, screen)
12        transform.screenToCanvas(screen[0], screen[1], canvas)
13        assertEquals(240f, canvas[0], 0.001f)
14        assertEquals(310f, canvas[1], 0.001f)
15    }
16
17    @Test fun zoomAroundAnchorKeepsAnchorStable() {
18        val initial = ViewTransform(translateX = 20f, translateY = 30f)
19        val before = FloatArray(2)
20        val after = FloatArray(2)
21        initial.screenToCanvas(400f, 300f, before)
22        initial.zoomAround(400f, 300f, 3f).screenToCanvas(400f, 300f, after)
23        assertEquals(before[0], after[0], 0.001f)
24        assertEquals(before[1], after[1], 0.001f)
25    }
26
27    @Test fun gestureDeltaKeepsTheAnchorStableWithoutPan() {
28        val initial = ViewTransform(translateX = 30f, translateY = -10f, scale = 1.4f, rotationRad = .2f)
29        val before = FloatArray(2)
30        val after = FloatArray(2)
31        initial.screenToCanvas(400f, 300f, before)
32        initial.applyGestureDelta(
33            panX = 0f,
34            panY = 0f,
35            anchorX = 400f,
36            anchorY = 300f,
37            zoomFactor = 1.8f,
38            rotationDelta = .3f,
39        ).screenToCanvas(400f, 300f, after)
40        assertEquals(before[0], after[0], .001f)
41        assertEquals(before[1], after[1], .001f)
42    }
43 }

```

118. app/src/test/java/com/wetinknext/engine/undo/TileCoordinateContractTest.kt

```
1 package com.wetinknext.engine.undo
2
3 import org.junit.Assert.assertEquals
4 import org.junit.Test
5
6 class TileCoordinateContractTest {
7     @Test
8     fun topCanvasTileStartsAtBottomGlRegion() {
9         val canvasTop = 0
10        val glBottom = canvasTop
11        assertEquals(0, glBottom)
12    }
13
14    @Test
15    fun bottomCanvasTileUsesItsCanvasYAsGlY() {
16        val canvasTop = 1792
17        val glBottom = canvasTop
18        assertEquals(1792, glBottom)
19    }
20 }
```

119. app/src/test/java/com/wetinknext/engine/undo/TileSnapshotTest.kt

```
1 package com.wetinknext.engine.undo
2
3 import org.junit.Assert.assertEquals
4 import org.junit.Assert.assertEquals
5 import org.junit.Test
6
7 class TileSnapshotTest {
8     @Test
9     fun rawIdentityTileKeepsItsExactBytes() {
10         val bytes = byteArrayOf(0, 10, 20, 30, 40, 50, 60, 70)
11         val tile = TileSnapshot(
12             coord = TileCoord(2, 3),
13             pixelLeft = 512,
14             pixelTop = 768,
15             pixelWidth = 1,
16             pixelHeight = 2,
17             bytesPerPixel = TileSnapshot.BYTES_PER_PIXEL_RGBA8,
18             storedData = bytes,
19         )
20
21         assertEquals(bytes.size, tile.memorySize)
22         assertEquals(bytes.size, tile.rawSize)
23         assertEquals(bytes, tile.decompress())
24     }
25
26     @Test
27     fun tileSizeIsTheSpecified256Pixels() {
28         assertEquals(256, TileSnapshot.TILE_SIZE)
29     }
30 }
```

120. app/src/test/java/com/wetinknext/engine/undo/UndoManagerTest.kt

```

1  package com.wetinknext.engine.undo
2
3  import org.junit.Assert.assertEquals
4  import org.junit.Assert.assertFalse
5  import org.junit.Assert.assertNotNull
6  import org.junit.Assert.assertSame
7  import org.junit.Assert.assertTrue
8  import org.junit.Test
9
10 class UndoManagerTest {
11     @Test
12     fun undoAndRedoMoveTheSameEntryBetweenStacks() {
13         val manager = UndoManager()
14         val entry = entry(1L)
15         manager.push(entry)
16
17         assertTrue(manager.canUndo)
18         assertFalse(manager.canRedo)
19         assertSame(entry, manager.peekUndo())
20         assertTrue(manager.commitUndo(entry))
21         assertFalse(manager.canUndo)
22         assertTrue(manager.canRedo)
23         assertSame(entry, manager.peekRedo())
24         assertTrue(manager.commitRedo(entry))
25         assertTrue(manager.canUndo)
26         assertFalse(manager.canRedo)
27     }
28
29     @Test
30     fun aNewEntryClearsRedoHistory() {
31         val manager = UndoManager()
32         val first = entry(1L)
33         manager.push(first)
34         val undo = manager.peekUndo()
35         assertNotNull(undo)
36         assertTrue(manager.commitUndo(checkNotNull(undo)))
37         assertTrue(manager.canRedo)
38
39         manager.push(entry(1L))
40
41         assertFalse(manager.canRedo)
42         assertEquals(1, manager.undoCount)
43         assertEquals(0, manager.redoCount)
44     }
45
46     @Test
47     fun stepLimitEvictsTheOldestHistoryEntry() {
48         val manager = UndoManager(maxSteps = 2)
49         manager.push(entry(1L))
50         manager.push(entry(2L))
51         manager.push(entry(3L))
52
53         assertEquals(2, manager.undoCount)
54     }
55
56     @Test
57     fun rejectedCommitLeavesUndoHistoryUntouched() {
58         val manager = UndoManager()
59         val stored = entry(1L)
60         manager.push(stored)
61
62         assertFalse(manager.commitUndo(entry(2L)))
63         assertSame(stored, manager.peekUndo())
64         assertTrue(manager.canUndo)
65         assertFalse(manager.canRedo)
66     }
67
68     @Test
69     fun removingLayerPurgesOnlyThatLayersUndoAndRedoEntries() {
70         val manager = UndoManager()
71         val firstLayer = entry(1L)
72         val secondLayer = entry(2L)
73         manager.push(firstLayer)
74         manager.push(secondLayer)
75         assertTrue(manager.commitUndo(secondLayer))
76
77         manager.removeEntriesForLayer(1L)
78
79         assertFalse(manager.canUndo)
80         assertTrue(manager.canRedo)
81         assertSame(secondLayer, manager.peekRedo())
82
83         manager.removeEntriesForLayer(2L)
84
85         assertFalse(manager.canRedo)
86     }
87
88     @Test
89     fun entryKeepsItsOperationMetadata() {
90         val entry = UndoEntry(

```

```

91         layerId = 1L,
92         beforeTiles = emptyList(),
93         afterTiles = emptyList(),
94         operation = UndoOperationType.CLEAR_LAYER,
95         tag = "clear_layer",
96     )
97
98     assertEquals(UndoOperationType.CLEAR_LAYER, entry.operation)
99     assertEquals("clear_layer", entry.tag)
100 }
101
102 @Test
103 fun removingLayerEntriesDoesNotBlockOtherHistory() {
104     val manager = UndoManager()
105     val removedLayerEntry = entry(layerId = 10L)
106     val validEntry = entry(layerId = 20L)
107     manager.push(removedLayerEntry)
108     manager.push(validEntry)
109
110     manager.removeEntriesForLayer(10L)
111
112     assertTrue(manager.canUndo)
113     assertEquals(validEntry, manager.peekUndo())
114 }
115
116 @Test
117 fun fiveEntriesCanBeCommittedInOrder() {
118     val manager = UndoManager()
119     repeat(5) { index -> manager.push(entry(layerId = index.toLong())) }
120
121     assertEquals(5, manager.undoCount)
122     repeat(5) {
123         val entry = manager.peekUndo()
124         assertNotNull(entry)
125         assertTrue(manager.commitUndo(checkNotNull(entry)))
126     }
127
128     assertEquals(0, manager.undoCount)
129     assertEquals(5, manager.redoCount)
130 }
131
132 private fun entry(layerId: Long): UndoEntry = UndoEntry(
133     layerId = layerId,
134     beforeTiles = listOf(tile(0)),
135     afterTiles = listOf(tile(1)),
136 )
137
138 private fun tile(x: Int): TileSnapshot = TileSnapshot(
139     coord = TileCoord(x, 0),
140     pixelLeft = x,
141     pixelTop = 0,
142     pixelWidth = 1,
143     pixelHeight = 1,
144     bytesPerPixel = TileSnapshot.BYTES_PER_PIXEL_RGBA8,
145     storedData = arrayOf(1, 2, 3, 4),
146 )
147 }

```

121. build.gradle.kts

```
1 plugins {  
2     id("com.android.application") version "8.6.1" apply false  
3     id("org.jetbrains.kotlin.android") version "2.0.21" apply false  
4     id("org.jetbrains.kotlin.plugin.compose") version "2.0.21" apply false  
5     id("org.jetbrains.kotlin.plugin.serialization") version "2.0.21" apply false  
6 }
```


122. gradle.properties

```
1 org.gradle.jvmargs=-Xmx2048m -Dfile.encoding=UTF-8
2 android.useAndroidX=true
3 android.overridePathCheck=true
4 kotlin.code.style=official
5 org.gradle.daemon=true
6 org.gradle.caching=true
```

123. gradle/wrapper/gradle-wrapper.properties

```
1 distributionBase=GRADLE_USER_HOME
2 distributionPath=wrapper/dists
3 distributionUrl=https\://services.gradle.org/distributions/gradle-8.7-bin.zip
4 networkTimeout=10000
5 validateDistributionUrl=true
6 zipStoreBase=GRADLE_USER_HOME
7 zipStorePath=wrapper/dists
```

124. gradlew

```
1 #!/bin/sh
2
3 APP_HOME=$(CDPATH= cd -- "$(dirname -- "$0")" && pwd -P) || exit 1
4 JAVA_CMD="{JAVA_HOME:+$JAVA_HOME/bin/}java"
5 exec "$JAVA_CMD" -classpath "$APP_HOME/gradle/wrapper/gradle-wrapper.jar" org.gradle.wrapper.GradleWrapperMain "$@"
```

125. gradlew.bat

```
1 @rem Minimal Gradle startup script for Windows.
2 @echo off
3 setlocal
4 set APP_HOME=%~dp0
5 if defined JAVA_HOME goto javaHome
6 java -classpath "%APP_HOME%gradle\wrapper\gradle-wrapper.jar" org.gradle.wrapper.GradleWrapperMain %*
7 goto end
8 :javaHome
9 "%JAVA_HOME%\bin\java.exe" -classpath "%APP_HOME%gradle\wrapper\gradle-wrapper.jar" org.gradle.wrapper.GradleWrapperMain %*
10 :end
```

126. local.properties

```
1 sdk.dir=C:\\Users\\iick\\AppData\\Local\\Android\\Sdk
```

127. settings.gradle.kts

```
1  pluginManagement {
2      repositories {
3          google()
4          mavenCentral()
5          gradlePluginPortal()
6      }
7  }
8
9  dependencyResolutionManagement {
10     repositoriesMode.set(RepositoriesMode.FAIL_ON_PROJECT_REPOS)
11     repositories {
12         google()
13         mavenCentral()
14     }
15 }
16
17 rootProject.name = "WetInk Next"
18 include(":app")
```